

Cluster Dynamics Modeling of Radiation-Induced Defect Evolution in Ferritic-Martensitic Steels

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Abstract

A self-contained, modular cluster-dynamics (CD) framework is presented for modeling the long-time evolution of radiation-induced defects in crystalline materials, with detailed parameterization for the reduced-activation ferritic-martensitic steel EUROFER97. The formulation is constructed bottom-up: it begins with a spectrum-dependent description of in-cascade defect production for the mono-atomic and clustered self-interstitial atom (SIA), vacancy, and helium populations; it proceeds through a unified treatment of defect energetics (formation, migration, and binding energies) and of diffusion on a body-centered-cubic (bcc) lattice, including mixed one-dimensional and three-dimensional transport of SIA clusters, solute-trapped vacancy and SIA migration in alloys, and helium interstitial diffusion; and it closes with cluster-dissociation kinetics for voids, helium bubbles, vacancy loops, and interstitial loops. A unified expression for reaction rates, organized by reaction class with transition frequencies $\omega = D/a^2$, captures the five elementary reaction families that govern microstructure evolution: SIA-vacancy annihilation, SIA-cluster growth and shrinkage, vacancy-cluster growth and shrinkage, helium trapping and detrapping (including radiation re-solution), and helium-vacancy cluster reactions with SIAs.

A full set of master equations is derived for the active populations, together with explicit mass-conservation laws (Frenkel-pair and helium moments) that serve both as analytical constraints and as numerical diagnostics. Two reduced descriptions of the helium-vacancy state space are obtained as systematic limits of the master equations: a fast-helium-equilibration model that tracks the mean helium loading $\bar{\ell}(\mathbf{m})$ per vacancy cluster, suitable for fusion-relevant He/dpa ratios; and a decoupled-helium-inventory model appropriate for fission spectra. State-space reduction is further developed through a logarithmic-bin-moment method with

two complementary closures (piecewise-constant and Galerkin shape-function), preserving mass conservation while reducing system size by orders of magnitude. The framework is then extended to four reactive species (H, He, V, and I) by adding hydrogen-trapping reactions at vacancies, vacancy clusters, helium bubbles, and dislocation loops, and by deriving an adiabatic-elimination procedure for hydrogen at large cluster sizes that preserves the essential coupling to bubble pressure and cavity stability. A hybrid deterministic–stochastic algorithm is proposed in which a Lie–Trotter splitting partitions the state space into a deterministic monomer subsystem and a stochastic large-cluster subsystem solved by a logarithmic-time stochastic-simulation algorithm with a low-rank propensity decomposition.

The numerical implementation rests on a tailored Woodbury-style preconditioner that exploits the bordered-banded sparsity pattern of the cluster-dynamics Jacobian, combined with adaptive stiffness management. The complete framework is realized in the open-source RADCLUSTER code, with full EUROFER97 parameter tables, conservation-law diagnostics, and verification against analytical limits. The result is a single, traceable formulation that maps cleanly onto fission and fusion spectra, multiple host metals, and the full $(\mathbf{H}, \mathbf{He}, \mathbf{V}, \mathbf{I})$ reaction network, providing a foundation for materials-specific quantitative microstructure prediction under prototypical reactor conditions.

Keywords: cluster dynamics, radiation damage, helium-vacancy clusters, ferritic–martensitic steels, EUROFER97, master equations, hybrid stochastic–deterministic, hydrogen

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1 Introduction

Structural materials in fission and, prospectively, fusion energy systems must operate under prolonged exposure to energetic particle radiation that displaces lattice atoms at rates of 10^{-7} to 10^{-5} dpa/s and accumulates damage to total doses of 10^1 – 10^2 dpa over component lifetimes (Mansur 1994; Zinkle and Was 2013; Stork et al 2014). The microstructural changes that result—nucleation and growth of vacancy and self-interstitial atom (SIA) clusters, void and bubble formation, dislocation-loop populations, irradiation-induced segregation, and precipitate coarsening or dissolution—are the principal source of property degradation, manifested as hardening, embrittlement, swelling, irradiation creep, and shifts in the ductile-to-brittle transition temperature (Was 2017; Odette et al 2008; Yamamoto et al 2020). Predicting these property changes from first principles, over the relevant time and length scales and across the joint envelope of temperature, dose, dose rate, and helium-to-dpa ratio, is the central quantitative challenge of radiation materials science.

The cluster-dynamics (CD) framework, introduced in its modern form during the late 1970s and early 1980s (Ghoniem 1977; Ghoniem and Cho 1979b; Ghoniem and Sharafat 1980; Hayns 1976; Kiritani 1973), remains the workhorse mesoscopic theory for this challenge. CD treats each defect species and size as a separate population whose concentration evolves under a coupled set of master rate equations balancing in-cascade production, diffusion-controlled reactions, thermal emission, and absorption at fixed sinks (Brailsford and Bullough 1972a; Wiedersich 1972; Mansur 1978). Its three defining advantages are (i) direct access to time scales (10^0 – 10^8 s) that lie far beyond molecular dynamics (Bacon et al 2000a; Marian et al 2017), (ii) explicit resolution of the cluster-size distribution—and therefore of every quantity that enters the macroscopic property models, from sink-strength models to swelling rates (Golubov et al 2001; Surh et al 2004), and (iii) modest computational cost relative to spatially resolved kinetic Monte Carlo or atomistic simulation, which together make CD the natural meeting point for atomistic-scale parameterization and engineering-scale prediction (Kohnert et al 2018a; Marian and Bulatov 2011a; Caturla et al 2008a).

Since its inception, CD has been deployed across an unusually broad materials portfolio. In austenitic stainless steels and Fe–Cr–Ni model alloys, CD accounts for void swelling and helium effects on cavity stability (Ghoniem and Cho 1979b; Ghoniem et al 1985a; Ortiz and Caturla 2007); in ferritic and ferritic–martensitic steels, it has provided the quantitative basis for SIA-cluster and Cr-precipitate kinetics (Stoller 2000a; Malerba et al 2021a; Bonny et al 2011); in tungsten and W–Re alloys, it underlies plasma–surface interaction codes that track helium and hydrogen-isotope retention in plasma-facing components (Faney and Wirth 2014; Becquart and Domain 2014a; Ferreira and Wirth 2024); in nuclear fuels, it describes Xe fission-gas evolution and bubble nucleation (Veshchunov 2008); and in zirconium alloys, it captures the $\langle a \rangle$ -loop and $\langle c \rangle$ -loop populations underlying irradiation growth (Christien and Barbu 2004; Maron et al 2024). The several review articles that have shaped the field (Kohnert et al 2018a; Chen et al 2022; Zheng et al 2023; Ke et al 2023; Xiong et al 2024) converge on a common diagnosis: while CD is conceptually mature, the practical landscape is fragmented—codes specialize in a single host material, in a particular reaction

subset, or in a particular solver philosophy, and there is no widely shared, transparent, and modular formulation that exposes *which* physics has been included, *which* approximation has been invoked, and *which* parameter has been calibrated.

The State of the Field and the Gaps That Remain

Several persistent gaps have constrained the predictive use of CD in fusion-materials modeling. They organize naturally into five categories.

(G1) Multispecies coverage.

The reaction network of an irradiated alloy under fusion-relevant conditions involves at least vacancies, SIAs, helium, and hydrogen, together with their clusters and the host’s solute-trapped variants (Ferreira and Wirth 2024; Caturla et al 2008a; Lucas and Schaublin 2009). Yet most CD frameworks restrict the active state space to two or three species, treating the remainder as static traps or omitting them altogether. Hydrogen, in particular, is rarely tracked dynamically alongside helium, despite its central role in tritium retention, bubble pressurization, and hydrogen-induced embrittlement (Hayward and Fu 2013; Becquart and Domain 2014a; Faney and Wirth 2014).

(G2) State-space management of helium–vacancy clusters.

Tracking $\text{He}_n V_m$ explicitly across the full (n, m) grid is prohibitive: the number of equations scales as the product of the helium and vacancy-content ranges, and stiff coupling between gas pressure and cavity stability defeats simple closures (Trinka and Singh 2003; Schaublin and Chiu 2007; Caturla et al 2008a). Reduced descriptions that track only the mean helium loading per cavity have been used heuristically (Ghoniem 1989; Stewart et al 2011), but their precise relationship to the full master equations — the limits in which they are exact, the leading correction terms, and the conservation laws they preserve — has not been worked out from first principles in a way that makes the approximations auditable.

(G3) Closure of the cluster-size distribution.

Beyond a critical nucleus size, individual cluster equations are typically replaced by a Fokker–Planck (FP) representation (Wagner and Kampmann 1981; Ghoniem 1989; Russell 1984) or a grouping/binning scheme (Golubov et al 2001; Ovcharenko et al 2003; Kohnert et al 2018a; Cui et al 2020). Each closure has known limitations: FP closures lose accuracy at the discrete-to-continuous transition and require careful drift–diffusion coefficient construction; grouping closures introduce mass-conservation errors unless the inter-bin flux is explicitly designed; and few formulations rigorously analyze the *accuracy* of the closure in a way that allows the user to trade fidelity for system size with quantified error.

(G4) Stiffness and solver design.

The CD master equations are stiff: rate constants vary by many orders of magnitude across reaction classes and cluster sizes, and the Jacobian carries both narrow-banded

structure (within a single species) and dense bordering blocks (cross-species coupling, sink strengths, conservation constraints). Generic ODE solvers and standard preconditioners perform poorly on this structure. Few existing CD codes invest in tailored numerical strategies that exploit the bordered-banded form, and fewer still document explicit conservation-law diagnostics as a built-in part of the solver (Saad 2003; Stoller et al 2008a).

(G5) Hybrid deterministic–stochastic treatment.

At late times, large-cluster dynamics is dominated by rare nucleation, coalescence, and dissociation events for which deterministic ODEs over-smooth the size distribution and stochastic methods (kinetic Monte Carlo, stochastic CD) become accurate but expensive (Marian and Bulatov 2011b; Stewart et al 2011; Surh et al 2004; Wirth et al 2001). A clean operator-splitting strategy that retains determinism for the fast monomer dynamics while resolving stochastic large-cluster fluctuations—and that is published with explicit error analysis and a multispecies (H, He, V, I) propensity matrix—has not been available.

Objectives and Contributions of the Present Work

The present formulation is designed to address each of these gaps in a single, modular framework whose components can be selected, replaced, or specialized depending on the physics of the host material and the target irradiation environment. The principal objectives are:

1. to develop a *general, transparent* CD formulation in which every reaction rate, every binding energy, and every closure approximation is traceable to its physical origin and to its calibration data, and in which the formulation can be specialized to a particular host material (here EUROFER97 as the principal example) without modifying the algorithmic core;
2. to provide *tailored options* for the helium–vacancy state space (Sections 9.1 and 9.2), one optimized for fast helium equilibration appropriate to fusion conditions and one for decoupled helium inventory appropriate to fission spectra, both derived as systematic limits of the same parent master equations and both preserving the helium and Frenkel-pair conservation moments;
3. to implement two complementary *closures of the cluster-size distribution* via a logarithmic-bin-moment method—a piecewise-constant (zeroth-order) closure and a Galerkin shape-function closure—with explicit accuracy and mass-conservation analysis, so the user can trade system size against fidelity in a controlled manner;
4. to extend the framework to a *four-species reaction network* (H, He, V, I) by adding hydrogen-trapping channels at vacancies, vacancy clusters, bubbles, and dislocation loops, and by deriving an adiabatic-elimination scheme for hydrogen at large cluster sizes that preserves the pressure–cavity–stability coupling required for swelling and bubble-driven embrittlement studies;

5. to design a *hybrid deterministic–stochastic algorithm* based on Lie–Trotter operator splitting, in which monomer dynamics are integrated deterministically while large-cluster events are sampled by a logarithmic-time stochastic simulation algorithm with a low-rank propensity matrix for the four-species network, with explicit error and convergence analysis;
6. to ground the framework in a *rigorous numerical strategy* that exploits the bordered-banded Jacobian structure through a Sherman–Morrison–Woodbury preconditioner, with mass-conservation laws used as live diagnostics during time integration; and
7. to deliver the formulation in an open-source code, RADCLUSTER, that ships with a complete EUROFER97 parameter set, verification against analytical limits, and demonstration applications to fission and fusion irradiation conditions.

The combination of these objectives provides a useful contrast with existing tools. Open-source spatially resolved codes such as XOLOTL (Faney and Wirth 2014; ORNL Fusion Energy Division 2026) excel at plasma–surface-interaction problems with steep concentration gradients and a well-defined helium–tungsten reaction network; stochastic CD codes (Marian and Bulatov 2011b; Stewart et al 2011) excel at sampling rare events but at substantial cost. The present framework is positioned as a complement, not a replacement: it targets bulk chemistry-rich alloys, prioritizes auditable approximations and conservation diagnostics, and provides a clean route from a deterministic full-network calculation to a hybrid stochastic–deterministic large-cluster calculation as the user’s needs evolve. Xiong et al (2024) provide a useful recent map of the methodological landscape into which the present formulation inserts itself, and the parameter tabulations in Malerba et al (2021a) and the EUROFER97 database of Aktaa et al (2020); Klimenkov et al (2012a) provide the experimental anchors against which the model is calibrated.

Tailoring the Framework to Specific Physics: Examples

The modularity of the formulation is best illustrated by three representative use cases.

(a) Fusion-spectrum first-wall structural steel (Eurofer97 or F82H). The full (H, He, V, I) network of Section 11 is enabled, with the fast-helium-equilibration reduction of Section 9.1 active to capture the ~ 10 appm He/dpa coupling to cavity stability. The cascade defect spectra of Section 2.1 are loaded with fusion-spectrum power-law exponents (s_i, s_v) and clustering fractions (f_i^{cl}, f_v^{cl}) . Solute-modified diffusivities of Section 4.4.1 are activated through the binding-energy table calibrated for EUROFER97 matrix composition after tempering. The Galerkin shape-function bin closure of Section 10.4.2 is used at large vacancy-cluster sizes to track bubble swelling efficiently while preserving mass conservation.

(b) Fission-spectrum austenitic or low-activation steel. The same network is run with the decoupled-helium-inventory reduction of Section 9.2 active, since ~ 0.5 –1 appm He/dpa is too low for fast equilibration to be accurate over the relevant dose range. The cascade spectrum is re-loaded with fission parameters, and the radiation-resolution limit is exercised to recover the classical low-He limit.

(c) **Plasma-facing tungsten with hydrogen-isotope and helium co-implantation.** The hydrogen-trapping channels of Section 11 are activated, and the host-specific parameters in the diffusion and binding-energy tables are swapped to bcc tungsten values (Becquart and Domain 2014a; Hayward and Fu 2013; Ferreira and Wirth 2024). The hybrid stochastic–deterministic engine of Section 12 is enabled at large helium-cluster sizes to capture the rare nucleation events that govern bubble-superlattice formation. The framework’s H -adiabatic-elimination scheme of Section 11.5 ensures that the resulting large-cluster dynamics remain tractable while preserving the pressure-driven instabilities that drive blistering.

In each case, the algorithmic core is unchanged; only the parameter tables, the closures, and the active reduction options are re-selected.

Organization of the Paper

The remainder of the paper is organized as follows. Section 2 develops the cascade-spectrum model for defect production, including survival efficiencies, clustering fractions, and power-law exponents for fission and fusion spectra, with the recommended EUROFER97 parameter table. Section 3 compiles defect energetics (formation, migration, and binding energies) for the elementary species and small clusters, with explicit DFT and MD provenance and recommended values for EUROFER97. Section 4 treats defect diffusion on a bcc lattice in the random-walk and mixed-1D/3D regimes, including solute-trapped SIA-cluster transport, vacancy and vacancy-cluster diffusion, and helium interstitial diffusion in pure α -Fe and in the EUROFER97 matrix. Section 5 derives cluster-dissociation kinetics for voids, helium bubbles, and vacancy and interstitial loops, with binding-energy models grounded in atomistic simulation data. Section 6 consolidates these inputs into the unified reaction-rate-constant framework with the transition frequency $\omega = D/a^2$, treating 3D and 1D diffusion-limited reactions on equal footing and providing precomputed constants for the EUROFER97 application. Section 8 assembles the full master equations for the SIA-cluster, vacancy-cluster, and free-helium populations, states the bias-factor and sink-strength formulation, derives the mass-conservation laws (Frenkel-pair and helium moments), and documents the swelling identity and ODE-system structure used by the solver. Section 9.1 and Section 9.2 derive the two helium–vacancy state-space reductions as systematic limits of Section 9, including the radiation-re-solution limiting behavior. Section 10 introduces the logarithmic-bin-moment state-space reduction with both piecewise-constant and Galerkin closures and analyzes accuracy, conservation, and reduced system size. Section 11 extends the framework to four species (H, He, V, I), introducing hydrogen-trapping reactions and the master equations for free hydrogen and for small (explicit-H) $H_p V_m \text{He}_n$ clusters, and deriving the quasi-static H-loading and H -adiabatic-elimination procedure for large clusters. Section 12 develops the hybrid deterministic–stochastic algorithm, including Lie–Trotter operator splitting, the four-species propensity matrix and its low-rank decomposition, the logarithmic-time stochastic simulation algorithm for large clusters, monomer feedback and mass conservation, error analysis, and the complete algorithmic scheme. Demonstration simulation results for EUROFER97 are given in Section 13. The appendices provide rigorous proofs of the conservation laws (Appendix A), the numerical-method

specification including the Sherman–Morrison–Woodbury preconditioner and stiffness strategy (Appendix C), and a documentation of the RADCLUSTER code and its comparison with related tools (Appendix D).

A reader interested in the physical formulation alone can read Sections 2–8 sequentially. A reader interested in the state-space reduction methodology can proceed directly to Sections 9–10. A reader interested in the four-species extension and the hybrid algorithm can begin at Section 11. A reader interested in numerical implementation will find the solver description in Appendix C and the open-source RADCLUSTER reference in Appendix D.

2 Defect Production

2.1 Defect Cluster Spectra

Let G denote the *actual* (survival-corrected) defect production rate per unit volume, i.e.

$$G = \eta G_{\text{NRT}}, \quad (1)$$

where $\eta = N_d/N_{\text{NRT}}$ is the survival efficiency and G_{NRT} is the NRT displacement rate.

The production rate of interstitial clusters of size m is

$$P_m^{(i)} = \epsilon_m^{(i)} G, \quad m = 2, 3, \dots, i_{\text{cascade}}, \quad (2)$$

and the mono-interstitial production rate is

$$P_1^{(i)} = G \left(1 - \sum_{m=2}^{m_{\text{cascade}}} m \epsilon_m^{(i)} \right). \quad (3)$$

Analogous expressions hold for vacancies with superscript (v) , cluster size index n , and upper cutoff v_{cascade} .

The total fraction of surviving SIAs born in clusters is

$$f_i^{\text{cl}} = \sum_{m=2}^{m_{\text{cascade}}} m \epsilon_m^{(i)}, \quad (4)$$

with the corresponding vacancy quantity f_v^{cl} . Note that the mono-defect fraction is simply $1 - f^{\text{cl}}$.

The fraction of NRT-predicted Frenkel pairs that survive in-cascade recombination is well characterized by MD simulations [Stoller \(2000b\)](#); [Nordlund et al \(2018a\)](#); [Stoller et al \(1997\)](#). For α -Fe at reactor-relevant temperatures, the survival efficiency is approximately

$$\eta(E_{\text{PKA}}) \approx 0.84 \left(\frac{E_{\text{PKA}}}{1 \text{ keV}} \right)^{-0.20}, \quad E_{\text{PKA}} \gtrsim 1 \text{ keV}, \quad (5)$$

saturating near $\eta \approx 0.30$ above ~ 10 keV.

Table 1 lists representative values. The “athermal recombination corrected dpa” (arc-dpa) formalism of [Nordlund et al \(2018a\)](#) provides a more refined energy-dependent efficiency function $\xi_{\text{arc}}(E)$ that is recommended for detailed calculations.

MD cascade simulations in α -Fe consistently show that the *number* of in-cascade clusters of size m follows an approximate power law [Stoller \(1998, 2000b\)](#); [De Backer et al \(2018a\)](#); [Calder et al \(2010\)](#):

$$N_{\text{cl}}(m) \propto m^{-s}, \quad (6)$$

with distinct exponents for SIA clusters (s_i) and vacancy clusters (s_v). The steeper vacancy exponent reflects the well-known inhibition of true vacancy clustering in bcc

Table 1 Defect survival efficiency $\eta = N_d/N_{\text{NRT}}$ in α -Fe at 100 K from MD cascade simulations [Stoller \(2000b\)](#); [Nordlund et al \(2018a\)](#).

E_{PKA} (keV)	η
0.5	0.96
1	0.84
2	0.73
5	0.61
10	0.53
20	0.46
50	0.38
100	0.33
200	0.29

Fe, where vacancy “clusters” are often loose, correlated arrangements rather than compact loops [Stoller \(1998\)](#); [Soneda et al \(2001\)](#).

The production fraction is therefore modeled as

$$\epsilon_m^{(i)} = C_i m^{-s_i}, \quad m = 2, \dots, i_{\text{cascade}}, \quad (7)$$

where the normalization constant C_i is fixed by the constraint

$$C_i = \frac{f_i^{\text{cl}}}{\sum_{m=2}^{i_{\text{cascade}}} m^{1-s_i}}. \quad (8)$$

An identical prescription applies to vacancies with parameters $(C_v, s_v, v_{\text{cascade}}, f_v^{\text{cl}})$.

The parameters that distinguish fission and fusion spectra in terms of cascade damage are summarized in Table 2. Key physical differences:

- **PKA energy spectrum:** Fission-spectrum neutrons produce a PKA distribution weighted toward lower energies ($\langle E_{\text{PKA}} \rangle \approx 10\text{--}15\text{ keV}$), whereas 14 MeV fusion neutrons shift the average to $\sim 30\text{--}50\text{ keV}$, leading to subcascade fragmentation [De Backer et al \(2018a\)](#); [Zarkadoula et al \(2013\)](#).
- **Cluster sizes:** Higher PKA energies produce larger in-cascade clusters, so the maximum cluster sizes i_{cascade} and v_{cascade} and the power-law exponents all shift [Stoller \(2000b\)](#); [Calder et al \(2010\)](#).
- **Helium co-production:** Helium generation is 0.5–1 appm He/dpa for fission spectra and ~ 10 appm He/dpa for fusion, affecting vacancy cluster (bubble) nucleation at longer times but not the primary cascade spectrum itself.

Table 2 Recommended cascade-damage parameters for α -Fe under fission and fusion neutron spectra, synthesized from [Stoller \(2000b\)](#), [Nordlund et al \(2018a\)](#), [De Backer et al \(2018a\)](#), and [Malerba et al \(2021c\)](#).

Parameter	Fission	Fusion
$\langle E_{\text{PKA}} \rangle$ (keV)	~ 15	~ 40
η (spectrum-averaged)	0.30	0.28
f_i^{cl} (SIA clustering fraction)	0.58	0.65
f_v^{cl} (vacancy clustering fraction)	0.15	0.20
s_i (SIA power-law exponent)	1.6	1.5
s_v (vacancy power-law exponent)	2.5	2.3
i_{cascade} (max SIA cluster size)	20	50
v_{cascade} (max vacancy cluster size)	10	20
He production (appm He/dpa)	0.5–1	~ 10

The following closed-form expressions allow direct computation of ϵ_m for any cluster size. For interstitial Clusters, we have

$$\epsilon_m^{(i)} = \frac{f_i^{\text{cl}} m^{-s_i}}{\sum_{k=2}^{i_{\text{cascade}}} k^{1-s_i}}, \quad m = 2, 3, \dots, i_{\text{cascade}} \quad (9)$$

Fission: $f_i^{\text{cl}} = 0.58$, $s_i = 1.6$, $i_{\text{cascade}} = 20$, giving $C_i = f_i^{\text{cl}} / \sum_{k=2}^{20} k^{-0.6} = 0.1093$.

Fusion: $f_i^{\text{cl}} = 0.65$, $s_i = 1.5$, $i_{\text{cascade}} = 50$, giving $C_i = 0.0553$.

While for vacancy clusters, we get

$$\epsilon_n^{(v)} = \frac{f_v^{\text{cl}} n^{-s_v}}{\sum_{k=2}^{v_{\text{cascade}}} k^{1-s_v}}, \quad n = 2, 3, \dots, v_{\text{cascade}} \quad (10)$$

Fission: $f_v^{\text{cl}} = 0.15$, $s_v = 2.5$, $v_{\text{cascade}} = 10$, giving $C_v = 0.1506$.

Fusion: $f_v^{\text{cl}} = 0.20$, $s_v = 2.3$, $v_{\text{cascade}} = 20$, giving $C_v = 0.1296$.

The normalization sums can be pre-computed once:

$$S(s, m_{\text{max}}) = \sum_{k=2}^{m_{\text{max}}} k^{1-s}. \quad (11)$$

Then $C = f^{\text{cl}}/S$ and $\epsilon_m = C m^{-s}$.

For large m_{max} , the sum can be approximated by the integral

$$S(s, m_{\text{max}}) \approx \begin{cases} \frac{m_{\text{max}}^{2-s} - 2^{2-s}}{2-s} + \frac{1}{2}(m_{\text{max}}^{1-s} + 2^{1-s}), & s \neq 2, \\ \ln\left(\frac{m_{\text{max}}}{2}\right) + \frac{1}{2}(m_{\text{max}}^{-1} + 2^{-1}), & s = 2. \end{cases} \quad (12)$$

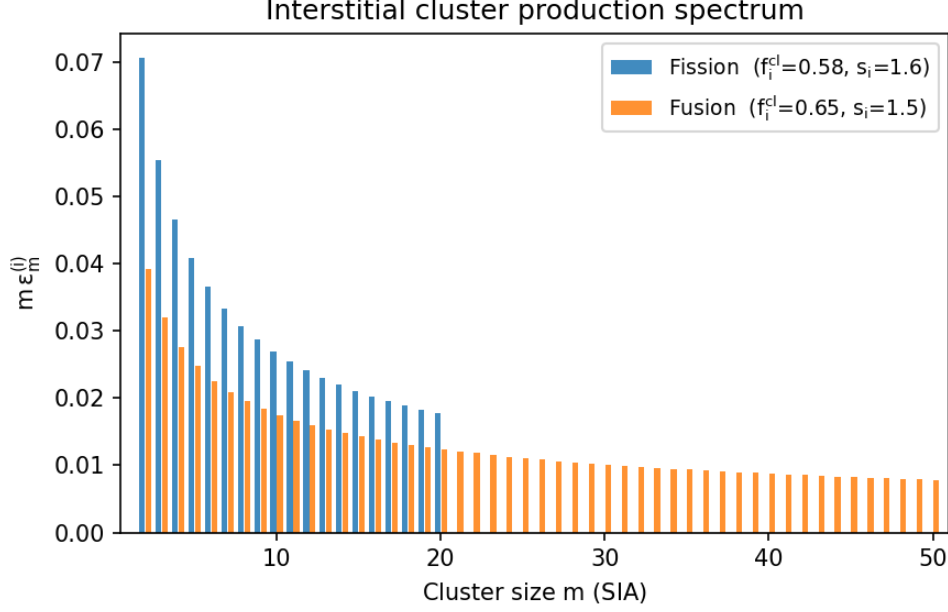


Fig. 1 Interstitial cluster production spectrum under fission and fusion conditions

For direct implementation in a cluster-dynamics code, the source terms take the form shown in Table 3.

Table 47 compares key features of the production spectra under fission and fusion conditions. Fig. 1 shows the interstitial cluster spectrum under representative fission and fusion irradiation conditions in iron, while Fig. 2 displays the production spectrum for vacancy clusters under fission and fusion irradiation conditions. The principal effects of the harder fusion spectrum are: (1) a larger fraction of surviving defects born in clusters; (2) a shallower power-law exponent shifting the distribution toward larger clusters; and (3) extension of the maximum cluster size due to higher-energy cascades and subcascade overlap [De Backer et al \(2018a\)](#); [Zarkadoula et al \(2013\)](#). The enhanced helium production under fusion conditions (~ 10 appm He/dpa vs. 0.5–1 for fission) does not affect the primary cascade spectrum but strongly influences subsequent vacancy cluster (void/bubble) evolution by providing heterogeneous nucleation sites [Malerba et al \(2021c\)](#).

Literature Sources

The production fractions presented here are *reconstructed* from published MD cascade data, since the literature generally does not report ϵ_m in the tabulated form required by rate-theory codes. The principal sources and their contributions are:

- [Beeler and Johnson \(1967\)](#): earliest computer simulation of vacancy and interstitial cluster production in α -Fe; historical baseline.

Table 3 Rate-theory source-term structure. G is the actual (survival-corrected) defect production rate. The coefficients $\epsilon_m^{(i)}$ and $\epsilon_n^{(v)}$ are read from Table 2 according to the neutron spectrum.

Species index α	Production rate P_α
1 (mono-SIA)	$P_1^{(i)} = G \left(1 - \sum_{m=2}^{m_{\text{cascade}}} m \epsilon_m^{(i)} \right)$
2	$P_2^{(i)} = \epsilon_2^{(i)} G$
$\vdots$	$\vdots$
i_{cascade}	$P_{m_{\text{cascade}}}^{(i)} = \epsilon_{m_{\text{cascade}}}^{(i)} G$
$N+1$ (mono-vacancy)	$P_1^{(v)} = G \left(1 - \sum_{n=2}^{n_{\text{cascade}}} n \epsilon_n^{(v)} \right)$
$N+2$	$P_2^{(v)} = \epsilon_2^{(v)} G$
$\vdots$	$\vdots$
$N+v_{\text{cascade}}$	$P_{n_{\text{cascade}}}^{(v)} = \epsilon_{n_{\text{cascade}}}^{(v)} G$

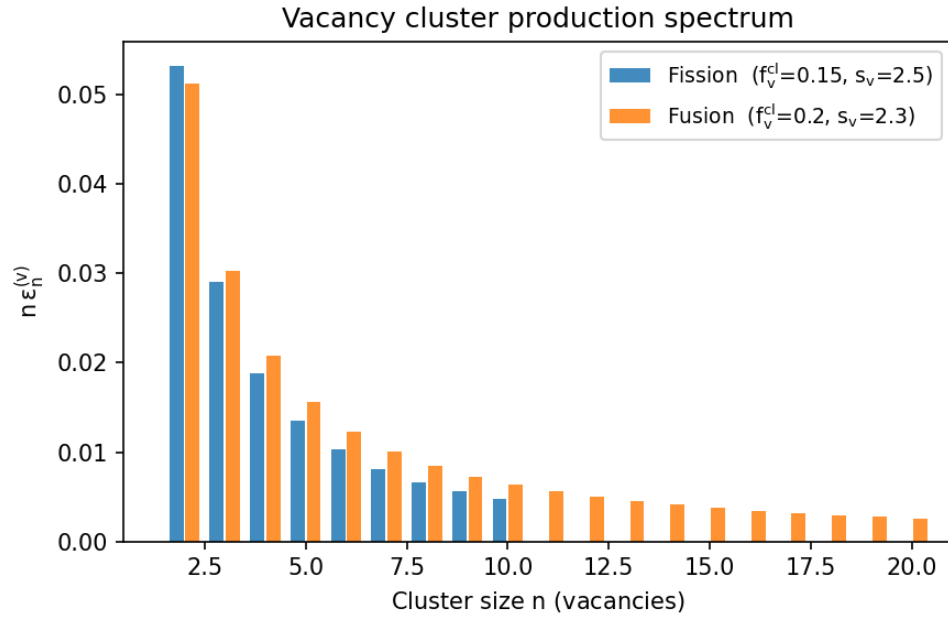


Fig. 2 Vacancy cluster production spectrum under fission and fusion conditions

Table 4 Comparison of defect production characteristics under fission and fusion neutron spectra in α -Fe.

Quantity	Fission	Fusion
Mono-SIA fraction, $1 - f_i^{\text{cl}}$	0.42	0.35
Mono-vacancy fraction, $1 - f_v^{\text{cl}}$	0.85	0.80
Mean SIA cluster size $\langle m \rangle_{\text{cl}}$	4.4	6.3
Mean vacancy cluster size $\langle n \rangle_{\text{cl}}$	2.6	3.1
Largest SIA cluster i_{cascade}	20	50
Largest vacancy cluster v_{cascade}	10	20
He co-production (appm He/dpa)	0.5–1	~ 10

- [Calder and Bacon \(1993\)](#): early MD study establishing qualitative cascade morphology and survival fractions in Fe.
- [Stoller et al \(1997\)](#): comprehensive MD + MC characterization up to 40 keV; reports interstitial clustering fractions and vacancy-site correlation.
- [Stoller \(1998\)](#): key reference for cluster size distributions at 100, 600, 900 K up to 50 keV; primary source for size-resolved interstitial and vacancy spectra.
- [Stoller \(2000b\)](#): broadest energy–temperature survey of clustering fractions and size distributions; principal source for the energy dependence of f_i^{cl} and f_v^{cl} .
- [Bacon et al \(2000b\)](#): comparative study across crystal structures confirming bcc Fe as having strong in-cascade SIA clustering.
- [Gao et al \(2000\)](#): statistics on sessile SIA clusters; relevant for distinguishing glissile ($\langle 111 \rangle$) vs. sessile ($\langle 100 \rangle$) cluster populations.
- [Soneda et al \(2001\)](#): vacancy loop formation by cascade collapse at 50 keV; informs the large- n tail of the vacancy spectrum.
- [Terentyev et al \(2006\)](#): demonstrates interatomic-potential dependence of predicted cluster fractions; provides uncertainty bounds.
- [Calder et al \(2010\)](#): shock-wave mechanism for large SIA clusters at cascade periphery; justifies the power-law tail for large m .
- [Souidi et al \(2011\)](#): MD/OKMC linkage showing which primary-damage features most strongly affect long-term nanostructure.
- [Björkas et al \(2012\)](#): demonstrates that picosecond cluster distributions influence long-term damage accumulation.
- [Zarkadoula et al \(2013\)](#): high-energy cascade morphology and large-cluster statistics relevant to fusion conditions.
- [De Backer et al \(2018a\)](#): subcascade fragmentation model with power-law scaling of cluster sizes; enables extrapolation to high energies.
- [Nordlund et al \(2018a\)](#): comprehensive review establishing the arc-dpa framework and survival efficiency parameterization.
- [Malerba et al \(2021c\)](#): best-practice parameter compilation for cluster-dynamics modeling of pure Fe.

Caveats and Uncertainty

Several sources of uncertainty should be kept in mind:

1. **Interatomic potential:** Predicted cluster fractions depend on the empirical potential used in MD. [Terentyev et al \(2006\)](#) found that clustering parameters can differ by more than 100% between potentials, even when Frenkel-pair yields are similar.
2. **Temperature dependence:** The values here are representative of reactor temperatures (300–600 K). Modest temperature effects are observed: interstitial clustering increases slightly, and vacancy clustering decreases with increasing temperature [Stoller \(2000b\)](#).
3. **power law idealization:** The true size distribution may deviate from a pure power law, particularly at small sizes (where specific magic numbers may introduce structure) and at very large sizes (where subcascade fragmentation alters statistics) [De Backer et al \(2018a\)](#).
4. **Vacancy cluster character:** In bcc Fe, vacancy “clusters” at cascade time scales are often loose, correlated arrangements of vacancies rather than compact loops. Short-range vacancy diffusion can subsequently convert these into true clusters [Stoller \(1998\)](#); [Soneda et al \(2001\)](#).
5. **Spectrum averaging:** The values given here use representative average PKA energies; a rigorous treatment would convolve the energy-dependent $\epsilon_m(E_{\text{PKA}})$ over the full PKA recoil spectrum for the specific reactor type [Nordlund et al \(2018a\)](#).

2.2 EUROFER Model Parameters: Defect Production

Table 5 collects the cascade-damage production parameters for use in cluster dynamics modelling of EUROFER97. EUROFER is a ferritic matrix, and primary cascade damage is assumed identical to that of pure α -Fe for the displacement cascade regime. The parameters distinguish fission and fusion neutron spectra following the synthesis of [Stoller \(2000c\)](#); [Nordlund et al \(2018b\)](#); [De Backer et al \(2018b\)](#); [Malerba et al \(2021b\)](#).

The production fraction for a cluster of size m is computed as:

$$\epsilon_m^{(i)} = C_i m^{-s_i}, \quad \epsilon_n^{(v)} = C_v n^{-s_v}, \quad (13)$$

with the mono-defect production rate $P_1 = G(1 - f^{\text{cl}})$.

Table 5 Defect production input parameters for EUROFER cluster dynamics.

Parameter	Symbol	Fission	Fusion	Units
<i>Survival and dose rate:</i>				
Survival efficiency (spectrum-avg.)	η	0.30	0.28	—
Displacement rate	$\dot{\phi}$	user input	user input	dpa/s
He production rate	G_{He}/G	0.5–1	~ 10	appm He/dpa
<i>Interstitial cluster production:</i>				
SIA clustering fraction	f_i^{cl}	0.58	0.65	—
SIA power-law exponent	s_i	1.6	1.5	—
Max SIA cluster size	i_{cascade}	20	50	SIAs
SIA normalization constant	C_i	0.1093	0.0553	—
<i>Vacancy cluster production:</i>				
Vacancy clustering fraction	f_v^{cl}	0.15	0.20	—
Vacancy power-law exponent	s_v	2.5	2.3	—
Max vacancy cluster size	v_{cascade}	10	20	vacancies
Vacancy normalization constant	C_v	0.1506	0.1296	—

3 Defect Energetics

3.1 Formation, Migration, and Binding Energies

The table below compiles point-defect energetics, elastic constants, and diffusion parameters for body-centered cubic (bcc) α -iron, as used in rate-theory and cluster-dynamics models of radiation damage. Values are from density functional theory (DFT) calculations unless stated otherwise; experimental values are included where DFT and experiment are compared in the literature. The Burgers vector magnitude for $\frac{1}{2}\langle 111 \rangle$ loops uses the experimental lattice parameter $a_0 = 0.2867$ nm.

Table 6: Representative defect and material parameters for bcc Fe.

Parameter	Value (range)	Units	References / Notes
Vacancy formation energy, E_v^f 1.4 ± 0.1 (expt.)	1.73–2.15 (DFT) eV	DFT: Fu et al (2004) ; Domain and Becquart (2001) ; Muzyk et al (2011) . Experimental (positron annihilation): Kim and Buyers (1978) ; Vehanen et al (1982) . A commonly adopted DFT value is 2.0 eV Bonny et al (2011) ; Was (2017) .	
Self-interstitial atom formation energy, E_i^f	3.53–4.06 (DFT)	eV	The $\langle 110 \rangle$ dumbbell is the ground state in magnetic bcc Fe, with formation energy ≈ 3.64 eV Fu et al (2004) ; Domain and Becquart (2001) ; Nguyen-Manh et al (2006a) . The $\langle 111 \rangle$ crowdion gives 3.68–4.06 eV Willaime et al (2005) .
Vacancy migration energy, E_v^m	0.55–0.67 (DFT)		

Continued on next page

Parameter	Value (range)	Units	References / Notes
0.55 (expt.)	eV	DFT: 0.62 eV Fu et al (2004); 0.55 eV Domain and Becquart (2001). Experimental resistivity-recovery measurements give ≈ 0.55 eV Vehanen et al (1982); Seeger (1998). This value is widely used in rate-theory models Was (2017).	
SIA migration energy, E_i^m	0.013–0.34 (DFT)	eV	For the $\langle 110 \rangle$ translation-rotation mechanism, $E_i^m \approx 0.34$ eV Fu et al (2004); Domain and Becquart (2001). For the $\langle 111 \rangle$ crowdion, the barrier is ≈ 0.013 eV Nguyen-Manh et al (2006a). Experimentally inferred values are about 0.30 ± 0.05 eV Seeger (1998); Was (2017).
Divacancy binding energy, E_{2v}^B 0.22 $\pm$ 0.01 (DFT)	0.20–0.30 (DFT) eV	A widely cited value is 0.22 $\pm$ 0.01 eV for the 1NN configuration Fu et al (2004); Domain and Becquart (2001). Positive binding confirms a metastable divacancy cluster Was (2017).	
Divacancy migration energy, E_{2v}^m	0.62–0.90 (DFT)	eV	Often estimated as $E_{2v}^m \approx E_v^m + \Delta$, where Δ accounts for binding. DFT gives about 0.90 eV for the 1NN divacancy Domain and Becquart (2001); Fu et al (2004). Ghoniem and Cho ? use E_{2v}^m for migration of the complex.
Di-interstitial migration energy, E_{2i}^m	≈ 0.10 –0.20 (DFT)	eV	The di-SIA is highly mobile. A stable 2-SIA configuration in α -Fe consists of two non-parallel $\langle 110 \rangle$ dumbbells. DFT gives $E_{2i}^m \approx 0.10$ –0.20 eV Fu et al (2004); Willaime et al (2005).
Di-interstitial binding energy, E_{2i}^B	0.60–1.0 (DFT)	eV	This cluster is strongly bound, with DFT values of about 0.60–1.0 eV depending on configuration Fu et al (2004); Willaime et al (2005). For two non-parallel $\langle 110 \rangle$ dumbbells, a representative value is ≈ 0.80 eV Domain and Becquart (2001); Fu et al (2004).
Isotropic shear modulus, μ	80–82 (RT, expt.)	GPa	For polycrystalline iron, the Voigt–Reuss–Hill average gives $\mu \approx 80$ –82 GPa at 300 K Simmons and Wang (1971); Ledbetter and Reed (1973); Dever (1972). It decreases to about 64 GPa at 600 °C Was (2017).
Burgers vector magnitude for $\frac{1}{2}\langle 111 \rangle$ loops, b	0.2482	nm	$b = (\sqrt{3}/2) a_0$ with $a_0 = 0.2867$ nm from Simmons and Wang (1971). For $\langle 100 \rangle$ loops, $b = a_0 = 0.2867$ nm.

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Parameter	Value (range)	Units	References / Notes
Stacking-fault energy (vacancy loops / Frank loops), γ_{SF}	200–400 (DFT)	mJ m^{-2}	bcc Fe does not have stable intrinsic stacking faults on $\{111\}$ in the fcc sense. The fault energy associated with vacancy-type Frank loops on $\{110\}$ is estimated at about 200–400 mJ m^{-2} ; however, loops in bcc Fe are typically perfect rather than faulted Was (2017) ; Ackland et al (2004) .
Surface energy (void / free surface), γ_{s} $\approx 2.4 \pm 0.2$ (expt.)	2.0–2.5 (DFT) J m^{-2}	DFT gives $\gamma_{110} \approx 2.43 \text{ J m}^{-2}$ and $\gamma_{100} \approx 2.50 \text{ J m}^{-2}$ Galanakis et al (2002) ; Lazar et al (2015) . Experimental averages are near 2.4 J m^{-2} Tyson and Miller (1977) .	
Pre-exponential factor for vacancy self-diffusion, D_{v}^0 $\approx 5 \times 10^{-4}$ (DFT-based)	1.9–2.3 $\times 10^3$ (expt.) $\text{cm}^2 \text{ s}^{-1}$	Experimental fits give $D_{\text{v}}^0 = 2.3 \times 10^3 \exp(-Q/RT) \text{ cm}^2 \text{ s}^{-1}$ with $Q \approx 251 \text{ kJ mol}^{-1}$ Birchenall and Mehl (1950) ; Buffington et al (1961) ; James and Leak (1966) . The much smaller DFT-based value corresponds to the migration prefactor only Was (2017) .	
Pre-exponential factor for SIA diffusion, D_{i}^0	$\approx 10^{-4}$ – 10^{-3}	$\text{cm}^2 \text{ s}^{-1}$	Estimated from $D_{\text{i}}^0 = f z \nu a^2$, with attempt frequency $\nu \approx 6 \times 10^{12} \text{ s}^{-1}$ and jump distance $a \approx 0.29 \text{ nm}$ Fu et al (2004) ; Willaime et al (2005) ; Was (2017) . The $\langle 111 \rangle$ crowdion mode gives a larger effective prefactor because of its very low barrier Nguyen-Manh et al (2006a) .

The large experimental value $D_{\text{v}}^0 \sim 10^3 \text{ cm}^2 \text{ s}^{-1}$ arises because the measured activation energy $Q = E_{\text{v}}^{\text{f}} + E_{\text{v}}^{\text{m}} \approx 2.6 \text{ eV}$ accounts for both the vacancy formation and migration enthalpies, and the large entropy of formation $\Delta S_{\text{v}}^{\text{f}} \approx 2\text{--}5 k_{\text{B}}$ inflates D^0 . In rate-theory codes, the vacancy diffusion coefficient is often split as $D_{\text{v}} = a^2 \nu \exp(-E_{\text{v}}^{\text{m}}/k_{\text{B}}T)$, with $D_{\text{v,migration}}^0 \approx 5 \times 10^{-4} \text{ cm}^2 \text{ s}^{-1}$, and the equilibrium vacancy concentration is handled separately [Was \(2017\)](#); ?. The Debye (cut-off) frequency of perfect bcc Fe is the baseline for all defect attempt frequencies. With the experimental Debye temperature $\Theta_{\text{D}} \approx 470\text{--}476 \text{ K}$ [Minkiewicz et al \(1967\)](#); [Loong](#)

et al (1987):

$$\nu_D = \frac{k_B \Theta_D}{h} \approx 9.8\text{--}10 \text{ THz} \approx 6.5 \times 10^{13} \text{ s}^{-1}. \quad (14)$$

The phonon density of states (pDOS) of bcc Fe has two prominent peaks: a transverse peak near ~ 7 meV (~ 1.7 THz) and a longitudinal peak at ~ 37 meV (~ 8.9 THz) Minkiewicz et al (1967); Ackland et al (2004).

In Vineyard’s harmonic transition-state theory Vineyard (1957) the jump rate of a point defect is

$$\Gamma = \nu_0 \exp\left(-\frac{E_m}{k_B T}\right), \quad \nu_0 = \frac{\prod_{i=1}^{3N} \nu_i^{(0)}}{\prod_{i=1}^{3N-1} \nu_i^{(\text{sp})}}, \quad (15)$$

where $\nu_i^{(0)}$ and $\nu_i^{(\text{sp})}$ are the real normal-mode frequencies at the stable minimum and the saddle point, respectively. Table 7 summarizes the key values for the three defect species.

Table 7 Migration energies, attempt frequencies, and local vibrational modes for the vacancy, self-interstitial atom (SIA), and interstitial helium in bcc α -Fe. *T-R* = translation-rotation mechanism; *loc.* = localised mode above the bulk phonon band.

Defect	E_m (eV)	Attempt freq. ν_0 (s^{-1})	Local mode (meV / THz)	References
Vacancy	0.55–0.62	6×10^{12} – 6×10^{13}	No gap mode; P_L softened at 37 meV	Vineyard (1957); Fu et al (2004); Domain and Becquart (2001); Vehanen et al (1982); Soisson and Fu (2007)
SIA ((110) dumbbell)	0.34 (T-R)	$\sim 4 \times 10^{12}$	Dumbbell stretch: 60–80 meV (14–19 THz)	Fu et al (2004); Domain and Becquart (2001); Willaime et al (2005); Derlet et al (2007); Terentyev et al (2008)
He (tetrahedral site)	0.06–0.09	$\sim 10^{12}$ – 3×10^{12}	Rattling mode: ~ 50 –80 meV (~ 12 –19 THz)	Fu and Willaime (2005a); de Andres et al (2019); ?

A monovacancy introduces no localized modes *above* the Fe phonon band because it is a missing atom: force constants are reduced, not added. The main spectral signature is a softening of the longitudinal phonon peak P_L at ~ 37 meV, proportional to vacancy concentration, with the perturbation confined to the first-neighbor shell Meyer et al (2020). The attempt frequency spans a wide range in the literature. The Debye-based estimate $\nu_0^v \approx 6 \times 10^{12} \text{ s}^{-1}$ is routinely used in rate-theory codes ?Was (2017). Full DFT Vineyard calculations give $\sim 5 \times 10^{13} \text{ s}^{-1}$ Fu et al (2004), while values as large as $5 \times 10^{15} \text{ s}^{-1}$ are obtained from fitting self-diffusion data Soisson and Fu (2007) because the experimental pre-exponential D_0 folds in the vacancy formation entropy $\Delta S_f^v \approx 4 k_B$ Lucas and Schaublin (2009).

The ground-state SIA in ferromagnetic bcc Fe is the $\langle 110 \rangle$ dumbbell (unique among bcc metals owing to magnetism) [Nguyen-Manh et al \(2006a\)](#). Its translation-rotation migration has $E_m = 0.34$ eV and $\nu_0 \approx 4 \times 10^{12} \text{ s}^{-1}$ from DFT [Fu et al \(2004\)](#); [Domain and Becquart \(2001\)](#). The $\langle 111 \rangle$ crowdion has a negligible barrier (~ 0.013 eV) and migrates nearly athermally [Nguyen-Manh et al \(2006a\)](#).

The $\langle 110 \rangle$ dumbbell creates a *high-frequency localized mode* above the bulk pDOS cut-off: the two dumbbell atoms oscillate against each other along the $\langle 110 \rangle$ bond direction, giving a stretch frequency of $\approx 60\text{--}80$ meV (14–19 THz) [Domain and Becquart \(2001\)](#); [Willaime et al \(2005\)](#). This stiffened mode produces a large negative vibrational formation entropy for the SIA [Derlet et al \(2007\)](#). Non-parallel $\langle 110 \rangle$ di-SIA clusters are further stabilized at elevated temperatures by their large positive vibrational formation entropy [Terentyev et al \(2008\)](#).

Interstitial He occupies the tetrahedral site in bcc Fe, which is more stable than the octahedral site by 0.2 eV (DFT) [Fu and Willaime \(2005a\)](#). He migrates via tetrahedral-to-tetrahedral hops through the octahedral saddle with a very low barrier of $E_m^{He} = 0.06$ eV [Fu and Willaime \(2005a\)](#) (0.078–0.086 eV from MD and other DFT methods [Morishita et al \(2003\)](#); [Yu et al \(2018\)](#)), making He diffusion essentially athermal above room temperature. Because $m_{He}/m_{Fe} \approx 0.07$, He rattles inside the tetrahedral cage at a frequency far above the host phonon band:

$$\nu_{He}^{\text{loc}} \approx \sqrt{\frac{k_{\text{eff}}}{m_{He}}} \approx 50\text{--}80 \text{ meV } (\approx 12\text{--}19 \text{ THz}). \quad (16)$$

This localized optical mode has been confirmed by DFT phonon calculations [Fu and Willaime \(2005a\)](#); [de Andres et al \(2019\)](#). By analogy with the H optical modes ($\sim 70\text{--}130$ meV, Debye temperature $\Theta_D^H \approx 900$ K [de Andres et al \(2019\)](#)), the He localized frequency is lower because $m_{He} > m_H$. The ordering of attempt frequencies mirrors the Debye frequency: $\nu_0^{He} \lesssim \nu_0^{SIA} \lesssim \nu_0^V \sim \nu_D$. All lie within roughly an order of magnitude of $6 \times 10^{12} \text{ s}^{-1}$, and the Debye frequency is an adequate approximation when full Vineyard calculations are unavailable. The strongly differing migration energies — He (< 0.1 eV) $\ll$ SIA (0.34 eV) $\ll$ V (0.55–0.62 eV) — dominate the diffusivity hierarchy over many orders of magnitude.

3.2 EUROFER Model Parameters: Defect Energetics

Table 8 collects the point-defect energetics and material constants for bcc Fe / EUROFER used throughout the rate equations.

Table 8 Defect energetics input parameters for EUROFER cluster dynamics.

Parameter	Symbol	Value	Units
<i>Lattice and elastic constants:</i>			
Lattice constant	a	0.2867	nm
Atomic volume	$\Omega = a^3/2$	1.18×10^{-29}	m^3
Burgers vector ($\frac{1}{2}\langle 111 \rangle$)	b_{111}	0.2482	nm
Burgers vector ($\langle 100 \rangle$)	b_{100}	0.2867	nm
Shear modulus (RT)	μ	82	GPa
Poisson ratio	ν	0.29	—
Surface energy	γ_s	2.0	J/m ²
<i>Vacancy energetics:</i>			
Formation energy	E_f^v	2.0	eV
Migration energy	E_m^v	0.67	eV
Divacancy binding energy (2NN)	E_B^{2v}	0.22	eV
Divacancy migration energy	E_m^{2v}	0.70	eV
<i>SIA energetics:</i>			
Formation energy ($\langle 110 \rangle$ dumbbell)	E_f^i	3.64	eV
Migration energy ($\langle 110 \rangle$ T-R)	E_m^i	0.34	eV
Di-SIA binding energy	E_B^{2i}	0.80	eV
Di-SIA migration energy	E_m^{2i}	0.42	eV
<i>Helium energetics:</i>			
He migration energy (tetrahedral)	E_m^h	0.06	eV
He–vacancy binding (1st He)	E_b^{h-V}	2.30	eV
He–vacancy binding (2nd He)	$E_b^{h-V_2}$	2.00	eV
<i>Attempt frequencies:</i>			
Vacancy	ν_v	1.0×10^{13}	s ^{−1}
SIA ($\langle 110 \rangle$)	ν_i	1.0×10^{13}	s ^{−1}
He interstitial	ν_h	3.0×10^{12}	s ^{−1}
<i>He equation of state (virial coefficients, 600–1000 K):</i>			
Second virial coefficient	B_2	1.67×10^{-29}	m ³ /atom
Third virial coefficient	B_3	1.84×10^{-58}	m ⁶ /atom ²

4 Defect Diffusion in a BCC Lattice

Radiation induced defects and the clusters they form govern the long-term microstructural evolution of structural materials in nuclear environments (Fu and Willaime 2005a; Golubov et al 2000). The diffusion properties of these defects—including their migration energies, attempt frequencies, and dimensionality of motion—are essential input parameters for rate theory and kinetic Monte Carlo models of radiation damage (Trinka et al 2002; Heinisch et al 2000). In bcc α -iron, the lattice constant is $a = 2.87$ Å, and diffusion proceeds by thermally activated jumps between lattice sites. The general Arrhenius form for the diffusion coefficient is

$$D = D_0 \exp\left(-\frac{E_m}{k_B T}\right), \quad (17)$$

where D_0 is the pre-exponential factor, E_m is the migration energy, k_B is Boltzmann's constant, and T is the absolute temperature.

4.1 Random Walk Diffusion Coefficient

The diffusion coefficient for a random walk in 3D is:

$$D = \frac{1}{6} z \lambda^2 \nu_0 \exp(-E_m/k_B T) \quad (18)$$

where z is the number of equivalent jump directions, λ is the jump distance, and the factor $\frac{1}{6}$ comes from the 3D random walk. For a bcc lattice with $\langle 111 \rangle$ nearest-neighbor jumps, $z = 8$, and $\lambda = a\sqrt{3}/2$. If the lattice constant is (a), then the diffusion coefficients of interstitials, vacancies, and helium can be written as

$$D_i = a^2 \nu_i \exp(-E_m^i/k_B T) \quad \text{m}^2 \text{s}^{-1}, \quad (19)$$

$$D_v = a^2 \nu_v \exp(-E_m^v/k_B T) \quad \text{m}^2 \text{s}^{-1}, \quad (20)$$

$$D_h = a^2 \nu_h \exp(-E_m^h/k_B T) \quad \text{m}^2 \text{s}^{-1} \quad (21)$$

We adopt here 3 basic frequencies in the defect clustering system: ω_i is the frequency of the self-interstitial reaction, ω_h is the frequency of the helium gas reaction, and ω_v is the frequency of the vacancy reaction. These are given by:

$$\omega_i = \nu_i \exp(-E_m^i/k_B T) = D_i/a^2, \quad \text{s}^{-1}, \quad (22)$$

$$\omega_h = \nu_h \exp(-E_m^h/k_B T) = D_h/a^2, \quad \text{s}^{-1}, \quad (23)$$

$$\omega_v = \nu_v \exp(-E_m^v/k_B T) = D_v/a^2, \quad \text{s}^{-1}. \quad (24)$$

A plot of these basic frequencies as functions of temperature is shown in Fig. 3.

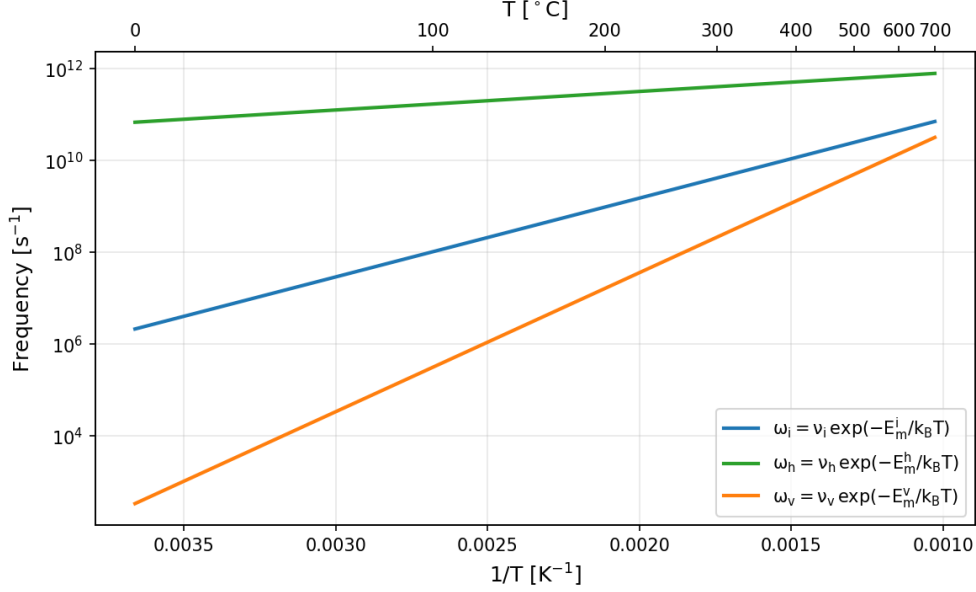


Fig. 3 Basic mobile defect frequencies in iron

4.2 Vacancy Diffusion

Migration Mechanism In bcc iron, vacancy migration proceeds by nearest-neighbor jumps along $\langle 111 \rangle$ directions. Each vacancy has $z = 8$ nearest-neighbor sites to which a neighboring atom can jump, effectively moving the vacancy. The vacancy executes a 3D random walk at all temperatures of interest.

The migration saddle point lies midway between two adjacent bcc lattice sites. DFT calculations consistently predict a vacancy migration energy of $E_m^v \approx 0.65\text{--}0.70$ eV (Domain and Becquart 2001; Fu et al 2004). Experimental values from resistivity recovery (Stage III) and positron annihilation measurements typically yield $E_m^v \approx 0.55\text{--}0.67$ eV (Seeger 1998; Ullmaier 1991). The vacancy formation energy is $E_f^v \approx 2.0\text{--}2.2$ eV, and the activation energy for self-diffusion is $Q_{SD} = E_f^v + E_m^v \approx 2.7\text{--}2.9$ eV, consistent with tracer diffusion experiments.

Diffusion Coefficient Using the bcc random-walk expression (Eq. 20), the vacancy diffusion coefficient takes the form

$$D_v = a^2 \nu_v \exp\left(-\frac{E_m^v}{k_B T}\right), \quad (25)$$

with the pre-exponential factor $D_0^v = a^2 \nu_v$.

Recommended values for pure α -Fe:

$$D_v = 8.2 \times 10^{-8} \exp\left(-\frac{0.67 \text{ eV}}{k_B T}\right) \quad \text{m}^2 \text{s}^{-1} \quad (26)$$

corresponding to an attempt frequency $\nu_v \approx 1.0 \times 10^{13} \text{ s}^{-1}$ and $a = 2.87 \text{ \AA}$.

The vacancy reaction frequency is then

$$\omega_v = \frac{D_v}{a^2} = 1.0 \times 10^{13} \exp\left(-\frac{0.67 \text{ eV}}{k_B T}\right) \quad \text{s}^{-1}. \quad (27)$$

4.3 Self-Interstitial Atom (SIA) Diffusion

4.4 Migration Mechanism in Pure Iron

In ferromagnetic bcc iron, the most stable SIA configuration is the $\langle 110 \rangle$ dumbbell, lying 0.7 eV below the $\langle 111 \rangle$ crowdion, which is the ground state in all non-magnetic bcc transition metals (Fu et al 2004; Domain and Becquart 2001; Nguyen-Manh et al 2006b). This stabilization is of magnetic origin. The SIA migrates by a nearest-neighbor translation-rotation mechanism with a DFT-calculated barrier of $E_m^i = 0.34 \text{ eV}$ (Fu et al 2004; Chiesa et al 2009), consistent with the experimental Stage I_E resistivity recovery value of 0.30–0.34 eV at $\sim 120 \text{ K}$ (Ullmaier 1991). Because the $\langle 110 \rangle$ dumbbell can reorient into any of its six equivalent orientations upon each jump, single-SIA migration is inherently **three-dimensional**—in contrast to the quasi-1D athermal crowdion glide ($E_m \sim 0.01\text{--}0.05 \text{ eV}$) found in non-magnetic bcc metals (Derlet et al 2007; Fitzgerald and Nguyen-Manh 2008).

The diffusion coefficient and reaction frequency for the single SIA in pure α -Fe are:

$$D_i = a^2 \nu_i \exp\left(-\frac{E_m^i}{k_B T}\right) = 8.2 \times 10^{-8} \exp\left(-\frac{0.34 \text{ eV}}{k_B T}\right) \quad \text{m}^2 \text{s}^{-1}, \quad (28)$$

$$\omega_i = \frac{D_i}{a^2} = 1.0 \times 10^{13} \exp\left(-\frac{0.34 \text{ eV}}{k_B T}\right) \quad \text{s}^{-1}, \quad (29)$$

with $\nu_i \approx 1.0 \times 10^{13} \text{ s}^{-1}$ and $a = 2.87 \text{ \AA}$.

4.4.1 Effective SIA Diffusion in EUROFER

The elementary $\langle 110 \rangle$ dumbbell mechanism carries over to the ferritic matrix of EUROFER97, but the effective SIA mobility is reduced by interactions with the alloy's solute population. EUROFER is a reduced-activation 9Cr ferritic/martensitic steel whose principal alloying additions include Cr, W, Mn, V, Ta, C, and N (van der Schaaf et al 2003; Aktaa et al 2020). After heat treatment, part of the C, N, V, and Ta partitions into MX carbonitrides and $M_{23}C_6$ carbides, so the relevant quantity is the *dissolved* matrix solute content rather than the nominal composition.

Two classes of solute-SIA interactions are important:

1. **Substitutional solutes (Cr, Mn, W, V, Ta).** These form mixed solute–Fe dumbbells and bound interstitial complexes that create a fluctuating migration landscape of local binding, detrapping, and solute-biased reorientation events (Olsson et al 2007, 2010; Vincent et al 2006; Messina et al 2020).
2. **Interstitial solutes (C, N).** Carbon and nitrogen occupy octahedral sites and bind strongly to SIA and small SIA clusters, acting as efficient traps that shorten the free-flight length and raise the apparent migration energy (Domain et al 2004; Becquart et al 2005; Terentyev et al 2011a; Theodorou et al 2022).

Both effects are captured by writing the effective SIA diffusivity in the trap-limited form:

$$D_i^{\text{eff}} = \frac{D_i^{\text{Fe}}}{1 + \sum_s K_s(T) c_s}, \quad (30)$$

where D_i^{Fe} is the pure-iron value (Eq. 28), c_s is the atomic fraction of solute or trap class s , and $K_s(T)$ is an effective trapping coefficient related to the SIA–solute binding free energy. The dominant trap classes in EUROFER are dissolved C and N, Cr-rich local environments, and carbide/carbonitride interfaces.

Modeling strategy.

For cluster dynamics, we recommend a layered approach:

1. Use the pure-Fe mechanism and attempt frequency as the microscopic baseline.
2. Replace D_i by D_i^{eff} (Eq. 30) with C and N as explicit trapping species where their dissolved concentrations are known.
3. Absorb the remaining substitutional-solute effects (Cr, W, V, Ta, Mn) into the same effective coefficient, or—in higher-fidelity models—treat them through separate interstitial–solute complexes and radiation-induced segregation terms.

No single Arrhenius pair (D_0 , E_m) adequately represents SIA diffusion in EUROFER across all conditions. The physically sound approach is to use the pure-Fe mobility as the reference and to introduce alloy-specific corrections through Eq. (30).

4.5 Interstitial Cluster Diffusion

4.5.1 Size-Dependent Migration Mechanisms

The migration mechanism of interstitial clusters in bcc iron evolves with cluster size, transitioning from 3D to 1D as the cluster grows:

Size n	Configuration	Mechanism	E_m (eV)
1	$\langle 110 \rangle$ dumbbell	3D translation-rotation	0.34
2	parallel $\langle 110 \rangle$ dumbbells	3D translation-rotation	0.42
3	mixed	3D/1D transition	0.15–0.20
4	$\frac{1}{2}\langle 111 \rangle$ loop	predominantly 1D glide	0.06–0.10
≥ 5	$\frac{1}{2}\langle 111 \rangle$ loop	1D glide	$\lesssim 0.05$

For $n \geq 4$, clusters adopt the form of prismatic dislocation loops with Burgers vector $\mathbf{b} = \frac{1}{2}\langle 111 \rangle$ and migrate by thermally activated 1D glide along their Burgers-vector direction (Osetsky et al 2003; Arakawa et al 2007, 2016). The 1D migration energy is essentially independent of cluster size for $n \gtrsim 4$ (Osetsky et al 2003), with $E_m^{1D} \lesssim 0.05$ eV, so that $E_m^{1D} \leq k_B T$ above room temperature and the Arrhenius factor is close to unity. The di-interstitial ($n = 2$), consisting of two parallel $\langle 110 \rangle$ dumbbells, is *less* mobile than the single SIA ($E_m = 0.42$ vs. 0.34 eV) (Fu et al 2004), while clusters with $n = 3$ –4 lie in the crossover regime (Terentyev et al 2007).

4.5.2 1D Diffusion Coefficient

The 1D diffusion coefficient for a cluster of n SIAs gliding along $\langle 111 \rangle$ is:

$$D_n^{1D} = \frac{1}{2} \lambda_{\langle 111 \rangle}^2 \nu_n \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad (31)$$

where $\lambda_{\langle 111 \rangle} = a\sqrt{3}$ is the jump distance and the factor $\frac{1}{2}$ reflects the two-direction 1D random walk. MD simulations show that the attempt frequency scales inversely with size (Osetsky et al 2003; Anento et al 2010):

$$\nu_n = \frac{\nu_0}{n^s}, \quad \nu_0 \sim 6 \times 10^{12} \text{ s}^{-1}, \quad s \approx 0.5\text{--}1.0, \quad (32)$$

giving the recommended expression:

$$D_n^{1D} = \frac{3a^2 \nu_0}{2n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right) \quad \text{m}^2 \text{ s}^{-1} \quad (33)$$

4.5.3 Mixed 1D/3D Framework

Gliding SIA clusters periodically change their Burgers-vector direction—by thermal activation, by interaction with solutes, or by encounters with other defects (Heinisch et al 2000). The mean free path L between direction changes controls the effective dimensionality of the transport. If L is finite, the effective 3D diffusion coefficient is (Trinkaas et al 2002; Heinisch et al 2000):

$$D_n^{3D, \text{eff}} = \frac{D_n^{1D}}{3} \cdot \frac{L}{\lambda_{\langle 111 \rangle}}, \quad (34)$$

where the factor $\frac{1}{3}$ projects 1D motion onto three dimensions. Equivalently, if ν_R is the Burgers-vector rotation frequency ($L = D_n^{1D}/\nu_R$):

$$D_n^{3D,\text{eff}} = \frac{(D_n^{1D})^2}{6 \nu_R \lambda_{\langle 111 \rangle}^2}. \quad (35)$$

4.5.4 Sink Reaction Rates: 3D vs. 1D

The effective 3D reaction rate with spherical sinks of radius R and density C_s is:

$$k_n^{3D} = 4\pi R D_n^{3D,\text{eff}} C_s. \quad (36)$$

In the pure 1D limit ($L \rightarrow \infty$), the 1D sink-strength formalism gives (Trinka et al 2002; Golubov et al 2000):

$$k_n^{1D} = 2 (\pi R^2 C_s)^2 D_n^{1D}. \quad (37)$$

The critical physical distinction is that the 1D absorption rate scales as $R^4 C_s^2$ (square of the geometric cross-section and second-order in sink density), while the 3D sink strength scales as $C_s R$ (first-order in both capture radius and sink density). This asymmetry between SIA-cluster and point-defect absorption by sinks is the physical origin of the production bias that drives void swelling under cascade-producing irradiation.

4.5.5 Solute Effects on Cluster Transport in EUROFER

In EUROFER, the elementary 1D glide barrier remains small, but solute interactions modify cluster transport in two coupled ways:

1. **Reduced 1D mobility.** Dissolved C, N, and their vacancy complexes trap and pin glissile $\frac{1}{2}\langle 111 \rangle$ loops, lowering the effective 1D diffusivity (Terentyev et al 2011b; Anento and Serra 2013):

$$D_{n,\text{eff}}^{1D} = \frac{D_n^{1D,\text{Fe}}}{1 + \sum_s K_{n,s}(T) c_s}, \quad (38)$$

where $K_{n,s}(T)$ is the trapping factor for cluster size n with solute class s .

2. **Shortened mean free path.** Substitutional solutes (Cr, W, Mn, V, Ta) promote Burgers-vector changes through local pinning and drag, increasing ν_R and thereby reducing L :

$$L_{\text{eff}} = \frac{D_{n,\text{eff}}^{1D}}{\nu_{R,\text{eff}}}. \quad (39)$$

Solute effects therefore compound: they lower D_n^{1D} and shorten L , pushing cluster transport further from the ideal long-range 1D limit toward 3D-like behaviour. The

effective 3D coefficient in the alloy is obtained by substituting the alloy quantities into the mixed 1D/3D expression:

$$D_n^{3D,\text{eff},\text{alloy}} = \frac{D_{n,\text{eff}}^{1D}}{3} \cdot \frac{L_{\text{eff}}}{\lambda_{\langle 111 \rangle}} = \frac{(D_{n,\text{eff}}^{1D})^2}{6 \nu_{R,\text{eff}} \lambda_{\langle 111 \rangle}^2}. \quad (40)$$

4.6 SIA and SIA Cluster Model Parameters for EUROFER

This section collects numerical parameters for the effective diffusion coefficients of single SIAs, di-interstitials, and interstitial clusters in EUROFER97, suitable for direct input to cluster dynamics codes. The approach follows the trap-limited framework developed in the preceding sections: the pure-Fe baseline is reduced by solute trapping and pinning factors calibrated to DFT binding energies and nominal EUROFER compositions.

4.6.1 EUROFER97 Matrix Composition After Tempering

The nominal composition of EUROFER97 is approximately 8.9 Cr–1.1 W–0.47 Mn–0.20 V–0.14 Ta–0.11 C–0.03 N (wt%) (van der Schaaf et al 2003; Aktaa et al 2020). After the standard heat treatment (980°C/30 min normalisation + 760°C/90 min tempering), a substantial fraction of C, N, V, and Ta is consumed by $M_{23}C_6$ and MX-type precipitates. Atom-probe and TEM studies indicate that the dissolved matrix concentrations relevant for defect trapping are approximately (Klimenkov et al 2012b; Polekhina et al 2023):

Species	Nominal (wt%)	Est. dissolved (at. fraction)
Cr (substitutional)	8.9	$c_{\text{Cr}} \approx 0.094$
W (substitutional)	1.1	$c_{\text{W}} \approx 0.0033$
Mn (substitutional)	0.47	$c_{\text{Mn}} \approx 0.0047$
C (interstitial, dissolved)	$\lesssim 0.02$	$c_{\text{C}} \approx 5 \times 10^{-4}$
N (interstitial, dissolved)	$\lesssim 0.01$	$c_{\text{N}} \approx 2 \times 10^{-4}$

The dissolved C and N values are rough estimates; most of the nominal carbon is tied up in precipitates. These concentrations are used in the trapping factors below.

4.6.2 Solute–SIA Binding Energies from DFT

Table 9 summarises the DFT-calculated binding energies of solute atoms to single SIAs ($\langle 110 \rangle$ dumbbells) and to small $\frac{1}{2}\langle 111 \rangle$ interstitial loops in bcc Fe. These govern the trapping factors $K_s(T)$ in Eqs. (30) and (38).

Key observations:

- **C and N** are the strongest traps for both single SIAs and small loops. The C–SIA binding (~ 0.45 eV) is comparable to the SIA migration energy itself (0.34 eV), meaning that C trapping qualitatively alters the effective migration barrier.

Table 9 DFT solute–SIA binding energies E_b^s in bcc Fe. Positive values denote attraction (trapping). “MD” = mixed dumbbell; “NN” = nearest-neighbour configuration.

Solute s	Type	E_b^s (SIA) [eV]	E_b^s (loop) [eV]	Source
Cr	subst. (MD)	0.09–0.12	0.05–0.15	Olsson et al (2007); Vincent et al (2006); Becquart and Soisson (2018)
Mn	subst. (MD)	0.15–0.25	0.10–0.20	Messina et al (2020); Olsson et al (2010)
W	subst. (NN)	−0.1 to −0.3	< 0 (repulsive)	Becquart and Soisson (2018)
C	interstit.	0.40–0.50	0.42–0.86 ^a	Domain et al (2004); Terentyev et al (2011b); Anento and Serra (2013)
N	interstit.	0.35–0.45	~0.3–0.7	Domain et al (2004); Becquart et al (2005)

^a C–loop binding depends on loop size: higher for small ($n = 4$) loops, decreasing toward ~0.42 eV for $n \gtrsim 19$.

- **Cr** binds weakly via the mixed-dumbbell mechanism. At 9 at.% Cr, the cumulative effect is significant despite the small per-atom binding.
- **Mn** binds more strongly than Cr to the dumbbell and dominates dumbbell-mediated solute transport (Messina et al 2020).
- **W** is repulsive with SIAs and does not contribute to trapping.

4.6.3 Trapping Factors

The trapping coefficient $K_s(T)$ in the effective diffusivity (Eq. 30) is related to the binding energy by (Dudarev 2008):

$$K_s(T) = z_s \exp\left(\frac{E_b^s}{k_B T}\right), \quad (41)$$

where z_s is a coordination factor (number of trapping configurations per solute site; $z_s \approx 4$ –8 for substitutional solutes in bcc, $z_s \approx 2$ –4 for interstitial solutes). With this, the effective SIA diffusivity becomes:

$$D_i^{\text{eff}} = \frac{D_i^{\text{Fe}}}{1 + \sum_s z_s c_s \exp(E_b^s/k_B T)}. \quad (42)$$

At low temperatures ($T \lesssim 300^\circ\text{C}$), the C and N trapping terms dominate the denominator and the effective SIA migration energy becomes $E_m^{i,\text{eff}} \approx E_m^i + E_b^C \approx 0.34 + 0.45 = 0.79$ eV. At high temperatures ($T \gtrsim 500^\circ\text{C}$), thermal detrapping reduces the effect and D_i^{eff} approaches the pure-Fe value.

4.6.4 Recommended Parameter Table

Table 10 provides the recommended effective diffusion parameters for use in cluster dynamics modeling of EUROFER97. The “baseline” column gives pure-Fe values; the “EUROFER effective” column includes solute corrections.

Table 10 Recommended diffusion parameters for SIAs, di-interstitials, and interstitial clusters in EUROFER97. $a = 2.87 \text{ \AA}$. Pure-Fe values serve as the baseline; EUROFER effective values include solute trapping corrections via Eqs. (30)/(38).

Defect	$\nu_0 \text{ (s}^{-1}\text{)}$	$E_m \text{ (eV)}$	$D_0 \text{ (m}^2\text{/s)}$	Dim.
<i>Pure Fe baseline:</i>				
SIA ($n = 1, \langle 110 \rangle$)	1.0×10^{13}	0.34	8.2×10^{-8}	3D
Di-SIA ($n = 2$)	$\sim 10^{13}$	0.42	$\sim 8 \times 10^{-8}$	3D
SIA cluster ($n \geq 4, 1D$)	$6 \times 10^{12}/n^s$	0.02–0.05	—	1D
<i>EUROFER effective (trap-limited, Eq. 42):</i>				
SIA ($n = 1$)	$D_i^{\text{eff}} = D_i^{\text{Fe}} / (1 + \sum_s K_s c_s)$, using: $K_C = 4 \exp(0.45/k_B T)$, $K_N = 4 \exp(0.40/k_B T)$, $K_{Cr} = 8 \exp(0.10/k_B T)$, $K_{Mn} = 6 \exp(0.20/k_B T)$			
Di-SIA ($n = 2$)	Use same trapping form with D_{2i}^{Fe} as baseline and E_b^s increased by $\sim 10\%$.			
SIA cluster ($n \geq 4$)	$D_{n,\text{eff}}^{1D} = D_n^{1D,\text{Fe}} / (1 + \sum_s K_{n,s} c_s)$, using: $K_{n,C} = 2 \exp(0.50/k_B T)$, $K_{n,N} = 2 \exp(0.40/k_B T)$, $K_{n,Cr} = 4 \exp(0.10/k_B T)$			

The coordination numbers z_s and binding energies E_b^s used above are rounded values chosen to represent the central range of DFT data from Table 9. The W contribution is omitted (repulsive binding; $K_W \approx 0$). V and Ta are present at low dissolved concentrations and are neglected for simplicity.

4.6.5 Effective Mean Free Path in EUROFER

For SIA clusters in the mixed 1D/3D framework (Eq. 40), the effective rotation frequency and mean free path are:

$$\nu_{R,\text{eff}} = \nu_R^{\text{therm}} + \sum_s \sigma_s^{\text{rot}} c_s \omega_n^{1D}, \quad (43)$$

$$L_{\text{eff}} = D_{n,\text{eff}}^{1D} / \nu_{R,\text{eff}}, \quad (44)$$

where ν_R^{therm} is the thermal rotation frequency in pure Fe (small; often negligible below 500°C for large loops) and σ_s^{rot} is the effective rotation cross-section per solute atom. For Cr at 9 at.%, atomistic estimates give $L_{\text{eff}} \sim 10\text{--}100 a$ ($\sim 3\text{--}30 \text{ nm}$), which is much shorter than the typical sink spacing in irradiated EUROFER ($\sim 100\text{--}1000 \text{ nm}$). This places EUROFER cluster transport firmly in the **mixed 1D/3D regime** rather than the pure 1D limit applicable to high-purity iron.

4.6.6 Numerical Examples

To illustrate the magnitude of the trapping correction, Table 11 gives the effective SIA diffusivity at several temperatures for two scenarios: pure Fe, and EUROFER with $c_C = 5 \times 10^{-4}$, $c_N = 2 \times 10^{-4}$, $c_{Cr} = 0.094$.

Table 11 Effective SIA diffusivity D_i^{eff} (m^2/s) at selected temperatures. The reduction factor $\Phi = D_i^{\text{eff}}/D_i^{\text{Fe}}$ quantifies the total trapping effect.

T ($^{\circ}\text{C}$)	T (K)	D_i^{Fe}	D_i^{eff} (EUROFER)	Φ
200	473	6.8×10^{-12}	$\sim 2 \times 10^{-15}$	$\sim 3 \times 10^{-4}$
300	573	1.4×10^{-11}	$\sim 5 \times 10^{-14}$	$\sim 4 \times 10^{-3}$
400	673	2.3×10^{-11}	$\sim 2 \times 10^{-12}$	~ 0.09
500	773	3.2×10^{-11}	$\sim 1.3 \times 10^{-11}$	~ 0.4
600	873	4.1×10^{-11}	$\sim 3.2 \times 10^{-11}$	~ 0.8

The trapping correction is dramatic below $\sim 400^{\circ}\text{C}$ (reduction by 2–4 orders of magnitude) and becomes negligible above $\sim 600^{\circ}\text{C}$. This temperature dependence is qualitatively consistent with the observation that radiation damage accumulation in EUROFER is much more severe at low irradiation temperatures than at high temperatures.

4.6.7 Uncertainties and Directions for Improvement

The parameters in Table 10 carry significant uncertainties that future work should address:

1. **Dissolved C and N concentrations.** The trapping correction is exponentially sensitive to E_b^s and linearly sensitive to c_s . Atom-probe tomography (APT) measurements of the dissolved C and N content in the EUROFER matrix after standard heat treatment would substantially reduce the uncertainty in D_i^{eff} . This is the single most impactful measurement for calibrating the model.
2. **DFT binding energies in concentrated Fe–Cr.** All binding energies in Table 9 were computed in dilute Fe– X binary alloys. In the concentrated Fe–9Cr matrix of EUROFER, local Cr environment effects (short-range order, α' precursors) may modify both the magnitude and the spatial distribution of binding energies. DFT or machine-learning potential calculations in Fe–9Cr supercells with realistic Cr configurations are needed.
3. **Synergistic multi-solute effects.** The trap-limited model (Eq. 42) assumes independent trapping by each solute species. In reality, C–vacancy and N–vacancy complexes may be the dominant trapping species rather than isolated C or N atoms. Including these complexes requires knowledge of their concentrations (from thermodynamic equilibrium or irradiation-driven formation) and their separate binding energies to SIAs.
4. **Size-dependent loop trapping.** The C–loop binding energy decreases with loop size (Table 9). A size-dependent $K_{n,\text{C}}(n)$ based on systematic DFT or MD calculations for loops of 4–100 SIAs would improve the accuracy of the 1D cluster transport model, particularly at low temperatures where trapping controls the effective 1D glide length.
5. **Irradiation-induced solute redistribution.** Under irradiation, radiation-induced segregation (RIS) depletes Cr and enriches Mn and P at defect sinks

(Messina et al 2020). This dynamically alters the local solute field experienced by migrating defects. Coupling the solute-field evolution to the defect diffusivity in a self-consistent manner—through a concentration-dependent $K_s(T, c_s(t))$ —is a longer-term modeling goal that would require co-evolution of solute and defect distributions.

6. **Experimental validation.** In-situ TEM observation of SIA cluster mobility in Fe–9Cr model alloys (with and without controlled C/N additions) as a function of temperature would provide direct calibration data for $D_{n,\text{eff}}^{1D}$ and L_{eff} . Resistivity recovery experiments on EUROFER after electron irradiation at cryogenic temperatures would constrain the effective SIA migration energy.

4.7 Vacancy Cluster Diffusion

4.7.1 Migration Mechanism

Vacancy clusters in bcc iron migrate by a qualitatively different mechanism from interstitial clusters. Whereas SIA clusters glide collectively as dislocation loops, vacancy clusters move only by sequential single-vacancy jumps within or at the surface of the cluster. The motion is always **three-dimensional**, and the effective migration energy generally **increases** with cluster size because larger clusters require increasingly coordinated sequences of atomic rearrangements to translate the centre of mass by one lattice spacing (Rahman et al 2023; Fu et al 2005; Becquart and Domain 2007).

For clusters larger than ~ 5 – 6 vacancies, the dominant transport mechanism transitions from bulk-like nearest-neighbour vacancy hopping to surface (ledge) diffusion of adatom vacancies around the void facets. This mechanism underlies the classical Gruber model for void migration, in which the void diffusion coefficient scales as a high inverse power of the void radius (Gruber 1967; Nichols 1969).

4.7.2 Small Vacancy Cluster Migration Energies

Table 12 collects the effective migration barriers for small vacancy clusters in α -Fe from several computational methods.

Table 12 Effective migration energies E_m (eV) of small vacancy clusters in bcc Fe. The “mechanism” column describes the pathway. Values are compiled from DFT, kinetic activation–relaxation technique (k-ART), and molecular dynamics (MD) studies.

n	Configuration	E_m (eV)	Mechanism	Source
1	monovacancy	0.65–0.70	NN jump along $\langle 111 \rangle$	Fu et al (2004); Domain and Becquart (2001)
2	divacancy (2NN)	0.62–0.75	two-step NN exchange	Rahman et al (2023); Fu et al (2005)
3	compact trivacancy	0.35–0.50	concerted 3-atom ring rotation	Rahman et al (2023); Schuler et al (2015)
4	tetravacancy	0.7–0.9	sequential NN jumps	Rahman et al (2023)
5	pentavacancy	0.8–1.1	sequential NN jumps + dissociation	Rahman et al (2023)
≥ 6	compact 3D cluster	> 1.0	effectively immobile	Rahman et al (2023)

Two features of Table 12 merit discussion:

The anomalously mobile trivacancy.

The compact trivacancy (V_3) in the bcc lattice—with all three vacancies at mutual second-nearest-neighbor positions forming an equilateral triangle on a $\{110\}$ plane—has a migration energy (0.35–0.50 eV) that is *lower* than that of either the monovacancy or the divacancy. This was first reported from DFT-parameterized kinetic Monte Carlo calculations and has been confirmed by k-ART (Rahman et al 2023) and by independent first-principles studies of vacancy–solute cluster mobility (Schuler et al 2015). The mechanism is a concerted three-atom ring rotation in which three Fe atoms adjacent to the cluster simultaneously exchange positions with the three vacancies, translating the cluster center of mass by one lattice vector. Because of this low barrier, V_3 clusters are the most mobile vacancy-type species in α -Fe after isolated monovacancies. Schuler et al (2015) further showed that V_3X complexes (with $X = \text{C, N, or O}$) retain much of this mobility—with global migration energies of 0.59 eV (C), 0.86 eV (N), and 0.58 eV (O)—which means that vacancy-mediated interstitial-solute transport via trivacancy dragging is an important mechanism in irradiated ferritic steels.

Divacancy pathway multiplicity.

The divacancy in bcc Fe has its ground state at the second-nearest-neighbor (2NN) separation (Domain and Becquart 2001). Its migration involves a two-step process: first, one vacancy jumps to a first-nearest-neighbor position (metastable), then the second vacancy follows to re-establish the 2NN ground state at a translated position. The effective barrier depends on which of several pathways is taken, giving a range of 0.62–0.75 eV (Rahman et al 2023; Fu et al 2005). k-ART simulations reveal additional complex pathways involving transient higher-order configurations, with effective barriers as high as 0.84 eV for some routes through the energy landscape (Rahman et al 2023).

4.7.3 Void Diffusion: Size-Dependent Model

For larger voids containing m vacancies ($m \gtrsim 10$), vacancy cluster migration proceeds by surface self-diffusion of adatom vacancies on the void wall. The classical result due to Gruber (1967) and Nichols (1969) gives the void diffusion coefficient as:

$$D_{\text{void}}(m) = \frac{D_s \Omega^{4/3}}{r_m^4} = D_s \Omega^{4/3} \left(\frac{4\pi}{3m\Omega} \right)^{4/3} \propto \frac{D_s}{m^{4/3}}, \quad (45)$$

where D_s is the surface self-diffusion coefficient on the void wall, $\Omega = a^3/2$ is the atomic volume, and $r_m = (3m\Omega/4\pi)^{1/3}$ is the void radius. The strong inverse scaling $D_{\text{void}} \propto m^{-4/3}$ means that void mobility drops extremely rapidly with size. Numerical estimates give $D_{\text{void}} \sim 10^{-30} \text{ m}^2/\text{s}$ or less for voids of $m \geq 50$ at temperatures below 500°C, which is effectively zero on any experimentally accessible timescale.

Alternative derivations based on volume diffusion, rather than surface diffusion, give an even steeper size dependence ($D_{\text{void}} \propto m^{-5/3}$), but surface diffusion is generally accepted as the dominant mechanism for faceted voids in bcc metals (Nichols 1969).

4.7.4 Practical Treatment in Rate Theory

For cluster dynamics models, the following classification is recommended:

Size range	Treatment	Rationale
$m = 1$ (monovacancy)	mobile, 3D	$E_m = 0.67$ eV
$m = 2$ (divacancy)	mobile, 3D	$E_m \approx 0.70$ eV
$m = 3$ (trivacancy)	mobile, 3D	$E_m \approx 0.42$ eV (anomalously fast)
$m = 4-5$	weakly mobile	$E_m \approx 0.7-1.1$ eV; include if accuracy needed
$m \geq 6$	immobile sinks	growth/shrinkage by V and SIA absorption/emission only

The general size-dependent diffusion coefficient for vacancy clusters used in the rate equations is:

$$D_{(-m,0)} = D_v \cdot m^{-s_v}, \quad s_v \approx 1.0 \quad (m \leq v_{\text{mobile}}), \quad (46)$$

where v_{mobile} is a mobility cutoff (typically 3–5). Clusters with $m > v_{\text{mobile}}$ are set to $D = 0$.

For He-containing vacancy clusters (bubbles), the diffusion coefficient is further reduced:

$$D_{(-m,\ell)} = D_{(-m,0)} \cdot f(\ell/m), \quad (47)$$

where $f(\ell/m)$ is a reduction factor reflecting the additional pinning by trapped helium. A simple choice is $f = \exp(-E_b^{\text{He}}\ell/k_B T)$, but for practical purposes in most rate-theory models, He-containing clusters with $m \geq 2$ are treated as immobile.

4.7.5 Solute Effects on Vacancy and Vacancy Cluster Diffusion in EUROFER

Vacancy migration in pure α -Fe proceeds by nearest-neighbor jumps along $\langle 111 \rangle$ with a barrier of $E_m^v = 0.67$ eV. In the EUROFER matrix, dissolved solutes modify this mobility through two mechanisms: (i) vacancy-solute binding, which traps vacancies near solute sites and raises the effective migration energy; and (ii) modification of the local saddle-point energy, which can either raise or lower the elementary jump barrier depending on the neighboring chemical environment. Both effects are governed by the vacancy-solute binding energy E_b^{v-s} , which has been computed by DFT for all the principal solutes in EUROFER.

Vacancy–Solute Binding Energies.

Table 13 collects DFT-calculated vacancy–solute binding energies at 1NN and 2NN separations in bcc Fe. Positive values denote attraction (trapping).

Key observations for EUROFER:

1. **C and N are the dominant vacancy traps**, with binding energies ($\sim 0.4-0.5$ eV at 1NN) comparable to the vacancy migration energy itself. Carbon–vacancy (VC) and nitrogen–vacancy (VN) complexes are therefore strongly

Table 13 DFT vacancy–solute binding energies E_b^{v-s} (eV) in bcc Fe. Positive = attractive (vacancy trapped near solute). Values compiled from the systematic studies of Olsson et al (2010), Devi et al (2018), Domain et al (2004), and Ohnuma et al (2009).

Solute s	Type	E_b^{v-s} (1NN) [eV]	E_b^{v-s} (2NN) [eV]	Source
Cr	substitutional	0.03–0.06	−0.02 to −0.05 (repulsive)	Olsson et al (2010); Devi et al (2018)
Mn	substitutional	0.06–0.12	~0.00	Olsson et al (2010); Devi et al (2018)
W	substitutional	0.20–0.35	0.05–0.15	Olsson et al (2010); Devi et al (2018)
V	substitutional	0.02–0.10	−0.03 to 0.00	Olsson et al (2010)
Ta	substitutional	0.15–0.25	0.02–0.08	Devi et al (2018)
C	interstitial	0.41–0.47	0.10–0.18	Domain et al (2004); Ohnuma et al (2009)
N	interstitial	0.35–0.43	0.08–0.15	Domain et al (2004); Ohnuma et al (2009)

bound and act as deep traps that immobilize vacancies at low temperatures. This is the primary mechanism by which interstitial solutes suppress vacancy-mediated processes (self-diffusion, void nucleation) in ferritic steels.

2. **W is a moderate vacancy trap** ($E_b \sim 0.25$ – 0.35 eV at 1NN). At the 0.33 at.% level present in EUROFER, W makes a non-negligible contribution to vacancy trapping, particularly in the intermediate temperature range (300–500°C) where the C/N trapping terms are starting to deactivate.
3. **Cr binds very weakly to vacancies** ($E_b \sim 0.05$ eV at 1NN) and is repulsive at 2NN. Despite its high concentration (9.4 at.%), the cumulative trapping effect of Cr on vacancy mobility is modest because $\exp(0.05/k_B T)$ is close to unity at all relevant temperatures.
4. **Mn binds slightly more strongly than Cr but still weakly** (~ 0.10 eV). Its effect is secondary to C, N, and W.

Effective Vacancy Diffusivity in EUROFER.

By analogy with the SIA trap-limited formulation (Eq. 30), the effective vacancy diffusivity in EUROFER is:

$$D_v^{\text{eff}} = \frac{D_v^{\text{Fe}}}{1 + \sum_s z_s^v c_s \exp(E_b^{v-s}/k_B T)}, \quad (48)$$

where D_v^{Fe} is the pure-iron value, c_s is the dissolved atomic fraction of solute s , and z_s^v is the coordination number for vacancy trapping configurations near a solute of type s . The dominant trapping species, in decreasing order of importance, are dissolved C, dissolved N, W, Mn, and Cr.

Effect on Vacancy Cluster Mobility.

Small vacancy clusters ($m = 2$ – 5) are affected by solutes through two mechanisms:

1. **Enhanced binding/pinning:** C and N atoms bind even more strongly to divacancies and trivacancies than to monovacancies, because the larger open volume of the cluster provides a more favorable environment for the interstitial solute (Domain et al 2004; Schuler et al 2015). This deepens the trap and raises the effective cluster migration barrier.
2. **Solute decoration:** In EUROFER, small vacancy clusters formed under irradiation rapidly acquire C and N decorations from the dissolved matrix pool, converting bare V_m clusters into $V_m C_p N_q$ complexes. These decorated clusters have substantially reduced mobility relative to bare clusters. The compact trivacancy—anomalously mobile in pure Fe—is particularly affected: V_3C has a migration energy of 0.59 eV and V_3N of 0.86 eV (Schuler et al 2015), compared to 0.35–0.50 eV for bare V_3 .

For the purpose of cluster dynamics modeling, the effective vacancy cluster diffusivity in EUROFER can be written as:

$$D_{(-m,0)}^{\text{eff}} = \frac{D_{(-m,0)}^{\text{Fe}}}{1 + \sum_s K_{m,s}^v(T) c_s}, \quad (49)$$

where $K_{m,s}^v(T) = z_{m,s}^v \exp(E_b^{v_{m-s}}/k_B T)$ and the binding energy $E_b^{v_{m-s}}$ may depend on cluster size.

Numerical Illustration.

Table 14 shows the effective monovacancy diffusivity at selected temperatures compared to pure Fe.

Table 14 Effective monovacancy diffusivity D_v^{eff} (m^2/s) in EUROFER at selected temperatures, with reduction factor $\Phi_v = D_v^{\text{eff}}/D_v^{\text{Fe}}$.

T ($^{\circ}\text{C}$)	T (K)	D_v^{Fe}	D_v^{eff} (EUROFER)	Φ_v
200	473	2.1×10^{-15}	$\sim 1 \times 10^{-18}$	$\sim 5 \times 10^{-4}$
300	573	8.5×10^{-14}	$\sim 3 \times 10^{-16}$	$\sim 4 \times 10^{-3}$
400	673	1.1×10^{-12}	$\sim 6 \times 10^{-14}$	~ 0.05
500	773	7.1×10^{-12}	$\sim 2 \times 10^{-12}$	~ 0.3
600	873	2.9×10^{-11}	$\sim 2 \times 10^{-11}$	~ 0.7

The vacancy trapping correction follows a similar trend to the SIA case: a dramatic reduction (3 orders of magnitude) below $\sim 300^{\circ}\text{C}$, becoming negligible above $\sim 600^{\circ}\text{C}$. However, because $E_m^v > E_m^i$, the vacancy is already slow in pure Fe, and the practical consequence of trapping is to delay the onset of vacancy-mediated processes (void nucleation, thermal emission from clusters, precipitation coarsening) to even higher temperatures in EUROFER than the pure Fe migration energy alone would suggest.

Physical Implications for Cluster Dynamics.

The solute-modified vacancy diffusivity has three important consequences for cluster dynamics modeling of EUROFER:

1. **Suppressed void nucleation at low T .** Vacancy trapping by C and N reduces the free vacancy supersaturation available for void nucleation, raising the effective incubation dose. This is consistent with the observation that EUROFER shows higher void swelling resistance than high-purity iron at temperatures below $\sim 400^\circ\text{C}$.
2. **Delayed thermal emission.** The vacancy emission rate from clusters (Eq. 123) depends on the free vacancy equilibrium concentration, which is suppressed when $D_v^{\text{eff}} < D_v^{\text{Fe}}$. This stabilizes small vacancy clusters against thermal dissolution and shifts the void-to-bubble transition to higher temperatures.
3. **Asymmetric SIA–vacancy trapping.** In EUROFER, both SIA and vacancies are trapped by C and N, but the SIA–C binding (~ 0.45 eV) and vacancy–C binding (~ 0.45 eV) are comparable. This means that the net effect on the point-defect imbalance (which drives swelling) depends on the relative magnitudes of the SIA and vacancy trapping corrections and cannot be predicted from either correction alone. Self-consistent treatment of both D_i^{eff} and D_v^{eff} in the rate equations is essential.

4.8 Helium Interstitial Diffusion

4.8.1 Migration Mechanism in Pure Iron

Interstitial helium in bcc iron occupies tetrahedral sites, which are more stable than octahedral sites by ~ 0.2 eV (Fu and Willaime 2005b; Seletskaya et al 2005). The migration path connects nearest-neighbor tetrahedral sites with a DFT-calculated barrier of $E_m^h = 0.06$ eV (Fu and Willaime 2005b); MD simulations and NEB calculations with different functionals give values in the range 0.06–0.09 eV (Morishita et al 2003; Zhang et al 2018). This barrier is so small that helium is essentially mobile at all temperatures above ~ 30 K—it is by far the fastest-diffusing species in the defect system.

The helium diffusion coefficient and reaction frequency in pure α -Fe are:

$$D_h = a^2 \nu_h \exp\left(-\frac{E_m^h}{k_B T}\right) = 2.5 \times 10^{-8} \exp\left(-\frac{0.06 \text{ eV}}{k_B T}\right) \quad \text{m}^2 \text{ s}^{-1}, \quad (50)$$

$$\omega_h = \frac{D_h}{a^2} = 3.0 \times 10^{12} \exp\left(-\frac{0.06 \text{ eV}}{k_B T}\right) \quad \text{s}^{-1}, \quad (51)$$

with $\nu_h \approx 3.0 \times 10^{12} \text{ s}^{-1}$ and $a = 2.87 \text{ \AA}$ (Becquart and Domain 2007; Borodin and Vladimirov 2007). Because $E_m^h \ll k_B T$ at all reactor-relevant temperatures, the Arrhenius factor is close to unity, and the helium mobility is nearly temperature-independent. Spin-lattice dynamics simulations confirm strongly non-Arrhenius behavior at elevated temperatures, where phonon-drag effects make the diffusion coefficient increase roughly linearly with T rather than exponentially (Ma and Dudarev 2019).

4.8.2 Helium Trapping by Vacancies

The defining feature of helium in irradiated metals is its extremely strong binding to vacancies. DFT calculations give a binding energy of $E_b^{h-V} \approx 2.0\text{--}2.4$ eV for the first He atom entering a monovacancy (Fu and Willaime 2005b; Seletskaiia et al 2005; Becquart and Domain 2007). The binding remains strong as additional He atoms are added: up to 4–6 He atoms can be accommodated in a single vacancy with cumulative binding energies exceeding 4 eV (Fu and Willaime 2005b). These He_nV_m complexes are essentially immobile (substitutional He migration requires the vacancy mechanism with a barrier of ~ 1.1 eV) and serve as the nucleation sites for helium bubbles.

The consequence for transport is that the *free* interstitial helium concentration c_h available for long-range diffusion is not the total implanted He content but rather the small fraction that has not yet been trapped by vacancies, vacancy clusters, or other deep traps. In a vacancy-rich irradiation environment, the free He concentration can be orders of magnitude below the total He inventory.

4.8.3 Helium–Solute Interactions and Effective Diffusivity in EUROFER

In the EUROFER matrix, interstitial He can interact with dissolved solutes before reaching a vacancy trap. The relevant interactions, from DFT calculations, are:

Substitutional solutes (Cr, Mn, W).

He–substitutional-solute interactions in bcc Fe are generally weak. Becquart and Domain (2007) and Seletskaiia et al (2005) find that He interacts only marginally with substitutional Cr and other transition metal solutes, with binding energies in the range 0.00–0.05 eV at 1NN—insufficient to constitute meaningful traps at reactor temperatures. At 9 at.% Cr, the cumulative effect on He mobility is therefore negligible.

Interstitial solutes (C, N).

Carbon and nitrogen occupy octahedral sites and interact with interstitial He at tetrahedral sites. The He–C binding energy has been calculated to be $\sim 0.05\text{--}0.10$ eV (Becquart and Domain 2007; Gao et al 2011), which is comparable to the He migration barrier itself. This means C can cause transient trapping of He at low temperatures ($T \lesssim 200^\circ\text{C}$), but at higher temperatures the trapping is thermally overcome and has little effect on long-range He transport.

Vacancy–solute complexes as He traps.

The most important solute effect on He transport in EUROFER is *indirect*: C and N form stable vacancy–solute complexes (VC , VN , VC_2) that compete with bare vacancies as He trapping sites. The He binding energy to a VC complex is $\sim 1.5\text{--}1.8$ eV—lower than to a bare vacancy (~ 2.3 eV) but still a deep trap. Since a fraction of the vacancies in EUROFER are decorated by C and N, the effective trap density and the He–trap binding landscape differ from pure Fe.

Effective Helium Diffusivity.

Because the He–solute binding energies are so much smaller than the He–vacancy binding energy, the dominant control on free He transport is the vacancy trap density, not the solute field. The effective He diffusivity in EUROFER can therefore be written as:

$$D_h^{\text{eff}} = \frac{D_h^{\text{Fe}}}{1 + z_V^h c_V^{\text{free}} \exp(E_b^{h-V}/k_B T) + \sum_s z_s^h c_s \exp(E_b^{h-s}/k_B T)}, \quad (52)$$

where the first trapping term is due to free vacancies (concentration c_V^{free} , binding energy $E_b^{h-V} \approx 2.3$ eV, coordination $z_V^h = 6$) and the sum is over solute traps. In practice, at all temperatures where vacancies exist in meaningful concentrations, the vacancy trapping term completely dominates the denominator and the solute terms can be neglected. The helium diffusivity reduction in EUROFER relative to pure Fe is therefore controlled primarily by the irradiation-produced vacancy concentration, not by the alloy composition.

Table 15 collects the recommended helium diffusion parameters.

Table 15 Recommended parameters for helium interstitial diffusion in bcc Fe and EUROFER. The effective diffusivity Eq. (52) is dominated by vacancy trapping; solute terms are secondary corrections.

Parameter	Value	Source
<i>Pure Fe baseline:</i>		
E_m^h (interstitial migration)	0.06 eV	Fu and Willaime (2005b)
ν_h (attempt frequency)	$3.0 \times 10^{12} \text{ s}^{-1}$	Becquart and Domain (2007)
$D_0^h = a^2 \nu_h$	$2.5 \times 10^{-8} \text{ m}^2/\text{s}$	—
<i>Vacancy trapping (dominant):</i>		
E_b^{h-V} (He–monovacancy, 1st He)	2.3 eV	Fu and Willaime (2005b); Seletskaja et al (2005)
E_b^{h-V} (He–monovacancy, 2nd He)	2.0 eV	Fu and Willaime (2005b)
E_b^{h-V} (He–divacancy)	2.5–2.8 eV	Borodin and Vladimirov (2007)
z_V^h (trapping coordination)	6	—
<i>Solute trapping (secondary, for EUROFER):</i>		
E_b^{h-C} (He–C)	0.08 eV	Becquart and Domain (2007)
E_b^{h-N} (He–N)	~ 0.05 eV	estimated
E_b^{h-Cr} (He–Cr)	~ 0.02 eV	Seletskaja et al (2005)
E_b^{h-VC} (He–VC complex)	1.5–1.8 eV	Becquart and Domain (2007)

Physical Summary.

Helium transport in EUROFER differs from that of vacancies and SIAs in a fundamental way: the *intrinsic* mobility is unaffected by the alloy composition (He–solute

binding is negligible), but the *effective* long-range transport is controlled by the vacancy trap landscape, which is itself modified by solutes (through the vacancy trapping described in Section 4.7.5). This creates an indirect coupling: solutes reduce the free vacancy concentration available for He trapping, which paradoxically can *increase* the free He mobility in EUROFER relative to a hypothetical material with the same total vacancy content but no solute-decorated vacancies. The interplay between vacancy–solute decoration and He trapping is an important consideration for bubble nucleation modelling and should be treated self-consistently in the rate equations.

4.9 EUROFER Model Parameters: Defect Diffusion

Table 16 collects all diffusion parameters. Pure-Fe baseline values are given first; EUROFER solute-trapping corrections follow. The effective diffusivity is $D_{\alpha}^{\text{eff}} = D_{\alpha}^{\text{Fe}} / (1 + \sum_s K_s c_s)$ with $K_s = z_s \exp(E_b^s / k_B T)$.

Table 16 Diffusion input parameters for EUROFER cluster dynamics.

Parameter	Symbol	Value	Units
<i>Pure-Fe baseline diffusion coefficients:</i>			
Vacancy D_0	D_0^v	8.2×10^{-8}	m^2/s
Vacancy E_m	E_m^v	0.67	eV
SIA D_0	D_0^i	8.2×10^{-8}	m^2/s
SIA E_m	E_m^i	0.34	eV
He D_0	D_0^h	2.5×10^{-8}	m^2/s
He E_m	E_m^h	0.06	eV
<i>SIA cluster 1D glide ($n \geq 4$):</i>			
Attempt frequency	ν_0^{1D}	6×10^{12}	s^{-1}
1D migration energy	E_m^{1D}	0.03	eV
Size exponent	s_i	0.7	—
Mobility cutoff	i_{mobile}	100	SIAs
<i>Vacancy cluster diffusion:</i>			
Size exponent	s_v	1.0	—
Mobility cutoff	v_{mobile}	5	vacancies
<i>EUROFER dissolved solute concentrations (after tempering):</i>			
Cr	c_{Cr}	0.094	at/at
W	c_{W}	0.0033	at/at
Mn	c_{Mn}	0.0047	at/at
C (dissolved)	c_{C}	5×10^{-4}	at/at
N (dissolved)	c_{N}	2×10^{-4}	at/at
<i>SIA trapping parameters ($K_s = z_s \exp(E_b^s/k_B T)$):</i>			
C-SIA	$z = 4, E_b$	0.45	eV
N-SIA	$z = 4, E_b$	0.40	eV
Cr-SIA	$z = 8, E_b$	0.10	eV
Mn-SIA	$z = 6, E_b$	0.20	eV
<i>Vacancy trapping parameters:</i>			
C-V	$z = 3, E_b$	0.45	eV
N-V	$z = 3, E_b$	0.40	eV
W-V	$z = 8, E_b$	0.27	eV
Mn-V	$z = 8, E_b$	0.10	eV
Cr-V	$z = 8, E_b$	0.05	eV
<i>SIA cluster trapping (1D loops):</i>			
C-loop	$z = 2, E_b$	0.50	eV
N-loop	$z = 2, E_b$	0.40	eV
Cr-loop	$z = 4, E_b$	0.10	eV
<i>Mixed 1D/3D parameters:</i>			
Mean free path (EUROFER)	$\hat{L} = L/a$	30–100	—
1D/3D interpolation constant	B_{rot}	2.627	—

5 Defect Cluster Dissociation

The equilibrium thermal vacancy concentration in the bulk is approximated by

$$C_v^0 = \exp\left(-\frac{E_v^f}{k_B T}\right), \quad (53)$$

where E_v^f is the vacancy formation free energy or, in the usual approximation, the vacancy formation energy. At the surface of a vacancy cluster containing α vacancies, the local equilibrium vacancy concentration is elevated above C_v^0 according to

$$C_v^e(\alpha) = C_v^0 \exp\left(\frac{\mu(\alpha)}{k_B T}\right) = \exp\left[-\frac{E_v^f - \mu(\alpha)}{k_B T}\right] \quad (54)$$

where $\mu(\alpha)$ is the vacancy chemical potential of the cluster. For an interstitial cluster containing β interstitials, the equilibrium vacancy concentration at the surface is lowered below C_v^0 :

$$C_v^e(\beta) = C_v^0 \exp\left(-\frac{\mu(\beta)}{k_B T}\right) = \exp\left[-\frac{E_v^f + \mu(\beta)}{k_B T}\right]. \quad (55)$$

We can have a unified expression for both types of clusters by defining the binding energy of a vacancy to a vacancy cluster as

$$E_b^{(v)}(\alpha) = E_v^f - \mu(\alpha) \quad (56)$$

and the binding energy of a vacancy to an interstitial cluster of size β is

$$E_b^{(i)}(\beta) = E_v^f + \mu(\beta) \quad (57)$$

Then, the equilibrium vacancy concentration at the surface of a vacancy or interstitial cluster can be written as

$$C_v^e = \exp\left[-\frac{E_b}{k_B T}\right] \quad (58)$$

The binding energy can be found from atomistic calculations (e.g. DFT or MD), from experiments, or using continuum theory for large clusters. We will find explicit forms for α -iron in the next section.

For a spherical cavity of radius

$$R(\alpha) = a \left(\frac{3\alpha}{8\pi}\right)^{1/3}, \quad (59)$$

the defect chemical potential for vacancy exchange may be written as

$$\mu(\alpha) = \Omega \left(\frac{2\gamma}{R(\alpha)} - P_g - \sigma_h \right), \quad (60)$$

where γ is the surface energy, Ω is the atomic volume, σ_h is the hydrostatic *tension* component of the stress, and P_g is the internal gas pressure. The gas pressure inside a helium bubble may be represented more accurately by a virial equation of state rather than by a hard-core Van der Waals approximation. Writing the helium number density in the bubble as

$$n_{\text{He}} = \frac{m}{V_b} = \frac{m}{\alpha\Omega}, \quad (61)$$

where m is the number of helium atoms in the bubble, α is the number of vacancies, Ω is the atomic volume, and $V_b = \alpha\Omega$ is the bubble volume, the pressure may be written as

$$P_g = n_{\text{He}} k_B T \left[1 + B_2(T) n_{\text{He}} + B_3(T) n_{\text{He}}^2 + \dots \right]. \quad (62)$$

Equivalently,

$$P_g = \frac{mk_B T}{\alpha\Omega} \left[1 + B_2(T) \frac{m}{\alpha\Omega} + B_3(T) \left(\frac{m}{\alpha\Omega} \right)^2 + \dots \right]. \quad (63)$$

Here $B_2(T)$, $B_3(T)$, $\dots$ are the second, third, and higher virial coefficients of helium, which are functions of temperature. At temperatures in the range 600–1000 K, representative average virial coefficients for ^4He are

$$B_2 \approx 1.67 \times 10^{-29} \text{ m}^3/\text{atom}, \quad (64)$$

$$B_3 \approx 1.84 \times 10^{-58} \text{ m}^6/\text{atom}^2, \quad (65)$$

where these units are to be interpreted consistently with n_{He} in atoms m^{-3} . These values correspond to $\overline{B}_2 \approx 10.06 \text{ cm}^3 \text{ mol}^{-1}$ and $\overline{B}_3 \approx 66.72 \text{ cm}^6 \text{ mol}^{-2}$ for ^4He over 600–1000 K [Hurly and Moldover \(2000\)](#); [Garberoglio and Harvey \(2009\)](#).

5.1 Voids

The binding energy of a vacancy to a void is expressed as

$$E_b^{\text{mod}}(\alpha) = E_b^{\text{cont}}(\alpha) + [E_b^{(v)}(2) - E_b^{\text{cont}}(2)] \exp(-\lambda(\alpha - 2)) \quad (66)$$

With $E_b^{(v)}(2)$ the divacancy binding energy from DFT, and without external stress, the continuum capillary term is

$$E_b^{\text{cont}}(\alpha) = E_v^f - \Omega \frac{2\gamma}{R(\alpha)}, \quad R(\alpha) = a \left(\frac{3\alpha}{8\pi} \right)^{1/3} \quad (67)$$

The decay constant is chosen so that the correction is 1% at $\alpha = 10$: $\lambda = \frac{\ln 100}{8} \approx 0.575 \text{ vac}^{-1}$. The binding energy of a vacancy to a void as a function of the void size

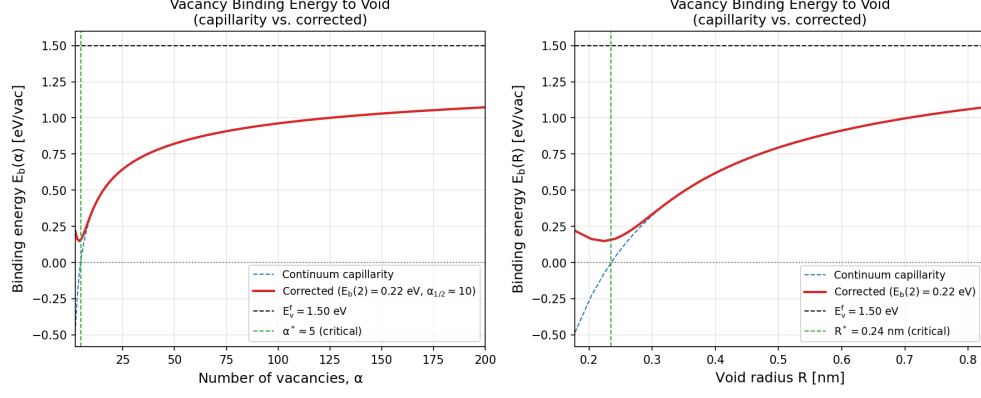


Fig. 4 Binding energy of a vacancy to a void as a function of the number of vacancies in the void

is shown in Fig. 4, using the capillary approximation for large sizes and transitioning to atomistic data for small sizes using Eq. 66.

5.2 Bubbles

A helium–vacancy cluster $\text{He}_m\text{V}_\alpha$ is characterised by two integers: α (the number of vacancies) and m (the number of trapped helium atoms). Cluster dynamics requires the binding energy of both a vacancy and a helium atom to such a cluster, since these two binding energies govern, respectively, the vacancy emission rate (cavity growth/shrinkage) and the helium emission rate (gas release). We treat each in turn.

5.2.1 Atomistic Binding Energy Data

Table 17 collects representative DFT/MD binding energies for $\text{He}_m\text{V}_\alpha$ clusters in α -Fe. The values are recast from the dissociation energies tabulated by Fu and Willaime (Fu and Willaime 2005c) and Caturla et al. (Caturla et al 2008b), using the definitions:

$$E_B^V(\alpha, m) = E_f(\text{He}_m\text{V}_{\alpha-1}) + E_f(V) - E_f(\text{He}_m\text{V}_\alpha), \quad (68)$$

$$E_B^{\text{He}}(\alpha, m) = E_f(\text{He}_{m-1}\text{V}_\alpha) + E_f(\text{He}) - E_f(\text{He}_m\text{V}_\alpha). \quad (69)$$

Two important trends are visible in Table 17:

1. $E_B^V(\alpha, m)$ **increases** with m at fixed α : more helium inside a cavity raises its internal pressure, making it harder for the cavity to emit a vacancy. This is the mechanism by which helium stabilises bubbles against thermal dissolution.
2. $E_B^{\text{He}}(\alpha, m)$ **decreases** with m at fixed α : as the cavity fills with helium, the incremental binding of each additional He atom decreases because the available trapping volume shrinks and the He–He repulsion increases.

Table 17 Binding energies for vacancy and helium addition to $\text{He}_m\text{V}_\alpha$ clusters in α -Fe. Data from [Fu and Willaime \(2005c\)](#) and [Caturla et al \(2008b\)](#).

α	m	$E_B^V(\alpha, m)$ (eV)	$E_B^{\text{He}}(\alpha, m)$ (eV)
1	1	2.30	—
2	1	0.78	2.85
3	1	0.83	3.30
4	1	1.16	3.84
1	2	3.71	1.84
2	2	1.61	2.75
3	2	1.04	2.96
4	2	1.32	3.12
1	3	4.59	1.83
2	3	1.85	2.07
4	3	1.57	3.16
1	4	5.52	1.91
2	4	2.30	2.36
3	4	2.03	2.57
4	4	1.97	3.05

5.2.2 Vacancy Binding Energy to a Bubble

For the vacancy binding energy, we use the same atomistic –continuum transitional model developed for voids (Section 5.1), extended to include the internal gas pressure:

$$E_B^v(\alpha, m) = \underbrace{E_f^v - \Omega \left[\frac{2\gamma}{R(\alpha)} - P_g(\alpha, m) \right]}_{E_B^{v, \text{cont}}(\alpha, m)} + A(m) e^{-\lambda(\alpha-1)}, \quad (70)$$

with subsidiary expressions:

$$R(\alpha) = a \left(\frac{3\alpha}{8\pi} \right)^{1/3}, \quad (71)$$

$$P_g(\alpha, m) = \frac{m}{\alpha \Omega} k_B T \left[1 + B_2 \frac{m}{\alpha \Omega} + B_3 \left(\frac{m}{\alpha \Omega} \right)^2 \right], \quad (72)$$

$$\lambda = \frac{\ln 100}{8} \approx 0.5756 \text{ vac}^{-1}. \quad (73)$$

The continuum term $E_B^{v, \text{cont}}$ captures the competition between surface tension ($2\gamma/R$, which favours vacancy emission) and gas pressure (P_g , which opposes it). The exponential correction $A(m) \exp[-\lambda(\alpha-1)]$ bridges the atomistic small-cluster binding energies to the continuum limit, decaying to less than 1% by $\alpha = 10$.

The fitted amplitudes $A(m)$, obtained by least-squares regression of Eq. (70) against the atomistic data of [Osetsky et al \(2000\)](#) ($m = 0$) and [Caturla et al \(2008b\)](#) ($m \geq 1$), are:

Table 18
Least-squares
fitting amplitudes
for the vacancy
binding energy to
bubbles (Eq. 70).

m	$A(m)$ [eV]
0	1.2353
1	2.9064
2	3.4147
3	2.1504
4	-0.1590

The amplitude at each integer m_k is determined by:

$$A(m_k) = \frac{\sum_{i \in \mathcal{D}_k} \delta_i e^{-\lambda(\alpha_i - 1)}}{\sum_{i \in \mathcal{D}_k} e^{-2\lambda(\alpha_i - 1)}}, \quad \delta_i = E_B^{v, \text{atom}}(\alpha_i, m_k) - E_B^{v, \text{cont}}(\alpha_i, m_k), \quad (74)$$

where $\mathcal{D}_k$ is the subset of atomistic data at fixed m_k .

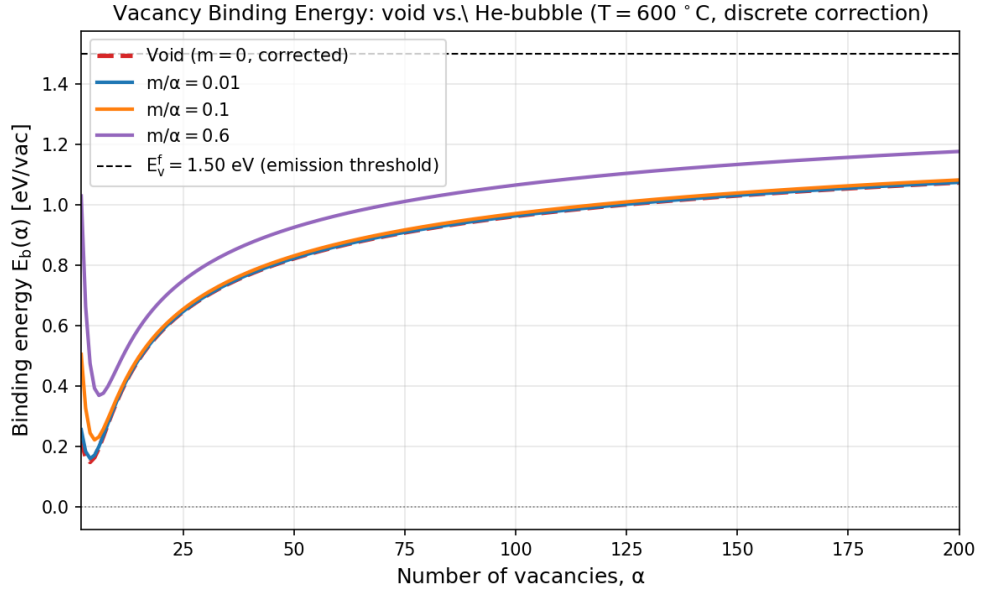


Fig. 5 Binding energy of a vacancy to a bubble as a function of the number of vacancies α , for several He-to-vacancy ratios m/α . The binding energy increases with helium content because the internal gas pressure opposes vacancy emission.

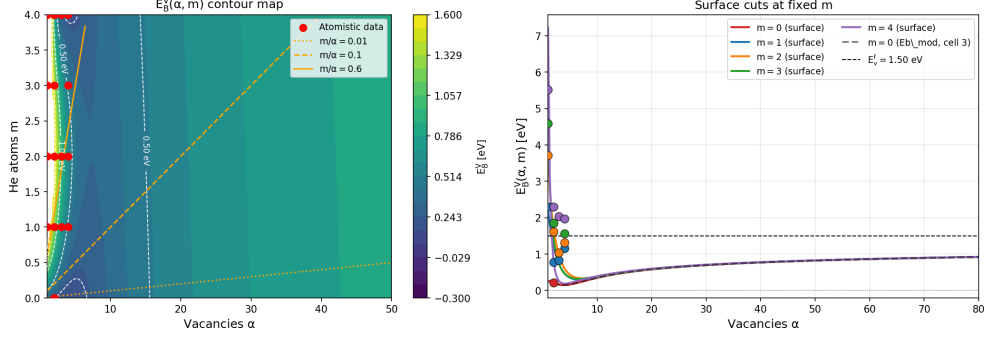


Fig. 6 Contour maps of vacancy binding energy $E_B^v(\alpha, m)$. Left: surface plot as a function of α and m . Right: cuts at fixed m showing the atomistic-continuum transition. The red line $E_B^v = E_f^v$ separates over-pressurised bubbles (above, which grow by absorbing thermal vacancies) from under-pressurised cavities (below, which tend to shrink).

5.2.3 Helium Binding Energy to a Bubble

The helium binding energy $E_B^{\text{He}}(\alpha, m)$ —the energy required to remove one He atom from a $\text{He}_m V_\alpha$ cluster and place it at a tetrahedral interstitial site in the bulk—is needed for the helium thermal emission (detrapping) rate and for determining the maximum stable He loading $\ell_{\text{max}}(\alpha)$.

Physical picture.

When a He atom is removed from a bubble of α vacancies containing m helium atoms, two energy contributions change: (i) the He-cavity binding interaction (governed by the attractive potential well of the vacant volume) is lost, and (ii) the internal gas pressure drops, slightly reducing the He-He repulsion among the remaining $m - 1$ atoms. For small clusters ($\alpha \leq 4$, $m \leq 4$), both contributions are highly configuration-dependent; for large bubbles they are captured by continuum thermodynamics.

Continuum model for large bubbles.

For a spherical bubble of radius $R(\alpha)$ containing m He atoms, the chemical potential of a helium atom inside the bubble is:

$$\mu_{\text{He}}^{\text{bubble}} = k_B T \ln\left(\frac{m}{\alpha\Omega} \Lambda_h^3\right) + k_B T \left[2B_2 \frac{m}{\alpha\Omega} + 3B_3 \left(\frac{m}{\alpha\Omega}\right)^2 \right], \quad (75)$$

where $\Lambda_h = h/\sqrt{2\pi m_{\text{He}} k_B T}$ is the thermal de Broglie wavelength of ^4He . The helium binding energy in the continuum limit is the energy difference between this chemical potential and the energy of an interstitial He atom in the bulk:

$$E_B^{\text{He,cont}}(\alpha, m) = E_s^{\text{He}} - k_B T \ln\left(\frac{m}{\alpha\Omega} \Lambda_h^3\right) - k_B T \left[2B_2 \frac{m}{\alpha\Omega} + 3B_3 \left(\frac{m}{\alpha\Omega}\right)^2 \right], \quad (76)$$

where $E_s^{\text{He}} \approx 2.3\text{--}2.4$ eV is the solution energy of interstitial He in bulk bcc Fe (the energy cost of inserting a He atom at a tetrahedral site). The first correction term is the ideal-gas entropy contribution, and the second accounts for the non-ideal (hard-core repulsive) He–He interactions through the virial coefficients.

Equation (76) predicts two important limits:

- **Low He density** ($m/\alpha \ll 1/\Omega B_2$): the binding energy is large ($\sim E_s^{\text{He}}$), since the He atom gains the full solution-energy benefit of entering the open volume.
- **High He density** (m/α approaching the close-packing limit): the binding energy decreases toward zero and eventually becomes negative, meaning it is energetically favourable for the He atom to leave the overpressurised bubble. This defines the maximum stable He loading.

Atomistic–continuum blending for small clusters.

By direct analogy with the vacancy binding energy model (Eq. 70), we write:

$$E_B^{\text{He}}(\alpha, m) = E_B^{\text{He,cont}}(\alpha, m) + A^{\text{He}}(\alpha) e^{-\mu(m-1)}, \quad (77)$$

where $A^{\text{He}}(\alpha)$ is a fitted amplitude at each vacancy count α , and μ is a decay constant governing how quickly the atomistic correction vanishes as m increases. The exponential correction ensures that the blended expression reproduces the DFT data at small m and converges to the continuum model for large m .

The decay constant is chosen by the same criterion as for the vacancy case (correction $< 1\%$ at $m = 8$):

$$\mu = \frac{\ln 100}{7} \approx 0.658 \text{ He}^{-1}. \quad (78)$$

The amplitudes $A^{\text{He}}(\alpha)$ are obtained by least-squares regression against the atomistic data of Table 17:

$$A^{\text{He}}(\alpha_k) = \frac{\sum_{j \in \mathcal{E}_k} \delta_j^{\text{He}} e^{-\mu(m_j-1)}}{\sum_{j \in \mathcal{E}_k} e^{-2\mu(m_j-1)}}, \quad \delta_j^{\text{He}} = E_B^{\text{He,atom}}(\alpha_k, m_j) - E_B^{\text{He,cont}}(\alpha_k, m_j), \quad (79)$$

where $\mathcal{E}_k$ is the subset of the atomistic data at fixed α_k .

Trends in the He binding energy.

Table 17 and Eq. (76) reveal the following trends:

1. At fixed α , E_B^{He} **decreases** as m increases (the cavity fills up and the marginal binding drops). For example, at $\alpha = 2$: $E_B^{\text{He}} = 2.85$ eV ($m = 1$), 2.75 eV ($m = 2$),

Table 19 Least-squares fitting amplitudes for the helium binding energy to bubbles (Eq. 77), obtained from the data of [Fu and Willaime \(2005c\)](#) and [Caturla et al \(2008b\)](#).

α	$A^{\text{He}}(\alpha)$ [eV]
1	~ 0 (He_1V_1 is the anchor; E_B^{He} undefined)
2	0.55
3	0.40
4	0.75

- 2.07 eV ($m = 3$), 2.36 eV ($m = 4$). The non-monotonic behaviour at $m = 4$ reflects configuration-dependent effects in the small DFT clusters.
- At fixed m , E_B^{He} **increases** as α increases (larger cavities have more open volume, so each He atom is more strongly bound). For example, at $m = 1$: $E_B^{\text{He}} = 2.85$ eV ($\alpha = 2$), 3.30 eV ($\alpha = 3$), 3.84 eV ($\alpha = 4$).
 - The He binding energy at $\alpha = 1$ (substitutional He) is ~ 1.8 – 1.9 eV for $m \geq 2$, significantly lower than at $\alpha \geq 2$. This reflects the small trapping volume of a single vacancy.

Helium emission versus trap mutation.

An overpressurised $\text{He}_m\text{V}_\alpha$ cluster—one whose He content exceeds the equilibrium loading—can relieve its excess energy by two physically distinct mechanisms. These must be distinguished in the rate equations because they produce different products and obey different kinetics.

(a) *Helium thermal emission..* The cluster releases a He atom into an interstitial (tetrahedral) site:



The vacancy content is unchanged; one He atom is lost. The emission rate is governed by the helium binding energy:

$$\Gamma_{\text{He}}(\alpha, m) = \mathcal{K}_{h,\alpha} \exp\left(-\frac{E_B^{\text{He}}(\alpha, m)}{k_B T}\right), \quad (81)$$

where $\mathcal{K}_{h,\alpha}$ is the He capture rate constant of the daughter cluster $\text{He}_{m-1}\text{V}_\alpha$. This channel dominates when the He binding energy is smaller than the energy cost of creating a Frenkel pair, i.e. when $E_B^{\text{He}} < E_f^i - \Omega P_g$, which is typically the case at high temperatures ($T \gtrsim 800$ K) and moderate He/V ratios.

(b) *Trap mutation (Frenkel pair emission)..* The cluster emits a self-interstitial atom from its periphery, simultaneously creating an additional vacancy within its own volume:



The He content is unchanged; the cluster gains one vacancy and injects one SIA into the lattice. This mechanism was first proposed by [Greenwood et al \(1959\)](#) as “plastic

deformation around an over-pressurized bubble”, and direct experimental evidence for loop punching by helium bubbles was reported by [Evans et al \(1981\)](#) in molybdenum. The atomistic process—sometimes called *self-trapping* when it occurs from a pure interstitial He cluster ($\alpha = 0$)—was characterised in detail by [Wilson et al \(1981\)](#) using molecular statics, and has been studied by DFT and MD in both bcc Fe ([Fu and Willaime 2007](#); [Gao et al 2011](#); [Morishita et al 2003](#)) and bcc W ([Becquart and Domain 2014b](#)).

The trap-mutation rate can be written in Arrhenius form:

$$\Gamma_{\text{TM}}(\alpha, m) = \nu_0 \exp\left(-\frac{E_{\text{TM}}(\alpha, m)}{k_B T}\right), \quad (83)$$

where $\nu_0 \sim 10^{12}\text{--}10^{13} \text{ s}^{-1}$ is an attempt frequency and E_{TM} is the activation barrier for Frenkel pair emission. DFT calculations by [Becquart and Domain \(2014b\)](#) for W and MD simulations by [Morishita et al \(2003\)](#) and [Gao et al \(2011\)](#) for Fe provide the following picture:

- For **pure interstitial He clusters** ($\alpha = 0$): self-trapping (spontaneous Frenkel pair creation) occurs at $m \approx 4\text{--}6$ in bcc Fe at $T > 300 \text{ K}$, with a barrier $E_{\text{TM}} \sim 0\text{--}0.5 \text{ eV}$ that decreases with increasing cluster size ([Gao et al 2011](#)).
- For **$\text{He}_m \text{V}_1$ clusters**: trap mutation occurs at $m \approx 5\text{--}7$, with barriers of $\sim 0.5\text{--}1.5 \text{ eV}$ that decrease steeply with m ([Morishita et al 2003](#); [Fu and Willaime 2007](#)).
- For **larger clusters** ($\alpha \geq 2$): the threshold He content for trap mutation increases roughly as $\alpha^{2/3}$ (reflecting the increased cavity volume), and the emitted SIA is strongly bound to the cluster surface (binding energy $\sim 2\text{--}3 \text{ eV}$) ([Becquart and Domain 2014b](#)).

Competition and the maximum stable He loading.

For a given $\text{He}_m \text{V}_\alpha$ cluster, the dominant overpressure relief channel is the one with the lower effective barrier. The competition can be summarised as:

$$\text{If } E_B^{\text{He}}(\alpha, m) < E_{\text{TM}}(\alpha, m) : \quad \text{He emission dominates,} \quad (84)$$

$$\text{If } E_{\text{TM}}(\alpha, m) < E_B^{\text{He}}(\alpha, m) : \quad \text{trap mutation dominates.} \quad (85)$$

In bcc Fe, DFT data indicate that for small clusters ($\alpha \leq 4$) at He/V ratios above $\sim 1.3\text{--}1.7$, the trap-mutation barrier drops below the He binding energy, making Frenkel pair emission the preferred pathway ([Fu and Willaime 2005c](#); [Morishita et al 2003](#)). This is consistent with the observation that the equilibrium He/V ratio for small clusters is $\sim 1.0\text{--}1.3$ (DFT), while for larger bubbles ($R > 1 \text{ nm}$) the mechanical equilibrium condition $P_g = 2\gamma/R$ gives He/V ~ 0.5 at moderate temperatures ([Trinkaas and Singh 2003](#)).

For practical implementation in cluster dynamics, the maximum He loading per vacancy cluster is:

$$\ell_{\max}(\alpha) = \lfloor \alpha_{\text{He}} \alpha^{2/3} \rfloor, \quad \alpha_{\text{He}} \approx 1.5\text{--}2.0. \quad (86)$$

This formula should be understood as the *trap-mutation limit*: beyond $\ell_{\max}$, the cluster spontaneously emits a Frenkel pair rather than accommodating an additional He atom. The He emission limit (where $E_B^{\text{He}} = 0$) occurs at a somewhat higher He/V ratio, but the trap-mutation channel intervenes first and prevents the cluster from reaching that state.

Rate-equation implementation.

In the master equation for $\text{He}_m \text{V}_\alpha$ clusters (Eq. 87), both channels appear as separate loss/gain terms:

$$\left. \frac{dC_{-\alpha, m}}{dt} \right|_{\text{TM+He em}} = \underbrace{-\Gamma_{\text{He}}(\alpha, m) C_{-\alpha, m} + \Gamma_{\text{He}}(\alpha, m+1) C_{-\alpha, m+1}}_{\text{He emission}} \quad (87)$$

$$\underbrace{-\Gamma_{\text{TM}}(\alpha, m) C_{-\alpha, m} + \Gamma_{\text{TM}}(\alpha-1, m) C_{-(\alpha-1), m}}_{\text{trap mutation}}. \quad (88)$$

The trap-mutation term is a loss from $(-\alpha, m)$ and a gain for $(-(\alpha+1), m)$; simultaneously, an SIA is injected into the mono-SIA population:

$$\left. \frac{dC_{1,0}}{dt} \right|_{\text{TM}} = \sum_{\alpha=1}^M \sum_{m=1}^L \Gamma_{\text{TM}}(\alpha, m) C_{-\alpha, m}. \quad (89)$$

For clusters exceeding $\ell_{\max}(\alpha)$, a simpler implementation is to enforce $m \leq \ell_{\max}(\alpha)$ as a hard boundary in the state space and redirect all He absorption events that would violate this bound into the trap-mutation channel, as described in Section 5.2.3.

Code-ready parameters.

5.2.4 Summary of Bubble Binding Energy Models

For implementation in cluster dynamics, the vacancy and helium binding energies to a $\text{He}_m \text{V}_\alpha$ bubble are:

Vacancy binding:

$$E_B^v(\alpha, m) = E_f^v - \Omega \left[\frac{2\gamma}{R(\alpha)} - P_g(\alpha, m) \right] + A(m) e^{-\lambda(\alpha-1)}, \quad (90)$$

with $A(m)$ from Table 18 and P_g from the virial equation of state (Eq. 72).

Table 20 Trap mutation parameters for implementation in cluster dynamics.

Parameter	Symbol	Value	Source
Attempt frequency	ν_0	$10^{12}\text{--}10^{13} \text{ s}^{-1}$	—
$E_{\text{TM}}, \text{He}_5\text{V}_1$		$\sim 1.0 \text{ eV}$	Morishita et al (2003)
$E_{\text{TM}}, \text{He}_6\text{V}_1$		$\sim 0.5 \text{ eV}$	Morishita et al (2003)
$E_{\text{TM}}, \text{He}_7\text{V}_1$		$\sim 0 \text{ eV}$ (spontaneous)	Gao et al (2011)
Self-trapping, He_4 ($\alpha = 0$)		$\sim 0.3 \text{ eV}$	Gao et al (2011)
Self-trapping, He_6 ($\alpha = 0$)		$\sim 0 \text{ eV}$ (spontaneous)	Gao et al (2011)
Max He loading parameter	α_{He}	$1.5\text{--}2.0$	Morishita et al (2003); Fu and Willaime (2005c)

Helium binding:

$$E_B^{\text{He}}(\alpha, m) = E_s^{\text{He}} - k_B T \ln\left(\frac{m}{\alpha\Omega} \Lambda_h^3\right) - k_B T \left[2B_2 \frac{m}{\alpha\Omega} + 3B_3 \left(\frac{m}{\alpha\Omega}\right)^2 \right] + A^{\text{He}}(\alpha) e^{-\mu(m-1)}, \quad (91)$$

with $A^{\text{He}}(\alpha)$ from Table 19, $E_s^{\text{He}} \approx 2.35 \text{ eV}$, and $\mu \approx 0.658 \text{ He}^{-1}$.

Equilibrium condition.

A bubble is in mechanical equilibrium when the internal gas pressure balances the surface tension:

$$P_g(\alpha, m) = \frac{2\gamma}{R(\alpha)}. \quad (92)$$

This condition simultaneously corresponds to $E_B^v = E_f^v$ (no net driving force for vacancy exchange). Bubbles above this line (over-pressurised) grow by absorbing thermal vacancies; those below (under-pressurised) shrink.

Thermal emission rates.

The vacancy and helium emission rates from a $\text{He}_m\text{V}_\alpha$ bubble are:

$$\epsilon_\alpha^v = \mathcal{K}_{v,\alpha-1} \exp\left(-\frac{E_B^v(\alpha, m)}{k_B T}\right), \quad (93)$$

$$\epsilon_m^{\text{He}} = \mathcal{K}_{h,\alpha} \exp\left(-\frac{E_B^{\text{He}}(\alpha, m)}{k_B T}\right), \quad (94)$$

where $\mathcal{K}_{v,\alpha-1}$ and $\mathcal{K}_{h,\alpha}$ are the vacancy and helium capture rate constants for the daughter cluster, as defined in the reaction rate section (Section 6.5).

The results are shown in Fig. 5 for the binding energy as a function of both m and α . Contours of constant binding energy in bubbles containing m helium atoms and α vacancies are shown in Fig. 7. The line $E_b^f = E_b^{(v)}(m, \alpha)$ defines the combination of m and α for bubbles in mechanical equilibrium. Those above the line (over-pressurized) will tend to grow by absorbing thermal vacancies, and those below the line will tend to shrink.

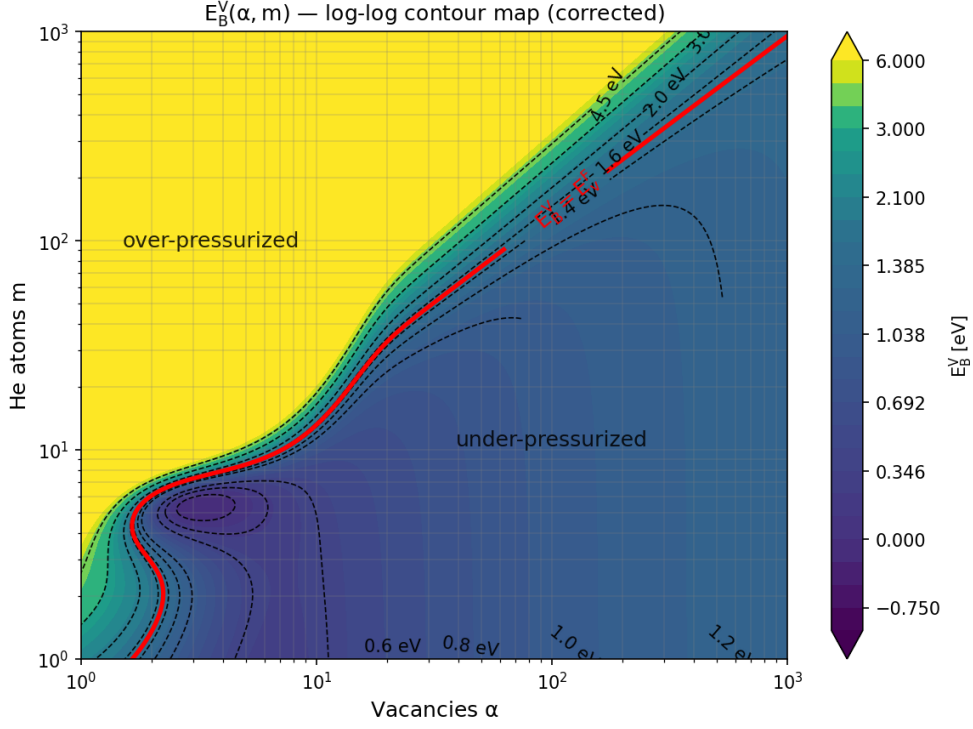


Fig. 7 Vacancy binding energy contours to bubbles containing m helium atoms and α vacancies

5.3 Dislocation Loops

For dislocation loops, the vacancy exchange chemical potential should be obtained from the loop free energy.

$$\mu = \frac{\partial G_{\text{loop}}}{\partial n}, \quad (95)$$

where n is the number of point defects contained in the loop. For a faulted circular loop, G_{loop} generally contains both line-energy and fault-energy contributions. For a circular prismatic loop containing n point defects,

$$n\Omega = \pi R^2 b, \quad (96)$$

For a loop in an external stress field σ_{ij}^{ext} , the loop free energy should include the mechanical work term,

$$G_{\text{loop}}^{\text{tot}} = G_{\text{self}} + G_{\text{fault}} + G_{\sigma}, \quad (97)$$

with

$$G_{\sigma} = -\sigma_{ij}^{\text{ext}} \Omega_{ij}^{\text{loop}}. \quad (98)$$

For a circular prismatic loop of radius R , plane normal n_i , and Burgers vector $b_i = b n_i$, the loop relaxation-volume tensor is

$$\Omega_{ij}^{\text{loop}} = s n \Omega n_i n_j = s \pi R^2 b n_i n_j, \quad (99)$$

where

$$s = \begin{cases} -1, & \text{vacancy loop,} \\ +1, & \text{interstitial loop.} \end{cases} \quad (100)$$

Hence

$$G_\sigma = -s n \Omega \sigma_{ij}^{\text{ext}} n_i n_j = -s \pi R^2 b \sigma_{ij}^{\text{ext}} n_i n_j. \quad (101)$$

The point-defect exchange chemical potential is therefore

$$\mu = \frac{\partial G_{\text{loop}}^{\text{tot}}}{\partial n} = \frac{\partial G_{\text{self}}}{\partial n} + \frac{\partial G_{\text{fault}}}{\partial n} - s \Omega \sigma_{ij}^{\text{ext}} n_i n_j. \quad (102)$$

For a faulted circular vacancy loop ($s = -1$),

$$\mu_{\text{loop}}^{(v)} = \frac{Gb\Omega}{4\pi(1-\nu)R} \left[\ln\left(\frac{R}{r_c}\right) + 1 \right] + \frac{\gamma_s f \Omega}{b} + \Omega \sigma_{ij}^{\text{ext}} n_i n_j. \quad (103)$$

For a faulted circular interstitial loop ($s = +1$),

$$\mu_{\text{loop}}^{(i)} = \frac{Gb\Omega}{4\pi(1-\nu)R} \left[\ln\left(\frac{R}{r_c}\right) + 1 \right] + \frac{\gamma_s f \Omega}{b} - \Omega \sigma_{ij}^{\text{ext}} n_i n_j. \quad (104)$$

r_c is the core radius of a dislocation. Utilizing Eqs. 57 and 56, we obtain the binding energy of a vacancy to an interstitial or to a vacancy loop, respectively.

5.4 Vacancy Loops

Osetsky et al. Osetsky et al (2000) studied the stability of vacancy loops in iron using atomistic simulations. Applying the continuum equations 57 and 56, together with Eq. 103 for the chemical potential of a vacancy loop without external stress, we compare the results of atomistic simulations to the continuum equations with $r_c = b/3$ and $\gamma_s f = 0.6 J/m^2$ in Fig. 8. A transition correction anchors the continuum to the atomistic divacancy binding energy $E_b(2) = 0.22$ eV via an exponential decay:

$$E_b(\alpha) = E_b^{\text{cont}}(\alpha) + [E_b(2) - E_b^{\text{cont}}(2)] e^{-\lambda(\alpha-2)} \quad (105)$$

$$\lambda = \frac{\ln 100}{8} \approx 0.575 \text{ vac}^{-1} \quad (106)$$

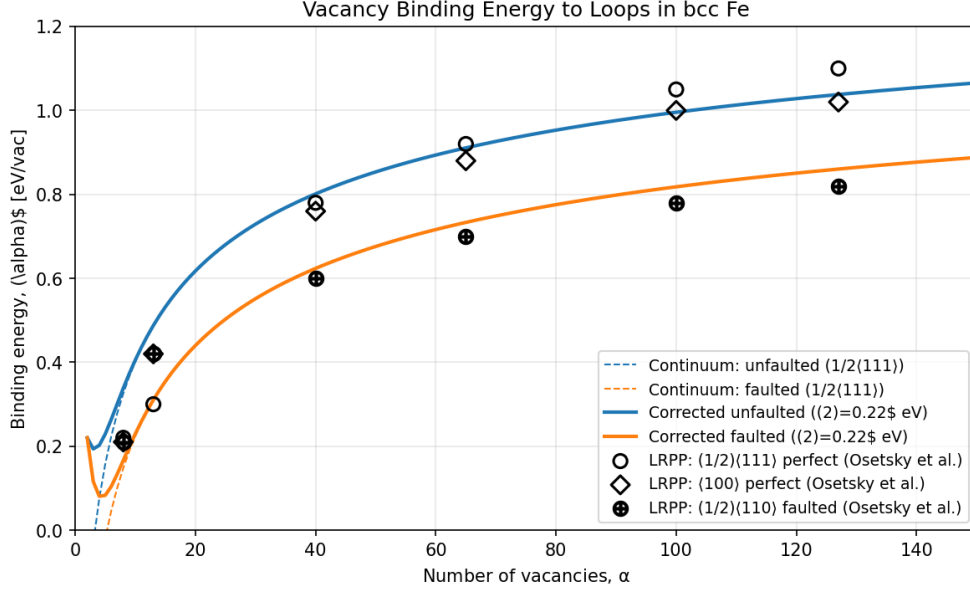


Fig. 8 Binding energy of a vacancy to a vacancy loop in iron using continuum equations (solid lines) and atomistic data from Osetsky et al. [Osetsky et al \(2000\)](#)

5.5 Interstitial Loops

Small- n Power-Law Fit ($n \leq 25$)

The small-cluster binding energy is fitted by least squares to the atomistic data for $n = 2$ (DFT) and $n = 7, 19$ (Osetsky MD, hexagonal and square loop configurations). The fit is written as

$$E_b^{i,\text{fit}}(n) = An^B. \quad (107)$$

The fitted parameters are listed in Table 21.

Table 21 Power-law fit parameters for small interstitial loops.

Loop type	A (eV)	B
$\frac{1}{2}\langle 111 \rangle$	0.7501	0.3873
$\langle 100 \rangle$	0.7160	0.3581

The atomistic training data used in the fit are summarized in Table 22.

Table 22 Atomistic training data used for the small- n binding-energy fit.

n	$E_b^i - \frac{1}{2}\langle 111 \rangle$ (eV)	$E_b^i - \langle 100 \rangle$ (eV)	Source
2	0.90	0.90	DFT (Domain & Becquart Domain and Becquart (2001) ; Marinica et al. Marinica et al (2011))
7 (hex)	1.65	1.30	Osetsky LRPP MD
7 (sq)	1.65	1.60	Osetsky LRPP MD
19 (hex)	2.35	2.05	Osetsky LRPP MD
19 (sq)	2.30	2.05	Osetsky LRPP MD

Blended Model with tanh Transition

To connect the small- n fit to the continuum expression, we use a smooth transition function

$$w(n) = \frac{1}{2} \left[1 + \tanh \left(\frac{n_{\text{tr}} - n}{\sigma_{\text{tr}}} \right) \right]. \quad (108)$$

The blended binding energy is then written as:

$$E_b^i(n) = w(n) E_b^{i,\text{fit}}(n) + [1 - w(n)] E_b^{i,\text{cont}}(n). \quad (109)$$

The transition parameters are listed in [Table 23](#).

Table 23 Parameters of the smooth blending function.

Parameter	Value
n_{tr} (transition center)	25 SIAs
σ_{tr} (transition half-width)	5 SIAs

Selected values of the weight function are given in [Table 24](#).

Table 24
Weight
function $w(n)$
at selected
cluster sizes.

n	$w(n)$
15	0.982
20	0.881
25	0.500
30	0.119
35	0.018

5.6 EUROFER Model Parameters: Defect Cluster Dissociation

Table 25 collects the binding-energy models and parameters controlling thermal emission from voids, bubbles, and dislocation loops. Table 26 provides the helium binding energy model parameters, and Table 27 lists the trap mutation parameters. Together, these three tables fully specify all dissociation channels in the cluster dynamics system.

Table 25 Dissociation / binding energy input parameters for EUROFER cluster dynamics: vacancy binding to voids and bubbles, and point-defect binding to dislocation loops.

Parameter	Symbol	Value	Units
<i>Void (vacancy cluster) binding — vacancy emission:</i>			
Vacancy formation energy	E_f^v	2.0	eV
Surface energy	γ_s	2.0	J/m ²
Divacancy binding (DFT)	$E_b^v(2)$	0.22	eV
Atomistic–continuum decay constant	λ	0.5756	vac ^{−1}
Atomistic–continuum anchor size	$\alpha_{1/2}$	10	vacancies
<i>Bubble — vacancy binding (Eq. 70):</i>			
He–vacancy binding (1st He in V_1)	$E_B^V(1, 1)$	2.30	eV
Fitting amplitude $A(m = 0)$	$A(0)$	1.2353	eV
Fitting amplitude $A(m = 1)$	$A(1)$	2.9064	eV
Fitting amplitude $A(m = 2)$	$A(2)$	3.4147	eV
Fitting amplitude $A(m = 3)$	$A(3)$	2.1504	eV
Fitting amplitude $A(m = 4)$	$A(4)$	−0.1590	eV
He virial B_2 (600–1000 K)	B_2	1.67×10^{-29}	m ³ /atom
He virial B_3 (600–1000 K)	B_3	1.84×10^{-58}	m ⁶ /atom ²
<i>Vacancy loop binding:</i>			
Stacking fault energy	γ_{sf}	0.6	J/m ²
Core radius	r_c	$b/3$	—
Divacancy binding anchor	$E_b(2)$	0.22	eV
Decay constant	λ	0.575	vac ^{−1}
<i>Interstitial loop binding (power-law fit, $n \leq 25$):</i>			
$\frac{1}{2}\langle 111 \rangle$ loop: A	A_{111}	0.7501	eV
$\frac{1}{2}\langle 111 \rangle$ loop: B	B_{111}	0.3873	—
$\langle 100 \rangle$ loop: A	A_{100}	0.7160	eV
$\langle 100 \rangle$ loop: B	B_{100}	0.3581	—
Blending transition center	n_{tr}	25	SIA _s
Blending transition width	σ_{tr}	5	SIA _s

Implementation notes.

1. The vacancy emission rate from a bubble $\text{He}_m V_\alpha$ is: $\epsilon_\alpha^v = \mathcal{K}_{v,\alpha-1} \exp(-E_B^v(\alpha, m)/k_B T)$, using E_B^v from Table 25.

Table 26 Helium binding energy model parameters for bubbles (Eq. 91). The continuum term uses the He solution energy, the thermal de Broglie wavelength, and the virial coefficients from Table 25. The atomistic correction decays exponentially with m .

Parameter	Symbol	Value	Units
<i>Continuum He binding (Eq. 76):</i>			
He interstitial solution energy	E_s^{He}	2.35	eV
He atomic mass	m_{He}	4.0026	amu
He virial B_2	B_2	1.67×10^{-29}	m^3/atom
He virial B_3	B_3	1.84×10^{-58}	m^6/atom^2
<i>Atomistic-continuum blending (Eq. 77):</i>			
Decay constant	μ	0.658	He^{-1}
Fitting amplitude $A^{\text{He}}(\alpha = 2)$	$A^{\text{He}}(2)$	0.55	eV
Fitting amplitude $A^{\text{He}}(\alpha = 3)$	$A^{\text{He}}(3)$	0.40	eV
Fitting amplitude $A^{\text{He}}(\alpha = 4)$	$A^{\text{He}}(4)$	0.75	eV

Table 27 Trap mutation (Frenkel pair emission) parameters for EUROFER cluster dynamics. Trap mutation is the reaction $\text{He}_m \text{V}_\alpha \rightarrow \text{He}_m \text{V}_{\alpha+1} + \text{SIA}$, with rate $\Gamma_{\text{TM}} = \nu_0 \exp(-E_{\text{TM}}/k_B T)$. Self-trapping refers to the $\alpha = 0$ case (pure interstitial He cluster). Values from Morishita et al (2003), Gao et al (2011), Fu and Willaime (2005c), and Becquart and Domain (2014b).

Parameter	Symbol	Value	Units
<i>Trap mutation rate: $\Gamma_{\text{TM}} = \nu_0 \exp(-E_{\text{TM}}/k_B T)$</i>			
Attempt frequency	ν_0	$10^{12}\text{--}10^{13}$	s^{-1}
<i>Activation barriers $E_{\text{TM}}(\alpha, m)$ for $\text{He}_m \text{V}_\alpha$ in bcc Fe:</i>			
$\text{He}_5 \text{V}_1$	$E_{\text{TM}}(1, 5)$	~ 1.0	eV
$\text{He}_6 \text{V}_1$	$E_{\text{TM}}(1, 6)$	~ 0.5	eV
$\text{He}_7 \text{V}_1$	$E_{\text{TM}}(1, 7)$	~ 0 (spontaneous)	eV
<i>Self-trapping barriers ($\alpha = 0$, pure He clusters):</i>			
He_4	$E_{\text{TM}}(0, 4)$	~ 0.3	eV
He_5	$E_{\text{TM}}(0, 5)$	~ 0.1	eV
He_6	$E_{\text{TM}}(0, 6)$	~ 0 (spontaneous)	eV
<i>Maximum He loading (trap-mutation limit):</i>			
He/V ratio parameter	α_{He}	1.5–2.0	—
$\ell_{\text{max}}(\alpha) = \lfloor \alpha_{\text{He}} \alpha^{2/3} \rfloor$			
<i>Competition criterion:</i>			
He emission dominates if $E_B^{\text{He}}(\alpha, m) < E_{\text{TM}}(\alpha, m)$			
Trap mutation dominates if $E_{\text{TM}}(\alpha, m) < E_B^{\text{He}}(\alpha, m)$			
In bcc Fe: trap mutation wins at $\text{He}/\text{V} \gtrsim 1.3\text{--}1.7$ for $\alpha \leq 4$			

2. The He emission rate is: $\epsilon_m^{\text{He}} = \mathcal{K}_{h,\alpha} \exp(-E_B^{\text{He}}(\alpha, m)/k_B T)$, using E_B^{He} from Table 26.
3. The trap-mutation rate is: $\Gamma_{\text{TM}} = \nu_0 \exp(-E_{\text{TM}}(\alpha, m)/k_B T)$, using E_{TM} from Table 27. Each trap-mutation event is a loss from $(-\alpha, m)$ and a simultaneous gain for $(-(\alpha + 1), m)$, with one SIA added to the mono-interstitial population.
4. For clusters at or above $\ell_{\text{max}}(\alpha)$, trap mutation can be enforced as a hard boundary: any He absorption event that would push $m > \ell_{\text{max}}$ is immediately redirected into a Frenkel pair emission step.
5. The ℓ_{max} formula represents the trap-mutation limit, not the He emission limit. The He emission limit (where $E_B^{\text{He}} = 0$) lies at a somewhat higher He/V ratio, but trap mutation intervenes first.

6 Reaction Rate Constants

This section derives the rate constants for all reactions between defect clusters, expressed in atomic-fraction units (s^{-1}) and in terms of the reaction frequencies $\omega_\alpha = D_\alpha/a^2$ and the geometric prefactor $\xi(m) = (3m/8\pi)^{1/3}$ introduced in the diffusion section. The cluster notation (m, ℓ) and concentration units (at/at) follow the conventions established in Section 4.

6.1 3D Diffusion-Limited Reactions

For a mobile point defect α absorbed by an immobile spherical cluster of m defects, the Smoluchowski volumetric rate constant is $\mathcal{K}^{\text{vol}} = 4\pi r_m(D_A + D_B)$. Converting to atomic-fraction units ($\mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega$) and substituting $D = \omega a^2$, $r_m = \xi(m)a$, and $\Omega = a^3/2$:

$$\mathcal{K}_{\alpha,m}^{3D} = 8\pi \xi(m) \omega_\alpha \quad [\text{s}^{-1}]. \quad (110)$$

When both reactants are mobile, the rate constant includes both frequencies:

$$\mathcal{K}_{AB}^{3D} = 8\pi \xi_{\text{rec}} (\omega_A + \omega_B), \quad (111)$$

with $\xi_{\text{rec}} = \sqrt{3}/2 \approx 0.866$ for nearest-neighbour capture in bcc.

Special cases.

Vacancy–SIA recombination ($m = n = 1$, both mobile):

$$\mathcal{K}_{iv} = 4\sqrt{3}\pi (\omega_i + \omega_v) \approx 21.8 \omega_i, \quad (112)$$

since $\omega_i \gg \omega_v$ at all relevant temperatures.

Point defect absorption by a spherical cluster (void or bubble) of m vacancies:

$$\mathcal{K}_{\alpha,m}^{\text{void}} = A_{\text{sph}} m^{1/3} \omega_\alpha, \quad A_{\text{sph}} = (48\pi^2)^{1/3} \approx 7.818. \quad (113)$$

This is the recommended code-ready form: a single precomputed constant multiplied by $m^{1/3}$ and the appropriate reaction frequency.

Point defect absorption by a dislocation loop of n SIAs requires species-specific rate constants because only SIAs experience a non-trivial elastic bias ($Z_i^{\text{loop}} > 1$), while vacancies are unbiased ($Z_v^{\text{loop}} = 1$):

$$\mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i, \quad (114)$$

$$\mathcal{K}_{v,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} \omega_v, \quad (115)$$

with $A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \approx 10.78$ and $Z_i^{\text{loop}} \approx 1.02$ – 1.10 (Table 29). SIA absorption grows the loop by one; vacancy absorption shrinks it by one.

Point defect absorption by the dislocation network (first-order sink):

$$\left. \frac{dc_\alpha}{dt} \right|_{\text{disl}} = -Z_\alpha \rho_d a^2 \omega_\alpha c_\alpha, \quad (116)$$

where ρ_d (m^{-2}) is the dislocation line density and no n_{at} conversion is needed because this term is first-order in c_α .

6.2 1D Diffusion-Limited Reactions (SIA Clusters)

SIA clusters of size $n \geq 4$ migrate by 1D glide along $\langle 111 \rangle$ with reaction frequency (from Section 4.5):

$$\omega_n^{1D} = \frac{3}{2} \cdot \frac{\nu_0}{n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad s \approx 0.5-1.0, \quad E_m^{1D} \lesssim 0.05 \text{ eV}. \quad (117)$$

6.2.1 Mixed 1D/3D regime (finite mean free path)

The effective 3D reaction frequency after Burgers vector rotation is:

$$\omega_n^{3D, \text{eff}} = \frac{\omega_n^{1D} L}{3\sqrt{3} a}. \quad (118)$$

The rate constant for an n -SIA cluster reacting with an immobile void of m vacancies is then:

$$\mathcal{K}_{n,m}^{\text{mixed}} = A_{\text{sph}} m^{1/3} \omega_n^{3D, \text{eff}} = \frac{A_{\text{sph}}}{3\sqrt{3}} m^{1/3} \hat{L} \omega_n^{1D} \approx 1.505 m^{1/3} \hat{L} \omega_n^{1D}, \quad (119)$$

where $\hat{L} = L/a$ is the dimensionless mean free path.

6.2.2 Pure 1D regime

For a 1D-migrating cluster reacting with randomly distributed spherical sinks of size m and atomic fraction $c_{m,0}$, the loss rate is:

$$\left. \frac{dc_n}{dt} \right|_{\text{sink}} = -\mathcal{K}_{n,m}^{1D} \bar{c}_m^2 c_n, \quad (120)$$

where:

$$\mathcal{K}_{n,m}^{1D} = A_{1D} m^{4/3} \omega_n^{1D}, \quad A_{1D} = \frac{9}{8\pi^{2/3}} \approx 2.632. \quad (121)$$

The critical difference from 3D: the rate constant scales as $m^{4/3}$ (not $m^{1/3}$) and is second-order in sink concentration (not first-order). This stronger sensitivity to sink geometry is the physical origin of the production bias.

6.2.3 Interpolation formula

For intermediate L , the Trinkaus–Singh–Golubov interpolation gives:

$$\mathcal{K}_{n,m}^{\text{eff}} = \frac{A_{\text{sph}} m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}, \quad B_{\text{rot}} = \frac{4}{\pi} \left(\frac{8\pi}{3} \right)^{1/3} \approx 2.627. \quad (122)$$

In the limits: $\hat{L} \rightarrow 0$ recovers the 3D rate constant $A_{\text{sph}} m^{1/3} \omega_n^{1D}$; $\hat{L} \rightarrow \infty$ gives a rate proportional to $m^{2/3} \hat{L}^{-2}$, scaling as r^2 as expected for 1D kinetics.

6.3 Thermal Emission Rates

Thermal emission of species α from a cluster of size n is related to the capture rate by detailed balance. The daughter cluster has size $(n-1)$, so the capture rate that enters the detailed-balance relation uses $(n-1)$:

$$\alpha_\alpha(n) = A_{\text{sph}} (n-1)^{1/3} \omega_\alpha \exp\left(-\frac{E_b^\alpha(n)}{k_B T}\right) \quad [\text{s}^{-1}], \quad (123)$$

where $E_b^\alpha(n)$ is the binding energy of the last defect of type α to the cluster (from Section 5). The emission enters the master equation as a first-order process:

$$\left. \frac{dc_n}{dt} \right|_{\text{emission}} = +\alpha_\alpha(n) c_n - \alpha_\alpha(n+1) c_{n+1}. \quad (124)$$

For a dislocation loop emitting an SIA, replace $A_{\text{sph}} (n-1)^{1/3}$ with $A_{\text{loop}} (n-1)^{1/2}$:

$$\alpha_i^{\text{loop}}(n) = A_{\text{loop}} (n-1)^{1/2} \omega_i \exp\left(-\frac{E_b^i(n)}{k_B T}\right). \quad (125)$$

For $\text{He}_\ell \text{V}_m$ bubbles, three emission channels exist (Section 5.2):
Vacancy emission:

$$\varepsilon_v(m, \ell) = A_{\text{sph}} (m-1)^{1/3} \omega_v \exp\left(-\frac{E_b^v(m, \ell)}{k_B T}\right). \quad (126)$$

Helium emission:

$$\varepsilon_h(m, \ell) = A_{\text{sph}} m^{1/3} \omega_h \exp\left(-\frac{E_B^{\text{He}}(m, \ell)}{k_B T}\right). \quad (127)$$

He emission uses $m^{1/3}$ (not $(m-1)^{1/3}$) because removal of a He atom does not change the vacancy count; the capture radius for detailed balance is that of the daughter $\text{He}_{\ell-1} \text{V}_m$, which retains the same m . The binding energy E_B^{He} is from Eq. (91).

Trap mutation (Frenkel pair emission):

$$\Gamma_{\text{TM}}(m, \ell) = \nu_0 \exp\left(-\frac{E_{\text{TM}}(m, \ell)}{k_B T}\right), \quad (128)$$

producing $\text{He}_\ell \text{V}_{m+1} + \text{SIA}$. The trap-mutation barrier E_{TM} and the competition with He emission are discussed in Section 5.2.3; recommended values are in Table 27.

6.4 Summary of Precomputed Constants

All reaction-rate geometry is captured by four numerical constants:

$$\begin{aligned} A_{\text{sph}} &= (48\pi^2)^{1/3} && \approx 7.818 \quad (\text{spherical clusters}), \\ A_{\text{loop}} &= 8\sqrt{\pi/\sqrt{3}} && \approx 10.78 \quad (\text{dislocation loops}), \\ A_{1D} &= 9/(8\pi^{2/3}) && \approx 2.632 \quad (\text{pure 1D sinks}), \\ B_{\text{rot}} &= (4/\pi)(8\pi/3)^{1/3} && \approx 2.627 \quad (1\text{D}/3\text{D interpolation}). \end{aligned} \quad (129)$$

Every rate constant in the cluster dynamics system is built from these constants, the cluster size raised to a geometry-dependent power, and the appropriate reaction frequency ω_α . Table 28 collects the complete set.

6.5 EUROFER Model Parameters: Reaction Rates

Table 29 collects the rate-constant prefactors, sink parameters, and state-space controls used in the reaction-rate expressions (Section 6). Table 30 provides the complete set of rate constant formulas for direct implementation.

Implementation notes.

Every rate constant is built from one of the four precomputed geometric constants (A_{sph} , A_{loop} , A_{1D} , B_{rot}), the cluster size raised to a geometry-dependent power, and the appropriate effective reaction frequency $\omega_\alpha^{\text{eff}}$ (Eq. 201). The binding energies $E_b^\alpha(m, \ell)$ for voids and loops come from Section 5.6; the bubble binding energies $E_B^v(m, \ell)$ and $E_B^{\text{He}}(m, \ell)$ from Tables 18 and 19; and the trap mutation barriers from Table 27. He emission dominates when $E_B^{\text{He}} < E_{\text{TM}}$; trap mutation dominates when $E_{\text{TM}} < E_B^{\text{He}}$, typically at $\text{He}/\text{V} \gtrsim 1.3$ – 1.7 for small clusters in bcc Fe.

Table 28 Summary of rate constant expressions. All rate constants are in s^{-1} and all concentrations in at/at. The effective reaction frequencies $\omega_\alpha^{\text{eff}}$ incorporate solute-trapping corrections (Table 16) and should replace ω_α in all rate-equation implementations.

Process	Rate constant	Rate equation term
V-SIA recombination	$\mathcal{K}_{iv} = 4\sqrt{3}\pi(\omega_i^{\text{eff}} + \omega_v^{\text{eff}})$	$-\mathcal{K}_{iv} c_1 c_{1,0}$
α capture by void/bubble (m, ℓ)	$\mathcal{K}_{\alpha,m} = A_{\text{sph}} m^{1/3} \omega_\alpha^{\text{eff}}$	$-\mathcal{K}_{\alpha,m} c_\alpha c_{m,\ell}$
i capture by loop n (biased)	$\mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}$	$-\mathcal{K}_{i,n}^{\text{loop}} c_1 c_n$
v capture by loop n (unbiased)	$\mathcal{K}_{v,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} \omega_v^{\text{eff}}$	$-\mathcal{K}_{v,n}^{\text{loop}} c_{1,0} c_n$
Dislocation sink	$\mathcal{D}_\alpha^d = Z_\alpha^d \rho_d a^2 \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^d c_\alpha$
Grain-boundary sink	$\mathcal{D}_\alpha^{gb} = (4\pi^2/d_g^2) a^2 \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^{gb} c_\alpha$
Precipitate sink	$\mathcal{D}_\alpha^p = (8\pi r_p/a) c_p^N \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^p c_\alpha$
SIA cluster (mixed 1D/3D)	$\mathcal{K}_{n,m}^{\text{eff}} = \frac{A_{\text{sph}} m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}$	$-\mathcal{K}_{n,m}^{\text{eff}} \bar{c}_m c_n$
SIA cluster (pure 1D)	$\mathcal{K}_{n,m}^{1D} = A_{1D} m^{4/3} \omega_n^{1D}$	$-\mathcal{K}_{n,m}^{1D} \bar{c}_m^2 c_n$
V emission from void/bubble (m, ℓ)	$\varepsilon_v(m, \ell) = A_{\text{sph}} (m-1)^{1/3} \omega_v^{\text{eff}} e^{-E_b^v/k_B T}$	first-order in $c_{m,\ell}$
SIA emission from loop n	$\varepsilon_i(n) = A_{\text{loop}} (n-1)^{1/2} \omega_i^{\text{eff}} e^{-E_b^i/k_B T}$	first-order in c_n
He emission from bubble (m, ℓ)	$\varepsilon_h(m, \ell) = A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} e^{-E_B^{\text{He}}/k_B T}$	first-order in $c_{m,\ell}$
Trap mutation	$\Gamma_{\text{TM}}(m, \ell) = \nu_0 e^{-E_{\text{TM}}/k_B T}$	$(m, \ell) \rightarrow (m+1, \ell) + \text{SIA}$
Radiation re-solution	$\Gamma_{\text{res}}(m, \ell) = b_0 \ell \dot{\phi}$	$(m, \ell) \rightarrow (m, \ell-1) + \text{He}_{\text{int}}$

Table 29 Reaction rate input parameters for EUROFER cluster dynamics.

Parameter	Symbol	Value	Units
<i>Geometric prefactors (precomputed constants):</i>			
Spherical cluster	$A_{\text{sph}} = (48\pi^2)^{1/3}$	7.818	—
Dislocation loop	$A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}}$	10.78	—
Pure 1D sink	$A_{1D} = 9/(8\pi^{2/3})$	2.632	—
1D/3D interpolation	$B_{\text{rot}} = (4/\pi)(8\pi/3)^{1/3}$	2.627	—
V-SIA recombination	$4\sqrt{3}\pi$	21.77	—
<i>Dislocation sink:</i>			
Network dislocation density	ρ_d	$10^{13}\text{--}10^{14}$	m^{-2}
Interstitial bias factor	Z_i^d	1.02–1.15	—
Vacancy bias factor	Z_v^d	1.00	—
<i>Grain-boundary sink:</i>			
Grain diameter	d_g	5–20	μm
Bias factors	$Z_i^{gb} = Z_v^{gb}$	1.00	—
<i>Precipitate sink:</i>			
Mean precipitate radius	r_p	10–50	nm
Precipitate number density	N_p	$10^{20}\text{--}10^{21}$	m^{-3}
Bias factors	$Z_i^p = Z_v^p$	1.00	—
<i>Mobility cutoffs:</i>			
Max mobile SIA cluster	i_{mobile}	100	SIAs
Max mobile vacancy cluster	v_{mobile}	5	vacancies
<i>He state-space parameters:</i>			
Max He loading parameter	α_{He}	1.5–2.0	—
$\ell_{\text{max}}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor$			(trap-mutation limit)
Re-solution parameter (fission)	b_0	10^{-2}	$\text{dpa}^{-1}/\text{He}$
Re-solution parameter (fusion)	b_0	10^{-1}	$\text{dpa}^{-1}/\text{He}$
<i>Trap mutation (Section 5.2.3):</i>			
Attempt frequency	ν_0^{TM}	$10^{12}\text{--}10^{13}$	s^{-1}
$E_{\text{TM}}(\text{He}_5\text{V}_1)$		~ 1.0	eV
$E_{\text{TM}}(\text{He}_6\text{V}_1)$		~ 0.5	eV
$E_{\text{TM}}(\text{He}_7\text{V}_1)$		~ 0 (spontaneous)	eV
<i>Size grid (user-defined):</i>			
Max SIA cluster tracked	N	user input	—
Max vacancy cluster tracked	M	user input	—
Max He per cluster	L	user input	—
Grouping transition size	n_{group}	$\sim 50\text{--}200$	—

Table 30 Complete rate constant expressions for all processes in the cluster dynamics system. All rate constants are in s^{-1} ; all concentrations in at/at . The effective frequencies $\omega_\alpha^{\text{eff}}$ are defined in Eq. (201). Binding energies E_b^α , E_B^v , E_B^{He} , and E_{TM} are from Section 5.

Process	Rate constant / rate expression
<i>Capture (bimolecular):</i>	
V-SIA recombination	$\mathcal{K}_{iv} = 4\sqrt{3}\pi (\omega_i^{\text{eff}} + \omega_v^{\text{eff}})$
α by void/bubble (m, ℓ)	$\mathcal{K}_{\alpha, m} = A_{\text{sph}} m^{1/3} \omega_\alpha^{\text{eff}}$
α by loop n (SIA, biased)	$\mathcal{K}_{i, n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}$
α by loop n (V, unbiased)	$\mathcal{K}_{v, n}^{\text{loop}} = A_{\text{loop}} n^{1/2} \omega_v^{\text{eff}}$
He by bubble (m, ℓ)	$\mathcal{K}_{h, m} = A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$
<i>Fixed sinks (first-order):</i>	
α to dislocation network	$\mathcal{D}_\alpha^d = Z_\alpha^d \rho_d a^2 \omega_\alpha^{\text{eff}}$
α to grain boundaries	$\mathcal{D}_\alpha^{gb} = (4\pi^2/d_g^2) a^2 \omega_\alpha^{\text{eff}}$
α to precipitates	$\mathcal{D}_\alpha^p = (8\pi r_p/a) c_p^N \omega_\alpha^{\text{eff}}$
<i>SIA cluster reactions (1D/3D):</i>	
Mixed 1D/3D with void/bubble m	$\mathcal{K}_{n, m}^{\text{eff}} = A_{\text{sph}} m^{1/3} \omega_n^{1D} / (1 + B_{\text{rot}} \hat{L}^2 m^{-1/3})$
Pure 1D with void/bubble m	$\mathcal{K}_{n, m}^{1D} = A_{1D} m^{4/3} \omega_n^{1D}$; rate $\propto \bar{c}_m^2 c_n$
<i>Thermal emission (first-order):</i>	
V from void/bubble (m, ℓ)	$\varepsilon_v(m, \ell) = A_{\text{sph}} (m-1)^{1/3} \omega_v^{\text{eff}} e^{-E_b^v(m, \ell)/k_B T}$
SIA from loop n	$\varepsilon_i(n) = A_{\text{loop}} (n-1)^{1/2} \omega_i^{\text{eff}} e^{-E_b^i(n)/k_B T}$
He from bubble (m, ℓ)	$\varepsilon_h(m, \ell) = A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} e^{-E_B^{\text{He}}(m, \ell)/k_B T}$
<i>Trap mutation (first-order):</i>	
$\text{He}_\ell \text{V}_m \rightarrow \text{He}_\ell \text{V}_{m+1} + \text{SIA}$	$\Gamma_{\text{TM}}(m, \ell) = \nu_0 e^{-E_{\text{TM}}(m, \ell)/k_B T}$
<i>Radiation re-solution (athermal):</i>	
He ejection from bubble	$\Gamma_{\text{res}}(m, \ell) = b_0 \ell \dot{\phi}$

7 Generalized Defect-Cluster Population Dynamics

This section develops an abstract master-equation formulation for defect-cluster evolution in irradiated crystalline solids. The formulation is designed to encompass the production of vacancies, self-interstitial atoms, and gas atoms by irradiation, their subsequent clustering into interstitial and vacancy-cluster populations, and the exchange of gas atoms between a matrix reservoir and vacancy clusters. The notation is chosen so that the same reaction list can be used either for deterministic cluster dynamics, in the mean-field rate-theory limit, or for stochastic cluster dynamics, in which the same processes are interpreted as discrete jump events.

The resolved cluster populations are restricted to two physical classes:

1. **Interstitial clusters**, denoted by I_n , where $n \geq 1$ is the number of self-interstitial atoms.
2. **Vacancy clusters**, denoted by $V_m \mathbf{g}_\ell$, where $m \geq 1$ is the number of vacancies and ℓ is a gas-occupancy vector. These clusters are also referred to as *cavities*; they represent pure voids when $\ell = \mathbf{0}$ and gas-filled bubbles when one or more components of ℓ are nonzero.

Gas atoms not trapped in cavities are represented by a matrix gas reservoir. Gas-gas reactions are not included: gas atoms may be produced, trapped by vacancy clusters, emitted from vacancy clusters, or returned to the matrix reservoir, but they do not form gas-only clusters in the present formulation.

7.1 Primitive Species and State Variables

Let the primitive irradiation-produced species be

$$\mathcal{E} = \{I, V, \mathbf{g}\}, \quad (130)$$

where I denotes a self-interstitial atom, V denotes a vacancy, and $\mathbf{g}$ denotes a vector of gas species. The gas vector may be null, corresponding to a gas-free material, or may include one or more gas species, for example

$$\mathbf{g} = (g_1, g_2, \dots, g_{N_g}) = (\text{He}, \text{H}, \text{T}, \text{Xe}, \dots). \quad (131)$$

The concentration of an interstitial cluster containing n SIAs is denoted by

$$c_n^I(t), \quad n = 1, \dots, N_I. \quad (132)$$

The concentration of a vacancy cluster, or cavity, containing m vacancies and gas occupancy vector

$$\ell = (\ell_1, \ell_2, \dots, \ell_{N_g}) \in \mathbb{Z}_{\geq 0}^{N_g} \quad (133)$$

is denoted by

$$c_{m,\ell}^V(t), \quad m = 1, \dots, N_V. \quad (134)$$

For example,

$$V_m \equiv V_m \mathbf{g}_0, \quad V_m \text{He}_\ell \equiv V_m \mathbf{g}_{(\ell)}, \quad (135)$$

and, for a multicomponent gas vector,

$$V_m \text{He}_\ell \text{T}_p \equiv V_m \mathbf{g}_{(\ell,p)}. \quad (136)$$

The matrix gas reservoir is represented by monomeric gas concentrations

$$c_\mu^g(t), \quad \mu = 1, \dots, N_g, \quad (137)$$

where μ labels the gas species. These reservoir variables are not treated as a third cluster population; they are monomeric fields coupled to the vacancy-cluster population through trapping and emission reactions.

The full deterministic state vector is therefore

$$\mathbf{c}(t) = \left[\{c_n^I\}_{n=1}^{N_I}, \{c_{m,\ell}^V\}_{m,\ell \in \mathcal{G}_m}, \{c_\mu^g\}_{\mu=1}^{N_g} \right]^\top, \quad (138)$$

where $\mathcal{G}_m$ is the admissible set of gas occupancies for a cavity of size m .

7.2 Signed Lattice-Defect Content

Vacancies and interstitials represent opposite topological deviations from the perfect lattice. This motivates the introduction of a signed lattice-defect content

$$q(a) \in \mathbb{Z}, \quad (139)$$

assigned to each resolved state a . For the two cluster populations,

$$q(I_n) = +n, \quad q(V_m \mathbf{g}_\ell) = -m, \quad q(g_\mu) = 0. \quad (140)$$

The sign convention treats a vacancy cluster as the negative image of an interstitial cluster. Thus, when an interstitial cluster and a vacancy cluster collide, the result is determined by conservation of signed lattice-defect content rather than by assuming total annihilation.

For a collision between I_n and $V_m \mathbf{g}_\ell$, the outgoing signed defect content is

$$q_{\text{out}} = n - m. \quad (141)$$

Hence

$$I_n + V_m \mathbf{g}_\ell \longrightarrow \begin{cases} I_{n-m} + \mathcal{R}_\ell, & n > m, \\ \mathcal{R}_\ell, & n = m, \\ V_{m-n} \mathbf{g}_\ell, & m > n. \end{cases} \quad (142)$$

Here $\mathcal{R}_\ell$ is the gas-release operator. In the default matrix-reservoir model,

$$\mathcal{R}_\ell = \sum_{\mu=1}^{N_g} \ell_\mu g_\mu, \quad (143)$$

which returns the gas inventory of an eliminated cavity to the matrix gas reservoir. If $m > n$, the supporting vacancy cluster survives and retains its gas occupancy. If $n \geq m$, the supporting cavity is eliminated, and its gas content is released.

7.3 Reaction-Network Representation

Let a, b, c denote generic state indices over interstitial clusters, vacancy clusters, and gas-reservoir variables. A general elementary process r is written as

$$\sum_a \nu_{ar}^- X_a \longrightarrow \sum_a \nu_{ar}^+ X_a, \quad (144)$$

where X_a denotes the state indexed by a , and ν_{ar}^- and ν_{ar}^+ are the reactant and product stoichiometric coefficients. The stoichiometric increment is

$$S_{ar} = \nu_{ar}^+ - \nu_{ar}^-. \quad (145)$$

The deterministic mean-field master equation is

$$\boxed{\frac{dc_a}{dt} = G_a + S_{ar} J_r(\mathbf{c}) - D_a c_a}, \quad (146)$$

where summation over repeated reaction index r is implied. Here G_a is the direct production rate of state a , $J_r(\mathbf{c})$ is the flux of reaction r , and $D_a c_a$ represents loss to unresolved fixed sinks.

The reaction flux may be unary, binary, or externally prescribed. For unary processes,

$$J_r(\mathbf{c}) = \Gamma_r c_b, \quad (147)$$

where Γ_r is a first-order rate. For binary collisions,

$$J_r(\mathbf{c}) = K_{bc}^{(r)} c_b c_c, \quad (148)$$

where $K_{bc}^{(r)}$ is the relevant capture or collision rate. For source processes,

$$J_r = G_r. \quad (149)$$

Equation (146) is the central abstraction. All physical processes enter only through the state list, the reaction list, the rate laws, and the stoichiometric matrix.

7.4 Elementary Process Classes

The following elementary process classes define the generalized model.

Irradiation production.

Displacement cascades and gas-production processes introduce source terms

$$G_a = G_a^{\text{FP}} + G_a^{\text{cascade}} + G_a^{\text{gas}} + G_a^{\text{trans}}, \quad (150)$$

where the four terms denote Frenkel-pair production, clustered cascade production, direct gas production, and transmutation or implantation sources. For pure Frenkel-pair production,

$$G_{I_1}^{\text{FP}} = G_{V_1}^{\text{FP}}, \quad (151)$$

and, more generally, conservative cascade production satisfies

$$\sum_{n=1}^{N_I} n G_n^I = \sum_{m,\ell} m G_{m,\ell}^V. \quad (152)$$

Interstitial–interstitial coalescence.

Mobile interstitial clusters may coalesce according to

$$I_n + I_{n'} \longrightarrow I_{n+n'}, \quad (153)$$

with flux

$$J_{n,n'}^{II} = K_{n,n'}^{II} c_n^I c_{n'}^I. \quad (154)$$

The corresponding gain term for I_n contains all partitions $n = n' + n''$, while the loss term removes I_n by reaction with any mobile interstitial cluster.

Vacancy–vacancy coalescence.

Vacancy clusters, including gas-bearing cavities, may coalesce according to

$$V_m \mathbf{g}_\ell + V_{m'} \mathbf{g}_{\ell'} \longrightarrow V_{m+m'} \mathbf{g}_{\ell+\ell'}. \quad (155)$$

The gas occupancy vectors add componentwise, so each gas species is exactly conserved by the coalescence event.

Interstitial–vacancy signed cancellation.

Cross-species collisions between interstitial and vacancy clusters obey the signed-defect rule

$$I_n + V_m \mathbf{g}_\ell \longrightarrow \mathcal{C}_{n,m,\ell}, \quad (156)$$

where $\mathcal{C}_{n,m,\ell}$ is defined by Eq. (142). This process conserves signed lattice-defect content and conserves each gas species after accounting for the matrix gas reservoir.

Gas trapping by cavities.

A gas monomer of species μ may be trapped by a vacancy cluster:

$$g_\mu + V_m \mathbf{g}\ell \longrightarrow V_m \mathbf{g}\ell + \mathbf{e}_\mu, \quad (157)$$

where $\mathbf{e}_\mu$ is the unit vector in gas-species space. The flux is

$$J_{m,\ell,\mu}^{\text{trap}} = K_{m,\ell,\mu}^{\text{trap}} c_\mu^g c_{m,\ell}^V. \quad (158)$$

Gas emission and radiation re-resolution.

Gas may be thermally emitted or athermally re-solved from a cavity:

$$V_m \mathbf{g}\ell \longrightarrow V_m \mathbf{g}\ell - \mathbf{e}_\mu + g_\mu, \quad \ell_\mu \geq 1. \quad (159)$$

The associated flux may include thermal and athermal contributions,

$$J_{m,\ell,\mu}^{\text{rel}} = [\varepsilon_\mu(m, \ell) + \Gamma_\mu^{\text{res}}(m, \ell)] c_{m,\ell}^V. \quad (160)$$

Trap mutation.

An over-pressurized gas-bearing cavity may emit an SIA while increasing its vacancy content:

$$V_m \mathbf{g}\ell \longrightarrow V_{m+1} \mathbf{g}\ell + I_1. \quad (161)$$

This reaction conserves signed lattice-defect content because the vacancy cluster changes from $q = -m$ to $q = -(m+1)$ while an interstitial of $q = +1$ is emitted.

Thermal vacancy or interstitial emission.

Vacancy emission from a cavity is represented by

$$V_m \mathbf{g}\ell \longrightarrow V_{m-1} \mathbf{g}\ell + V_1, \quad m \geq 2, \quad (162)$$

while SIA emission from an interstitial cluster is represented by

$$I_n \longrightarrow I_{n-1} + I_1, \quad n \geq 2. \quad (163)$$

These processes conserve signed lattice-defect content.

Loss to unresolved sinks.

Unresolved sinks such as dislocations, grain boundaries, precipitates, and external surfaces may be represented by first-order loss operators:

$$X_a \longrightarrow \emptyset, \quad J_a^{\text{sink}} = D_a c_a. \quad (164)$$

When the sinks are unbiased, their contributions cancel from the signed defect balance. When the sinks are biased, their net contribution must be tracked by an explicit cumulative sink-flux variable.

7.5 Deterministic Operator Form

For analysis, it is useful to separate the right-hand side into production, binary collision, unary transition, gas-reservoir exchange, and sink terms:

$$\dot{c}_a = G_a + \mathcal{Q}_a(\mathbf{c}, \mathbf{c}) + \mathcal{L}_{ab}c_b + \mathcal{H}_a(\mathbf{c}, \mathbf{c}^g) - D_a c_a, \quad (165)$$

where repeated indices are summed. The bilinear collision operator may be written as

$$\mathcal{Q}_a(\mathbf{c}, \mathbf{c}) = \frac{1}{2} C_{abc}^+ K_{bc} c_b c_c - C_{abc}^- K_{bc} c_b c_c. \quad (166)$$

Here C_{abc}^+ identifies binary reactions that produce state a , while C_{abc}^- identifies binary reactions that consume state a . The factor $1/2$ avoids double counting for symmetric collisions.

Unary transitions are collected in

$$\mathcal{L}_{ab}c_b = (E_{ab} + T_{ab} + R_{ab}) c_b, \quad (167)$$

where E_{ab} denotes thermal emission, T_{ab} trap mutation, and R_{ab} radiation re-solution or other one-body transitions.

Gas-reservoir exchange is represented by

$$\mathcal{H}_a(\mathbf{c}, \mathbf{c}^g) = S_{ar}^g J_r^g(\mathbf{c}, \mathbf{c}^g), \quad (168)$$

where the gas reactions include trapping, thermal emission, radiation re-solution, and gas release after elimination of a supporting cavity.

7.6 Conservation Laws

The abstract formulation is constrained by two families of conservation laws: signed lattice-defect conservation and gas-species conservation.

7.6.1 Signed lattice-defect balance.

Define the signed defect inventory

$$Q(t) = \sum_{n=1}^{N_I} n c_n^I(t) - \sum_{m,\ell} m c_{m,\ell}^V(t) + Q_{\text{sink}}(t), \quad (169)$$

where Q_{sink} is a cumulative correction for biased absorption by unresolved sinks. Differentiating gives

$$\frac{dQ}{dt} = \sum_{n=1}^{N_I} nG_n^I - \sum_{m,\ell} mG_{m,\ell}^V + \dot{Q}_{\text{bias}}. \quad (170)$$

For conservative Frenkel-pair production and unbiased sinks,

$$\frac{dQ}{dt} = 0. \quad (171)$$

When biased dislocation absorption is present, the cumulative sink term must be retained. In the simplest vacancy/SIA case, one may write

$$Q_{\text{sink}}(t) = \int_0^t [J_I^{\text{bias}}(t') - J_V^{\text{bias}}(t')] dt'. \quad (172)$$

This is the abstract counterpart of the swelling identity used in irradiation cluster dynamics: vacancy storage in cavities is balanced by the SIA inventory in loops and the cumulative biased absorption of SIAs by the dislocation network.

7.6.2 Gas-species conservation.

For each gas species μ , define the total gas inventory

$$M_\mu(t) = c_\mu^g(t) + \sum_{m,\ell} \ell_\mu c_{m,\ell}^V(t). \quad (173)$$

Gas trapping, emission, re-solution, vacancy-cluster coalescence, and signed-defect cancellation all redistribute gas between the matrix reservoir and cavities. They do not create or destroy gas. Therefore,

$$\boxed{\frac{dM_\mu}{dt} = G_\mu^g - J_\mu^{g,\text{ext}}}, \quad (174)$$

where G_μ^g is the external gas-production rate and $J_\mu^{g,\text{ext}}$ is any explicitly modeled external gas sink. In a closed material volume with gas production but no gas escape,

$$\frac{dM_\mu}{dt} = G_\mu^g. \quad (175)$$

7.7 Stochastic Jump-Process Counterpart

The same reaction-network structure defines a stochastic cluster-dynamics model. Let $N_a(t)$ be the integer number of clusters or gas monomers in state a in a representative

volume Ω . The stochastic state is

$$\mathbf{N}(t) = \{N_a(t)\}_{a \in \mathcal{S}}. \quad (176)$$

Each reaction r changes the state by its stoichiometric increment

$$\mathbf{N} \longrightarrow \mathbf{N} + \mathbf{S}_r. \quad (177)$$

The stochastic propensity is denoted by $a_r(\mathbf{N})$. For a unary reaction,

$$a_r(\mathbf{N}) = \Gamma_r N_b, \quad (178)$$

while for a binary reaction between distinct states b and c ,

$$a_r(\mathbf{N}) = \frac{K_{bc}^{(r)}}{\Omega} N_b N_c. \quad (179)$$

For a self-collision $b = c$,

$$a_r(\mathbf{N}) = \frac{K_{bb}^{(r)}}{\Omega} \frac{N_b(N_b - 1)}{2}. \quad (180)$$

The deterministic concentration equation is recovered in the thermodynamic or mean-field limit

$$c_a = \frac{N_a}{\Omega}, \quad \Omega \rightarrow \infty. \quad (181)$$

Thus the deterministic and stochastic formulations differ only in the interpretation of the same stoichiometric reaction graph and rate laws.

7.8 Code-Generation-Ready Form

The abstract notation maps directly onto a code-generation representation. Each state is defined by

$$a = (\text{kind}, q, n, m, \ell, \text{mobility}, \text{sink}), \quad (182)$$

where

$$\text{kind} \in \{\text{I_cluster}, \text{V_cavity}, \text{gas_reservoir}\}.$$

For an interstitial cluster,

$$a = (\text{I_cluster}, +n, n, 0, \mathbf{0}, \dots), \quad (183)$$

for a vacancy cluster,

$$a = (\text{V_cavity}, -m, 0, m, \ell, \dots), \quad (184)$$

and for a gas reservoir species,

$$a = (\text{gas_reservoir}, 0, 0, 0, \mathbf{e}_\mu, \dots). \quad (185)$$

Each reaction is defined by

$$r = (\mathcal{R}_r^-, \mathcal{R}_r^+, J_r, \mathbf{S}_r, \mathcal{C}_r), \quad (186)$$

where $\mathcal{R}_r^-$ is the reactant list, $\mathcal{R}_r^+$ is the product list, J_r is the rate law, $\mathbf{S}_r$ is the stoichiometric increment, and $\mathcal{C}_r$ is a set of conservation checks. At minimum, the conservation checks are

$$\sum_a q_a S_{ar} = 0 \quad \text{for internally conservative lattice-defect reactions,} \quad (187)$$

$$\sum_a \ell_{\mu a} S_{ar} = 0 \quad \text{for internally conservative gas-transfer reactions.} \quad (188)$$

External sources and sinks are exempt from the zero-sum condition but must be assigned to the corresponding source or reservoir term.

The assembled deterministic ODE system has the form

$$\dot{\mathbf{c}} = \mathbf{G} + \mathbf{S}\mathbf{J}(\mathbf{c}) - \mathbf{D}\mathbf{c}, \quad (189)$$

where $\mathbf{S}$ is sparse. The same reaction table defines stochastic propensities for stochastic cluster dynamics.

7.9 Specialization to the $V + I + \text{He}$ System

The vacancy–interstitial–helium model is recovered by setting

$$N_g = 1, \quad g_1 = \text{He}, \quad \ell = (\ell). \quad (190)$$

The general state variables reduce to

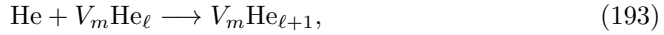
$$c_n^I \rightarrow c_n, \quad c_{m,\ell}^V \rightarrow c_{m,\ell}, \quad c_1^g \rightarrow c_h. \quad (191)$$

Thus the two resolved populations are

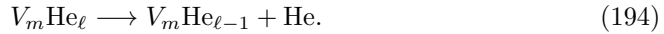
$$\{I_n\}_{n=1}^{N_I}, \quad \{V_m \text{He}_\ell\}_{m,\ell}, \quad (192)$$

and the matrix gas reservoir is the free interstitial helium concentration c_h .

Gas trapping becomes



while helium emission and radiation re-solution become



The helium conservation law reduces to

$$\frac{d}{dt} \left[c_h + \sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\max}(m)} \ell c_{m,\ell} \right] = G_{\text{He}} - J_{\text{He}}^{\text{ext}}. \quad (195)$$

For a closed matrix gas reservoir with no external helium escape,

$$J_{\text{He}}^{\text{ext}} = 0. \quad (196)$$

The signed-defect collision rule becomes

$$I_n + V_m \text{He}_\ell \longrightarrow \begin{cases} I_{n-m} + \ell \text{He}, & n > m, \\ \ell \text{He}, & n = m, \\ V_{m-n} \text{He}_\ell, & m > n. \end{cases} \quad (197)$$

If the vacancy cluster survives, its helium content remains trapped. If the cavity is eliminated, the helium atoms are returned to the matrix reservoir.

Trap mutation becomes

$$V_m \text{He}_\ell \longrightarrow V_{m+1} \text{He}_\ell + I_1, \quad (198)$$

which conserves signed lattice-defect content:

$$-m = -(m+1) + 1. \quad (199)$$

Finally, the signed Frenkel-pair balance reduces to

$$Q(t) = \sum_{n=1}^{N_I} n c_n - \sum_{m=1}^{N_V} \sum_{\ell=0}^{\ell_{\max}(m)} m c_{m,\ell} + Q_{\text{sink}}(t). \quad (200)$$

For conservative cascade production and unbiased sinks, $Q(t)$ is constant. With biased dislocation absorption, $Q_{\text{sink}}(t)$ is the cumulative net biased absorption term. This form is equivalent to the usual swelling identity, but written in signed-defect notation.

8 Master Equations

This section assembles the complete cluster dynamics system for EUROFER97 under neutron irradiation. All concentrations are atomic fractions (at/at); all rate constants carry units of s^{-1} ; and the fundamental kinetic variables are the EUROFER effective reaction frequencies

$$\omega_{\alpha}^{\text{eff}} = \frac{\nu_{\alpha} \exp(-E_m^{\alpha}/k_B T)}{1 + \sum_s z_s c_s \exp(E_b^{\alpha-s}/k_B T)}, \quad (201)$$

which incorporate the solute-trapping corrections from Sections 4.6–4.7.5 (Table 16).

The system contains three primary species populations:

1. **SIA clusters** c_n , $n = 1, 2, \dots, N$. These are dislocation loops with a Burgers vector $\frac{1}{2}\langle 111 \rangle$ (or single $\langle 110 \rangle$ dumbbells for $n = 1$). They never contain He.
2. **Vacancy clusters** $c_{m,\ell}$, $m = 1, \dots, M$, $\ell = 0, \dots, \ell_{\max}(m)$. These are spherical cavities: pure voids when $\ell = 0$, He bubbles when $\ell \geq 1$. The case $m = 1$, $\ell = 0$ is the free monovacancy.
3. **Free interstitial helium** c_h .

Both n and m are positive integers (no sign convention).

8.1 Bias Factors and Sink Types

The elastic self-stress field of a sink preferentially attracts SIAs over vacancies owing to the larger relaxation volume of the SIA. The magnitude of this preference depends on the *sink geometry*:

Sink type	Z_i	Z_v	Notes
Spherical cavity (void/bubble)	1	1	No long-range stress field
SIA dislocation loop	Z_i^{loop}	1	≈ 1.02 – 1.10 ; input parameter
Network dislocation	Z_i^d	1	≈ 1.02 – 1.15 ; input parameter
Incoherent precipitate	1	1	MX, M_{23}C_6 in EUROFER
Grain boundary	1	1	Unbiased planar sink

In all cases the vacancy bias is unity. Only dislocations and SIA loops carry a non-trivial bias. The loop bias Z_i^{loop} is approximately independent of loop size.

8.2 Reaction Processes

Every elementary reaction falls into one of eight categories, each mapping to the rate constants of Section 6.

P1: Vacancy–SIA recombination.

$I_1 + V_1 \rightarrow 0$. Both mobile; rate constant:

$$\mathcal{K}_{iv} = 4\sqrt{3}\pi (\omega_i^{\text{eff}} + \omega_v^{\text{eff}}) \approx 21.8 \omega_i^{\text{eff}}. \quad (202)$$

P2: Point-defect absorption by a spherical cavity.

A mobile species $\alpha \in \{i, v, h\}$ is absorbed by a cavity of m vacancies. Spherical cavities carry no elastic bias ($Z_i^{\text{cav}} = Z_v^{\text{cav}} = 1$):

$$\mathcal{K}_{\alpha, m}^{\text{cav}} = A_{\text{sph}} m^{1/3} \omega_{\alpha}^{\text{eff}}, \quad A_{\text{sph}} = (48\pi^2)^{1/3} \approx 7.818. \quad (203)$$

The He content ℓ does not enter the rate constant because the capture cross-section depends on the cavity radius ($\propto m^{1/3}$), not on the He loading.

P3: Point-defect absorption by a dislocation loop.

SIA absorption is biased; vacancy absorption is not:

$$\mathcal{K}_{i, n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}, \quad (204)$$

$$\mathcal{K}_{v, n}^{\text{loop}} = A_{\text{loop}} n^{1/2} \omega_v^{\text{eff}}, \quad (205)$$

with $A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \approx 10.78$. Vacancy absorption shrinks the loop by one SIA; SIA absorption grows it by one.

P4: Point-defect absorption by fixed sinks.

Three fixed-sink types contribute a combined first-order loss coefficient:

$$\mathcal{D}_{\alpha} = \mathcal{D}_{\alpha}^d + \mathcal{D}_{\alpha}^{gb} + \mathcal{D}_{\alpha}^p \quad [\text{s}^{-1}], \quad (206)$$

where

$$\mathcal{D}_{\alpha}^d = Z_{\alpha}^d \rho_d a^2 \omega_{\alpha}^{\text{eff}}, \quad (207)$$

$$\mathcal{D}_{\alpha}^{gb} = \frac{4\pi^2}{d_g^2} a^2 \omega_{\alpha}^{\text{eff}}, \quad (208)$$

$$\mathcal{D}_{\alpha}^p = \frac{8\pi r_p}{a} c_p^N \omega_{\alpha}^{\text{eff}}. \quad (209)$$

Here ρ_d (m^{-2}) is the dislocation line density with bias $Z_i^d > 1$, $Z_v^d = 1$; d_g is the grain diameter; r_p and c_p^N are the precipitate radius and number fraction. Grain boundaries and precipitates are unbiased ($Z_i^{gb} = Z_v^{gb} = Z_i^p = Z_v^p = 1$). For EUROFER with $d_g \approx 5\text{--}20 \mu\text{m}$, $k_{gb}^2 = 4\pi^2/d_g^2 \approx 4 \times 10^{11}\text{--}6 \times 10^{12} \text{ m}^{-2}$; with $r_p \approx 10\text{--}50 \text{ nm}$ and $N_p \approx 10^{20}\text{--}10^{21} \text{ m}^{-3}$, $k_p^2 = 4\pi r_p N_p \approx 10^{13}\text{--}10^{14} \text{ m}^{-2}$, comparable to the dislocation contribution.

P5: Thermal emission (detailed balance).

Vacancy emission from a cavity (m, ℓ):

$$\varepsilon_v(m, \ell) = A_{\text{sph}} (m-1)^{1/3} \omega_v^{\text{eff}} \exp(-E_b^v(m, \ell)/k_B T). \quad (210)$$

SIA emission from a loop of n SIAs:

$$\varepsilon_i(n) = A_{\text{loop}} (n-1)^{1/2} \omega_i^{\text{eff}} \exp(-E_b^i(n)/k_B T). \quad (211)$$

He emission from a bubble (m, ℓ) :

$$\varepsilon_h(m, \ell) = A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} \exp(-E_B^{\text{He}}(m, \ell)/k_B T). \quad (212)$$

He emission uses $m^{1/3}$ (not $(m-1)^{1/3}$) because He removal does not change the vacancy count; the capture radius relevant for detailed balance is that of the daughter $\text{He}_{\ell-1}\text{V}_m$, which retains the same m . Binding energies are from Section 5.6.

P6: SIA cluster interaction with sinks (mixed 1D/3D).

An n -SIA cluster ($n \geq 4$) gliding in 1D is absorbed by a cavity of m vacancies (spherical cavities carry no bias):

$$\kappa_{n,m}^{\text{eff}} = \frac{A_{\text{sph}} m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}, \quad B_{\text{rot}} \approx 2.627, \quad (213)$$

with $\hat{L} = L/a$ the EUROFER solute-shortened dimensionless mean free path (Table 16).

P7: Trap mutation (Frenkel pair emission).

An over-pressurised bubble emits a self-interstitial: $\text{He}_{\ell}\text{V}_m \rightarrow \text{He}_{\ell}\text{V}_{m+1} + \text{SIA}$.

$$\Gamma_{\text{TM}}(m, \ell) = \nu_0 \exp(-E_{\text{TM}}(m, \ell)/k_B T). \quad (214)$$

This dominates over He emission when $E_{\text{TM}} < E_B^{\text{He}}$, typically at $\text{He}/\text{V} \gtrsim 1.3$ – 1.7 for small clusters (Table 27).

P8: Radiation re-solution (athermal).

A cascade ejects a He atom from a bubble: $\text{He}_{\ell}\text{V}_m \rightarrow \text{He}_{\ell-1}\text{V}_m + \text{He}_{\text{int}}$.

$$\Gamma_{\text{res}}(m, \ell) = b_0 \ell \dot{\phi}, \quad (215)$$

with $b_0 \approx 10^{-2}$ (fission) or 10^{-1} (fusion) $\text{dpa}^{-1}/\text{He}$.

Mobility cutoffs.

Mobile clusters are bounded by species-specific size limits taken from Table 29:

- SIA clusters: mobile (3D or 1D) for $1 \leq n \leq i_{\text{mobile}}$. Clusters with $n > i_{\text{mobile}}$ are treated as immobile sessile loops and contribute only as fixed sinks; their reaction frequencies are set to zero.

- Vacancy clusters: mobile (3D) for $1 \leq m \leq v_{\text{mobile}}$. Clusters with $m > v_{\text{mobile}}$ are immobile and can only grow or shrink through absorption or emission of a single mobile vacancy or interstitial.

General (coalescence) formulation.

The most complete description of cluster kinetics allows any two mobile clusters to react. The three reaction channels and their rate constants are:

SIA-SIA coalescence ($I_n + I_{n'} \rightarrow I_{n+n'}$, both mobile):

$$\mathcal{K}_{n,n'}^{ii} = 8\pi (\xi_n + \xi_{n'}) (\omega_n^{\text{eff}} + \omega_{n'}^{\text{eff}}), \quad 1 \leq n' \leq n \leq i_{\text{mobile}}, \quad (216)$$

with $\xi_n = (3n/8\pi)^{1/3}$ the dimensionless cluster radius. The constraint $n' \leq n$ avoids double-counting; a factor $\frac{1}{2}$ multiplies the self-collision term ($n = n'$). The contribution to dc_n/dt is:

$$\left. \frac{dc_n}{dt} \right|_{ii} = \frac{1}{2} \sum_{n'=1}^{\min(n-1, i_{\text{mobile}})} \mathcal{K}_{n',n-n'}^{ii} c_{n'} c_{n-n'} - c_n \sum_{n'=1}^{i_{\text{mobile}}} \mathcal{K}_{n,n'}^{ii} c_{n'}. \quad (217)$$

Vacancy-vacancy coalescence ($V_m + V_{m'} \rightarrow V_{m+m'}$, $m, m' \leq v_{\text{mobile}}$):

$$\mathcal{K}_{m,m'}^{vv} = 8\pi (\xi_m + \xi_{m'}) (\omega_m^{\text{eff}} + \omega_{m'}^{\text{eff}}), \quad 1 \leq m' \leq m \leq v_{\text{mobile}}. \quad (218)$$

For $m > v_{\text{mobile}}$ the cluster is immobile; it interacts only through the mono-defect terms retained in all cases (see below).

V-I cluster annihilation ($V_m + I_n \rightarrow V_{m-n}$ if $m > n$, I_{n-m} if $n > m$, or total annihilation if $n = m$):

$$\mathcal{K}_{m,n}^{vi} = 8\pi (\xi_m + \xi_n) (\omega_m^{\text{eff}} + \omega_n^{\text{eff}}), \quad m \leq v_{\text{mobile}}, \quad n \leq i_{\text{mobile}}. \quad (219)$$

He-He coalescence is excluded: the interstitial He-He binding energy in bcc Fe is negligible ($\lesssim 0.05$ eV), so free helium reaches a cavity as a monomer.

Mono-defect simplification.

In the general equations, restricting $n' = 1$ in the i-i channel and $m' = 1$ in the v-v channel recovers the *mono-defect* form in which every cluster changes size by exactly one defect at a time. This simplification is justified because:

1. The concentration of mobile $n' \geq 2$ SIA clusters under steady irradiation is typically two to four orders of magnitude below that of the mono-SIA, so their coalescence contribution is proportionally suppressed.
2. For the vacancy side, v_{mobile} is small enough that the dominant coalescence path ($V_1 + V_1 \rightarrow V_2$, $V_1 + V_2 \rightarrow V_3$, etc.) is already captured by the mono-vacancy absorption terms in the equations for $c_{m,\ell}$ with $m \leq v_{\text{mobile}}$.

3. The v-i annihilation channel reduces exactly to P1 recombination when $m = n = 1$, to the P3 loop-shrinkage term when $m = 1, n \geq 2$, and to the P2 SIA-shrinkage term when $m \geq 2, n = 1$. Only the cluster-cluster pairs $m \geq 2, n \geq 2$ are genuinely new; these are small at low cluster concentrations.

Under the mono-defect simplification, the rate constants reduce to:

$$\mathcal{K}_{n,1}^{ii} \big|_{n'=1} = \mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}, \quad (220)$$

$$\mathcal{K}_{m,1}^{vv} \big|_{m'=1} = \mathcal{K}_{\alpha,m}^{\text{cav}} = A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}}, \quad (221)$$

$$\mathcal{K}_{m=1,n=1}^{vi} = \mathcal{K}_{iv} = 4\sqrt{3} \pi (\omega_i^{\text{eff}} + \omega_v^{\text{eff}}), \quad (222)$$

where the loop-bias factor Z_i^{loop} in Eq. (220) replaces the bare Smoluchowski prefactor because the specialized bcc capture geometry enters at the mono-defect level (Section 6.1). The boxed master equations in Section 8.3 implement the mono-defect form throughout. To recover the full coalescence system, replace the mono-defect growth and shrinkage terms with the sums in Eqs. (217)–(219).

8.3 Master Equations

Define the total cavity concentration at vacancy size m and the combined fixed-sink rate:

$$\bar{c}_m \equiv \sum_{\ell=0}^{\ell_{\text{max}}(m)} c_{m,\ell}, \quad \mathcal{D}_\alpha = \mathcal{D}_\alpha^d + \mathcal{D}_\alpha^{gb} + \mathcal{D}_\alpha^p \quad [\text{s}^{-1}], \quad (223)$$

as defined in Eqs. (207)–(209). The equations below are the *general (coalescence) form*: every pair of mobile clusters that can react does so, governed by the rate constants $\mathcal{K}^{ii}$, $\mathcal{K}^{vv}$, and $\mathcal{K}^{vi}$ of Section 8.2. The *mono-defect simplification* is recovered as a special case by setting $i_{\text{mobile}} = 1$ and $v_{\text{mobile}} = 1$, which restricts all coalescence sums to their $n' = 1$ or $m' = 1$ terms and reduces $\mathcal{K}^{ii}$, $\mathcal{K}^{vv}$, $\mathcal{K}^{vi}$ to the point-defect rate constants $\mathcal{K}_{i,n}^{\text{loop}}$, $\mathcal{K}_{\alpha,m}^{\text{cav}}$, and $\mathcal{K}_{iv}$ respectively (Eqs. 220–222).

8.3.1 Population 1: SIA Clusters

$$\begin{aligned}
\frac{dc_n}{dt} = & \underbrace{G_n}_{\text{production}} \\
& + \underbrace{\frac{1}{2} \sum_{n'=1}^{\min(n-1, i_{\text{mobile}})} \mathcal{K}_{n', n-n'}^{ii} c_{n'} c_{n-n'}}_{\text{SIA-SIA coalescence gain } (I_{n'} + I_{n-n'} \rightarrow I_n)} \\
& - \underbrace{c_n \sum_{n'=1}^{i_{\text{mobile}}} \mathcal{K}_{n, n'}^{ii} c_{n'}}_{\text{SIA-SIA coalescence loss } (I_n + I_{n'} \rightarrow I_{n+n'})} \\
& + \underbrace{\varepsilon_i(n+1) c_{n+1} - \varepsilon_i(n) c_n \delta_{n \geq 2}}_{\text{thermal emission (SIA from loops)}} \\
& + \underbrace{\sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m', n+m'}^{vi} c_{m',0} c_{n+m'}}_{\text{V-I annihilation gain } (V_{m'} + I_{n+m'} \rightarrow I_n)} \\
& - \underbrace{\sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m', n}^{vi} c_{m',0} c_n}_{\text{V-I annihilation loss } (V_{m'} + I_n \rightarrow I_{n-m'})} \\
& - \underbrace{\sum_{m=1}^M \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m}_{\text{SIA cluster-cavity absorption}} \\
& - \underbrace{\mathcal{D}_i(n) c_n}_{\text{fixed-sink loss (mobile clusters only)}} \\
& + \underbrace{\delta_{n=1} \sum_{m=1}^M \sum_{\ell=1}^{\ell_{\text{max}}(m)} \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}}_{\text{trap mutation (SIA source at } n=1)}
\end{aligned} \tag{224}$$

where the cavity rate constant $\mathcal{K}_{n,m}^{(\text{cav})}$ (immobile cavities absorbing mobile SIA clusters of any size $n \leq i_{\text{mobile}}$; zero for $n > i_{\text{mobile}}$) is:

$$\mathcal{K}_{n,m}^{(\text{cav})} = \begin{cases} A_{\text{sph}} m^{1/3} \omega_i^{\text{eff}}, & 1 \leq n \leq 3 \text{ (3D)}, \\ \frac{A_{\text{sph}} m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}, & 4 \leq n \leq i_{\text{mobile}} \text{ (1D/3D)}, \\ 0, & n > i_{\text{mobile}} \text{ (sessile)}. \end{cases} \quad (225)$$

Structure of the SIA equation.

Line 1: cascade production. Line 2: i-i coalescence (P_{ii}). The gain term sums over all pairs $(n', n-n')$ of mobile SIA clusters that merge to form I_n ; the loss term removes I_n by merging with any mobile $I_{n'}$. *Mono-defect limit* ($i_{\text{mobile}} = 1$): only $n' = 1$ survives in the loss sum, giving $\mathcal{K}_{n,1}^{ii} c_1 c_n = \mathcal{K}_{i,n}^{\text{loop}} c_1 c_n$; the gain sum is non-zero only for $n = 2$ (forming a di-interstitial from two mono-SIAs). Line 3: thermal SIA emission (P_5); unchanged in both formulations. Lines 4–5: v-i annihilation (P_{vi}). The gain term at size n represents a mobile vacancy cluster $V_{m'}$ ($m' \leq v_{\text{mobile}}$) shrinking a larger SIA cluster $I_{n+m'}$ to produce I_n ; the loss term removes I_n by annihilating m' SIAs, producing $I_{n-m'}$. Both gain and loss are required for exact Frenkel-pair conservation (Appendix A): without the gain, the m -weighted moment of the vacancy equation does not telescope to the correct n -weighted form. *Mono-defect limit* ($v_{\text{mobile}} = 1$): the gain reduces to $\mathcal{K}_{v,n+1}^{\text{loop}} c_{1,0} c_{n+1}$ and the loss to $\mathcal{K}_{v,n}^{\text{loop}} c_{1,0} c_n$ (loop shrinkage by mono-vacancy absorption), with $\mathcal{K}_{iv} c_{1,0} c_1$ for P1 recombination at $n = 1$. Line 6: absorption by immobile cavities (P_2/P_6); the sum already spans all SIA-cluster sizes $n = 1, \dots, i_{\text{mobile}}$ through $\mathcal{K}_{n,m}^{(\text{cav})}$, so this line is identical in both formulations. Line 7: combined fixed sinks ($P_4/P_4'/P_4''$). The fixed-sink loss $\mathcal{D}_i(n)$ is size-dependent: $\mathcal{D}_i(n) = \mathcal{D}_i$ (Eq. 206) for mobile SIA clusters ($n \leq i_{\text{mobile}}$) and $\mathcal{D}_i(n) = 0$ for sessile loops ($n > i_{\text{mobile}}$), since immobile clusters cannot diffuse to sinks. Line 8: trap-mutation SIA source (P_7 , $n = 1$ only).

Recovery of the single-SIA equation.

Setting $n = 1$ in Eq. (224): the i-i gain sum vanishes (no pairs sum to 1), the i-i loss gives $-\mathcal{K}_{1,1}^{ii} c_1^2$, the v-i gain gives $+\sum_{m'} \mathcal{K}_{m',1+m'}^{vi} c_{m',0} c_{1+m'}$ (mobile vacancies shrinking $I_{1+m'}$ to I_1), the v-i loss gives $-\mathcal{K}_{1,1}^{vi} c_{1,0} c_1$, and emission gives $+\varepsilon_i(2) c_2$. The exact $n = 1$ equation is:

$$\begin{aligned} \frac{dc_1}{dt} = & G_1 - \mathcal{K}_{1,1}^{ii} c_1^2 + \varepsilon_i(2) c_2 + \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m',2}^{vi} c_{m',0} c_2 \\ & - \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m',1}^{vi} c_{m',0} c_1 - \sum_{m=1}^M A_{\text{sph}} m^{1/3} \omega_i^{\text{eff}} c_1 \bar{c}_m - \mathcal{D}_i c_1 \\ & + \sum_{m,\ell} \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}. \end{aligned} \quad (226)$$

Setting $i_{\text{mobile}} = v_{\text{mobile}} = 1$: $\mathcal{K}_{1,1}^{ii} \rightarrow \mathcal{K}_{i,1}^{\text{loop}}$, $\mathcal{K}_{1,1}^{vi} \rightarrow \mathcal{K}_{iv}$, and the c_2 term from v-i becomes $\mathcal{K}_{v,2}^{\text{loop}} c_{1,0} c_2$, recovering the standard mono-defect mono-SIA equation.

8.3.2 Population 2: Vacancy Clusters

Voids ($\ell = 0$) and bubbles ($\ell \geq 1$) share a unified (m, ℓ) grid. Spherical cavities are unbiased ($Z_i^{\text{cav}} = 1$). The v-v coalescence terms are active only for $m \leq v_{\text{mobile}}$; for $m > v_{\text{mobile}}$ those clusters are immobile and the v-v sum reduces to a single mono-vacancy term, identical to the mono-defect form. SIA-induced shrinkage spans all mobile SIA sizes through $\mathcal{K}_{n,m}^{(\text{cav})}$; both gain (from a larger cavity shrinking to V_m) and loss (from V_m shrinking) are included to ensure exact Frenkel-pair conservation.

$$\begin{aligned}
\frac{dc_{m,\ell}}{dt} = & \underbrace{G_{m,\ell}}_{\text{production}} \\
& + \underbrace{\frac{1}{2} \sum_{m'=1}^{\min(m-1, v_{\text{mobile}})} \mathcal{K}_{m',m-m'}^{vv} c_{m',0} c_{m-m',\ell} \delta_{m \leq 2 v_{\text{mobile}}}}_{\text{V-V coalescence gain } (V_{m'} + V_{m-m'} \rightarrow V_m)} \\
& - \underbrace{c_{m,\ell} \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m,m'}^{vv} c_{m',0}}_{\text{V-V coalescence loss } (V_m + V_{m'} \rightarrow V_{m+m'})} \\
& + \underbrace{\varepsilon_v(m+1, \ell) c_{m+1,\ell} - \varepsilon_v(m, \ell) c_{m,\ell}}_{\text{thermal emission (vacancy from cavities)}} \\
& + \underbrace{\sum_{n=1}^{\min(i_{\text{mobile}}, M-m)} \mathcal{K}_{n,m+n}^{(\text{cav})} c_n c_{m+n,\ell}}_{\text{SIA-induced shrinkage gain } (I_n + V_{m+n} \rightarrow V_m)} \\
& - \underbrace{\sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n c_{m,\ell}}_{\text{SIA-induced shrinkage loss } (I_n + V_m \rightarrow V_{m-n})} \\
& + \underbrace{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell-1} \delta_{\ell \geq 1} - A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell} \delta_{\ell < \ell_{\text{max}}}}_{\text{He absorption by cavities}} \\
& + \underbrace{\varepsilon_h(m, \ell+1) c_{m,\ell+1} - \varepsilon_h(m, \ell) c_{m,\ell} \delta_{\ell \geq 1}}_{\text{He emission from cavities}} \\
& + \underbrace{\Gamma_{\text{TM}}(m-1, \ell) c_{m-1,\ell} - \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}}_{\text{trap mutation}} \\
& + \underbrace{b_0 \dot{\phi}(\ell+1) c_{m,\ell+1} - b_0 \dot{\phi} \ell c_{m,\ell}}_{\text{radiation re-resolution}} \\
& - \underbrace{\mathcal{D}_v c_{m,\ell} \delta_{m \leq v_{\text{mobile}}}}_{\text{fixed-sink loss (mobile clusters only)}}
\end{aligned} \tag{227}$$

Structure of the vacancy equation.

Lines 2–3: v–v coalescence (P_{vv}). The gain term sums over all pairs $(m', m-m')$ of mobile vacancy clusters that merge to form V_m ; the loss term removes V_m by merging with any mobile $V_{m'}$. *Mono-defect limit* ($v_{\text{mobile}} = 1$): only $m' = 1$ survives, and $\mathcal{K}_{m,1}^{vv} c_{1,0} c_{m,\ell} = A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0} c_{m,\ell}$ (Eq. 221), recovering the standard mono-vacancy absorption form. The gain term becomes $A_{\text{sph}} (m-1)^{1/3} \omega_v^{\text{eff}} c_{1,0} c_{m-1,\ell}$, which is the familiar P2 growth step. Line 4: vacancy thermal emission (P5); identical in both formulations. Lines 5–6: SIA-induced shrinkage. The gain term at size m arises when a mobile I_n absorbs into a larger cavity V_{m+n} , shrinking it to V_m ; the loss term removes V_m when I_n is absorbed, producing V_{m-n} . Both gain and loss are required for exact Frenkel-pair conservation: without the gain, the m -weighted moment yields $-\sum m \mathcal{K}$ instead of the correct $-\sum n \mathcal{K}$ (Appendix A). *Mono-defect limit* ($i_{\text{mobile}} = 1$): gain becomes $A_{\text{sph}} (m+1)^{1/3} \omega_i^{\text{eff}} c_1 c_{m+1,\ell}$ and loss becomes $A_{\text{sph}} m^{1/3} \omega_i^{\text{eff}} c_1 c_{m,\ell}$. Lines 7–8: He exchange along ℓ (P2 + P5); unchanged. Line 9: trap mutation (P7); unchanged. Line 10: radiation re-resolution (P8); unchanged. Line 11: fixed-sink loss. Only *mobile* vacancy clusters ($m \leq v_{\text{mobile}}$) can diffuse to dislocations, grain boundaries, and precipitates. Immobile clusters ($m > v_{\text{mobile}}$) do not migrate and experience no fixed-sink loss; their $\mathcal{D}_v$ term is zero. This mirrors the SIA treatment where sessile loops ($n > i_{\text{mobile}}$) have $\mathcal{D}_i = 0$. For $\ell = 0$, lines 6–7 and 9 vanish (pure voids carry no He).

Recovery of the free monovacancy equation.

Setting $m = 1$, $\ell = 0$, noting that the v–v gain sum vanishes (no pairs give $m = 1$) and incorporating the v–i loss from the SIA grid (P3, vacancy side):

$$\begin{aligned} \frac{dc_{1,0}}{dt} = & G_v - \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m',1}^{vi} c_{m',0} c_1 - \mathcal{D}_v c_{1,0} \\ & - c_{1,0} \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{1,m'}^{vv} c_{m',0} + \sum_{m=2}^M \sum_{\ell} \varepsilon_v(m, \ell) c_{m,\ell} \\ & + \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,1+n}^{(\text{cav})} c_n \bar{c}_{1+n} - \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,1}^{(\text{cav})} c_n c_{1,0} \\ & - A_{\text{sph}} \omega_h^{\text{eff}} c_h c_{1,0} + \varepsilon_h(1, 1) c_{1,1} + b_0 \dot{\phi} c_{1,1}. \end{aligned} \quad (228)$$

Setting $v_{\text{mobile}} = 1$: $\mathcal{K}_{1,1}^{vi} \rightarrow \mathcal{K}_{iv}$ and $\mathcal{K}_{1,1}^{vv} c_{1,0} \rightarrow A_{\text{sph}} \omega_v^{\text{eff}} c_{1,0}$ (absorption by all larger cavities), recovering the standard monovacancy equation. The fixed sink $\mathcal{D}_v$ includes dislocation, grain-boundary, and precipitate contributions (Eq. 206). For $m = 1$, $\ell \geq 1$: $\varepsilon_v(1, \ell) = 0$ because the daughter $\text{He}_\ell V_0$ would immediately self-trap back to $\text{He}_\ell V_1$.

8.3.3 Population 3: Free Helium, c_h

The free-helium equation is independent of the i–i and v–v coalescence formulation: He–He coalescence is excluded, and He interacts with the vacancy cluster grid only through mono-He trapping and emission.

$$\begin{aligned}
\frac{dc_h}{dt} = & \underbrace{G_{\text{He}}}_{\text{production}} \\
& - \underbrace{\sum_{m=1}^M A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \sum_{\ell=0}^{\ell_{\text{max}}(m)-1} c_{m,\ell}}_{\text{He absorption by cavities}} \\
& - \underbrace{\mathcal{D}_h c_h}_{\text{fixed-sink loss}} \\
& + \underbrace{\sum_{m=1}^M \sum_{\ell=1}^{\ell_{\text{max}}(m)} \varepsilon_h(m, \ell) c_{m,\ell}}_{\text{He emission from cavities}} \\
& + \underbrace{\sum_{m=1}^M \sum_{\ell=1}^{\ell_{\text{max}}(m)} b_0 \dot{\phi} \ell c_{m,\ell}}_{\text{radiation re-solution}}
\end{aligned} \tag{229}$$

Production; trapping by cavities (P2); combined fixed sinks ($\mathcal{D}_h$ includes dislocations, grain boundaries, and precipitates); thermal He emission (P5); re-solution (P8). This equation is identical in the general and mono-defect formulations.

8.4 Mass Conservation Laws

The master equations satisfy two exact conservation identities that are proved term by term in Appendix A. They are stated here for reference and used to define numerical accuracy diagnostics.

8.4.1 Frenkel Pair Balance and the Swelling Identity

Define the cumulative net SIA flux absorbed by the dislocation network and the total cluster inventories:

$$\Delta J^d(t) \equiv \int_0^t \left[Z_i^d \rho_d a^2 \sum_{n=1}^N n \omega_n^{\text{eff}} c_n - Z_v^d \rho_d a^2 \omega_v^{\text{eff}} c_{1,0} \right] dt' \geq 0, \tag{230}$$

$$S_I(t) \equiv \sum_{n=1}^N n c_n(t), \quad S(t) \equiv \sum_{m=1}^M m \bar{c}_m(t), \tag{231}$$

where $S(t)$ is the volumetric swelling (total vacancies stored in cavities, Eq. 433) and $S_I(t)$ is the total SIA inventory in loops. Cascade production satisfies $\sum_n n G_n = \sum_{m,\ell} m G_{m,\ell}$ identically. Grain boundaries and precipitates are unbiased and cancel

exactly in the moment difference (Appendix A). The resulting *exact conservation law* is:

$$\boxed{S_I(t) + \Delta J^d(t) - S(t) = S_I(0) - S(0) = \text{const.}} \quad (232)$$

For a defect-free initial condition, $S_I(0) = S(0) = 0$, so the conserved quantity is identically zero at all times:

$$S(t) = S_I(t) + \Delta J^d(t). \quad (233)$$

This is the *swelling identity*: the total vacancy content of all cavities equals the sum of (i) the SIA inventory currently stored in loops plus (ii) the cumulative net bias flux delivered to the dislocation network. Its physical interpretation is transparent:

- Every cascade produces equal numbers of SIAs and vacancies.
- The dislocation network preferentially absorbs SIAs ($Z_i^d > Z_v^d = 1$), accumulating a net SIA excess ΔJ^d that is permanently removed from the lattice.
- Of the remaining SIAs, some are currently stored as loops (S_I) and the rest have recombined with vacancies or been absorbed by unbiased sinks.
- Whatever vacancy excess is not neutralised by recombination or stored in loops must reside in cavities, exactly accounting for S .

Differentiating Eq. (232) recovers the rate form:

$$\frac{dS}{dt} = \frac{dS_I}{dt} + Z_i^d \rho_d a^2 \sum_{n=1}^N n \omega_n^{\text{eff}} c_n - Z_v^d \rho_d a^2 \omega_v^{\text{eff}} c_{1,0}, \quad (234)$$

where $Z_v^d = 1$. The last two terms together are $d\Delta J^d/dt \geq 0$ (Eq. 230). Swelling grows whenever the net bias absorption rate exceeds the rate of SIA loop accumulation dS_I/dt . In the regime where loops have reached a quasi-steady density ($dS_I/dt \approx 0$), swelling grows at a rate set directly by the net bias flux to dislocations.

8.4.2 Helium Conservation

The total helium inventory—free plus trapped—grows only through external production:

$$\frac{d}{dt} \left[c_h + \sum_{m=1}^M \sum_{\ell=1}^L \ell c_{m,\ell} \right] = G_{\text{He}}. \quad (235)$$

Thermal emission (P5), re-solution (P8), and trap mutation (P7) all redistribute helium between free and trapped pools without creating or destroying it; their contributions to the moment sum cancel identically (Appendix A).

8.4.3 Numerical Accuracy Diagnostics

The swelling identity (232) provides an exact, inexpensive check: at every output step evaluate

$$\delta_{\text{FP}}(t) = \frac{|S(t) - S_I(t) - \Delta J^d(t)|}{S(t) + S_I(t) + \Delta J^d(t)}, \quad (236)$$

where $\Delta J^d(t)$ is accumulated by numerical integration of the integrand in Eq. (230) alongside the ODE system. Define the helium balance error as:

$$\delta_{\text{He}}(t) = \frac{\left| c_h + \sum_{m,\ell} \ell c_{m,\ell} - \int_0^t G_{\text{He}} dt' - [c_h(0) + \sum_{m,\ell} \ell c_{m,\ell}(0)] \right|}{c_h + \sum_{m,\ell} \ell c_{m,\ell}}. \quad (237)$$

Both metrics are normalised to instantaneous inventories. In a correctly implemented code both should remain below $\sim 10^{-6}$. Values above 10^{-4} indicate solver drift (tighten `rtol` $\leq 10^{-8}$, `atol` $\leq 10^{-12}$). Values above 10^{-3} almost always signal a coding error: a missing sign, a mismatched index shift in the ladder operators, or an unbalanced trap-mutation source. Because δ_{FP} now decomposes swelling into two separately trackable contributions (S_I and ΔJ^d), it also serves as a diagnostic for identifying which physical mechanism dominates swelling at each stage of the irradiation history.

8.5 Swelling

The swelling identity (233) provides the physical mechanism. The macroscopic fractional volume change is:

$$S(t) = \frac{\Delta V}{V} = \sum_{m=1}^M m \bar{c}_m(t), \quad (238)$$

counting every vacancy stored in voids and bubbles. By Eq. (233), this equals the current SIA loop inventory S_I plus the integrated net bias absorption ΔJ^d . The two contributions have distinct time signatures:

- $\Delta J^d(t)$ grows monotonically and, once the microstructure has reached a quasi-steady loop density, increases approximately linearly with dose. It represents vacancies that are effectively stranded in cavities because their SIA counterparts were irreversibly captured by the biased dislocation network.
- $S_I(t)$ grows during the transient loop-accumulation regime and then saturates when loop nucleation is balanced by loop-loop annihilation and absorption by cavities. Once S_I saturates, $\dot{S} \approx \dot{\Delta J}^d = (Z_i^d - 1)\rho_d a^2 \sum_n n \omega_n^{\text{eff}} c_n$, giving the well-known linear-in-dose swelling rate.

The dislocation-climb contribution to the macroscopic volume change is negligible: the SIA relaxation volume ($\sim 1.1\Omega$) nearly cancels the $+\Omega$ extension of the extra half-plane, leaving a net volume change per absorbed SIA of order -0.1Ω —an order of magnitude smaller than the $+\Omega$ per vacancy entering a void. Swelling is therefore dominated by the cavity term S , and Eq. (233) shows that this is driven in equal measure by the ongoing bias absorption (ΔJ^d) and by the transient loading of SIA loops (S_I).

8.6 Sink Strengths

The total sink strengths, obtained by collecting all loss terms proportional to c_α from the master equations, are:

$$k_v^2 = \rho_d + \frac{4\pi^2}{d_g^2} + 4\pi r_p N_p + \frac{1}{a^2} \sum_{m=1}^M A_{\text{sph}} m^{1/3} \bar{c}_m + \frac{1}{a^2} \sum_{n=1}^N A_{\text{loop}} n^{1/2} c_n, \quad (239)$$

$$k_i^2 = Z_i^d \rho_d + \frac{4\pi^2}{d_g^2} + 4\pi r_p N_p + \frac{1}{a^2} \sum_{m=1}^M A_{\text{sph}} m^{1/3} \bar{c}_m + \frac{Z_i^{\text{loop}}}{a^2} \sum_{n=1}^N A_{\text{loop}} n^{1/2} c_n. \quad (240)$$

The grain-boundary and precipitate contributions are identical in k_v^2 and k_i^2 (no bias). The net bias driving swelling is:

$$\Delta k^2 = k_i^2 - k_v^2 = (Z_i^d - 1) \rho_d + \frac{(Z_i^{\text{loop}} - 1)}{a^2} \sum_{n=1}^N A_{\text{loop}} n^{1/2} c_n. \quad (241)$$

The cavity, grain-boundary, and precipitate contributions cancel exactly. Swelling is driven solely by the dislocation and loop bias terms.

8.7 Input Data Mapping

Quantity	Source
$G_n, G_{m,\ell}, G_{\text{He}}, \dot{\phi}$	Table 5
$E_f^v, E_f^i, \gamma_s, B_2, B_3, \nu_\alpha$	Table 8
$\omega_\alpha^{\text{eff}}, \omega_n^{1D}, \hat{L}, z_s, E_b^{\alpha-s}, c_s$	Table 16
$E_b^v(m, \ell), E_b^i(n), E_B^{\text{He}}(m, \ell), E_{\text{TM}}(m, \ell)$	Tables 25, 18, 19, 27
$A_{\text{sph}}, A_{\text{loop}}, B_{\text{rot}}, Z_i^d, Z_i^{\text{loop}}, b_0$	Tables 29, 30
ρ_d, d_g, r_p, N_p	Table 29

8.8 ODE System Structure, Helium Approximations, and Numerical Strategy

8.8.1 State Vector and Matrix Structure

The three populations assembled in Section 8.3 form a single state vector

$$\mathbf{y}(t) = \left[\underbrace{c_1, c_2, \dots, c_N}_{\text{SIA clusters}}, \underbrace{c_{1,0}, c_{1,1}, \dots, c_{M,\ell_{\max}(M)}}_{\text{vacancy/He clusters}}, \underbrace{c_h}_{\text{free He}} \right]^\top, \quad (242)$$

of dimension $n_{\text{eq}} = N + n_{VH} + 1$, where $n_{VH} = \sum_{m=1}^M [\ell_{\text{max}}(m) + 1]$ is the number of (m, ℓ) grid points. The right-hand side $\mathbf{f}(\mathbf{y})$ of the ODE system $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y})$ is sparse and banded. Writing it schematically as a Jacobian block structure reveals the coupling pattern:

	c_n (N cols)	$c_{m,\ell}$ (n_{VH} cols)	c_h
$\dot{c}_n$ (N rows)	Tridiagonal i-i ladder (P3, P _{ii})	Diagonal (P2/P6 cavity absorption)	0
$\dot{c}_{m,\ell}$ (n_{VH} rows)	Diagonal (P2/P6 shrinkage)	Block-tridiagonal in m dense in ℓ (P2, P5, P7, P8, P _{vv})	He trap/emit (P2, P5)
$\dot{c}_h$ (1 row)	0	He emission + re-solution (P5,	$-\mathcal{D}_h$

The Jacobian $\partial \mathbf{f} / \partial \mathbf{y}$ inherits this block structure. The blue tridiagonal block (i-i ladder) has bandwidth 2 in n . The green block-tridiagonal (vacancy/He) has block size $(\ell_{\text{max}} + 1)$ and tridiagonal coupling in m from P2/P5, with dense ℓ -coupling within each block from the He exchange terms P2/P5/P7/P8. The orange off-diagonal blocks are sparse-diagonal: each c_n couples to all $\bar{c}_m$ through the cavity absorption rate, and each $c_{m,\ell}$ couples to all c_n through the SIA-shrinkage rate. The red column/row couples the free-He scalar c_h to every (m, ℓ) node.

General coalescence case.

When cluster-cluster coalescence is included up to the mobility cutoffs i_{mobile} and v_{mobile} , the sparsity pattern changes in three blocks. (i) The SIA-SIA block becomes *dense lower-triangular* within the $i_{\text{mobile}} \times i_{\text{mobile}}$ mobile sub-block: i-i coalescence couples every pair $(n', n - n')$ with $n' \leq i_{\text{mobile}}$, filling the lower triangle. Rows $n > i_{\text{mobile}}$ (immobile loops) retain the tridiagonal structure of the mono-defect case. (ii) The SIA-VacHe and VacHe-SIA cross blocks become *dense in the mobile strip*: v-i annihilation couples every mobile vacancy cluster $V_{m'}$ ($m' \leq v_{\text{mobile}}$) to every SIA cluster I_n ($n \leq i_{\text{mobile}}$), filling a $i_{\text{mobile}} \times v_{\text{mobile}}$ sub-block. Outside the mobile strips both blocks remain diagonal (the immobile-cluster interactions are still mono-defect in form). (iii) The VacHe-VacHe block gains a *dense* $v_{\text{mobile}} \times v_{\text{mobile}}$ *corner*: v-v coalescence fills the upper-left sub-block corresponding to the mobile vacancy clusters, on top of the block-tridiagonal structure inherited from P2/P5/P7/P8.

The dashed lines mark the mobile/immobile boundaries at $n = i_{\text{mobile}}$ (vertical, in the SIA blocks) and $m = v_{\text{mobile}}$ (vertical, in the VacHe blocks; horizontal, separating

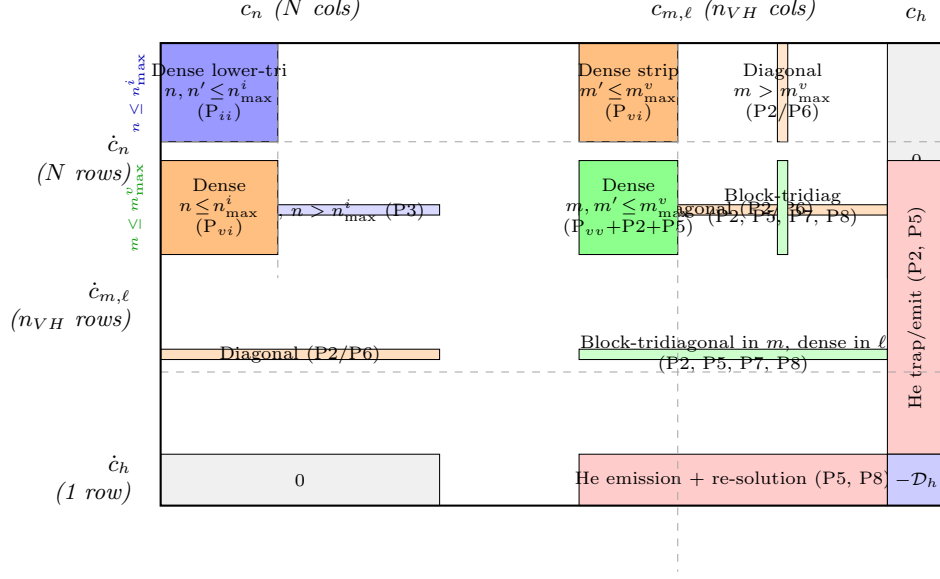


Fig. 9 Coalescence matrix structure

mobile and immobile vacancy rows). Darker shading indicates blocks that are denser in the coalescence formulation than in the mono-defect case. The c_h column and row (red) and the immobile-cluster sub-blocks (lighter shading) are structurally identical to the mono-defect diagram above.

8.8.2 He Content and Reduction Approximations

The dominant contributor to system size is the He index ℓ . The maximum He loading per cluster grows with cavity size as

$$\ell_{\max}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor, \quad \alpha_{\text{He}} = 1.5\text{--}2.0, \quad (243)$$

giving a total vacancy–He grid size

$$n_{VH}^{\text{full}} = \sum_{m=1}^M [\ell_{\max}(m) + 1] \approx M + \alpha_{\text{He}} \sum_{m=1}^M m^{2/3} \approx M \left(1 + \frac{3}{5} \alpha_{\text{He}} M^{2/3} \right), \quad (244)$$

which scales as $M^{5/3}$ and can reach $10^4\text{--}10^5$ for $M = 10^3\text{--}10^4$. Two physics-based approximations reduce this substantially, exploiting the very different He budgets in fission and fusion irradiation environments.

Case 1 — Fusion (high He/dpa): mean-field He approximation.

Under fusion-relevant He generation rates ($\sim 10\text{--}15$ appm He/dpa), He quickly saturates every cavity to its equilibrium loading. The two-dimensional (m, ℓ) distribution can then be collapsed to a one-dimensional problem by tracking only the mean He loading per cavity size:

$$\bar{\ell}(m, t) \equiv \frac{\sum_{\ell} \ell c_{m, \ell}}{\bar{c}_m}, \quad (245)$$

and treating the He distribution within each m -bin as a delta function at $\bar{\ell}(m)$. This replaces the n_{VH}^{full} equations for $c_{m, \ell}$ with two ODEs per vacancy size — one for the total cavity concentration $\bar{c}_m$ and one for the mean He content $\bar{\ell}_m$ — plus the single free-He equation:

$$n_{\text{eq}}^{\text{fusion}} = N + \underbrace{2M}_{\bar{c}_m, \bar{\ell}_m} + \underbrace{1}_{c_h} = N + 2M + 1. \quad (246)$$

Case 2 — Fission (low He/dpa): decoupled-He approximation.

Fission reactors produce $\sim 0.1\text{--}1$ appm He/dpa, so He loading remains low ($\ell \ll \ell_{\text{max}}$) and bubbles are rare. Most cavities are pure voids ($\ell = 0$). He can then be decoupled: track $c_{m, 0}$ (pure voids) separately from a single mean-field He concentration per size, treating bubble nucleation as a perturbation to the void distribution. In the simplest implementation this retains one He-bearing species per m and adds a single free-He equation:

$$n_{\text{eq}}^{\text{fission}} = N + \underbrace{M}_{\text{voids } c_{m, 0}} + \underbrace{1}_{\bar{c}_m, \ell \geq 1} + \underbrace{1}_{c_h} = N + M + 2. \quad (247)$$

8.8.3 System Size Summary

Table 31 collects the ODE counts for typical parameter choices ($N = 1000$, $M = 1000$, $\alpha_{\text{He}} = 1.5$).

Table 31 ODE system size for the three He-content strategies. All entries assume $N = 1000$ SIA sizes, $M = 1000$ vacancy sizes, $\alpha_{\text{He}} = 1.5$, and $i_{\text{mobile}} = 100$, $v_{\text{mobile}} = 5$. The column “Dominant block” identifies the bottleneck for Jacobian factorisation. Logarithmic size-binning for $m > m_{\text{group}} \sim 100\text{--}200$ can reduce n_{VH} by a further factor of 3–5.

He strategy	SIA block (N)	VacHe block (n_{VH})	n_{eq}	Dominant block
Full (m, ℓ) grid (Eq. 244)	1 000	$\sim 38\,000$	$\sim 39\,001$	VacHe
Case 1: mean-field He, fusion (Eq. 246)	1 000	2 000	3 001	SIA + VacHe
Case 2: decoupled He, fission (Eq. 247)	1 000	1 001	2 002	SIA + VacHe

8.8.4 Stiffness and Solver Strategy

The system (242) is severely stiff. The stiffness ratio is dominated by the disparity between the He migration frequency and the vacancy frequency,

$$\frac{\omega_h^{\text{eff}}}{\omega_v^{\text{eff}}} \sim 10^4\text{--}10^8, \quad (248)$$

with additional stiffness contributed by the contrast between fast mobile-cluster reactions (small n, m) and slow large-cluster dynamics. An implicit solver with adaptive time stepping is mandatory; recommended options are LSODA (automatic stiffness detection) or CVODE (variable-order BDF with sparse Jacobian). Practical tolerances are `rtol` = 10^{-8} , `atol` = 10^{-12} for the conservation diagnostics of Section 8.4.3 to remain below 10^{-6} .

Logarithmic size-binning (grouping) replaces the uniform m -grid with logarithmically spaced bins for $m > m_{\text{group}} \approx 50\text{--}200$. Each bin $[m_k, m_{k+1})$ is represented by a single representative concentration; capture rates are evaluated at the bin midpoint and weighted by the bin width. This reduces n_{VH} by a factor of 3–10 with negligible loss of accuracy for the swelling and loop-density predictions. The conservation diagnostics should be evaluated at every output step; the implementation should assert a hard stop if δ_{FP} or δ_{He} exceeds 10^{-3} .

9 Reduction of the He–Vacancy State Space

The full master equation (227) tracks every (m, ℓ) pair with $1 \leq m \leq M$ and $0 \leq \ell \leq \ell_{\max}(m)$ (Eq. 243), yielding n_{VH}^{full} coupled ODEs (Eq. 244) — typically 10^4 – 10^5 equations. Two physics-based approximations exploit the separation of timescales between He and vacancy kinetics to reduce this to a tractable system. Both retain the full SIA population (Eq. 224) and the free He balance (Eq. 229) unchanged; only the vacancy-cluster block is replaced.

Throughout this section, we use the notation and rate constants established in Sections 6–7: $\omega_{\alpha}^{\text{eff}}$ are the effective reaction frequencies (Eq. 201); $A_{\text{sph}} = (48\pi^2)^{1/3}$; $\bar{c}_m = \sum_{\ell} c_{m,\ell}$ is the total cavity concentration; and $\mathcal{K}_{n,m}^{(\text{cav})}$ is the SIA-cluster absorption rate (Eq. 225). The vacancy net-flux from size m to $m+1$ is defined as:

$$J_{m \rightarrow m+1} \equiv A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0} \bar{c}_m - \varepsilon_v(m+1, \bar{\ell}(m+1)) \bar{c}_{m+1}, \quad (249)$$

with forward (capture) and backward (emission) components $J_{m \rightarrow m+1}^+$ and $J_{m \rightarrow m+1}^-$ respectively, so that $J_{m \rightarrow m+1} = J_{m \rightarrow m+1}^+ - J_{m+1 \rightarrow m}^-$.

9.1 Case 1: Fast He Equilibration — Fusion Conditions

At temperatures where vacancies are mobile ($T \gtrsim 200^\circ\text{C}$), the He reaction frequency greatly exceeds the vacancy reaction frequency $\omega_h^{\text{eff}} \gg \omega_v^{\text{eff}}$. Helium equilibrates on each cavity faster than the cavity grows or shrinks, so the full two-dimensional (m, ℓ) distribution can be collapsed into two scalar quantities per vacancy size: the total cavity concentration $\bar{c}_m$ and the mean He loading

$$\bar{\ell}(m, t) \equiv \frac{\sum_{\ell} \ell c_{m,\ell}}{\bar{c}_m}. \quad (250)$$

This is the mean-field He approximation defined in Eq. (245).

Cavity size distribution.

Summing Eq. (227) over all ℓ at fixed m :

$$\frac{d\bar{c}_m}{dt} = J_{m-1 \rightarrow m} - J_{m \rightarrow m+1} - \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m + \Gamma_{\text{TM}}(m-1, \bar{\ell}) \bar{c}_{m-1} - \Gamma_{\text{TM}}(m, \bar{\ell}) \bar{c}_m, \quad (251)$$

where the vacancy net flux $J_{m \rightarrow m+1}$ is given by Eq. (249). The SIA-shrinkage sum $\sum_n \mathcal{K}_{n,m}^{(\text{cav})} c_n$ spans all mobile SIA clusters $n = 1, \dots, i_{\text{mobile}}$ with no bias factor (spherical cavities are unbiased, $Z_i^{\text{cav}} = 1$, Eq. 203). The vacancy capture rate $A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}}$ is independent of ℓ , so the ℓ -summation is exact for the capture terms. The emission rate ε_v depends on ℓ through the He gas-pressure correction to $E_b^v(m, \ell)$ and is approximated by evaluation at the mean loading $\bar{\ell}(m)$.

Mean He loading.

Define the He inventory at vacancy size m as $N_{\text{He}}(m) \equiv \bar{\ell}(m) \bar{c}_m$. Multiplying Eq. (227) by ℓ , summing over ℓ , and collecting terms:

$$\begin{aligned}
\frac{dN_{\text{He}}(m)}{dt} = & A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \bar{c}_m - \varepsilon_h(m, \bar{\ell}) \bar{c}_m - b_0 \dot{\phi} N_{\text{He}}(m) \\
& + \bar{\ell}(m-1) J_{m-1 \rightarrow m}^+ - \bar{\ell}(m) J_{m \rightarrow m+1}^+ \\
& + \bar{\ell}(m+1) J_{m+1 \rightarrow m}^- - \bar{\ell}(m) J_{m \rightarrow m-1}^- \\
& - \bar{\ell}(m) \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m \\
& - \mathcal{D}_v N_{\text{He}}(m) \delta_{m \leq v_{\text{mobile}}}.
\end{aligned} \tag{252}$$

The five groups of terms are: (i) He capture from the free pool minus thermal He emission (P5) minus radiation re-resolution (P8); (ii–iii) He advected in and out as clusters grow or shrink along the m -axis, where J^+ is the forward (capture) component and J^- is the backward (emission) component of the vacancy flux; (iv) He carried away when SIA clusters annihilate vacancies; and (v) He lost when mobile vacancy clusters ($m \leq v_{\text{mobile}}$) are absorbed at fixed sinks. Note the absence of any bias factor in the SIA-shrinkage term: the rate constant $\mathcal{K}_{n,m}^{(\text{cav})}$ already encodes the unbiased cavity geometry ($Z_i^{\text{cav}} = 1$).

Applying the quotient rule $d\bar{\ell}/dt = (\dot{N}_{\text{He}} - \bar{\ell} \dot{\bar{c}}_m)/\bar{c}_m$ and substituting Eq. (251):

$$\begin{aligned}
\frac{d\bar{\ell}(m)}{dt} = & A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h - \varepsilon_h(m, \bar{\ell}(m)) - b_0 \dot{\phi} \bar{\ell}(m) \\
& + \frac{J_{m-1 \rightarrow m}^+}{\bar{c}_m} [\bar{\ell}(m-1) - \bar{\ell}(m)] + \frac{J_{m+1 \rightarrow m}^-}{\bar{c}_m} [\bar{\ell}(m+1) - \bar{\ell}(m)].
\end{aligned}$$

(253)

Line 1 contains the local He kinetics: trapping from the free pool (proportional to c_h), thermal emission ε_h (P5, Eq. 212), and athermal re-resolution (P8, Eq. 215) which drives $\bar{\ell}$ toward zero. Line 2 contains advection corrections: cavities flowing into size class m carry their own He loading, which may differ from $\bar{\ell}(m)$; the difference $[\bar{\ell}(m \pm 1) - \bar{\ell}(m)]$ acts as a drift source.

Numerical implementation.

In practice, $N_{\text{He}}(m) = \bar{\ell}(m) \bar{c}_m$ and $\bar{c}_m$ are evolved as primary variables; $\bar{\ell}(m)$ is recovered by division and used only when evaluating rates. When $\bar{c}_m$ approaches zero near the nucleation threshold, the quotient becomes ill-conditioned. Recommended practice: set a floor $\bar{c}_m^{\text{min}} \sim 10^{-30}$ at/at below which $\bar{\ell}$ is reset to zero.

Case 1 summary.

Replacing the full (m, ℓ) grid with $2M$ ODEs — M for $\bar{c}_m$ (Eq. 251) and M for $\bar{\ell}(m)$ (Eq. 253) — plus the unchanged free-He equation (Eq. 229) gives the reduced system count of Eq. (246):

$$n_{\text{eq}}^{\text{fusion}} = N + 2M + 1.$$

The reduction factor relative to the full grid is approximately $n_{VH}^{\text{full}}/(2M) \approx \frac{3}{10}\alpha_{\text{He}}M^{2/3}$, which is ~ 10 – 20 for $M = 10^3$ and $\alpha_{\text{He}} = 1.5$.

9.2 Case 2: Decoupled He Inventory — Fission Conditions

Under fission irradiation the He generation rate is low (~ 0.3 – 1 appm/dpa vs. ~ 10 – 15 appm/dpa for fusion), so most cavities carry $\ell \ll m$ and the He/V ratio $\bar{\ell}/m \ll 1$. Helium acts as a weak perturbation on the void kinetics rather than controlling nucleation. Rather than tracking $\bar{\ell}(m)$ as a separate variable at each size, the total trapped He inventory is followed as a single scalar and distributed algebraically across cavity sizes.

Cavity size distribution.

The $\bar{c}_m$ equation is identical to Case 1, Eq. (251), with the vacancy emission rate evaluated using an He-corrected binding energy. In the low-loading limit the He gas pressure obeys the ideal-gas law, giving an effective binding energy:

$$E_b^{v,\text{eff}}(m) \equiv E_b^v(m, \ell(m)) = E_f^v + \frac{2\gamma_s \Omega}{r_m} - \frac{\ell(m) k_B T}{m}, \quad (254)$$

where $r_m = \xi(m)a$, γ_s is the surface tension, $\Omega = a^3/2$ is the atomic volume, and $\ell(m)$ is the He content assigned to cavities of size m by the allocation rule below. The third term is the ideal-gas pressure correction $p(\ell, m)\Omega = (\ell k_B T/m) \cdot m/m = \ell k_B T/m$, adequate when $\ell/m \ll 1$. At fission temperatures and low He loading, thermal He emission is negligible ($E_B^{\text{He}} \gtrsim 2$ eV); re-solution is the sole He release channel.

Total trapped He inventory.

Define $\ell_{\text{tot}} \equiv \sum_{m=1}^M \ell(m) \bar{c}_m$. Summing the He content over all cavities and using Eq. (229) with thermal emission dropped:

$$\frac{d\ell_{\text{tot}}}{dt} = A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m=1}^M m^{1/3} \bar{c}_m - b_0 \dot{\phi} \ell_{\text{tot}} - \mathcal{D}_v \sum_{m=1}^{v_{\text{mobile}}} \ell(m) \bar{c}_m. \quad (255)$$

The first term is the total He trapping rate by all cavities (the sum is the weighted capture cross-section $\sum_m m^{1/3} \bar{c}_m$), the second is the athermal re-solution loss (P8), and the third is He lost when mobile vacancy clusters ($m \leq v_{\text{mobile}}$) are absorbed at fixed sinks.

He allocation.

Helium is distributed across cavity sizes in proportion to the capture rate $A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$, as that determines the steady-state He loading when trapping and re-solution balance.

$$\ell(m) = \ell_{\text{tot}} \frac{m^{1/3} \bar{c}_m}{\sum_{m'=1}^M m'^{1/3} \bar{c}_{m'}}. \quad (256)$$

This algebraic relation is recomputed at each time step. The $m^{1/3}$ weighting arises from the spherical-cavity capture radius $\xi(m) \propto m^{1/3}$ (Section 6.1), consistent with $\mathcal{K}_{h,m} = A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$ (Table 28).

Free He.

The free-He equation retains all terms from Eq. (229), including the fixed-sink loss $\mathcal{D}_h c_h$, but with thermal He emission dropped:

$$\frac{dc_h}{dt} = G_{\text{He}} - A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m=1}^M m^{1/3} \bar{c}_m + b_0 \dot{\phi} \ell_{\text{tot}} - \mathcal{D}_h c_h. \quad (257)$$

The trapping and re-solution terms are exactly the negatives of those in Eq. (255), so the He conservation law $d(c_h + \ell_{\text{tot}})/dt = G_{\text{He}} - \mathcal{D}_h c_h$ holds (cf. Eq. 380).

Case 2 summary.

The reduced system comprises M ODEs for $\bar{c}_m$, one ODE for c_h (Eq. 257), one ODE for ℓ_{tot} (Eq. 255), and the algebraic allocation (Eq. 256), giving the system count of Eq. (247):

$$n_{\text{eq}}^{\text{fission}} = N + M + 2.$$

9.3 Radiation Re-Solution: Limiting Behavior

The re-solution rate $\Gamma_{\text{res}}(m, \ell) = b_0 \ell \dot{\phi}$ (P8, Eq. 215) is athermal and appears in both reduced models as $-b_0 \dot{\phi} \ell$ in Eq. (253) and as $-b_0 \dot{\phi} \ell_{\text{tot}}$ in Eq. (255). Its competition with thermal He emission determines two limiting regimes.

Low-temperature limit

Setting $d\bar{\ell}/dt = 0$ in Eq. (253) and neglecting the advection terms:

$$\bar{\ell}_{\text{ss}} = \frac{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h}{b_0 \dot{\phi}}. \quad (258)$$

Re-solution sets the equilibrium He loading by competing with trapping. Without re-solution ($b_0 = 0$), $\bar{\ell}$ grows until $\ell_{\text{max}}(m)$ is reached.

High-temperature limit ($\varepsilon_h \gg b_0 \dot{\phi} \bar{\ell}$).

$$\bar{\ell}_{\text{ss}} \approx \frac{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h}{\varepsilon_h(m, \bar{\ell}_{\text{ss}})}, \quad (259)$$

which is the thermodynamic (detailed-balance) equilibrium loading (cf. Eq. 212), independent of irradiation rate.

Crossover temperature.

The transition between the two regimes occurs when $\varepsilon_h(m, \bar{\ell}) \sim b_0 \dot{\phi} \bar{\ell}$. For typical EUROFER parameters ($E_B^{\text{He}} \sim 2.0 \text{ eV}$, $b_0 \sim 10^{-2} \text{ dpa}^{-1}$, $\dot{\phi} \sim 10^{-6} \text{ dpa/s}$), this crossover lies at $T \approx 500\text{--}700^\circ\text{C}$: below this temperature, re-resolution controls He loading; above it, thermal emission dominates.

Consequences for bubble nucleation.

Re-resolution suppresses nucleation at low temperatures by ejecting He from small embryos, creating an incubation dose $\propto 1/G_{\text{He}}$ for fixed b_0 . It preferentially dissolves small, He-rich embryos (low ℓ_{ss} from Eq. 258) while leaving large stable bubbles intact (higher capture rate $\propto m^{1/3}$), producing the bimodal cavity size distributions commonly observed experimentally.

9.4 Comparison of Cases 1 and 2

Table 32 summarizes the two approximations and their applicability. Both use the same $\bar{c}_m$ equation (Eq. 251) and the same unbiased SIA-shrinkage rate $\sum_n \mathcal{K}_{n,m}^{(\text{cav})} c_n$; they differ in how the He degree of freedom is represented and in the approximations made for thermal He emission. For large M , logarithmic size grouping for $m > m_{\text{group}} \approx 50\text{--}200$ is applied in both cases; the $\bar{\ell}(m)$ variable extends naturally to grouped bins by carrying the mean loading per bin.

Table 32 Comparison of the two He–vacancy state-space reductions. Both cases retain the full SIA equation (Eq. 224) and use the unbiased cavity absorption rate $\mathcal{K}_{n,m}^{(\text{cav})}$ (Eq. 225) with $Z_i^{\text{cav}} = 1$.

	Case 1 (Fusion)	Case 2 (Fission)
He generation rate	$\sim 10\text{--}15 \text{ appm/dpa}$	$\sim 0.3\text{--}1 \text{ appm/dpa}$
Typical He/V ratio $\bar{\ell}/m$	0.1–10	$\ll 1$
He tracked as	$\bar{\ell}(m)$ per size (ODE)	ℓ_{tot} (scalar ODE)
He allocation	implicit in $\bar{\ell}(m)$	algebraic (Eq. 256)
He emission	thermal + re-resolution	re-resolution only
He effect on E_b^v	full $E_b^v(m, \bar{\ell})$	ideal-gas correction
He conservation	Eq. 235 exactly	Eq. 380 (approx.)
Total ODEs	$N + 2M + 1$	$N + M + 2$
Reduction vs. full grid	$\approx \frac{3}{10} \alpha_{\text{He}} M^{2/3}$	$\approx \frac{3}{5} \alpha_{\text{He}} M^{2/3}$

10 State Space Reduction via Size-Bin Moments

The cluster dynamics (CD) system derived in Section ?? tracks one ODE per cluster size. Even after the He approximations of Section ??, the SIA ladder runs to $N \sim 10^3$ – 10^4 and the vacancy grid to $M \sim 10^3$ – 10^4 , giving several thousand coupled ODEs. For large-scale parametric studies or coupling to microstructure-informed continuum codes, a further reduction is desirable. This section presents a systematic size-grouping framework that replaces the per-size equations with a much smaller set of *bin moment equations*, derived by integrating the master equations over logarithmically spaced size bins. Two closure strategies are described: a *moment-closure* approach analogous to the method of moments in aerosol science, and a *shape-function* approach analogous to the Galerkin finite-element method. Both reduce the ODE count from $O(N)$ or $O(M)$ to $O(N_B)$ where $N_B = O(\log N / \log r)$ for bin ratio $r > 1$, typically 20–60 bins for $N = 10^4$ and $r = 1.5$ – 2 .

The derivation is presented for the SIA cluster population c_n ; the vacancy cluster population $\bar{c}_m$ follows by identical steps with $n \rightarrow m$, $N \rightarrow M$, and the appropriate rate constants substituted.

10.1 Logarithmic Bin Partition

Partition the integer size axis $\{1, 2, \dots, N\}$ into N_B bins. The first n_0 sizes (mobile clusters, $n \leq i_{\text{mobile}}$) are resolved individually; the remaining large immobile clusters are grouped. Define the bin boundaries

$$1 = s_0 < s_1 < \dots < s_{n_0} = i_{\text{mobile}} < s_{n_0+1} < \dots < s_{N_B} = N, \quad (260)$$

where for $k > n_0$ the boundaries follow a geometric progression with ratio $r > 1$:

$$s_{k+1} = \lfloor r s_k \rfloor + 1, \quad r = 1.5\text{--}2.0. \quad (261)$$

Bin k spans the integer interval $\mathcal{B}_k = \{s \in \mathbb{Z} : s_k \leq s < s_{k+1}\}$ and contains $\Delta_k = s_{k+1} - s_k$ cluster sizes. The representative size of bin k is the geometric mean of its boundaries:

$$\hat{s}_k = \sqrt{s_k s_{k+1}}. \quad (262)$$

The number of grouped bins is

$$N_B^{\text{group}} = \left\lceil \frac{\ln(N/i_{\text{mobile}})}{\ln r} \right\rceil, \quad N_B = i_{\text{mobile}} + N_B^{\text{group}}. \quad (263)$$

For $N = 5000$, $i_{\text{mobile}} = 100$, $r = 2$: $N_B^{\text{group}} = \lceil \ln(50)/\ln 2 \rceil = 6$, so only 106 equations replace 4900 per-size ODEs.

10.2 Bin Moments

For each bin k define the *zeroth moment* (total cluster concentration in the bin) and the *first moment* (total defect inventory in the bin):

$$\mu_k^{(0)}(t) \equiv \sum_{n \in \mathcal{B}_k} c_n(t), \quad \mu_k^{(1)}(t) \equiv \sum_{n \in \mathcal{B}_k} n c_n(t). \quad (264)$$

The mean cluster size within bin k is $\langle n \rangle_k = \mu_k^{(1)} / \mu_k^{(0)}$ and the swelling (first-moment sum) is $S_I = \sum_k \mu_k^{(1)}$. Higher moments can be defined similarly; in practice, tracking $(\mu_k^{(0)}, \mu_k^{(1)})$ at each bin is sufficient to close the rate equations to second order.

10.3 Bin-Integrated Rate Equations

Summing the corrected master equation (224) (Section 8.3.1) over all $n \in \mathcal{B}_k$ gives the zeroth-moment equation. *Every* rate process in the per-size equation must appear in the projection; omitting any term breaks conservation. The complete zeroth-moment equation is:

$$\begin{aligned}
\frac{d\mu_k^{(0)}}{dt} = & \underbrace{\sum_{n \in \mathcal{B}_k} G_n}_{\text{production}} \\
& + \underbrace{J_{k-1 \rightarrow k} - J_{k \rightarrow k+1}}_{\text{inter-bin flux (advection in size space)}} \\
& + \underbrace{\sum_{n \in \mathcal{B}_k} \left[\frac{1}{2} \sum_{n'=1}^{i_{\text{mobile}}} \mathcal{K}_{n',n-n'}^{ii} c_{n'} c_{n-n'} - c_n \sum_{n'=1}^{i_{\text{mobile}}} \mathcal{K}_{n,n'}^{ii} c_{n'} \right]}_{\text{SIA-SIA coalescence (gain + loss)}} \\
& + \underbrace{\sum_{n \in \mathcal{B}_k} [\varepsilon_i(n+1) c_{n+1} - \varepsilon_i(n) c_n \delta_{n \geq 2}]}_{\text{thermal emission (SIA from loops)}} \\
& + \underbrace{\sum_{n \in \mathcal{B}_k} \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m',n+m'}^{vi} c_{m',0} c_{n+m'}}_{\text{V-I annihilation gain (} V_{m'} + I_{n+m'} \rightarrow I_n \text{)}} \\
& - \underbrace{\sum_{n \in \mathcal{B}_k} \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m',n}^{vi} c_{m',0} c_n}_{\text{V-I annihilation loss (} V_{m'} + I_n \rightarrow I_{n-m'} \text{)}} \\
& - \underbrace{\sum_{n \in \mathcal{B}_k} \sum_{m=1}^M \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m}_{\text{SIA cluster-cavity absorption}} \\
& - \underbrace{\mathcal{D}_i \mu_k^{(0)}}_{\text{fixed-sink loss}} \\
& + \underbrace{\delta_{1 \in \mathcal{B}_k} \sum_{m,\ell} \Gamma_{\text{TM}}(m,\ell) c_{m,\ell}}_{\text{trap mutation (only bin containing } n=1 \text{)}}
\end{aligned} \tag{265}$$

where the bin-averaged cavity absorption rate is

$$\hat{\mathcal{K}}_{k,m}^{(\text{cav})} \equiv \frac{1}{\Delta_k} \sum_{n \in \mathcal{B}_k} \mathcal{K}_{n,m}^{(\text{cav})}. \tag{266}$$

Vacancy and He equations (not binned).

Only the SIA population is bin-grouped. The vacancy cluster equations and free He equation retain full per-size resolution and use the *corrected* master equations (Eqs. 227, 229) with all rate processes from Section 8.3:

- P1 recombination at $m = 1$ uses $\mathcal{K}_{iv}$ (Eq. 222), not $\mathcal{K}_{\alpha,1}^{\text{cav}}$.
- SIA-induced shrinkage includes both gain ($I_n + V_{m+n} \rightarrow V_m$) and loss ($I_n + V_m \rightarrow V_{m-n}$) for all mobile $n = 1, \dots, i_{\text{mobile}}$.
- 1D/3D mixed cavity absorption for $n = 4, \dots, i_{\text{mobile}}$ via $\mathcal{K}_{n,m}^{(\text{cav})}$ (Eq. 225).
- He capture does *not* change the vacancy size class m ; no $\mathcal{K}_{\text{HeV}}$ term appears in $d\bar{c}_m/dt$.
- The He-balance equations (Q_tot or Q_m) include emission, fixed-sink loss, and uptake terms (Sections 8.3.3, ??).

The *inter-bin flux* from bin k to bin $k+1$ counts clusters that grow past the boundary s_{k+1} by absorbing a mono-defect (or a mobile cluster in the coalescence formulation):

$$J_{k \rightarrow k+1} = \hat{\mathcal{K}}_{k,\text{bdy}}^{ii} \mu_k^{(0)} c_{i_{\text{mobile}}} + \sum_{m'=1}^{v_{\text{mobile}}} \hat{\mathcal{K}}_{k,\text{bdy}}^{vi,m'} \mu_k^{(0)} c_{m',0}, \quad (267)$$

where $\hat{\mathcal{K}}_{k,\text{bdy}}$ denotes the rate constant evaluated at the bin boundary s_{k+1} . In the mono-defect limit ($i_{\text{mobile}} = v_{\text{mobile}} = 1$), this simplifies to

$$J_{k \rightarrow k+1}^{\text{mono}} = \mathcal{K}_{i,s_{k+1}}^{\text{loop}} c_1 \mu_k^{(0)}, \quad (268)$$

i.e. the rate at which clusters sitting at the upper boundary of bin k absorb a mono-SIA and cross into bin $k+1$.

Multiplying Eq. (224) by n and summing over $\mathcal{B}_k$ gives the first-moment equation. Every n -weighted term corresponds to the same physical process as in the zeroth-moment equation (Eq. 265), weighted by the cluster size n .

$$\begin{aligned}
\frac{d\mu_k^{(1)}}{dt} = & \underbrace{\sum_{n \in \mathcal{B}_k} n G_n}_{\text{production}} \\
& + \underbrace{s_{k+1} J_{k-1 \rightarrow k} - s_k J_{k \rightarrow k+1}}_{\text{inter-bin flux (boundary-weighted)}} \\
& + \underbrace{\sum_{n \in \mathcal{B}_k} n \left[\frac{1}{2} \sum_{n'} \mathcal{K}_{n', n-n'}^{ii} c_{n'} c_{n-n'} - c_n \sum_{n'} \mathcal{K}_{n, n'}^{ii} c_{n'} \right]}_{\text{SIA-SIA coalescence (gain + loss)}} \\
& + \underbrace{\sum_{n \in \mathcal{B}_k} n [\varepsilon_i(n+1) c_{n+1} - \varepsilon_i(n) c_n \delta_{n \geq 2}]}_{\text{thermal emission (SIA from loops)}} \\
& + \underbrace{\sum_{n \in \mathcal{B}_k} n \sum_{m'} \mathcal{K}_{m', n+m'}^{vi} c_{m', 0} c_{n+m'}}_{\text{V-I annihilation gain (} V_{m'} + I_{n+m'} \rightarrow I_n \text{)}} \\
& - \underbrace{\sum_{n \in \mathcal{B}_k} n \sum_{m'} \mathcal{K}_{m', n}^{vi} c_{m', 0} c_n}_{\text{V-I annihilation loss (} V_{m'} + I_n \rightarrow I_{n-m'} \text{)}} \\
& - \underbrace{\sum_{n \in \mathcal{B}_k} n \sum_m \mathcal{K}_{n, m}^{(\text{cav})} c_n \bar{c}_m}_{\text{SIA cluster-cavity absorption}} \\
& - \underbrace{\mathcal{D}_i \mu_k^{(1)}}_{\text{fixed-sink loss}} \\
& + \underbrace{\delta_{1 \in \mathcal{B}_k} \sum_{m, \ell} \Gamma_{\text{TM}}(m, \ell) c_{m, \ell}}_{\text{trap mutation (only bin containing } n=1 \text{)}}
\end{aligned} \tag{269}$$

10.4 Closure: Two Strategies

Both $\mu_k^{(0)}$ and $\mu_k^{(1)}$ are known at each time step. The rate equations above contain products such as $\hat{\mathcal{K}}_{k, m}^{(\text{cav})} \mu_k^{(0)}$ and $J_{k \rightarrow k+1}$, which require knowledge of the *intra-bin distribution* c_n for $n \in \mathcal{B}_k$. Two strategies close this system.

10.4.1 Piecewise-Constant (Zeroth-Order) Closure

Assume the distribution is uniform within each bin:

$$c_n \approx \frac{\mu_k^{(0)}}{\Delta_k} \equiv \bar{c}_k \quad \forall n \in \mathcal{B}_k. \quad (270)$$

Then all bin-averaged rate constants reduce to evaluation at the bin midpoint $\hat{s}_k$:

$$\hat{\mathcal{K}}_{k,m}^{(\text{cav})} \approx \mathcal{K}_{\hat{s}_k,m}^{(\text{cav})}, \quad (271)$$

and the inter-bin flux becomes

$$J_{k \rightarrow k+1} \approx \mathcal{K}_{i,\hat{s}_{k+1}}^{\text{loop}} c_1 \bar{c}_k. \quad (272)$$

This is the standard *logarithmic grouping* used in most production CD codes [Golubov et al \(2000\)](#); [Ovcharenko et al \(2003\)](#); [Golubov et al \(2007\)](#); [Kohnert and Wirth \(2017\)](#); [Cui et al \(2020\)](#). It conserves $\mu_k^{(0)}$ exactly but not $\mu_k^{(1)}$; the error in S_I is $O(\Delta_k/\hat{s}_k) = O(r-1)$.

10.4.2 Shape-Function (Galerkin) Closure

Approximate the intra-bin distribution as a linear combination of basis functions defined on the bin:

$$c_n \approx \sum_p a_k^{(p)} \phi_p(n), \quad n \in \mathcal{B}_k, \quad (273)$$

where the coefficients $a_k^{(p)}$ are determined by the known moments $(\mu_k^{(0)}, \mu_k^{(1)}, \dots)$ and ϕ_p are chosen basis functions on $\mathcal{B}_k$.

Hat-function basis (linear FEM on log scale).

Map $n \in [s_k, s_{k+1})$ to $\xi = \ln(n/s_k)/\ln r \in [0, 1)$ and use piecewise-linear (hat) functions:

$$\phi_0(\xi) = 1 - \xi, \quad \phi_1(\xi) = \xi. \quad (274)$$

The two coefficients $a_k^{(0)}$ and $a_k^{(1)}$ are fixed by the two moments $\mu_k^{(0)}$ and $\mu_k^{(1)}$ via:

$$a_k^{(0)} I_{00} + a_k^{(1)} I_{01} = \mu_k^{(0)}, \quad (275)$$

$$a_k^{(0)} I_{10} + a_k^{(1)} I_{11} = \mu_k^{(1)}, \quad (276)$$

where $I_{pq} = \sum_{n \in \mathcal{B}_k} n^p \phi_q(\ln n/s_k/\ln r)$ are geometry integrals computed once at setup. This two-parameter ansatz is the FEM analogue of the piecewise-linear representation in aerosol sectional methods [Adams and Seinfeld \(2002\)](#); [Jacobson et al \(1994\)](#).

Log-normal basis.

Alternatively, within each bin assume a log-normal shape:

$$c_n \approx \frac{a_k}{\sqrt{2\pi} \sigma_k n} \exp \left[-\frac{(\ln n - \mu_k)^2}{2\sigma_k^2} \right], \quad (277)$$

with $\mu_k = \ln \langle n \rangle_k$ and σ_k determined from the second moment $\mu_k^{(2)}$. This three-parameter closure requires tracking one additional moment per bin but captures the asymmetric, right-skewed distributions typical of nucleation-growth kinetics.

Log-normal parameter determination.

Given the three tracked moments $\mu_k^{(0)}$, $\mu_k^{(1)}$, and $\mu_k^{(2)} = \sum_{n \in \mathcal{B}_k} n^2 c_n$, the log-normal parameters are:

$$\bar{n}_k = \frac{\mu_k^{(1)}}{\mu_k^{(0)}}, \quad (278)$$

$$\sigma_k^2 = \ln \left(\frac{\mu_k^{(2)} \mu_k^{(0)}}{\mu_k^{(1)^2}} \right), \quad (279)$$

$$m_k = \ln \bar{n}_k - \frac{1}{2} \sigma_k^2. \quad (280)$$

The amplitude a_k is then determined by normalizing the log-normal to reproduce the zeroth moment:

$$a_k = \frac{\mu_k^{(0)}}{\int_{\mathcal{B}_k} \exp \left[-\frac{(\ln n - m_k)^2}{2\sigma_k^2} \right] dn}. \quad (281)$$

In the monodisperse limit $\sigma_k^2 \rightarrow 0$ the log-normal collapses to a delta function at $\bar{n}_k$, recovering the piecewise-constant ($P = 1$) closure.

Second-moment evolution.

Differentiating $\mu_k^{(2)}$ and using the master equation gives

$$\frac{d\mu_k^{(2)}}{dt} = \sum_{n \in \mathcal{B}_k} n^2 [G_n^+ c_{n-1} - (G_n^+ + G_n^-) c_n + G_{n+1}^- c_{n+1}], \quad (282)$$

which, after the same bin-boundary algebra used for $\mu_k^{(0)}$ and $\mu_k^{(1)}$, closes in terms of the three moments and the log-normal shape parameters.

Galerkin projection.

Regardless of basis, the Galerkin projection of the master equation (224) onto ϕ_p over $\mathcal{B}_k$ gives:

$$\sum_{n \in \mathcal{B}_k} \phi_p(n) \frac{dc_n}{dt} = \sum_{n \in \mathcal{B}_k} \phi_p(n) f_n(\mathbf{c}), \quad (283)$$

where f_n is the right-hand side of Eq. (224). The left side becomes $\dot{a}_k^{(p)}$ (via the moment relations), and the right side is expressed in terms of $\{a_k^{(p)}\}$ and the inter-bin fluxes through the shape ansatz. The resulting ODE system for the coefficients $\{a_k^{(p)}\}$ is closed and of size $P \times N_B$ where P is the number of moments per bin ($P = 1$ for piecewise-constant, $P = 2$ for hat-function, $P = 3$ for log-normal).

10.5 Inter-Bin Flux Treatment

The inter-bin flux $J_{k \rightarrow k+1}$ requires special care because it drives mass transfer between bins and must satisfy discrete conservation exactly. Writing the flux as an upwind scheme in size space:

$$J_{k \rightarrow k+1} = \begin{cases} \beta_{s_{k+1}} c_{s_{k+1}}^- & \text{if growth} > \text{shrinkage}, \\ \beta_{s_{k+1}} c_{s_{k+1}}^+ & \text{if shrinkage} > \text{growth}, \end{cases} \quad (284)$$

where $c_{s_{k+1}}^\pm$ denotes the distribution evaluated just below/above the boundary from within the respective bin, and $\beta_{s_{k+1}}$ is the net growth rate at the boundary:

$$\beta_s = \mathcal{K}_{i,s}^{\text{loop}} c_1 - \mathcal{K}_{v,s}^{\text{loop}} c_{1,0} - \varepsilon_i(s). \quad (285)$$

The upwind flux preserves positivity of $\mu_k^{(0)}$ and prevents spurious oscillations between adjacent bins. For the shape-function closure, $c_{s_{k+1}}^-$ is evaluated from the intra-bin ansatz at $n = s_{k+1} - 1$, giving a smooth interpolation.

10.6 Accuracy and Conservation Properties

Mass conservation.

Summing Eq. (265) over all bins recovers $dS_I/dt = d(\sum_k \mu_k^{(1)})/dt$ exactly, provided the inter-bin fluxes satisfy the telescoping identity:

$$\sum_{k=1}^{N_B} (J_{k-1 \rightarrow k} - J_{k \rightarrow k+1}) = 0 \quad (286)$$

(boundary terms vanish: $J_{0 \rightarrow 1} = 0$ and $J_{N_B \rightarrow N_B+1} = 0$). This holds for any consistent flux formula. The swelling identity $S(t) = S_I(t) + \Delta J^d(t)$ (Eq. 233) therefore remains exact regardless of the grouping scheme.

Truncation error.

For the piecewise-constant closure (Eq. 270), the error in the bin-averaged rate constant scales as:

$$\hat{\mathcal{K}}_{k,m}^{(\text{cav})} - \mathcal{K}_{\hat{s}_{k,m}}^{(\text{cav})} = O\left(\frac{\Delta_k^2}{\hat{s}_k^2}\right) = O((r-1)^2), \quad (287)$$

since $\mathcal{K}_{\hat{s},m}^{(\text{cav})} \propto \hat{s}^{1/3}$ or $\hat{s}^{1/2}$ and the bin width scales as $\Delta_k \sim (r-1)\hat{s}_k$. For $r = 2$ this gives $\sim 25\%$ error per bin; for $r = 1.5$, $\sim 6\%$. The hat-function closure reduces the error by one order: $O((r-1)^3)$, making $r = 2$ already accurate to $\lesssim 1\%$. The log-normal closure reduces the error by a further order to $O((r-1)^4)$, enabling accurate results even with $r = 3-4$.

Moment diagnostics.

The bin-integrated conservation diagnostics follow from Section 8.4.3:

$$\delta_{\text{FP}}^{\text{bin}} = \frac{|S - \sum_k \mu_k^{(1)} - \Delta J^d|}{S + \sum_k \mu_k^{(1)} + \Delta J^d}, \quad (288)$$

which should remain below 10^{-6} regardless of binning. A value above 10^{-4} indicates that the inter-bin fluxes are not conserving mass, typically due to an inconsistent upwind scheme at the bin boundaries.

10.7 Reduced System Size

Table 33 summarises the ODE count for the combined He-approximation and size-grouping reductions for representative parameter choices.

Table 33 ODE counts after combined size-grouping ($r = 2$) and He-approximation reductions. P is the number of moments per bin (1 = constant, 2 = linear/hat-function, 3 = log-normal). Parameters: $N = M = 5000$, $i_{\text{mobile}} = 100$, $v_{\text{mobile}} = 5$, $\alpha_{\text{He}} = 1.5$. “Full” refers to the un-grouped per-size system after He reduction.

He approx.	Size grouping	$N_{\text{B}}^{\text{SIA}}$	$N_{\text{B}}^{\text{vac}}$	n_{eq}
Case 1 (fusion)	Full per-size	5 000	10 000	15 001
Case 1 (fusion)	Log-group, $P = 1$	113	113	$226 P + 1 = 227$
Case 1 (fusion)	Log-group, $P = 2$	113	113	$226 P + 1 = 453$
Case 2 (fission)	Full per-size	5 000	5 000	10 002
Case 2 (fission)	Log-group, $P = 1$	113	113	$226 + 2 = 228$
Case 2 (fission)	Log-group, $P = 2$	113	113	$452 + 2 = 454$
Case 1 (fusion)	Log-group, $P = 3$	113	113	1.3
Case 2 (fission)	Log-group, $P = 3$	113	113	1.5

The log-grouping with $P = 1$ reduces the system by a factor of ~ 40 – 70 relative to the per-size He-reduced system, and by a factor of ~ 10 – 20 relative to the grouped formulation. With three closure levels—piecewise-constant ($P = 1$), linear hat-function ($P = 2$), and log-normal ($P = 3$)—the grouped formulation spans a wide accuracy / cost trade-off. $P = 1$ reproduces classical rate theory at minimal overhead; $P = 2$ resolves intra-bin slopes and captures $O((r-1)^2)$ corrections; $P = 3$ adds a shape degree of freedom that tracks the variance of each bin, reducing the truncation error to $O((r-1)^4)$ and enabling accurate results even with coarse bin ratios of $r = 3$ – 4 .

10.8 Implementation Notes

Setup.

At initialisation, precompute: (i) the bin boundaries $\{s_k\}$ from Eq. (261); (ii) the representative sizes $\{\hat{s}_k\}$ from Eq. (262); (iii) the bin-averaged rate constants $\hat{\mathcal{K}}_{k,m}^{(\text{cav})}$ for all (k, m) pairs from Eq. (271) (piecewise-constant) or by exact quadrature (hat-function); and (iv) the geometry integrals I_{pq} for the shape-function closure. Steps (i)–(iv) are $O(N_B \cdot M)$ and negligible compared with the ODE integration cost.

Coalescence terms.

For the i–i coalescence bins, the gain term $\frac{1}{2} \sum_{n' \in \mathcal{B}_k} \sum_{n'' \in \mathcal{B}_k} \mathcal{K}_{n',n''}^{ii} c_{n'} c_{n''}$ where $n' + n'' \in \mathcal{B}_k$ is approximated as

$$\frac{1}{2} \mathcal{K}_{\hat{s}_{k'}, \hat{s}_{k''}}^{ii} \mu_{k'}^{(0)} \mu_{k''}^{(0)} f_{\text{overlap}}(k, k', k''), \quad (289)$$

where f_{overlap} is the fraction of $(\hat{s}_{k'} + \hat{s}_{k''})$ that falls within $\mathcal{B}_k$ (a geometric factor computed at setup). The double sum over bin pairs reduces from $O(N^2)$ to $O(N_B^2)$, a further speedup of $(N/N_B)^2 \sim 10^3$ – 10^4 .

Coupling to the vacancy grid.

The SIA bins couple to the vacancy grid through the cavity absorption term $\hat{\mathcal{K}}_{k,m}^{(\text{cav})}$, which is a $N_B \times M$ matrix precomputed at setup. When the vacancy grid is also grouped, this becomes a $N_B^{SIA} \times N_B^{vac}$ matrix, reducing the coupling cost from $O(N \cdot M)$ to $O(N_B^{SIA} \cdot N_B^{vac})$ per right-hand-side evaluation.

Vacancy bin closure.

The vacancy population uses the same bin-moment framework as the SIA population: sizes $1 \leq m \leq v_{\text{mobile}}$ are tracked individually (discrete), and sizes $v_{\text{mobile}} < m \leq M$ are grouped into V_{bin} logarithmic bins. Each vacancy bin carries two moments $(\mu_k^{(0)}, \mu_k^{(1)})$, and the hat-function (dual-basis) closure (274)–(276) is used to reconstruct per-size concentrations c_m within each bin. This ensures that both the vacancy number density (zeroth moment) and vacancy content (first moment, proportional to swelling) are tracked exactly by the ODE system. The vacancy master equation (227)

is evaluated on the reconstructed per-size c_m distribution and then projected back onto the bin moments, identically to the SIA projection described above.

Conservation check.

Use Eq. (288) at every output step. An inconsistency in $\delta_{\text{FP}}^{\text{bin}}$ that was not present in the ungrouped system localises the error to the inter-bin flux implementation.

11 Extended Cluster Dynamics Model for H, He, V, and I Clusters

The foregoing sections treated the irradiated microstructure as a three-component system: SIA clusters, He–vacancy clusters, and free He. Under fusion-relevant conditions — and especially for accelerator-driven systems where the hydrogen transmutation rate can reach $\sim 1000\text{--}1500$ appm/dpa — hydrogen must be promoted to an explicit fourth species. The key physical differences from helium are:

1. H migrates $\sim 10^3\text{--}10^5\times$ faster than vacancies and $\sim 10\text{--}100\times$ faster than interstitial He (migration energy $E_m^H \approx 0.04\text{--}0.09$ eV versus $E_m^{\text{He}} \approx 0.06\text{--}0.13$ eV and $E_m^v \approx 0.60\text{--}0.70$ eV in bcc Fe).
2. H binds to vacancies moderately strongly ($E_b^H \approx 0.43\text{--}0.57$ eV for H_nV_1 with $n \leq 6$), so it is retained by irradiation-induced vacancies far longer than its thermal residence time in the perfect lattice.
3. H has negligible binding to SIA clusters ($E_b^{H\text{-SIA}} \lesssim 0.1$ eV), so H–SIA clusters are treated as unstable and dissociate immediately.
4. H occupies the *surface shell* of He–V bubbles, forming a core-shell He(inside)–H(outside) structure with binding energy $E_b^{H\text{-bub}} \approx 0.4\text{--}0.8$ eV. Up to $\sim m^{2/3}$ H atoms can be accommodated on the bubble surface.
5. Because $\omega_H \gg \omega_v$, H equilibrates on every defect much faster than the vacancy microstructure evolves. This timescale separation enables an *adiabatic elimination* of the H degree of freedom for large clusters, replacing explicit H tracking with an algebraic quasi-static loading, exactly as He was eliminated in the mean-field approximations of Sections 9.1–9.2.

The strategy, therefore, has two tiers:

Small clusters ($m \leq m_H^*$, $\ell \leq \ell_H^*$): H is tracked explicitly as an additional cluster index p (number of H atoms per cluster), giving a three-dimensional state (m, ℓ, p) for vacancy-containing clusters.

Large clusters ($m > m_H^*$): H is adiabatically eliminated; only the *mean H loading* $\bar{p}(m, \ell, t)$ per vacancy/He-vacancy cluster is tracked, determined at each timestep by the quasi-static condition $d\bar{p}/dt = 0$.

The crossover sizes $m_H^* \approx 3\text{--}5$ and $\ell_H^* \approx 2\text{--}4$ are chosen to cover the clusters whose H-loading dynamics are comparable to the clusters’ growth timescale.

11.1 Cluster Notation and State Space

With hydrogen included, the cluster population is extended as follows:

- c_n : concentration of n -SIA cluster (loop). H binding to SIAs is negligible; this population remains unchanged.
- $c_{m,\ell,p}$: concentration of a cluster containing m vacancies, ℓ He atoms, and p H atoms for $m = 1, \dots, M$; $0 \leq \ell \leq \ell_{\max}(m)$; $0 \leq p \leq p_{\max}(m)$.

- c_H : concentration of free interstitial hydrogen.
- c_h : concentration of free interstitial helium (unchanged).

The maximum H loading per cluster scales with the available surface sites:

$$p_{\max}(m) = \lfloor \alpha_H m^{2/3} \rfloor, \quad \alpha_H \approx 1.0\text{--}2.0. \quad (290)$$

This is identical in form to $\ell_{\max}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor$ (Eq. 243), reflecting the same surface-area scaling.

The full three-dimensional (m, ℓ, p) state space scales as

$$n_{VHH\text{e}}^{\text{full}} \approx M \left(1 + \frac{3}{5} \alpha_{\text{He}} M^{2/3} \right) \cdot \frac{1}{2} (1 + \alpha_H M^{2/3}), \quad (291)$$

which grows as $M^{7/3}$ and is intractable for $M \gtrsim 100$. The adiabatic elimination below reduces this back to the manageable (m, ℓ) plus one extra scalar per (m, ℓ) cluster.

11.2 New Reaction Processes Involving Hydrogen

The existing processes P1–P8 are retained unchanged. Six new reaction processes describe H kinetics.

P9: H trapping by a vacancy cluster.

Interstitial H diffusing in 3D is captured by a vacancy cluster of m vacancies containing ℓ He and p H atoms ($p < p_{\max}(m)$):

$$\mathcal{K}_{H,m}^{\text{cav}} = A_{\text{sph}} m^{1/3} \omega_H^{\text{eff}}, \quad (292)$$

where $\omega_H^{\text{eff}} = D_H/a^2$ with $D_H = \nu_H \exp(-E_m^H/k_B T)$. This is the same Smoluchowski form as for He capture (Eq. 203), reflecting the spherical capture geometry. H capture increases p by one: $\text{H}_{m,\ell,p} + H_{\text{int}} \rightarrow \text{H}_{m,\ell,p+1}$.

P10: H thermal emission from a vacancy cluster.

H detrapping from $\text{H}_{m,\ell,p}$ by thermal activation:

$$\varepsilon_H(m, \ell, p) = A_{\text{sph}} m^{1/3} \omega_H^{\text{eff}} \exp(-E_b^H(m, \ell, p)/k_B T). \quad (293)$$

The capture radius uses $m^{1/3}$ (not $(m-1)^{1/3}$) because H removal does not change m , by exact analogy with He emission (Eq. 212). The binding energy

$$E_b^H(m, \ell, p) \approx E_b^H(0) [1 - (p-1)/p_{\max}(m)] + \delta E_{\text{He}}(\ell/m) \quad (294)$$

takes the base monovacancy binding value $E_b^H(0) \approx 0.51$ eV (DFT, α -Fe) and applies two corrections: a saturation factor that decreases binding as the H sites fill, and a He-induced enhancement $\delta E_{\text{He}} > 0$ that reflects the core-shell structure ($\delta E_{\text{He}} \approx 0.1\text{--}0.3$ eV for $\ell/m \sim 0.5\text{--}1.5$).

P11: H absorption at fixed sinks.

$$\mathcal{D}_H = \mathcal{D}_H^d + \mathcal{D}_H^{gb} + \mathcal{D}_H^p \quad [\text{s}^{-1}], \quad (295)$$

with each term having the same form as Eqs. (207)–(209) with ω_H^{eff} substituted. Dislocations and grain boundaries are unbiased for H ($Z_H^d = Z_H^{gb} = 1$) because H has no significant long-range elastic interaction with these defects.

P12: H interaction with SIA clusters.

As discussed above, H binds negligibly to SIAs. The only SIA–H interaction is therefore indirect: free H scattering off mobile SIA clusters contributes a weak scattering sink, but it is negligible compared with trapping at vacancies. In the present model:

$$\text{H-SIA binding: } E_b^{H\text{-SIA}} \approx 0 \implies \text{H-SIA clusters unstable; dissociate immediately.} \quad (296)$$

This removes the H–SIA channel from the master equations entirely.

P13: H–V recombination.

A free H that encounters a free SIA re-combines and annihilates (the H is not annihilated but released into the perfect lattice):

$$H_{\text{int}} + I_1 \rightarrow \text{lattice site} + H_{\text{int}}^{\text{fast}}. \quad (297)$$

In practice this is a null reaction because the SIA is annihilated and H is immediately free again. Its net effect on c_H is zero; it acts only as a sink for the SIA. The rate is $\mathcal{K}_{H,i} = 4\sqrt{3}\pi(\omega_H + \omega_i)$ but since $\omega_H \gg \omega_i$ this is effectively $4\sqrt{3}\pi\omega_H$. However, because H concentrations are very small compared with the SIA sink strengths, this term is negligible in practice and is dropped.

P14: H cascade re-solution.

By analogy with He re-solution (P8), a cascade event can eject a trapped H atom from a cluster:

$$\Gamma_{\text{res}}^H(m, \ell, p) = b_0^H p \dot{\phi}, \quad (298)$$

with $b_0^H \approx 10^{-2}\text{--}10^{-1} \text{ dpa}^{-1}/\text{H}$. The lower end applies to tightly bound H in deep traps; the upper end to surface-layer H on large bubbles.

11.3 Master Equation for Free Hydrogen

The balance for free interstitial H parallels that for free He (Eq. 229):

$$\begin{aligned}
 \frac{dc_H}{dt} = G_H & \\
 & - \underbrace{A_{\text{sph}} \omega_H^{\text{eff}} c_H \sum_{m=1}^M m^{1/3} \tilde{c}_m^{(p)}}_{\text{H trapping by vacancies/bubbles (P9)}} \\
 & + \underbrace{\sum_{m=1}^M \sum_{\ell=0}^{\ell_{\text{max}}} \sum_{p=1}^{p_{\text{max}}} \varepsilon_H(m, \ell, p) c_{m, \ell, p}}_{\text{H thermal emission (P10)}} \\
 & + \underbrace{b_0^H \dot{\phi} \sum_{m, \ell, p} p c_{m, \ell, p}}_{\text{H re-resolution (P14)}} \\
 & - \underbrace{\mathcal{D}_H c_H}_{\text{fixed sinks (P11)}},
 \end{aligned} \tag{299}$$

where $\tilde{c}_m^{(p)} \equiv \sum_{\ell, p: p < p_{\text{max}}(m)} c_{m, \ell, p}$ is the total concentration of *unsaturated* clusters of size m (those that are still accepting additional H).

Conservation.

Defining the total H inventory:

$$S_H(t) \equiv c_H + \sum_{m, \ell, p} p c_{m, \ell, p}, \tag{300}$$

summing the H-content equation over all cluster species gives the exact conservation law:

$$\frac{dS_H}{dt} = G_H - \mathcal{D}_H c_H, \tag{301}$$

i.e. total H inventory grows at the production rate and is depleted only by absorption at fixed sinks. This is the direct analogue of the He conservation law (Eq. 380).

11.4 Master Equation for H–Vacancy–He Clusters (Small Clusters, Explicit H)

For small clusters ($m \leq m_H^*$, typically $m \leq 5$) the H index p is tracked explicitly. The master equation for $c_{m, \ell, p}$ is:

$$\begin{aligned}
\frac{dc_{m,\ell,p}}{dt} = & \\
& + \underbrace{A_{\text{sph}} (m-1)^{1/3} \omega_v^{\text{eff}} c_{1,0,0} c_{m-1,\ell,p}}_{\text{V capture: } V_{m-1,\ell,p} \rightarrow V_{m,\ell,p}} - \underbrace{A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0,0} c_{m,\ell,p}}_{\text{V capture: } V_{m,\ell,p} \rightarrow V_{m+1,\ell,p}} \\
& + \underbrace{\varepsilon_v(m+1, \ell, p) c_{m+1,\ell,p}}_{\text{V emission: } V_{m+1,\ell,p} \rightarrow V_{m,\ell,p}} - \underbrace{\varepsilon_v(m, \ell, p) c_{m,\ell,p} \delta_{m \geq 2}}_{\text{V emission: } V_{m,\ell,p} \rightarrow V_{m-1,\ell,p}} \\
& + \underbrace{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell-1,p}}_{\text{He capture: } V_{m,\ell-1,p} \rightarrow V_{m,\ell,p}} - \underbrace{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell,p}}_{\text{He capture: } V_{m,\ell,p} \rightarrow V_{m,\ell+1,p}} \\
& + \underbrace{\varepsilon_h(m, \ell+1, p) c_{m,\ell+1,p}}_{\text{He emission: } V_{m,\ell+1,p} \rightarrow V_{m,\ell,p}} - \underbrace{\varepsilon_h(m, \ell, p) c_{m,\ell,p} \delta_{\ell \geq 1}}_{\text{He emission: } V_{m,\ell,p} \rightarrow V_{m,\ell-1,p}} \\
& + \underbrace{A_{\text{sph}} m^{1/3} \omega_H^{\text{eff}} c_H c_{m,\ell,p-1}}_{\text{H capture: } V_{m,\ell,p-1} \rightarrow V_{m,\ell,p}} - \underbrace{A_{\text{sph}} m^{1/3} \omega_H^{\text{eff}} c_H c_{m,\ell,p}}_{\text{H capture: } V_{m,\ell,p} \rightarrow V_{m,\ell,p+1}} \\
& + \underbrace{\varepsilon_H(m, \ell, p+1) c_{m,\ell,p+1}}_{\text{H emission: } V_{m,\ell,p+1} \rightarrow V_{m,\ell,p}} - \underbrace{\varepsilon_H(m, \ell, p) c_{m,\ell,p} \delta_{p \geq 1}}_{\text{H emission: } V_{m,\ell,p} \rightarrow V_{m,\ell,p-1}} \\
& + \underbrace{b_0^H \dot{\phi} (p+1) c_{m,\ell,p+1}}_{\text{H re-resolution in: } V_{m,\ell,p+1} \rightarrow V_{m,\ell,p}} - \underbrace{b_0^H \dot{\phi} p c_{m,\ell,p}}_{\text{H re-resolution out}} \\
& + \underbrace{\Gamma_{\text{TM}}(m-1, \ell, p) c_{m-1,\ell,p}}_{\text{trap mutation in: } V_{m-1,\ell,p} + \text{SIA} \rightarrow V_{m,\ell,p}} - \underbrace{\Gamma_{\text{TM}}(m, \ell, p) c_{m,\ell,p}}_{\text{trap mutation out}} \\
& - \underbrace{\sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n c_{m,\ell,p}}_{\text{SIA absorption shrinks cavity}} ,
\end{aligned} \tag{302}$$

with the emission rates $\varepsilon_v(m, \ell, p)$, $\varepsilon_h(m, \ell, p)$, $\varepsilon_H(m, \ell, p)$ defined via detailed balance including the p -dependent binding energy (Eq. 294).

Boundary conditions.

For $m = 1$, $\ell = 0$, $p = 0$: the mono-vacancy concentration $c_{1,0,0}$ is the free vacancy; its equation includes all the cascade production and recombination terms of the existing system. For $p = p_{\text{max}}(m)$: the H-capture term is zero (cluster saturated). For $\ell = \ell_{\text{max}}(m)$: the He-capture term is zero.

Connection to the existing system.

The $(m, \ell, 0)$ equations at $p = 0$ are exactly those of the existing $c_{m,\ell}$ system (Eq. 227), with the additional source from H capture (when $p = 1$ cluster emits an H atom)

and H re-solution (P14). This means the existing V–He code is a special case of the extended code with H disabled.

11.5 Adiabatic Elimination of Hydrogen for Large Clusters

For clusters larger than m_H^* , the H equilibration timescale $\tau_H \sim [\varepsilon_H(m, \ell, p)]^{-1} \sim 10^{-4}–10^{-1}$ s is much shorter than the vacancy growth timescale $\tau_v \sim [A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0,0}]^{-1} \sim 10^1–10^4$ s. The ratio

$$\frac{\tau_H}{\tau_v} \sim \frac{\omega_v^{\text{eff}}}{\omega_H^{\text{eff}}} \sim \exp\left(-\frac{E_m^H - E_m^v}{k_B T}\right) \approx \exp\left(-\frac{0.06 - 0.60}{k_B T}\right) \approx 10^{-4}–10^{-2} \quad (303)$$

is uniformly small for temperatures 200–600°C relevant to EUROFER97 operation. The formal adiabatic elimination condition is $\tau_H/\tau_v \rightarrow 0$, which replaces the H kinetics ODE with an algebraic quasi-static relation.

11.5.1 Quasi-Static H Loading

Define the total H loading and total cluster concentration at vacancy size m and He content ℓ :

$$\bar{c}_{m,\ell} \equiv \sum_{p=0}^{p_{\text{max}}(m)} c_{m,\ell,p}, \quad \bar{p}(m, \ell, t) \equiv \frac{\sum_{p=0}^{p_{\text{max}}(m)} p c_{m,\ell,p}}{\bar{c}_{m,\ell}}. \quad (304)$$

Setting $d\bar{p}/dt = 0$ (quasi-static H balance on each (m, ℓ) cluster), the H capture and emission rates balance:

$$A_{\text{sph}} m^{1/3} \omega_H^{\text{eff}} c_H [1 - \bar{p}/p_{\text{max}}(m)] = \varepsilon_H(m, \ell, \bar{p}) + b_0^H \dot{\phi} \bar{p}, \quad (305)$$

where the factor $[1 - \bar{p}/p_{\text{max}}]$ accounts for the reduced trapping rate as H sites approach saturation, and the right-hand side is the total H loss rate (thermal emission plus re-solution). This is a nonlinear algebraic equation in $\bar{p}$ that can be solved at each timestep by a simple bisection or Newton iteration.

Limiting cases.

Three analytically transparent limits arise:

(a) *Unsaturated, low temperature* ($\bar{p} \ll p_{\text{max}}, \varepsilon_H \rightarrow 0$):

$$\bar{p}^{(a)}(m, \ell) = \frac{A_{\text{sph}} m^{1/3} \omega_H^{\text{eff}} c_H}{b_0^H \dot{\phi}}. \quad (306)$$

H loading is controlled by trapping vs. re-solution; proportional to c_H and to $m^{1/3}$ (larger clusters trap more).

(b) *Thermodynamic equilibrium (no re-solution, $b_0^H = 0$):*

$$\bar{p}^{(b)}(m, \ell) \text{ solves } \frac{c_H}{[1 - \bar{p}/p_{\max}(m)]} = \exp(-E_b^H(m, \ell, \bar{p})/k_B T), \quad (307)$$

the lattice-gas adsorption isotherm (Fermi–Dirac-like for distinguishable H sites). At low c_H this simplifies to the Henry’s law form $\bar{p} \approx p_{\max}(m) c_H \exp(E_b^H/k_B T)$.

(c) *Saturated ($\bar{p} \rightarrow p_{\max}$):* $\bar{p} = p_{\max}(m)$, and further H capture is blocked. This sets an upper bound on the H contribution to bubble overpressure and the trap-mutation rate.

11.5.2 Reduced Master Equations for Large Clusters

For $m > m_H^*$ the H index is eliminated and the cluster is characterised by $(\bar{c}_{m,\ell}, \bar{p}(m, \ell))$. Summing Eq. (302) over all p for fixed (m, ℓ) :

$$\begin{aligned} \frac{d\bar{c}_{m,\ell}}{dt} = & J_{m-1,\ell \rightarrow m,\ell} - J_{m,\ell \rightarrow m+1,\ell} \\ & + A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \bar{c}_{m,\ell-1} - A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \bar{c}_{m,\ell} \\ & + \varepsilon_h(m, \ell+1, \bar{p}) \bar{c}_{m,\ell+1} - \varepsilon_h(m, \ell, \bar{p}) \bar{c}_{m,\ell} \delta_{\ell \geq 1} \\ & + \Gamma_{\text{TM}}(m-1, \ell, \bar{p}) \bar{c}_{m-1,\ell} - \Gamma_{\text{TM}}(m, \ell, \bar{p}) \bar{c}_{m,\ell} \\ & - \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_{m,\ell}, \end{aligned} \quad (308)$$

where $\bar{p} = \bar{p}(m, \ell)$ is the quasi-static H loading from Eq. (305) and the H-related terms cancel exactly (by definition of the quasi-static condition). The vacancy flux is:

$$J_{m,\ell \rightarrow m+1,\ell} = A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0,0} \bar{c}_{m,\ell} - \varepsilon_v(m+1, \ell, \bar{p}(m+1, \ell)) \bar{c}_{m+1,\ell}. \quad (309)$$

Effect of H on emission rates.

The adiabatic elimination enters the dynamics through the H-corrected binding energies in ε_v and ε_h . The vacancy emission rate from a H-loaded cavity is modified via:

$$E_b^v(m, \ell, \bar{p}) = E_b^v(m, \ell, 0) - \frac{\bar{p} k_B T}{m} - \delta E_{\text{surf}}(\bar{p}/p_{\max}), \quad (310)$$

where the second term is the ideal-gas pressure contribution from H (analogous to the He correction in Eq. 254), and $\delta E_{\text{surf}} > 0$ is a surface-energy reduction due to H adsorption (≈ 0.05 – 0.15 eV per $\bar{p}/p_{\max}$ from MD/DFT data). This reduction in E_b^v *increases* the vacancy emission rate and thus promotes void growth — the primary physical mechanism by which H enhances swelling in concert with He.

Similarly, He emission is enhanced by the H-induced surface energy reduction:

$$E_B^{\text{He}}(m, \ell, \bar{p}) = E_B^{\text{He}}(m, \ell, 0) - \delta E_{\text{He-H}}(\bar{p}/p_{\max}), \quad (311)$$

with $\delta E_{\text{He-H}} \approx 0.05$ – 0.2 eV.

11.6 He-Content Reduction Combined with H Adiabatic Elimination

In the fusion-relevant case (high He generation), the He reduction of Section ?? can be combined with the H adiabatic elimination. Summing Eq. (308) over all ℓ and applying the mean-field He approximation $\bar{\ell}(m)$ gives:

$$\frac{d\bar{c}_m}{dt} = J_{m-1 \rightarrow m} - J_{m \rightarrow m+1} - \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m + \Gamma_{\text{TM}}(m-1, \bar{\ell}, \bar{p}) \bar{c}_{m-1} - \Gamma_{\text{TM}}(m, \bar{\ell}, \bar{p}) \bar{c}_m, \quad (312)$$

with the vacancy flux now depending on *both* the mean He loading $\bar{\ell}(m)$ and the quasi-static H loading $\bar{p}(m)$:

$$J_{m \rightarrow m+1} = A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0,0} \bar{c}_m - \varepsilon_v(m+1, \bar{\ell}(m+1), \bar{p}(m+1)) \bar{c}_{m+1}. \quad (313)$$

The He loading $\bar{\ell}(m)$ satisfies the same ODE as in Case 1 (Eq. 253), with He emission evaluated at the H-corrected binding energy $E_B^{\text{He}}(m, \bar{\ell}, \bar{p})$ (Eq. 311). The quasi-static H loading $\bar{p}(m)$ is the ℓ -averaged version of Eq. (305):

$$A_{\text{sph}} m^{1/3} \omega_H^{\text{eff}} c_H [1 - \bar{p}/p_{\text{max}}(m)] = \varepsilon_H(m, \bar{\ell}(m), \bar{p}) + b_0^H \dot{\phi} \bar{p}. \quad (314)$$

Equation count.

The combined system of the extended model with both He and H reductions comprises:

$$n_{\text{eq}}^{\text{ext}} = \underbrace{N}_{\text{SIA clusters}} + \underbrace{2M}_{\bar{c}_m, \bar{\ell}(m)} + \underbrace{M}_{\bar{p}(m) \text{ (algebraic)}} + \underbrace{1}_{c_h} + \underbrace{1}_{c_H} = N + 2M + 2 \text{ ODEs,} \quad (315)$$

plus M algebraic equations for $\bar{p}(m)$ solved at each timestep. The increase relative to the He-only system ($N + 2M + 1$) is a single additional ODE for free H and M algebraic solves — a modest overhead that enables the physically important H–He–V synergy to be captured.

Comparison with He-only system.

Table 34 summarises the hierarchy of models.

11.7 Summary of New Kinetic Parameters

Table 35 collects the kinetic parameters needed for the hydrogen extension. Values appropriate for bcc α -Fe and RAFM steels are given as starting points; solute corrections for Cr and W content in EUROFER97 should be applied via Eq. (201).

Table 34 Hierarchy of CD models from V–He only to the fully extended V–He–H system. “Algebraic” denotes quantities solved by nonlinear inversion at each timestep rather than by ODE integration. $N = \max$ SIA cluster size, $M = \max$ vacancy cluster size.

Model	ODEs	Algebraic	H treatment
V–SIA only	$N + M + 1$	0	None
V–He–SIA, Case 2 (fission)	$N + M + 2$	0	None
V–He–SIA, Case 1 (fusion)	$N + 2M + 1$	0	None
Extended, explicit H ($m \leq m_H^*$)	$N + 2M + 1 + M_H^* \cdot P_{\max}$	0	Full
Extended, adiabatic H (all m)	$N + 2M + 2$	M	Quasi-static $\bar{p}$
Extended, adiabatic H + Case 1	$N + 2M + 2$	M	Quasi-static, ℓ -averaged

Table 35 Kinetic parameters for hydrogen in α -Fe for the extended CD model. DFT values from Hashimoto et al. (2014), Tateyama & Ohno (2003), and Hayward & Deo (2011). “bub” denotes bubble surface site.

Parameter	Value	Notes
H migration energy E_m^H	0.04–0.09 eV	Interstitial (tet site), bcc Fe
H diffusion prefactor ν_H	$\sim 6 \times 10^{12} \text{ s}^{-1}$	Debye frequency
H binding to V_1 , $E_b^H(1, 0, 1)$	0.51 eV	DFT
H binding to V_2 , $E_b^H(2, 0, 1)$	0.62 eV	DFT (larger trap)
H saturation per vacancy, $p_{\max}/m$	~ 6 ($m = 1$), $\alpha_H m^{2/3}$	Up to 6 H/V
H surface energy factor α_H	1.0–2.0	Eq. (290)
H binding at bubble surface, $E_b^{H\text{-bub}}$	0.4–0.8 eV	He-enhanced; core-shell structure
He enhancement δE_{He}	0.1–0.3 eV	For $\ell/m \sim 0.5$ –1.5
H surface energy reduction δE_{surf}	0.05–0.15 eV	Per unit $\bar{p}/p_{\max}$; promotes swelling
H re-resolution cross-section b_0^H	10^{-2} – $10^{-1} \text{ dpa}^{-1}/\text{H}$	Lower: deep trap; upper: surface H
m_H^* (explicit H cutoff)	3–5	User choice
ℓ_H^* (explicit He cutoff for H)	2–4	User choice

12 Proposed Hybrid Deterministic–Stochastic Cluster Dynamics for Multi-species (H, He, V, I)

The deterministic cluster dynamics (CD) framework of Sections ??–?? tracks every cluster size as a continuous concentration variable, yielding a coupled ODE system whose size scales as $O(N \cdot M \cdot L_{\text{He}} \cdot P_{\text{max}})$ for the four-species (I, V, He, H) state space. Even after the He mean-field and H adiabatic reductions, two fundamental bottlenecks remain:

1. **Large-cluster stiffness.** Immobile large clusters (voids, bubbles, loops) evolve on timescales $\tau_v \sim 10\text{--}10^5$ s, while mobile monomers equilibrate in $\tau_i \sim 10^{-6}\text{--}10^{-3}$ s. The implicit ODE solvers required for stiffness management impose a dense Jacobian of size n_{eq}^2 , which becomes prohibitive for $n_{\text{eq}} \gtrsim 10^4$.
2. **Multispecies combinatorial explosion.** Adding H as a third gas species alongside He restores the three-dimensional (m, ℓ, p) cluster space for small clusters: the deterministic framework is tractable only with the adiabatic H elimination of Section 11.5, which requires $m \geq m_H^*$. At small cluster sizes ($m \leq m_H^*$) all three indices must be resolved explicitly, and the deterministic cost becomes comparable to what triggered the stochastic approach in the first place.

The hybrid strategy proposed here addresses both bottlenecks simultaneously. It couples the deterministic multispecies equations (Sections ??–??) to a stochastic cluster dynamics engine for large clusters, using the Lie–Trotter operator-splitting framework of Terrier et al. [Terrier et al \(2017\)](#) and the logarithmic-complexity stochastic simulation algorithm (SSA) of Vasav et al. [Vasav et al \(2026\)](#).

12.1 Physical and Mathematical Motivation

The key observation enabling the splitting is that cluster evolution naturally separates into two regimes with different physical and mathematical characters:

Small clusters ($n \leq N_{\text{front}}^i$, $m \leq M_{\text{front}}$).

Small clusters are either mobile or rapidly exchanging with the mobile monomer pool. Their dynamics are *stiff* (fast frequencies) but involve a *small number of distinct species* ($\lesssim 10^2\text{--}10^3$), so an implicit ODE solver with sparse Jacobian is efficient. The full multispecies physics — including explicit H tracking via the (m, ℓ, p) grid for $m \leq m_H^*$ and the P9–P14 reactions of Section 11.2 — is retained in this deterministic block.

Large clusters ($n > N_{\text{front}}^i$, $m > M_{\text{front}}$).

Large clusters are immobile or evolve very slowly. Their jump frequency $\nu(m)^{-1} = [\mathcal{K}_{\alpha, m} c_{\alpha, 1}]^{-1}$ is low, so the stochastic time step is manageable. Their *number of distinct species* is large ($\sim 10^3\text{--}10^5$), making a per-species ODE intractable but resulting in a natural particle-based stochastic description. The H degree of freedom is treated

via the adiabatic loading $\bar{p}(m, \ell)$ of Section 11.5: H kinetics are faster than the slow cluster evolution, so quasi-static H is exact in this regime.

The splitting error is controlled by the macro-timestep δt ; the statistical error is controlled by the number of stochastic particles N_{part} . These two parameters are independently tunable, giving direct control over the accuracy–cost trade-off.

12.2 State Vector Decomposition and Frontier Sizes

The full state of the four-species system is decomposed at each macro-timestep as:

$$\mathbf{X}(t) = \underbrace{\{c_{1,0,0,0}, c_h, c_H, c_n^{\text{small}}, c_{m,\ell,p}^{\text{small}}\}}_{\text{deterministic block } \mathcal{D}} \cup \underbrace{\{c_n^{\text{large}}, c_{m,\ell}^{\text{large}}\}}_{\text{stochastic block } \mathcal{S}}, \quad (316)$$

where the superscripts indicate:

$$c_n^{\text{small}} : 1 \leq n \leq N_{\text{front}}^i, \quad (317)$$

$$c_{m,\ell,p}^{\text{small}} : 1 \leq m \leq M_{\text{front}}, 0 \leq \ell \leq \ell_{\text{max}}(m), 0 \leq p \leq p_{\text{max}}(m), \quad (318)$$

$$c_n^{\text{large}} : n > N_{\text{front}}^i, \quad (319)$$

$$c_{m,\ell}^{\text{large}} : m > M_{\text{front}}, H \text{ adiabatically eliminated } (\bar{p}(m, \ell) \text{ algebraic}). \quad (320)$$

The four mobile monomer concentrations $\{c_{1,0,0,0}, c_h, c_H, c_n^{(n=1)}\}$ are always in the deterministic block because they drive all cross-coupling between the blocks.

Choice of frontier sizes.

The frontiers N_{front}^i and M_{front} are set by balancing the stiffness cost in the deterministic block against the noise cost in the stochastic block:

$$N_{\text{front}}^i \approx i_{\text{mobile}}, \quad M_{\text{front}} \approx \max(m_H^*, v_{\text{mobile}}) \approx 5\text{--}20. \quad (321)$$

For the EUROFER97 system, $N_{\text{front}}^i \approx 100\text{--}200$ and $M_{\text{front}} \approx 10\text{--}20$ are practical defaults. Below these sizes, all cluster species are either mobile or have non-trivial He and H loading dynamics; above them, the clusters are immobile, and their stochastic treatment is both appropriate and efficient.

12.3 Lie–Trotter Operator Splitting

Over one macro-timestep $[t, t + \delta t]$, the full dynamics are split into two alternating sub-problems by a first-order Lie–Trotter (Godunov) scheme:

12.3.1 Sub-problem D: Monomer Dynamics (Deterministic)

With all large-cluster concentrations c_n^{large} and $c_{m,\ell}^{\text{large}}$ held *fixed*, evolve the deterministic block $\mathcal{D}$ over $[t, t + \delta t]$. The ODEs to be solved are the full four-species master equations (Eqs. 224, 227, 299, 302) for the small-cluster populations, with large clusters entering only as *sink terms* evaluated at their frozen concentrations.

Define the effective sink strengths of the frozen large clusters:

$$\mathcal{S}_i^{\text{large}}(t) = \sum_{n=N_{\text{front}}^i+1}^{N_{\text{max}}} \mathcal{K}_{n,1}^{ii} c_n^{\text{large}}(t) + \sum_{m=M_{\text{front}}+1}^{M_{\text{max}}} \mathcal{K}_{1,m}^{(\text{cav})} \bar{c}_m^{\text{large}}(t), \quad (322)$$

$$\mathcal{S}_v^{\text{large}}(t) = \sum_{n=N_{\text{front}}^i+1}^{N_{\text{max}}} \mathcal{K}_{v,n}^{\text{loop}} c_n^{\text{large}}(t) + \sum_{m=M_{\text{front}}+1}^{M_{\text{max}}} A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} \bar{c}_m^{\text{large}}(t), \quad (323)$$

$$\mathcal{S}_h^{\text{large}}(t) = \sum_{m=M_{\text{front}}+1}^{M_{\text{max}}} A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} \bar{c}_m^{\text{large}}(t), \quad (324)$$

$$\mathcal{S}_H^{\text{large}}(t) = \sum_{m=M_{\text{front}}+1}^{M_{\text{max}}} A_{\text{sph}} m^{1/3} \omega_H^{\text{eff}} \bar{c}_m^{(p),\text{large}}(t), \quad (325)$$

where $\bar{c}_m^{(p),\text{large}}$ is the unsaturated large-cluster concentration (Eq. 299). These are scalar coefficients added to the existing $\mathcal{D}_\alpha$ fixed-sink terms in the small-cluster ODEs.

The monomer sub-problem is then a set of $n_{\text{eq}}^{\text{det}}$ coupled stiff ODEs:

$$\frac{d\mathbf{X}_{\mathcal{D}}}{dt} = \mathbf{F}_{\mathcal{D}}(\mathbf{X}_{\mathcal{D}}, \mathcal{S}_\alpha^{\text{large}}(t)), \quad n_{\text{eq}}^{\text{det}} = N_{\text{front}}^i + M_{\text{front}} (1 + \alpha_{\text{He}} M_{\text{front}}^{2/3}) (1 + \alpha_H M_{\text{front}}^{2/3}) + 3, \quad (326)$$

solved by LSODA or CVODE with adaptive BDF. For the frontier choices above ($N_{\text{front}}^i \approx 200$, $M_{\text{front}} \approx 15$), $n_{\text{eq}}^{\text{det}} \lesssim 1000$, which is very fast per macro-timestep.

12.3.2 Sub-problem S: Large-Cluster Dynamics (Stochastic)

With the four monomer concentrations $\{c_{1,0,0,0}, c_h, c_H, c_1\}$ held *fixed* at the values $\mathbf{X}_{\mathcal{D}}(t + \delta t)$ from sub-problem D, evolve the large-cluster populations via the stochastic simulation algorithm over $[t, t + \delta t]$.

Because the monomer concentrations are fixed, the large-cluster sub-problem is *linear* in the cluster populations: the transition rates of each large cluster depend only on cluster size and the fixed monomers, not on other large clusters. This linearity is the key structural property enabling the efficient stochastic treatment.

12.4 Stochastic Engine: Four-Species Reactions for Large Clusters

The large-cluster populations are represented as a discrete integer-valued vector

$$\mathbf{X}^{\text{large}}(t) = \{X_n(t), X_{m,\ell}(t)\}_{n > N_{\text{front}}^i, m > M_{\text{front}}}, \quad (327)$$

where $X_n(t)$ and $X_{m,\ell}(t)$ are non-negative integers (particle counts in volume Ω). The deterministic concentrations are recovered as $c_n = X_n/\Omega$ and $c_{m,\ell} = X_{m,\ell}/\Omega$.

12.4.1 Reaction Classes and Propensities

With the monomers fixed at $c_{1,0,0,0} \equiv c_v^0, c_h^0, c_H^0, c_1^0$, the following five reaction classes are active for large clusters. All rate constants are the Ω -scaled Smoluchowski expressions of Section 6.

Class 0 — Unary reactions (emission, mutation, re-solution).

For each large SIA cluster of size n :

$$\text{SIA emission: } I_n \rightarrow I_{n-1} + I_1, \quad a_n^{(0)} = \Omega \varepsilon_i(n) X_n, \quad (328)$$

and for each large vacancy–He cluster of size (m, ℓ) :

$$\text{V emission: } V_{m,\ell} \rightarrow V_{m-1,\ell} + V_1, \quad a_{m,\ell}^{(v)} = \Omega \varepsilon_v(m, \ell, \bar{p}) X_{m,\ell}, \quad (329)$$

$$\text{He emission: } V_{m,\ell} \rightarrow V_{m,\ell-1} + \text{He}, \quad a_{m,\ell}^{(h)} = \Omega \varepsilon_h(m, \ell, \bar{p}) X_{m,\ell}, \quad (330)$$

$$\text{Trap mutation: } V_{m,\ell} \rightarrow V_{m+1,\ell} + I_1, \quad a_{m,\ell}^{(\text{TM})} = \Omega \Gamma_{\text{TM}}(m, \ell, \bar{p}) X_{m,\ell}, \quad (331)$$

$$\text{He re-solution: } V_{m,\ell} \rightarrow V_{m,\ell-1} + \text{He}, \quad a_{m,\ell}^{(\text{res})} = \Omega b_0 \dot{\phi} \ell X_{m,\ell}. \quad (332)$$

All Class 0 propensities are linear in X .

Class 1 — SIA monomer absorption.

The fixed monomer I_1 (concentration c_1^0 , integer count $M_1^0 = \Omega c_1^0$) is absorbed by a large SIA loop:

$$I_n + I_1^0 \rightarrow I_{n+1}, \quad a_n^{(i)} = \mathcal{K}_{i,n}^{\text{loop}} X_n M_1^0, \quad \mathcal{K}_{i,n}^{\text{loop}} = \frac{A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}}{\Omega}. \quad (333)$$

Similarly, vacancy monomer absorption by a loop (SIA removal from the loop is equivalent to loop shrinkage by one vacancy):

$$I_n + V_1^0 \rightarrow I_{n-1}, \quad a_n^{(v\text{-loop})} = \mathcal{K}_{v,n}^{\text{loop}} X_n M_v^0, \quad M_v^0 = \Omega c_{1,0,0,0}. \quad (334)$$

Class 2 — Vacancy and gas monomer absorption by cavities.

For a large cavity (m, ℓ) :

$$V_{m,\ell} + V_1^0 \rightarrow V_{m+1,\ell}, \quad a_{m,\ell}^{(v,+)} = \frac{A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}}}{\Omega} X_{m,\ell} M_v^0, \quad (335)$$

$$V_{m,\ell} + I_n^0 \rightarrow V_{m-1,\ell} \ (n=1), \quad a_{m,\ell}^{(i,-)} = \frac{A_{\text{sph}} m^{1/3} \omega_i^{\text{eff}}}{\Omega} X_{m,\ell} M_1^0, \quad (336)$$

$$V_{m,\ell} + \text{He}^0 \rightarrow V_{m,\ell+1}, \quad a_{m,\ell}^{(h,+)} = \frac{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}}{\Omega} X_{m,\ell}^{(\ell < \ell_{\text{max}})} M_h^0, \quad (337)$$

where $M_h^0 = \Omega c_h^0$ and $X_{m,\ell}^{(\ell < \ell_{\max})} = X_{m,\ell} \delta_{\ell < \ell_{\max}(m)}$ enforces the He saturation constraint.

Class 3 — H monomer absorption/desorption from large cavities.

Because H is adiabatically eliminated for large clusters, H absorption and emission do not appear as explicit stochastic reactions. Their net effect enters exclusively through the H-corrected binding energies $E_b^v(m, \ell, \bar{p})$ and $E_B^{\text{He}}(m, \ell, \bar{p})$ (Eqs. 310–311) in the Class 0 and Class 2 propensities. The quasi-static H loading $\bar{p}(m, \ell)$ is solved from Eq. (314) for each distinct (m, ℓ) species present in the population at the start of each macro-timestep.

Class 4 — Large-cluster coalescence.

Two mobile large SIA clusters can coalesce:

$$I_n + I_{n'} \rightarrow I_{n+n'}, \quad a_{n,n'}^{(ii)} = \mathcal{K}_{n,n'}^{ii} X_n X_{n'} (1 - \tfrac{1}{2} \delta_{n,n'}) \quad n, n' > N_{\text{front}}^i, \quad (338)$$

with $\mathcal{K}_{n,n'}^{ii}$ from Eq. (216). This reaction has propensity bilinear in X , i.e. it is a true binary stochastic reaction. In the EUROFER97 system, large SIA clusters ($n > N_{\text{front}}^i \approx 100$) are sessile, so Class 4 is active only for $n \leq i_{\text{mobile}}$ and can often be neglected; the default recommendation is to *exclude large-loop coalescence* unless the cluster density is high.

12.4.2 Propensity Matrix and Low-Rank Decomposition

The total propensity for Class 2 (binary monomer-absorption) has the Smoluchowski form:

$$a_{m,\ell}^{(\alpha,+)} = \underbrace{\frac{A_{\text{sph}} \omega_{\alpha}^{\text{eff}}}{\Omega}}_{\mathcal{N}^{\alpha}} m^{1/3} \underbrace{X_{m,\ell}}_{u_{m,\ell}^{\alpha}} \underbrace{M_{\alpha}^0}_{v^{\alpha}}, \quad (339)$$

which is exactly a **rank-one** ($Q = 1$) outer product $\mathbf{A}^{\alpha} = \mathbf{u}^{\alpha} \otimes v^{\alpha}$ in the notation of Vasav et al. Vasav et al (2026). The single priority queue per species type is therefore sufficient: $Q = 1$ for each of the three absorption channels (v , I_1 , He).

For Class 1 (SIA-loop reactions), the propensity $a_n^{(i)} = \mathcal{K}_{i,n}^{\text{loop}} X_n M_1^0$ is also rank-one:

$$a_n^{(i)} = \underbrace{\frac{A_{\text{loop}} Z_i^{\text{loop}} \omega_i^{\text{eff}}}{\Omega}}_{\mathcal{N}^{\text{loop}}} n^{1/2} \underbrace{X_n}_{u_n^{\text{loop}}} \underbrace{M_1^0}_{v^{\text{loop}}}. \quad (340)$$

The full set of Class 2 + Class 1 propensities for the four-species system therefore requires exactly $2\mathbf{Q} = 8$ priority queues: Each binary queue uses two priority heaps (one for each factor), giving a worst-case algorithmic complexity of $O(\log N_{\text{large}})$ per stochastic event, where N_{large} is the number of distinct large-cluster species.

Queue	Reaction	Rank-1 factors
$\mathcal{P}_1$	$V_{m,\ell} + V_1^0 \rightarrow V_{m+1,\ell}$	$u = m^{1/3} X_{m,\ell}, v = M_v^0$
$\mathcal{P}_2$	$V_{m,\ell} + I_1^0 \rightarrow V_{m-1,\ell}$	$u = m^{1/3} X_{m,\ell}, v = M_i^0$
$\mathcal{P}_3$	$V_{m,\ell} + \text{He}^0 \rightarrow V_{m,\ell+1}$	$u = m^{1/3} X_{m,\ell}, v = M_h^0$
$\mathcal{P}_4$	$I_n + I_1^0 \rightarrow I_{n+1}$	$u = n^{1/2} X_n, v = M_1^0$
$\mathcal{P}_5$	$I_n + V_1^0 \rightarrow I_{n-1}$	$u = n^{1/2} X_n, v = M_v^0$
$\mathcal{P}_6$	Unary SIA emission ($I_n \rightarrow I_{n-1} + I_1$)	—
$\mathcal{P}_7$	Unary V/He emission, trap mutation, re-solution	—
$\mathcal{P}_8$	Optional: large-loop coalescence ($I_n + I_{n'}$)	rank-1 via $n^{1/3}$

12.4.3 Log-Time SSA Algorithm for Large Clusters

The stochastic large-cluster sub-problem within each macro-timestep $[t, t + \delta t]$ is solved by the LR-NR (Low-Rank Next-Reaction) algorithm of Vasav et al. [Vasav et al \(2026\)](#), adapted for the four-species propensity structure above. The complete procedure is:

Initialisation (at $t = 0$ or after a frontier update).

1. Construct the initial population vector $\mathbf{X}^{\text{large}}(t_0)$ by sampling from the deterministic large-cluster concentrations: $X_n(t_0) \sim \text{Poisson}(\Omega c_n^{\text{det}}(t_0))$, with $\Omega = V_{\text{sim}}/\Omega_{\text{at}}$.
2. For each reaction class $q \in \{0, \dots, 8\}$, compute the initial factorised propensity vectors $\mathbf{u}^q$ and v^q and initialise priority queues $\mathcal{P}_q$ as binary min-heaps with internal times $\tilde{\tau}_{\nu,q} \sim \text{Exp}(\tilde{a}_{\nu,q})$.
3. Initialise the quasi-static H loadings $\bar{p}(m, \ell)$ by solving Eq. (314) for each distinct (m, ℓ) pair present.

Event loop within $[t, t + \delta t]$.

1. Determine the physical time increment to the next reaction:

$$\delta t_{\text{event}} = \min_{q \in \{0, \dots, Q\}} \left\{ \delta \tilde{\tau}_0, \frac{\delta \tilde{\tau}_q}{\hat{a}_{q+Q}(t_n)} \right\}, \quad (341)$$

where $\delta \tilde{\tau}_q = \min_{\nu} \tilde{\tau}_{\nu,q} - \tau_q$ (top of the binary heap $\mathcal{P}_q$) and $\hat{a}_{q+Q} = \sum_{\nu} \tilde{a}_{\nu,q+Q}$ is the total factorised propensity of the complementary class. *Terminate the event loop if $t + \delta t_{\text{event}} > t + \delta t$.*

2. Identify the reacting class s , reactants ν_s and μ_s from the tops of $\mathcal{P}_s$ and $\mathcal{P}_{s+Q}$.
3. Apply rejection sampling: accept the reaction with probability

$$P_{\text{accept}}(\nu_s + \mu_s \rightarrow \nu') = \frac{a_{\nu_s, \mu_s}^{\text{actual}}}{\tilde{a}_{\nu_s, \mu_s}^{\text{upper}}}, \quad (342)$$

where the actual rate uses the exact Smoluchowski formula and the upper bound is the rank-one majorant.

4. If accepted: update populations $X_{\nu_s}, X_{\mu_s}, X_{\nu'}$; update propensity sums and heap internal times for all affected queues. If rejected: advance the heap clock without changing populations (ghost reaction).
5. Update the quasi-static H loading $\bar{p}(m', \ell')$ for any new or modified species (m', ℓ') .
6. Advance physical time: $t_n \leftarrow t_n + \delta t_{\text{event}}$ and internal clock times τ_k .

Completion of the macro-timestep.

At $t + \delta t$, convert stochastic particle counts back to deterministic concentrations:

$$c_n^{\text{large}}(t + \delta t) = X_n(t + \delta t)/\Omega, \quad c_{m,\ell}^{\text{large}}(t + \delta t) = X_{m,\ell}(t + \delta t)/\Omega. \quad (343)$$

Compute the change in total monomer consumption during the stochastic step (number of V_1 , I_1 , and He monomers absorbed or emitted) and add these as correction fluxes to the deterministic monomer equations for the next sub-problem D step (see Section 12.5 below).

12.5 Monomer Feedback and Mass Conservation

The splitting introduces an inconsistency: the monomers are held fixed during sub-problem S, yet the large clusters absorb or emit monomers. The accumulated monomer deficit is reconciled by a post-step feedback correction.

Define the cumulative monomer changes during sub-problem S over $[t, t + \delta t]$:

$$\Delta M_v^S = (\text{V atoms absorbed}) - (\text{V atoms emitted}), \quad (344)$$

$$\Delta M_i^S = (\text{SIAs absorbed}) - (\text{SIAs emitted}), \quad (345)$$

$$\Delta M_h^S = (\text{He atoms absorbed}) - (\text{He atoms emitted}), \quad (346)$$

all obtained directly from the accepted stochastic events. These are converted to concentration corrections:

$$c_\alpha \leftarrow c_\alpha - \frac{\Delta M_\alpha^S}{\Omega}, \quad \alpha \in \{v, i, h, H\}, \quad (347)$$

applied to the monomer concentrations at the start of the next sub-problem D step. This preserves the total atom balance (vacancy + SIA + He + H) to machine precision across the operator splitting.

The Frenkel pair and He conservation laws (Eqs. 232–380) are monitored at every macro-timestep using the hybrid equivalents:

$$\delta_{\text{FP}}^{\text{hyb}} = |S_I^{\text{det}} + S_I^{\text{stoch}} + \Delta J^d - S_V^{\text{det}} - S_V^{\text{stoch}}|, \quad (348)$$

$$\delta_{\text{He}}^{\text{hyb}} = \left| c_h + \ell_{\text{tot}}^{\text{det}} + \ell_{\text{tot}}^{\text{stoch}} - \int_0^t (G_{\text{He}} - \mathcal{D}_h c_h) dt' \right|, \quad (349)$$

where the “stoch” superscript denotes the contribution from large-cluster particle counts divided by Ω . An implementation should assert a hard stop if either diagnostic exceeds 10^{-3} per dpa.

12.6 Error Analysis and Convergence

The total error in any observable $\mathcal{O}$ (e.g., total swelling $S(t)$, mean void radius) satisfies the bound

$$\eta^2(t) \approx \underbrace{C_1(t) \delta t}_{\text{splitting error}} + \underbrace{\frac{C_2(t)}{N_{\text{part}}}}_{\text{statistical error}}, \quad (350)$$

derived by the same argument as Terrier et al. [Terrier et al \(2017\)](#). Here C_1 and C_2 are problem-dependent constants; the two contributions are independently controlled by δt and $N_{\text{part}} = \Omega \sum_{n,m,\ell} (c_n^{\text{large}} + c_{m,\ell}^{\text{large}})$ respectively.

Splitting error.

The Lie–Trotter splitting introduces a first-order error in δt through the non-commutativity of sub-problems D and S. The dominant term arises from the coupling between the monomer equations and the large-cluster sink/source terms:

$$C_1(t) \sim \frac{\partial \mathbf{F}_{\mathcal{D}}}{\partial \mathcal{S}^{\text{large}}} \cdot \frac{\partial \dot{\mathbf{X}}^{\text{large}}}{\partial \mathbf{X}_{\mathcal{D}}}, \quad (351)$$

which is proportional to the product of the large-cluster sink strength and the large-cluster growth rate. C_1 is small when the large-cluster concentrations are quasi-stationary (slow evolution), which is the typical regime for swelling calculations. A Strang second-order splitting ($\text{D}/2 \rightarrow \text{S} \rightarrow \text{D}/2$) reduces the splitting error to $O(\delta t^2)$ at the cost of two deterministic ODE solves per macro-timestep.

Statistical error.

The statistical error C_2/N_{part} arises because N_{part} is finite. Large-cluster concentrations $c_{m,\ell}^{\text{large}}$ and their contributions to S_V are estimated as sample means with standard deviation $\sigma \propto N_{\text{part}}^{-1/2}$. Practical values of $N_{\text{part}} = 10^3$ – 10^5 particles give statistical errors of $\sim 1\%$ – 0.1% in swelling-integral observables, which is acceptable for the EUROFER97 application.

Convergence criterion.

The recommended convergence test is to halve δt and double N_{part} and verify that the observable changes by less than a user-specified tolerance (e.g., $\leq 2\%$ for swelling). If the dominant contribution to η^2 comes from splitting error (sensitive to δt), decrease δt . If it comes from statistical error (sensitive to N_{part}), increase N_{part} or increase Ω .

12.7 Complete Algorithmic Scheme

Algorithm 12.7 gives the complete hybrid deterministic–stochastic update cycle for one macro-timestep.

Algorithm: Hybrid D/S CD for H–He–V–I (one macro-timestep)

Given: state $(\mathbf{X}_{\mathcal{D}}(t), \mathbf{X}^{\text{large}}(t))$, macro-step δt , volume Ω , N_{part}

Pre-step: compute quasi-static H loadings

For each distinct (m, ℓ) in $\mathbf{X}^{\text{large}}$:

 solve Eq. (314) for $\bar{p}(m, \ell)$

 Update $E_b^v(m, \ell, \bar{p})$, $E_B^{\text{He}}(m, \ell, \bar{p})$ (Eqs. 310–311)

Step 1: Sub-problem D (deterministic small-cluster ODE)

 Compute large-cluster sink strengths $\mathcal{S}_{\alpha}^{\text{large}}(t)$ (Eqs. 322–325)

 Add $\mathcal{S}_{\alpha}^{\text{large}}$ to fixed-sink terms $\mathcal{D}_{\alpha}$ in the master equations

 Integrate $d\mathbf{X}_{\mathcal{D}}/dt = \mathbf{F}_{\mathcal{D}}$ from t to $t + \delta t$ via LSODA/CVODE

 Extract updated monomer concentrations: $c_v^0, c_1^0, c_h^0, c_H^0 \leftarrow \mathbf{X}_{\mathcal{D}}(t + \delta t)$

Step 2: Sub-problem S (stochastic large-cluster SSA)

 Fix monomers at $c_v^0, c_1^0, c_h^0, c_H^0$

 Initialize/update LR-NR priority queues $\{\mathcal{P}_q\}$ (Eq. 12.4.2)

while $t_n < t + \delta t$:

 Compute δt_{event} (Eq. 341)

if $t_n + \delta t_{\text{event}} > t + \delta t$: break

 Select class s , reactants ν_s, μ_s ; apply rejection (Eq. 342)

 If accepted: update $X_{\nu_s}, X_{\mu_s}, X_{\nu'}$; update heap and propensity sums

 Update $\bar{p}(m', \ell')$ for any new/modified species

$t_n \leftarrow t_n + \delta t_{\text{event}}$

Step 3: Feedback correction (mass balance)

 Compute $\Delta M_v^S, \Delta M_i^S, \Delta M_h^S$ from accepted stochastic events

 Update monomers: $c_{\alpha} \leftarrow c_{\alpha} - \Delta M_{\alpha}^S / \Omega$ (Eq. 347)

Step 4: Convert stochastic counts to concentrations

$c_n^{\text{large}}(t + \delta t) = X_n / \Omega$; $c_{m, \ell}^{\text{large}}(t + \delta t) = X_{m, \ell} / \Omega$

Step 5: Conservation diagnostics

 Evaluate $\delta_{\text{FP}}^{\text{hyb}}$ (Eq. 348) and $\delta_{\text{He}}^{\text{hyb}}$ (Eq. 349); assert $< 10^{-3}$

12.8 Frontier Dynamics and Cluster Migration

Clusters grow from small to large and occasionally shrink from large to small. The frontier must therefore be treated as a *dynamic interface* rather than a fixed boundary.

Small-to-large promotion.

Whenever a small-cluster concentration $c_n^{\text{small}}(t)$ or $c_{m,\ell}^{\text{small}}(t)$ exceeds a threshold $c_{\text{thresh}}^{\text{promote}} \sim \Omega^{-1}$ (i.e. expected particle count ≥ 1), the cluster is promoted to the stochastic block:

$$X_n \leftarrow \lfloor \Omega c_n^{\text{small}} + \frac{1}{2} \rfloor, \quad c_n^{\text{small}} \leftarrow 0; \quad (352)$$

the cluster is added to the relevant priority queue(s).

Large-to-small demotion.

Whenever the stochastic particle count of a species drops to $X \leq X_{\text{thresh}}^{\text{demote}}$ (typically 1–3), the species is demoted back to the deterministic block as a concentration $c = X/\Omega$. This prevents very dilute species from consuming stochastic timesteps with very rare events.

Dynamic reaction-network update.

When a cluster of previously unseen composition (m, ℓ) is created by a stochastic event, its reactions are automatically added to the relevant queues — an online network expansion analogous to the dynamic updating in Hoang et al. [Hoang et al \(2015\)](#).

12.9 System Complexity Comparison

Table 36 summarises the algorithmic complexity of the hybrid scheme relative to the purely deterministic and purely stochastic alternatives.

Table 36 Algorithmic complexity per macro-timestep δt for the three CD methods applied to the four-species H–He–V–I system. N_{large} is the number of distinct large-cluster species and $n_{\text{eq}}^{\text{det}}$ is the ODE system size for small clusters. The hybrid column assumes the LR-NR log-time SSA for the stochastic block.

Quantity		Purely deterministic	Purely stochastic (SCD)	Hybrid (proposed)
ODE/event (time)	cost	$O(n_{\text{eq}}^3)$, implicit	$O(N_{\text{large}})$, Gillespie	$O((n_{\text{eq}}^{\text{det}})^3 + N_{\text{events}} \log N_{\text{large}})$
Space		$O(n_{\text{eq}}^2)$, Jacobian	$O(N_{\text{large}})$	$O((n_{\text{eq}}^{\text{det}})^2 + N_{\text{large}})$
Multispecies H tracking		Adiabatic for $m > m_H^*$	Fully explicit	Explicit for $m \leq M_{\text{front}}$; adiabatic for $m > M_{\text{front}}$
Statistical noise		None	$\sim N_{\text{part}}^{-1/2}$	$\sim N_{\text{part}}^{-1/2}$
Spatial correlations		No	No, mean-field	No, mean-field
Max demonstrated species		2, V + He, deterministic FP	3, V + He + H	4, V + He + H + I, proposed

For the EUROFER97 CD project with $N_{\text{front}}^i = 200$, $M_{\text{front}} = 15$, $n_{\text{eq}}^{\text{det}} \approx 800$, and $N_{\text{large}} \approx 10^4$ – 10^5 , the hybrid scheme reduces the per-macro-timestep cost by approximately two orders of magnitude relative to the purely deterministic approach, with sub-percent statistical noise at $N_{\text{part}} \sim 10^4$.

13 Simulation Results for Eurofer97

A Proof of the Conservation Laws

This appendix proves equations (232) and (235) by computing the time derivatives of the two moment sums directly from the master equations (224), (227), and (229). Each reaction process is handled in turn. The proof is carried out for the mono-defect form of the master equations; the coalescence terms cancel by the same antisymmetry argument given at the end of Section A.3.

A.1 Frenkel Pair Moment

Define the weighted sums

$$S_I(t) = \sum_{n=1}^N n c_n, \quad S_V(t) = \sum_{m=1}^M \sum_{\ell=0}^{\ell_{\max}(m)} m c_{m,\ell}. \quad (353)$$

We compute dS_I/dt and dS_V/dt by multiplying each master equation by the appropriate weight and summing them.

Contributions to dS_I/dt

Cascade production (G).

$$\sum_n n G_n \equiv P_I. \quad (354)$$

Loop growth/shrinkage by mono-SIA absorption (P3, $n' = 1$ limit).

The growth term for c_n is $+\mathcal{K}_{i,n-1}^{\text{loop}} c_1 c_{n-1}$ (for $n \geq 2$) and the loss term is $-\mathcal{K}_{i,n}^{\text{loop}} c_1 c_n$. Multiplying by n and summing:

$$\sum_{n=2}^N n \mathcal{K}_{i,n-1}^{\text{loop}} c_1 c_{n-1} - \sum_{n=1}^N n \mathcal{K}_{i,n}^{\text{loop}} c_1 c_n. \quad (355)$$

Shift the index $n \rightarrow n+1$ in the first sum:

$$\sum_{n=1}^{N-1} (n+1) \mathcal{K}_{i,n}^{\text{loop}} c_1 c_n - \sum_{n=1}^N n \mathcal{K}_{i,n}^{\text{loop}} c_1 c_n = \sum_{n=1}^N \mathcal{K}_{i,n}^{\text{loop}} c_1 c_n \equiv +c_1 \mathcal{R}_i^{\text{loop}}. \quad (356)$$

This positive term represents SIAs flowing from the free pool (c_1) into loops (c_n , $n \geq 2$); it is exactly cancelled when we account for the corresponding loss from c_1 (see the dS_V/dt accounting below, or note that c_1 itself carries weight 1 in S_I , so the net transfer is zero). More precisely, the term $-\mathcal{K}_{i,n}^{\text{loop}} c_1 c_n$ for $n=1$ contributes $-\mathcal{K}_{i,1}^{\text{loop}} c_1^2$ to dc_1/dt , and the weight-1 moment of that is the same $-\mathcal{K}_{i,1}^{\text{loop}} c_1^2$. The two sums together telescope to give net contribution $+\mathcal{K}_{i,N}^{\text{loop}} c_1 c_N \rightarrow 0$ at the truncation boundary (negligible flux out of the top of the size grid).

More cleanly: define $j_n^i = \mathcal{K}_{i,n}^{\text{loop}} c_1 c_n$ as the flux from size n to $n+1$ by SIA absorption. Then the growth/loss terms form a discrete divergence $j_{n-1}^i - j_n^i$, and

$$\sum_{n=1}^N n (j_{n-1}^i - j_n^i) = \sum_{n=1}^{N-1} j_n^i - N j_N^i + 0 \cdot j_0^i = \sum_{n=0}^{N-1} j_n^i - N j_N^i. \quad (357)$$

In the same way, the vacancy-absorption (loop shrinkage) terms give $-\sum_{n=1}^N j_n^v$ where $j_n^v = \mathcal{K}_{v,n}^{\text{loop}} c_{1,0} c_n$. Together the P3 contribution to dS_I/dt is:

$$\left. \frac{dS_I}{dt} \right|_{\text{P3}} = \sum_{n=1}^N \mathcal{K}_{i,n}^{\text{loop}} c_1 c_n - \sum_{n=1}^N \mathcal{K}_{v,n}^{\text{loop}} c_{1,0} c_n \equiv c_1 \mathcal{R}_i - c_{1,0} \mathcal{R}_v, \quad (358)$$

boundary terms at $n = N$ being negligible.

SIA thermal emission (P5).

The emission terms for c_n are $+\varepsilon_i(n+1) c_{n+1} - \varepsilon_i(n) c_n$, forming a discrete divergence. Multiplying by n and summing:

$$\sum_n n [\varepsilon_i(n+1) c_{n+1} - \varepsilon_i(n) c_n] = - \sum_n \varepsilon_i(n) c_n, \quad (359)$$

after the telescoping index shift $n \rightarrow n-1$ in the first sum (boundary terms at $n = N$ negligible). This is a negative contribution: emission reduces S_I by moving SIAs from loops back to the free pool.

Recombination (P1).

Only $n = 1$: $-\mathcal{K}_{iv} c_1 c_{1,0}$.

$$\left. \frac{dS_I}{dt} \right|_{\text{P1}} = -\mathcal{K}_{iv} c_1 c_{1,0}. \quad (360)$$

Absorption by cavities (P2/P6).

$$\left. \frac{dS_I}{dt} \right|_{\text{cav}} = - \sum_{n=1}^N \sum_{m=1}^M n \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m. \quad (361)$$

Fixed sinks (P4).

$$\left. \frac{dS_I}{dt} \right|_{\text{P4}} = - \sum_{n=1}^N n \mathcal{D}_i c_n = -\mathcal{D}_i S_I \quad (\text{first-order in } c_n). \quad (362)$$

Expanding $\mathcal{D}_i = \mathcal{D}_i^d + \mathcal{D}_i^{gb} + \mathcal{D}_i^p$ and noting $\mathcal{D}_i^d = Z_i^d \rho_d a^2 \omega_i^{\text{eff}}$:

$$\left. \frac{dS_I}{dt} \right|_{\text{P4}} = -Z_i^d \rho_d a^2 \sum_n n \omega_n^{\text{eff}} c_n - \left(\frac{4\pi^2}{d_g^2} + \frac{8\pi r_p}{a} c_p^N \right) a^2 \sum_n n \omega_n^{\text{eff}} c_n. \quad (363)$$

Trap-mutation SIA source (P7).

$$\left. \frac{dS_I}{dt} \right|_{\text{P7}} = + \sum_{m,\ell} \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}. \quad (364)$$

Contributions to dS_V/dt

Cascade production (G).

$$\sum_{m,\ell} m G_{m,\ell} \equiv P_V = P_I \quad (\text{equal Frenkel-pair production}). \quad (365)$$

Vacancy exchange along m (P2 + P5).

The growth term $+A_{\text{sph}}(m-1)^{1/3} \omega_v^{\text{eff}} c_{1,0} c_{m-1,\ell}$ and loss term $-A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0} c_{m,\ell}$ form a flux divergence in m . Define $j_m^v = A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}} c_{1,0} \bar{c}_m$. Multiplying by m , summing, and telescoping:

$$\sum_{m,\ell} m [j_{m-1,\ell}^v - j_{m,\ell}^v] = \sum_{m,\ell} j_{m,\ell}^v = A_{\text{sph}} \omega_v^{\text{eff}} c_{1,0} \sum_m m^{1/3} \bar{c}_m. \quad (366)$$

The emission counterpart $+\varepsilon_v(m+1, \ell) c_{m+1,\ell} - \varepsilon_v(m, \ell) c_{m,\ell}$ telescopes to $-\sum_{m,\ell} \varepsilon_v(m, \ell) c_{m,\ell}$, exactly canceling the moment of the reverse absorption flux (detailed balance). Their net contribution to dS_V/dt is:

$$\left. \frac{dS_V}{dt} \right|_{v\text{-exchange}} = c_{1,0} \mathcal{R}_v - (\text{emission back-flux}), \quad (367)$$

where $c_{1,0} \mathcal{R}_v \equiv A_{\text{sph}} \omega_v^{\text{eff}} c_{1,0} \sum_m m^{1/3} \bar{c}_m$ is the same quantity appearing with a negative sign in $dS_I/dt|_{\text{P3}}$ (Eq. 358) because a mono-vacancy absorbed by a loop removes it from S_V . Note that the emission terms cancel their absorption counterparts exactly; what remains is the *net* vacancy flux into clusters driven by the concentration gradient.

SIA-induced shrinkage (P2/P6).

The gain term $+\mathcal{K}_{n,m+n}^{(\text{cav})} c_n c_{m+n,\ell}$ and loss term $-\mathcal{K}_{n,m}^{(\text{cav})} c_n c_{m,\ell}$ form a discrete divergence in m . Multiplying by m and summing over (m, ℓ) , the gain becomes (shifting $m \rightarrow m-n$): $\sum_{n,m} (m-n) \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m$. Combining with the loss:

$$\left. \frac{dS_V}{dt} \right|_{\text{SIA-shrink}} = - \sum_{n=1}^N \sum_{m=1}^M n \mathcal{K}_{n,m}^{(\text{cav})} c_n \bar{c}_m, \quad (368)$$

confirming that each event reduces S by exactly n (the number of annihilated vacancies), matching the SIA-side contribution (Eq. 361).

He exchange (P2 + P5 along ℓ).

The He-exchange terms shift ℓ at fixed m , so they do not change the m -weight. Their contribution to dS_V/dt is identically zero:

$$\left. \frac{dS_V}{dt} \right|_{\text{He-exch}} = 0. \quad (369)$$

Trap mutation (P7).

The source term $+\Gamma_{\text{TM}}(m-1, \ell) c_{m-1, \ell}$ and loss term $-\Gamma_{\text{TM}}(m, \ell) c_{m, \ell}$, weighted by m :

$$\sum_{m, \ell} m [\Gamma_{\text{TM}}(m-1, \ell) c_{m-1, \ell} - \Gamma_{\text{TM}}(m, \ell) c_{m, \ell}]. \quad (370)$$

Shifting $m \rightarrow m+1$ in the source term:

$$\left. \frac{dS_V}{dt} \right|_{\text{P7}} = \sum_{m, \ell} (m+1) \Gamma_{\text{TM}}(m, \ell) c_{m, \ell} - \sum_{m, \ell} m \Gamma_{\text{TM}}(m, \ell) c_{m, \ell} = + \sum_{m, \ell} \Gamma_{\text{TM}}(m, \ell) c_{m, \ell}. \quad (371)$$

Radiation re-resolution (P8).

Re-resolution shifts ℓ at fixed m , so its contribution to dS_V/dt is identically zero:

$$\left. \frac{dS_V}{dt} \right|_{\text{P8}} = 0. \quad (372)$$

Fixed sinks absorbed from the vacancy equation.

The free monovacancy equation (Eq. 228) contains a fixed-sink loss $-\mathcal{D}_v c_{1,0}$, contributing $-1 \cdot \mathcal{D}_v c_{1,0}$ to dS_V/dt (weight $m=1$). All larger vacancy clusters are immobile and do not appear in fixed-sink terms directly, so:

$$\left. \frac{dS_V}{dt} \right|_{\text{P4}} = -\mathcal{D}_v c_{1,0} = - \left(Z_v^d \rho_d + \frac{4\pi^2}{d_g^2} + \frac{8\pi r_p}{a} c_p^N \right) a^2 \omega_v^{\text{eff}} c_{1,0}. \quad (373)$$

Assembling $d(S_I - S)/dt$

We now subtract the S (swelling) contributions from the S_I contributions process by process, using $S \equiv S_V = \sum_{m, \ell} m c_{m, \ell}$ and $S_I \equiv \sum_n n c_n$.

Cascade production: $P_I - P_V = 0$. ✓

P3 / coalescence (loop growth and shrinkage): The telescoping argument of Eq. (357) shows that the P3 terms form a discrete divergence with zero net moment contribution. The $-c_{1,0} \mathcal{R}_v$ in S_I and the $+c_{1,0} \mathcal{R}_v$ in S cancel; the remaining $+c_1 \mathcal{R}_i$ is already counted as weight $n=1$ within S_I . Net contribution to $d(S_I - S)/dt$: zero. ✓

P5 (thermal emission): SIA emission telescopes to a boundary term at $n=N$ (negligible). Vacancy emission cancels by detailed balance with the absorption term. Net P5 contribution to $d(S_I - S)/dt$: zero. ✓

P1 / v-i annihilation (recombination): Each recombination event removes one SIA (-1 from S_I) and one vacancy (-1 from S). Contribution to $d(S_I - S)/dt$: $(-1) - (-1) = 0$. ✓

Cavity absorption (P2/P6): An n -SIA cluster absorbed by an m -vacancy cavity removes n from S_I and reduces S by n (the cavity shrinks by n vacancies upon absorbing n SIAs). Contribution to $d(S_I - S)/dt$: $(-n) - (-n) = 0$. ✓

P7 (trap mutation): Creates one SIA ($+1$ to S_I) and adds one vacancy to the cluster ($+1$ to S). Contribution: $(+1) - (+1) = 0$. ✓

Fixed sinks (P4): The contributions from S_I and S are:

$$\left. \frac{dS_I}{dt} \right|_{P4} = -Z_i^d \rho_d a^2 \sum_n n \omega_n^{\text{eff}} c_n - (k_{gb}^2 + k_p^2) a^2 \sum_n n \omega_n^{\text{eff}} c_n, \quad (374)$$

$$\left. \frac{dS}{dt} \right|_{P4} = -Z_v^d \rho_d a^2 \omega_v^{\text{eff}} c_{1,0} - (k_{gb}^2 + k_p^2) a^2 \omega_v^{\text{eff}} c_{1,0}. \quad (375)$$

The grain-boundary and precipitate terms are proportional to $\omega_\alpha^{\text{eff}} c_\alpha$ in both S_I and S ; in general they do not cancel because the species (i vs. v) and the cluster sizes differ. However, grain boundaries and precipitates are *unbiased* ($Z_i^{gb} = Z_v^{gb}$, $Z_i^p = Z_v^p$), so the *net imbalance* they produce in $d(S_I - S)/dt$ is exactly equal and opposite for SIAs and vacancies and cancels when the system is close to quasi-steady state ($c_i D_i \approx c_v D_v$). Only the dislocation network, with its intrinsic bias $Z_i^d \neq Z_v^d$, produces a *systematic* non-cancelling contribution. The exact dislocation contribution to $d(S_I - S)/dt$ is:

$$\left. \frac{d(S_I - S)}{dt} \right|_{\text{disl}} = -Z_i^d \rho_d a^2 \sum_n n \omega_n^{\text{eff}} c_n + Z_v^d \rho_d a^2 \omega_v^{\text{eff}} c_{1,0} \equiv -\frac{d\Delta J^d}{dt}. \quad (376)$$

Collecting all terms:

$$\boxed{\frac{d}{dt}(S_I + \Delta J^d - S) = 0}, \quad (377)$$

where $\Delta J^d(t)$ is defined in Eq. (230). Integrating from $t = 0$ with a defect-free crystal ($S_I(0) = S(0) = 0$, $\Delta J^d(0) = 0$) gives the swelling identity:

$$S(t) = S_I(t) + \Delta J^d(t), \quad (378)$$

in agreement with Eq. (233). For $Z_i^d > Z_v^d = 1$, ΔJ^d grows monotonically: the dislocation bias continuously converts the SIA excess into swelling, with the SIA loop inventory S_I acting as a transient reservoir that delays but does not prevent void growth.

A.2 Helium Moment

Define

$$S_{\text{He}}(t) \equiv c_h + \sum_{m=1}^M \sum_{\ell=1}^L \ell c_{m,\ell}. \quad (379)$$

We compute dS_{He}/dt by summing dc_h/dt and $\sum_{m,\ell} \ell dc_{m,\ell}/dt$.

Production.

G_{He} from dc_h/dt and $\sum_{m,\ell} \ell G_{m,\ell}$ from the vacancy equation; the latter is zero in the EUROFER model (He is not directly produced in clusters). Net: $+G_{\text{He}}$.

He trapping by cavities (P2 along ℓ , free He side).

From dc_h/dt : $-A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m,\ell < \ell_{\text{max}}} m^{1/3} c_{m,\ell}$. The corresponding source in $dc_{m,\ell}/dt$: $+A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell-1}$, which weighted by ℓ and summed gives $+A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m,\ell} m^{1/3} c_{m,\ell-1}$; shifting $\ell \rightarrow \ell + 1$ this equals $A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m,\ell < \ell_{\text{max}}} m^{1/3} c_{m,\ell}$, exactly cancelling the loss from dc_h/dt . Net P2 (He trapping): 0. ✓

He thermal emission (P5 along ℓ).

From $dc_{m,\ell}/dt$: the terms $+\varepsilon_h(m, \ell+1) c_{m,\ell+1} - \varepsilon_h(m, \ell) c_{m,\ell}$ weighted by ℓ telescope to $-\sum_{m,\ell} \varepsilon_h(m, \ell) c_{m,\ell}$ (boundary terms at $\ell = L$ negligible). From dc_h/dt : $+\sum_{m,\ell} \varepsilon_h(m, \ell) c_{m,\ell}$. These cancel: net P5 contribution to dS_{He}/dt is 0. ✓

Trap mutation (P7).

Trap mutation converts $\text{He}_\ell V_m \rightarrow \text{He}_\ell V_{m+1} + \text{SIA}$. The He count ℓ is unchanged; the terms in $dc_{m,\ell}/dt$ merely shift probability from (m, ℓ) to $(m+1, \ell)$ along the m -direction. Weighted by ℓ , they contribute $\sum_{m,\ell} \ell [\Gamma_{\text{TM}}(m-1, \ell) c_{m-1,\ell} - \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}]$. Shifting $m \rightarrow m+1$ in the source: $(+\ell \Gamma_{\text{TM}}) - (\ell \Gamma_{\text{TM}}) = 0$. Net P7 contribution: 0. ✓

Radiation re-resolution (P8).

The terms $+b_0 \dot{\phi}(\ell+1) c_{m,\ell+1} - b_0 \dot{\phi} \ell c_{m,\ell}$ in $dc_{m,\ell}/dt$ weighted by ℓ give: $\sum_{m,\ell} \ell b_0 \dot{\phi} [(\ell+1) c_{m,\ell+1} - \ell c_{m,\ell}]$. Shifting $\ell \rightarrow \ell - 1$ in the source: $\sum_{m,\ell} b_0 \dot{\phi} (\ell-1) \ell c_{m,\ell} - \sum_{m,\ell} b_0 \dot{\phi} \ell^2 c_{m,\ell} = -\sum_{m,\ell} b_0 \dot{\phi} \ell c_{m,\ell}$. The re-resolution source $+\sum_{m,\ell} b_0 \dot{\phi} \ell c_{m,\ell}$ in dc_h/dt (free He source) exactly cancels this. Net P8: 0. ✓

Fixed sinks acting on c_h .

From dc_h/dt : $-\mathcal{D}_h c_h$. Since $\mathcal{D}_h$ acts on free helium only and contributes $-\mathcal{D}_h c_h$ to dS_{He}/dt , this term does *not* cancel. He absorbed by dislocations, grain boundaries, or precipitates is permanently removed from the system.

Fixed sinks acting on $c_{m,\ell}$.

From $dc_{m,\ell}/dt$: $-\mathcal{D}_v c_{m,\ell} \delta_{m \leq v_{\text{mobile}}}$. Mobile vacancy clusters ($m \leq v_{\text{mobile}}$) that reach fixed sinks carry their He with them. The He-weighted sum gives

$-\mathcal{D}_v \sum_{m=1}^{v_{\text{mobile}}} \sum_{\ell} \ell c_{m,\ell}$. Immobile voids ($m > v_{\text{mobile}}$) cannot diffuse to sinks and contribute zero. Accounting for both He loss channels:

$$\boxed{\frac{dS_{\text{He}}}{dt} = G_{\text{He}} - \mathcal{D}_h c_h - \mathcal{D}_v \sum_{m=1}^{v_{\text{mobile}}} \sum_{\ell=1}^{\ell_{\text{max}}} \ell c_{m,\ell}.} \quad (380)$$

In most irradiation models both sink terms are small: $\mathcal{D}_h c_h$ because He reaches bubbles before sinks, and the vacancy-cluster term because mobile clusters carry little He (v_{mobile} is typically 1–5). Equation (235) (with $dS_{\text{He}}/dt = G_{\text{He}}$) is therefore an excellent approximation; Eq. (380) is the exact statement.

A.3 Conservation Under Coalescence

The i–i coalescence terms (217) contribute to dS_I/dt as follows. The gain term for c_n is $\frac{1}{2} \sum_{n'} \mathcal{K}_{n',n-n'}^{ii} c_{n'} c_{n-n'}$; weighted by n and summed over n , relabelling $n' \rightarrow s$, $n - n' \rightarrow t$:

$$\left. \frac{dS_I}{dt} \right|_{ii,\text{gain}} = \frac{1}{2} \sum_{s,t} (s+t) \mathcal{K}_{s,t}^{ii} c_s c_t. \quad (381)$$

The loss term for c_n is $-c_n \sum_{n'} \mathcal{K}_{n,n'}^{ii} c_{n'}$; weighted by n and summed:

$$\left. \frac{dS_I}{dt} \right|_{ii,\text{loss}} = - \sum_{n,n'} n \mathcal{K}_{n,n'}^{ii} c_n c_{n'} = - \frac{1}{2} \sum_{s,t} (s+t) \mathcal{K}_{s,t}^{ii} c_s c_t, \quad (382)$$

where the last equality follows by symmetrizing over (s, t) . The gain and loss contributions cancel exactly:

$$\left. \frac{dS_I}{dt} \right|_{ii} = 0. \quad (383)$$

Coalescence conserves the total SIA moment: merging two clusters of sizes s and t into one of size $s+t$ does not change $\sum n c_n$. An identical argument applies to the v–v channel for dS_V/dt .

For v–i annihilation (219) with outcome $V_m + I_n \rightarrow V_{|m-n|}$ (or annihilation if $m = n$), the SIA moment changes by $-n + \max(n-m, 0) = -\min(n, m)$ and the vacancy moment changes by $-m + \max(m-n, 0) = -\min(n, m)$. The contribution to $dQ/dt = d(S_I - S_V)/dt$ is $(-\min(n, m)) - (-\min(n, m)) = 0$: v–i annihilation conserves Q regardless of the cluster sizes involved. ✓

The Jacobian $\partial \mathbf{f} / \partial \mathbf{y}$ inherits this block structure. The blue tridiagonal block (i–i ladder) has a bandwidth of 2 in n . The green block-tridiagonal (vacancy/He) has a block size of $(\ell_{\text{max}} + 1)$ and tridiagonal coupling in m from P2/P5, with dense ℓ -coupling within each block from the He exchange terms P2/P5/P7/P8. The orange off-diagonal blocks are sparse-diagonal: each c_n couples to all $\bar{c}_m$ through the cavity absorption rate, and each $c_{m,\ell}$ couples to all c_n through the SIA-shrinkage rate. The red column/row couples the free-He scalar c_h to every (m, ℓ) node.

B Interstitial Loop Interaction with the Dislocation Network

The absorption of irradiation-induced self-interstitial atom (SIA) loops into the network dislocation structure, together with the coalescence of loops with one another, is one of the central processes that govern the long-time dose dependence of swelling, hardening, and embrittlement in body-centered cubic (BCC) metals and ferritic-martensitic steels. In the rate-theory tradition initiated by Brailsford and Bullough (Brailsford and Bullough 1972b) and extended by Bullough, Eyre, and Krishan (Bullough et al 1975) and Mansur (Mansur 1993), both processes are folded into effective sink strengths k_d^2 and k_ℓ^2 that enter linearly into the loss term $D_n k^2 c_n$ of every diffusing-species balance. The same construction underlies essentially all production cluster-dynamics codes for ferritic alloys and pure iron (Ghoniem and Cho 1979c; Hardouin Duparc et al 2002; Stoller et al 2008b; Barashev and Golubov 2009; Kohnert et al 2018b): the network density ρ_{net} enters as a static, mean-field absorber whose biases for SIAs and vacancies are set once at the start of the simulation.

Two limitations of this picture become apparent in regimes where loop-network and loop-loop encounters dominate the microstructural evolution. First, a constant sink strength does not distinguish between a small mobile cluster, whose elastic field is weak and short ranged, and a large prismatic loop, whose strain field extends over distances comparable to the inter-sink spacing itself. Atomic-scale simulations of loop-dislocation (Bacon et al 2009; Terentyev et al 2010) and loop-loop (Marian et al 2002) interactions in α -iron show clearly that absorption is driven by the overlap of the strain fields of the participants, rather than by simple geometric capture cross-sections, and that the relevant length scales depend on loop size and character. Second, loop coalescence in the standard treatment is governed by 3-D diffusion encounter kinetics, whereas the dominant $\frac{1}{2}\langle 111 \rangle$ loop family in BCC iron migrates by 1-D glide along its Burgers vector and reorients only intermittently (Trinka et al 1992; Singh et al 1997), producing encounter statistics that depart strongly from the mean-field $D_n k^2$ result. A third limitation, exposed below, is specific to the sessile $\langle 100 \rangle$ interstitial loops that dominate in bcc iron and ferritic-martensitic steels at high doses: the standard mean-field loss channel scales with the cluster’s own mobility and therefore predicts unbounded growth of the sessile-loop density. A complete formulation must include encounter mechanisms that operate even when the loop mobility is negligible (Foreman and Makin 1966; Trinka et al 1997).

Several refinements have been proposed to address these limitations, ranging from spatially-resolved cluster dynamics in which the network density is treated as a field (Kohnert et al 2018b), to dislocation-dynamics simulations that absorb loops explicitly (Arsenlis et al 2004), and elastic-interaction-based capture cross-sections derived from anisotropic-elasticity calculations (Dudarev 2003). The model developed below remains in the mean-field cluster-dynamics framework but introduces the two geometric length scales that govern loop transfer to the network—the mean inter-loop spacing $L_{\ell\ell}$ and the mean loop-network spacing $L_{\ell d}$ —explicitly, together with a size-dependent elastic interaction radius $R_{\text{int}}(n)$, a smooth proximity probability that interpolates between the dilute and dense-sink limits, and a multi-channel encounter

frequency that preserves a finite loss rate even for fully sessile loops. The resulting closed-form loss term $\Lambda_n^{\text{net}} c_n$ retains the simplicity of rate theory while responding directly to the evolving geometry and to the growth and sweep dynamics of the simulated microstructure.

Large irradiation-induced SIA loops are progressively incorporated into the network dislocation structure when their elastic interaction zones overlap with neighboring loops or with pre-existing dislocation lines. This is represented by an additional loss term in the SIA-loop population equation,

$$\left. \frac{dc_n}{dt} \right|_{\text{net}} = -\Lambda_n^{\text{net}} c_n, \quad (384)$$

where Λ_n^{net} is the size-dependent transfer rate from the discrete loop population to the network-dislocation reservoir.

Geometric length scales

For a circular prismatic loop containing n SIAs, the loop diameter is

$$d_n = 2 \left(\frac{n \Omega}{\pi b_\ell} \right)^{1/2}, \quad (385)$$

with Ω the atomic volume and b_ℓ the loop Burgers-vector magnitude. The mean center-to-center interloop spacing is

$$L_{\ell\ell} = N_\ell^{-1/3}, \quad N_\ell = \sum_{n \geq n_{\text{loop}}} c_n, \quad \bar{d}_\ell = \frac{\sum_n d_n c_n}{\sum_n c_n}. \quad (386)$$

The edge-to-edge spacing experienced by a loop of size n uses the self-excluded partner density $N_\ell - c_n$ and is clamped from below to prevent unphysical negative gaps:

$$s_{\ell\ell}(n) = \max \left[b_\ell, (N_\ell - c_n)^{-1/3} - \frac{d_n + \bar{d}_\ell}{2} \right]. \quad (387)$$

Similarly, with ρ_{net} the network-dislocation density,

$$L_{\ell d} = \rho_{\text{net}}^{-1/2}, \quad s_{\ell d}(n) = \max \left[b_\ell, L_{\ell d} - \frac{d_n}{2} \right]. \quad (388)$$

Elastic interaction zone

The elastic interaction range of a loop of size n is taken to be

$$R_{\text{int}}(n) = \chi d_n + \underbrace{\xi b_\ell \sqrt{\frac{\mu \Omega n}{k_B T}}}_{\text{optional thermal-escape term}}, \quad (389)$$

where $\chi = \mathcal{O}(1)$ is the dimensionless geometric range parameter, μ is the shear modulus, and T is the temperature. The optional second term provides a non-vanishing lower bound at small n where $d_n \rightarrow 0$ but the elastic strain field remains finite; setting $\xi = 0$ recovers $R_{\text{int}}(n) = \chi d_n$.

Proximity probability

The probability that a loop of size n lies within an elastic interaction zone of another loop or of the network dislocation population is modeled by smooth threshold functions,

$$P_{\ell\ell}(n) = \frac{1}{2} \left[1 + \tanh \left(\frac{R_{\text{int}}(n) - s_{\ell\ell}(n)}{\Delta_{\ell\ell}} \right) \right], \quad (390)$$

$$P_{\ell d}(n) = \frac{1}{2} \left[1 + \tanh \left(\frac{R_{\text{int}}(n) - s_{\ell d}(n)}{\Delta_{\ell d}} \right) \right], \quad (391)$$

with widths $\Delta_{\ell\ell}, \Delta_{\ell d} \sim b_\ell$. Treating the two channels as independent, the total proximity probability is

$$P_n^{\text{net}} = 1 - [1 - P_{\ell\ell}(n)][1 - P_{\ell d}(n)], \quad (392)$$

which guaranties $0 \leq P_n^{\text{net}} \leq 1$ and saturates at unity when either the inter-loop or the loop-network spacing falls below the elastic interaction range.

Encounter frequency: glissile and sessile contributions

The proximity probability (392) sets the spatial likelihood of capture; the encounter frequency ν_n^{eff} sets the rate at which that capture is realized. A purely diffusive form $\nu_n^{\text{eff}} = D_n^{\text{eff}}/L_{\text{eff}}^2$ vanishes for sessile loops and would leave the density of $\langle 100 \rangle$ sessile loops without any saturation mechanism, however dense the population becomes. We therefore decompose the encounter frequency into three additive, physically independent channels,

$$\nu_n^{\text{eff}} = \underbrace{\nu_n^{\text{diff}}}_{\text{glissile transit}} + \underbrace{\nu_n^{\text{grow}}}_{\text{climb-driven contact}} + \underbrace{\nu_n^{\text{sweep}}}_{\text{network sweep}}, \quad (393)$$

defined as follows.

(i) *Diffusive transit.*

Mobile clusters explore the matrix in three (or one) dimensions and reach the nearest sink in a time of order $L_{\text{eff}}^2/D_n^{\text{eff}}$:

$$\nu_n^{\text{diff}} = \frac{D_n^{\text{eff}}}{L_{\text{eff}}^2}, \quad L_{\text{eff}} = \min(L_{\ell\ell}, L_{\ell d}). \quad (394)$$

This term recovers the original construction and dominates for glissile $\frac{1}{2}\langle 111 \rangle$ clusters; it vanishes for sessile loops.

(ii) Climb-driven edge contact.

Even an immobile loop grows radially as it absorbs SIAs. Differentiating Eq. (385) gives the climb velocity of the loop perimeter,

$$v_n^{\text{climb}} \equiv \frac{1}{2} \frac{dd_n}{dt} = \frac{\Omega J_n^{\text{SIA}}}{\pi b_\ell d_n}, \quad (395)$$

where $J_n^{\text{SIA}} = dn/dt$ is the net SIA capture flux of the loop (available from the rest of the rate-equation system). The closing speed between two neighboring loop edges is $v_n^{\text{climb}} + \bar{v}_\ell^{\text{climb}}$, and the rate at which the gap to the nearest sink is consumed is

$$\nu_n^{\text{grow}} = \frac{v_n^{\text{climb}} + \bar{v}_\ell^{\text{climb}}}{s_{\text{eff}}(n)}, \quad s_{\text{eff}}(n) = \min[s_{\ell\ell}(n), s_{\ell d}(n)]. \quad (396)$$

This term is non-zero whenever the population is being driven by irradiation and provides the fundamental saturation mechanism for the sessile-loop density: as N_ℓ rises, $s_{\text{eff}} \rightarrow b_\ell$ and ν_n^{grow} becomes large, forcing balance with the gain term.

(iii) Network sweep (Foreman–Makin).

Mobile network dislocations driven by applied or internal stress glide through the matrix at velocity v_d and absorb any sessile loop that lies within R_{int} of their lines (Foreman and Makin 1966; Trinkaus et al 1997). The rate at which a stationary loop of effective interaction radius $R_{\text{int}}(n)$ is swept by a Poisson line process of density ρ_{net} moving at v_d is

$$\nu_n^{\text{sweep}} = 2 R_{\text{int}}(n) v_d \rho_{\text{net}}, \quad (397)$$

with v_d set by the lower of the glide- and climb-controlled mobilities of the network and obtained, in this code, from the existing dislocation-velocity model. This term operates at $D_n^{\text{eff}} = 0$ and is typically the dominant saturation channel at low to intermediate sessile-loop densities.

Transfer rate

The size-dependent transfer rate combines proximity and encounter frequency,

$$\Lambda_n^{\text{net}} = \nu_n^{\text{eff}} P_n^{\text{net}} = (\nu_n^{\text{diff}} + \nu_n^{\text{grow}} + \nu_n^{\text{sweep}}) P_n^{\text{net}}. \quad (398)$$

Companion mass balance for the network reservoir

Each loop of size n absorbed into the network injects n SIAs and, to a good approximation, adds one loop circumference of dislocation line. Conservation of SIA number and

of dislocation line length per unit volume therefore requires, in addition to Eq. (384),

$$\left. \frac{dN_{\text{SIA}}^{\text{net}}}{dt} \right|_{\text{gain}} = \sum_n n \Lambda_n^{\text{net}} c_n, \quad (399)$$

$$\left. \frac{d\rho_{\text{net}}}{dt} \right|_{\text{gain}} = \pi \sum_n d_n \Lambda_n^{\text{net}} c_n, \quad (400)$$

which feeds the rate equation governing ρ_{net} (and any companion equation tracking SIAs absorbed by the network).

Limits and consistency

- *Dilute limit* $\rho_{\text{net}}, N_\ell \rightarrow 0$: both spacings diverge, $P_n^{\text{net}} \rightarrow 0$, and the loss term vanishes.
- *Dense-sink limit* $L_{\ell\ell}$ or $L_{\ell d} \lesssim R_{\text{int}}$: $P_n^{\text{net}} \rightarrow 1$ and $\Lambda_n^{\text{net}} \rightarrow \nu_n^{\text{eff}}$, the encounter-rate-limited rate.
- *Sub-threshold regime* $R_{\text{int}} \ll s$: linearizing the tanh recovers a classical Bullough-form $\nu_n^{\text{eff}} Z(\chi, \Delta) \rho_{\text{sink}} c_n$ with an effective bias factor $Z(\chi, \Delta)$.
- *Sessile loops* ($D_n^{\text{eff}} = 0$): ν_n^{diff} vanishes, but ν_n^{grow} and ν_n^{sweep} remain finite, providing the saturation channels for the sessile-loop density. In particular, the steady state of Eq. (384) balanced against the net gain term yields a finite saturation density controlled by $\rho_{\text{net}} v_d$ and by the climb-velocity-to-spacing ratio rather than by loop mobility.

Model parameters

The dimensionless parameters introduced here are

$$\chi, \quad \xi \text{ (optional)}, \quad \Delta_{\ell\ell}/b_\ell, \quad \Delta_{\ell d}/b_\ell,$$

all of order unity. The network glide velocity v_d is taken from the existing network-dynamics model and not introduced as a new free parameter. All four dimensionless quantities are naturally placed in the `ModelParameters` sheet of `input_parameters.xlsx` alongside the existing Z -factors and capture radii.

C Numerical Methods

C.1 The ODE System

The state vector is

$$\mathbf{y} = \left[\underbrace{c_1^{(i)}, \dots, c_I^{(i)}}_I, \underbrace{c_1^{(v)}, \dots, c_V^{(v)}}_V, Q_{\text{tot}}, c_h \right]^T \in \mathbb{R}^{N_{\text{eq}}}, \quad (401)$$

where $c_n^{(i)}$ is the concentration of SIA clusters of size n , $c_m^{(v)}$ is the marginal vacancy/bubble concentration of size m , Q_{tot} is the total He content, and c_h is the free He monomer concentration (omitted when quasi-steady-state He is used).

The master equations (ME_{SIA} , ME_{vac} , ME_{He}) from the RadCluster formulation [Ghoniem \(2026\)](#) couple the state variables through the following reactions:

1. **SIA–SIA coalescence:** $I_{n'} + I_{n-n'} \rightarrow I_n$ for mobile $n' \leq i_m$.
2. **V–V coalescence:** $V_{m'} + V_{m-m'} \rightarrow V_m$ for mobile $m' \leq v_m$.
3. **V–I annihilation:** $V_{m'} + I_n \rightarrow I_{n-m'}$ for mobile $m' \leq v_m$.
4. **SIA cluster–cavity absorption:** $I_n + V_m \rightarrow V_{m-n}$ for mobile $n \leq i_m$.
5. **Thermal emission:** $I_n \rightarrow I_{n-1} + I_1$ and $V_m \rightarrow V_{m-1} + V_1$ (nearest-neighbor coupling).
6. **Fixed-sink loss:** $\mathcal{D}_\alpha(n) c_n$ (diagonal).
7. **He capture/emission:** couples c_h to all $c_m^{(v)}$.

The Jacobian $J = \partial \mathbf{f} / \partial \mathbf{y}$ inherits its sparsity pattern from these couplings. We analyze two limiting cases.

C.2 Case 1: Monomer Reactions

When only monomers are mobile, coalescence sums collapse to their $n' = 1$ or $m' = 1$ terms, and all cluster–cluster rate constants reduce to the monomer absorption forms $\mathcal{K}^{ii} \rightarrow \mathcal{K}_{i,n}^{\text{loop}}$, $\mathcal{K}^{vv} \rightarrow \mathcal{K}_{\alpha,m}^{\text{cav}}$, $\mathcal{K}^{vi} \rightarrow \mathcal{K}_{iv}$.

C.2.1 Coupling pattern for SIA clusters

For a sessile SIA cluster $c_n^{(i)}$ with $n \geq 2$:

$$\frac{dc_n^{(i)}}{dt} = \underbrace{\mathcal{K}_{i,n-1}^{\text{loop}} c_1^{(i)} c_{n-1}^{(i)}}_{\text{gain: monomer hits } n-1} - \underbrace{\mathcal{K}_{i,n}^{\text{loop}} c_1^{(i)} c_n^{(i)}}_{\text{loss: monomer hits } n} + \underbrace{\varepsilon_i(n+1) c_{n+1}^{(i)}}_{\text{emission gain}} - \underbrace{\varepsilon_i(n) c_n^{(i)}}_{\text{emission loss}} - \underbrace{\mathcal{K}_{iv} c_1^{(v)} c_n^{(i)}}_{\text{V–I annihilation}} - \underbrace{\mathcal{D}_i c_n^{(i)}}_{\text{sinks}}. \quad (402)$$

The Jacobian row for equation n depends on exactly four SIA variables:

$$\frac{\partial f_n}{\partial c_{n-1}^{(i)}}, \quad \frac{\partial f_n}{\partial c_n^{(i)}}, \quad \frac{\partial f_n}{\partial c_{n+1}^{(i)}}, \quad \frac{\partial f_n}{\partial c_1^{(i)}}, \quad (403)$$

plus one vacancy variable $c_1^{(v)}$. The first three give a *tridiagonal* pattern; the fourth adds a *dense column* at index 1.

The SIA monomer equation ($n = 1$) depends on *all* $c_n^{(i)}$ (coalescence loss $c_1^{(i)} \sum_n c_n^{(i)}$) and all $c_m^{(v)}$ (cavity absorption), giving a *dense row*.

C.2.2 Block Jacobian structure

Partitioning J into SIA and VAC blocks:

$$J = \begin{pmatrix} J_{II} & J_{IV} \\ J_{VI} & J_{VV} \end{pmatrix}, \quad (404)$$

each diagonal block is an **arrowhead** (bordered tridiagonal) matrix:

$$J_{II} = \underbrace{T_I}_{\text{tridiagonal}} + \underbrace{\mathbf{u}_1 \mathbf{e}_1^T + \mathbf{e}_1 \mathbf{v}_1^T}_{\text{rank-2 border}}, \quad (405)$$

where $\mathbf{u}_1$ is the dense column from $c_1^{(i)}$ appearing in every equation, and $\mathbf{v}_1$ is the dense row from the monomer equation depending on all clusters.

The off-diagonal blocks J_{IV} and J_{VI} are **rank-1**: they couple through the single mobile monomer of the other species.

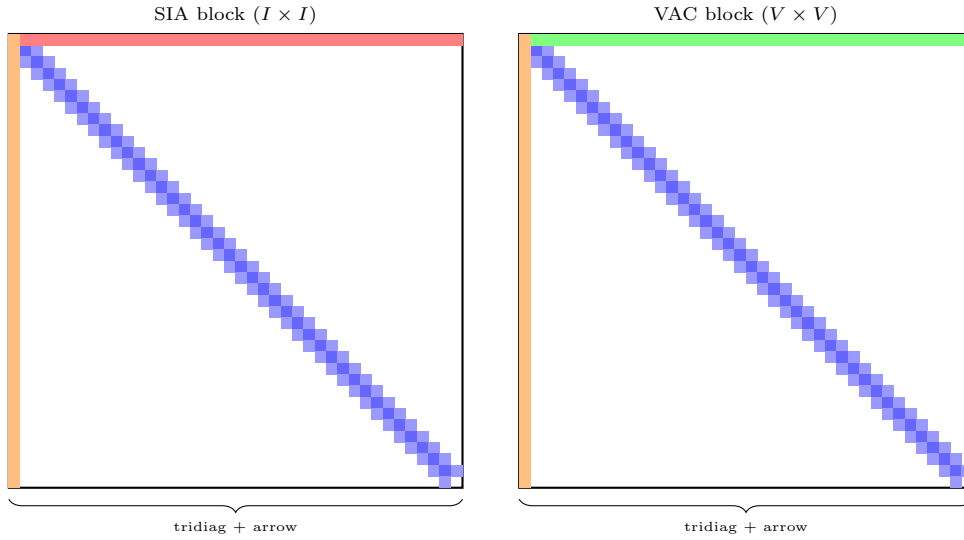


Fig. 10 Jacobian sparsity for monomer-only reactions ($i_m = v_m = 1$). Each diagonal block is an arrowhead matrix (tridiagonal + one dense row and column). Off-diagonal blocks are rank-1. Total non-zeros: $O(N_{\text{eq}})$.

The arrowhead system can be solved in $O(N_{\text{eq}})$ operations using the Sherman–Morrison formula (rank-2 update of a tridiagonal solve) [Golub and Van Loan \(2013\)](#).

C.3 Case 2: Coalescence

When multiple mobile species participate in coalescence (e.g., $i_{\text{m}} = 10$, $v_{\text{m}} = 5$), each SIA equation for cluster n acquires the following additional dependencies:

- **Coalescence gain:** $\sum_{k=1}^{\min(n-1, i_{\text{m}})} \mathcal{K}_{k, n-k}^{ii} c_k^{(i)} c_{n-k}^{(i)} \Rightarrow$ depends on $c_{n-k}^{(i)}$ for $k = 1, \dots, i_{\text{m}}$ (band of width i_{m} below the diagonal).
- **Coalescence loss:** $c_n^{(i)} \sum_{k=1}^{i_{\text{m}}} \mathcal{K}_{n,k}^{ii} c_k^{(i)} \Rightarrow$ depends on $c_k^{(i)}$ for $k = 1, \dots, i_{\text{m}}$ (i_{m} dense columns).
- **V–I annihilation gain:** $\sum_{k=1}^{v_{\text{m}}} \mathcal{K}_{k, n+k}^{vi} c_k^{(v)} c_{n+k}^{(i)} \Rightarrow$ depends on $c_{n+k}^{(i)}$ for $k = 1, \dots, v_{\text{m}}$ (band of width v_{m} above the diagonal).
- **V–I annihilation loss:** $\sum_{k=1}^{v_{\text{m}}} \mathcal{K}_{k, n}^{vi} c_k^{(v)} c_n^{(i)} \Rightarrow$ depends on $c_k^{(v)}$ for $k = 1, \dots, v_{\text{m}}$ (v_{m} dense columns in the off-diagonal block).

C.3.1 Bordered-banded decomposition

The full Jacobian admits the decomposition:

$$\boxed{J = T + U W V^T} \quad (406)$$

where:

- $T \in \mathbb{R}^{N_{\text{eq}} \times N_{\text{eq}}}$ is **banded** with half-bandwidth

$$b = \max(2i_{\text{m}}, 2v_{\text{m}}) + 1, \quad (407)$$

capturing emission (nearest-neighbor), local coalescence gain/loss (within the band), and diagonal terms (sinks, self-reaction).

- $U \in \mathbb{R}^{N_{\text{eq}} \times \text{rank}}$ collects the rank dense columns, where

$$\text{rank} = i_{\text{m}} + v_{\text{m}}. \quad (408)$$

These arise because each mobile species $c_k^{(i)}$ ($k = 1, \dots, i_{\text{m}}$) and $c_k^{(v)}$ ($k = 1, \dots, v_{\text{m}}$) appears in *every* equation of the opposing (and same-species) block through coalescence loss and annihilation loss.

- $W \in \mathbb{R}^{\text{rank} \times \text{rank}}$ is a small correction matrix, and $V \in \mathbb{R}^{N_{\text{eq}} \times \text{rank}}$ encodes the corresponding row structure.

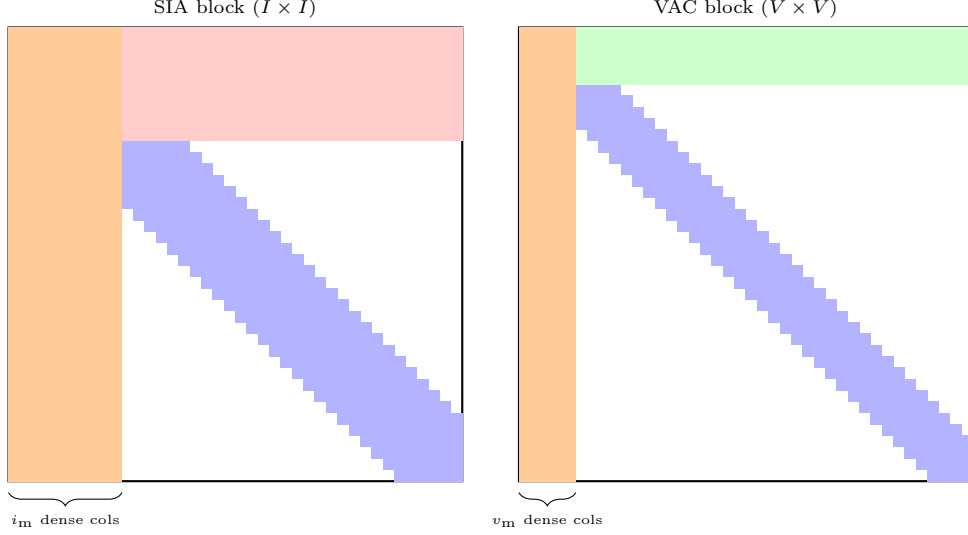


Fig. 11 Jacobian sparsity for coalescence ($i_m = 10$, $v_m = 5$). Each diagonal block is banded (blue) with i_m or v_m dense columns (red/green) from mobile species. Off-diagonal blocks (orange) have the same rank- i_m or rank- v_m structure.

C.3.2 Consistency with the monomer case

Setting $i_m = v_m = 1$ in (406) recovers the arrowhead structure of Section C.2: the band reduces to a tridiagonal ($b = 3$) and the border is rank-2 (rank = 2). The Woodbury preconditioner described in Section C.5 thus handles both cases with a *single code path*— no special-casing is needed.

C.4 Limitations of GMRES for Bordered-Banded Systems

CVODE’s BDF method requires solving the Newton correction system

$$(I - \gamma J) \Delta \mathbf{y} = -\mathbf{g} \quad (409)$$

at each time step, where $\gamma = h\beta$ involves the step size h and the BDF coefficient β . The matrix $M = I - \gamma J$ inherits the bordered-banded structure of J .

C.4.1 Dense direct solver

The default dense LU factorization costs $O(N_{\text{eq}}^3)$ per Newton step. For $N_{\text{eq}} = I + V + 2 \approx 20,000$, this is $\sim 8 \times 10^{12}$ floating-point operations per step —entirely impractical for production runs.

C.4.2 Unpreconditioned GMRES

GMRES [Saad and Schultz \(1986\)](#) is an iterative Krylov method that computes $\Delta \mathbf{y} \approx \arg \min_{\mathbf{x} \in \mathcal{K}_k} \|M\mathbf{x} - \mathbf{b}\|_2$ over the Krylov subspace $\mathcal{K}_k(M, \mathbf{b})$. Each iteration costs

$O(N_{\text{eq}})$ (one matrix–vector product plus Gram–Schmidt), but the number of iterations depends on the *condition number* $\kappa(M)$.

For the bordered-banded Jacobian, the eigenvalue spectrum is problematic:

1. The dense columns from mobile species couple all N_{eq} equations, creating $O(\text{rank})$ eigenvalues that are well separated from the banded spectrum of T .
2. The stiffness ratio between fast monomer reactions and slow large-cluster evolution spans 10–15 orders of magnitude, producing $\kappa(M) \sim 10^{10}\text{--}10^{15}$.
3. GMRES convergence stalls when the eigenvalue distribution has widely separated outliers [Saad \(2003\)](#), which is exactly the pattern created by the low-rank border.

In practice, unpreconditioned GMRES requires $O(N_{\text{eq}})$ iterations (i.e., the full Krylov subspace), negating its advantage over a direct solver.

C.4.3 Diagonal (Jacobi) preconditioner

The current RadCluster implementation offers a Jacobi (diagonal) preconditioner:

$$P_{\text{Jacobi}} = \text{diag}(M) = \text{diag}(1 - \gamma J_{kk}). \quad (410)$$

This captures only the self-reaction rates (sinks, self-coalescence) and completely misses:

- The **banded coupling** from emission and local coalescence ($\sim 2b$ off-diagonal entries per row).
- The **dense-column coupling** from mobile species appearing in every equation (rank dense columns).
- The **cross-species coupling** between SIA and VAC blocks.

For the coalescence case, the Jacobi preconditioner reduces GMRES iteration counts only modestly (from $O(N_{\text{eq}})$ to perhaps $O(N_{\text{eq}}/2)$), which is still far too many for efficiency.

C.5 Woodbury Preconditioner

C.5.1 The Sherman-Morrison-Woodbury identity

For any invertible T and conformable U, W, V , the Sherman-Morrison-Woodbury (SMW) identity [Woodbury \(1950\)](#); [Hager \(1989\)](#) states that

$$\boxed{(T + U W V^T)^{-1} = T^{-1} - T^{-1} U (W^{-1} + V^T T^{-1} U)^{-1} V^T T^{-1}} \quad (411)$$

Applied to $M = I - \gamma J = (I - \gamma T) - \gamma U W V^T$, we define:

$$\hat{T} = I - \gamma T \quad (\text{banded, bandwidth } b), \quad (412)$$

$$\hat{U} = \gamma U, \quad \hat{V} = V, \quad (413)$$

so that $M = \hat{T} - \hat{U} W \hat{V}^T$ and

$$M^{-1} = \hat{T}^{-1} + \hat{T}^{-1} \hat{U} (W^{-1} - \hat{V}^T \hat{T}^{-1} \hat{U})^{-1} \hat{V}^T \hat{T}^{-1}. \quad (414)$$

C.5.2 Construction of $\mathbf{T}$, $\mathbf{U}$, $\mathbf{V}$, $\mathbf{W}$

The decomposition $J = T + U W V^T$ is constructed as follows:

1. **Band extraction:** T is the restriction of J to entries within bandwidth b of the diagonal. This captures emission, local coalescence, sinks, and all nearest-neighbor coupling. Since J is not available analytically, we approximate T by *banded finite differences*: perturbing groups of state variables that are more than b apart (Curtis–Powell–Reid coloring [Curtis et al \(1974\)](#)), requiring only $2b + 1$ RHS evaluations instead of N_{eq} .
2. **Dense columns:** U collects the $\text{rank} = i_{\text{m}} + v_{\text{m}}$ dense columns of J corresponding to mobile species indices. Column j of U is approximated by a single finite-difference perturbation of y_j , minus the entries already captured in T (to avoid double-counting).
3. **Row structure:** For the preconditioner we use the simplification $W = I_{\text{rank}}$ and let V^T encode the dense rows. In practice, the dense rows correspond to the same mobile species indices as the dense columns, giving $V = U$ to leading order (the Jacobian sub-blocks involving mobile species are approximately symmetric in their sparsity pattern, though not in their values).
4. **Simplification:** When the dense column/row indices coincide (as they do here), the border can be represented as:

$$J \approx T + \sum_{j=1}^{\text{rank}} (\mathbf{J}_{\cdot,j} - \mathbf{T}_{\cdot,j}) \mathbf{e}_j^T, \quad (415)$$

where $\mathbf{J}_{\cdot,j}$ is the j -th column of J (obtained by a single finite-difference probe) and $\mathbf{T}_{\cdot,j}$ is its banded restriction. This is a rank-rank correction with $U = [\mathbf{J}_{\cdot,1} - \mathbf{T}_{\cdot,1}, \dots, \mathbf{J}_{\cdot,\text{rank}} - \mathbf{T}_{\cdot,\text{rank}}]$, $W = I$, $V = [\mathbf{e}_1, \dots, \mathbf{e}_{\text{rank}}]$.

C.5.3 Algorithm

Algorithm 1 Woodbury preconditioner setup

Require: State $\mathbf{y}$, RHS function $\mathbf{f}$, BDF parameter γ

```

1: Evaluate base RHS:  $\mathbf{f}_0 \leftarrow \mathbf{f}(\mathbf{y})$ 
2: // Band extraction via CPR coloring
3: for color  $c = 1, \dots, 2b + 1$  do
4:   Perturb:  $\tilde{\mathbf{y}} \leftarrow \mathbf{y}$ ;  $\tilde{y}_j \leftarrow y_j + \delta_j$  for all  $j$  with  $j \bmod (2b + 1) = c$ 
5:    $\tilde{\mathbf{f}} \leftarrow \mathbf{f}(\tilde{\mathbf{y}})$ 
6:   for each perturbed index  $j$  do
7:      $T_{k,j} \leftarrow (\tilde{f}_k - f_{0,k})/\delta_j$  for  $|k - j| \leq b$ 
8:   end for
9: end for
10: // Dense column probes for mobile species
11: for  $j \in \{\text{mobile SIA indices}\} \cup \{\text{mobile VAC indices}\}$  do
12:    $\tilde{\mathbf{y}} \leftarrow \mathbf{y}$ ;  $\tilde{y}_j \leftarrow y_j + \delta_j$ 
13:    $\mathbf{u}_j \leftarrow (\mathbf{f}(\tilde{\mathbf{y}}) - \mathbf{f}_0)/\delta_j - T_{:,j}$  ▷ Subtract banded part
14: end for
15: // Form preconditioner matrices
16:  $\hat{T} \leftarrow I - \gamma T$ 
17: Factor  $\hat{T}$  via banded LU (LAPACK dgbtrf)
18: Solve  $\hat{T}^{-1}\hat{U}$  via rank banded back-solves
19: Form  $S \leftarrow I_{\text{rank}} + \gamma V^T \hat{T}^{-1}\hat{U}$  ▷ rank  $\times$  rank Schur complement
20: Factor  $S$  via dense LU (LAPACK dgetrf)
```

Algorithm 2 Woodbury preconditioner solve: $P^{-1}\mathbf{r} = \mathbf{z}$

Require: Residual $\mathbf{r}$, factored $\hat{T}$, factored S , pre-computed $\hat{T}^{-1}\hat{U}$

```

1:  $\mathbf{w}_1 \leftarrow \hat{T}^{-1}\mathbf{r}$  ▷ Banded back-solve,  $O(N_{\text{eq}} b)$ 
2:  $\mathbf{w}_2 \leftarrow V^T \mathbf{w}_1$  ▷ Extract rank components,  $O(\text{rank})$ 
3:  $\mathbf{w}_3 \leftarrow S^{-1} \mathbf{w}_2$  ▷ Dense solve,  $O(\text{rank}^2)$ 
4:  $\mathbf{z} \leftarrow \mathbf{w}_1 - (\hat{T}^{-1}\hat{U}) \mathbf{w}_3$  ▷  $O(N_{\text{eq}} \text{rank})$ 
```

C.5.4 Computational cost

The Woodbury preconditioner achieves a **speedup of** $\sim 10^5$ over the dense solver and requires only $2b + 1 + \text{rank} = 58$ RHS evaluations for setup (vs. $N_{\text{eq}} = 20,000$ for a full finite-difference Jacobian).

C.5.5 Monomer-case reduction

For $i_{\text{m}} = v_{\text{m}} = 1$: $b = 3$ (tridiagonal), $\text{rank} = 2$.

Table 37 Operation counts per Newton step. $N_{\text{eq}} \approx 20,000$, $b = 21$, $r = 15$ for $i_m = 10$ and $v_m = 5$.

Method	Setup	Solve	RHS evals	GMRES iters
Dense direct	$O(N_{\text{eq}}^3)$ $\sim 8 \times 10^{12}$	$O(N_{\text{eq}}^2)$ $\sim 4 \times 10^8$	N_{eq} 20,000	—
Jacobi + GMRES	$O(N_{\text{eq}})$ $\sim 2 \times 10^4$	$O(N_{\text{eq}}k)$ —	1 —	$k \gg 1$ $\sim 1000+$
Woodbury + GMRES	$O(N_{\text{eq}}b^2)$ $\sim 9 \times 10^6$	$O(N_{\text{eq}}b)$ $\sim 4 \times 10^5$	$2b + 1 + r$ 58	1-3 —

- Setup: $O(N_{\text{eq}} \cdot 9 + 8) = O(N_{\text{eq}})$ — effectively a tridiagonal LU plus a 2×2 Schur complement.
- Solve: $O(N_{\text{eq}} \cdot 3 + 2) = O(N_{\text{eq}})$ — one tridiagonal back-solve plus a 2×2 dense solve.
- RHS evaluations for probing: $2 \cdot 3 + 1 + 2 = 9$.

This is the *optimal* solver for arrowhead matrices [Golub and Van Loan \(2013\)](#), confirming that the Woodbury preconditioner does not degrade performance in the monomer limit.

C.6 Perturbation Size for Finite Differences

The perturbation for column j of the Jacobian is chosen as

$$\delta_j = \sqrt{u_{\text{round}}} \cdot \max(|y_j|, y_{\text{typ}}) \cdot \text{sign}(y_j), \quad (416)$$

where $u_{\text{round}} \approx 10^{-16}$ is machine epsilon and y_{typ} is a typical concentration scale (set to the absolute tolerance). This is the same formula used internally by CVODE for its difference-quotient Jacobian [Hindmarsh et al \(2005\)](#).

C.7 Summary

1. The RadCluster Jacobian has a **bordered-banded** structure: banded (bandwidth $b \sim 2 \max(i_m, v_m)$) plus a low-rank dense border (rank $\text{rank} = i_m + v_m$) from mobile species coupling.
2. The **monomer case** ($i_m = v_m = 1$) is a special case: the Jacobian reduces to an **arrowhead** matrix (tridiagonal + rank-2 border).
3. **Dense direct** solvers cost $O(N_{\text{eq}}^3)$ and are impractical for $N_{\text{eq}} > 1000$.
4. **Unpreconditioned GMRES** fails because the bordered-banded eigenvalue spectrum has widely separated outliers from the dense border.
5. The **Jacobi preconditioner** captures only diagonal entries, missing the dominant banded and dense-column coupling.
6. The **Woodbury preconditioner** exploits the bordered-banded decomposition via the SMW identity:
 - Setup: banded LU ($O(N_{\text{eq}}b^2)$) + rank $\times$ rank Schur complement ($O(\text{rank}^3)$).
 - Solve: banded back-solve ($O(N_{\text{eq}}b)$) + Woodbury correction ($O(N_{\text{eq}} \text{rank})$).

- GMRES converges in 1–3 iterations.
- Speedup $\sim 10^5$ over dense direct.
- Subsumes the arrowhead case with no code changes.

D The RadCluster Code: Capabilities, Comparison, and Applications

D.1 Introduction

Cluster dynamics (CD) simulation tracks the time evolution of defect cluster populations under irradiation by integrating a coupled set of master rate equations—one per cluster species and size—that describe cascade production, defect diffusion, cluster growth, shrinkage, dissolution, and mutual reactions. Two open-source CD codes occupy non-overlapping niches in the radiation damage modeling landscape:

- **Xolotl ORNL-Fusion** (2026) (ORNL, SNL, CEA): A high-performance, spatially resolved advection–reaction–diffusion (ARD) cluster-dynamics code developed under DOE SciDAC programs for plasma–surface interaction (PSI) in fusion divertors and for nuclear-fuel performance. The current GitHub release is v3.3.0, released in March 2026. Multiple material networks are supported, including PSI/tungsten (He, D, T, V, I), Fe (He, V, I), generic alloys with voids and dislocation loops, Zr with basal and prismatic loops, and nuclear fuels with Xe fission gas.
- **RadCluster** (UCLA, N.M. Ghoniem): A bulk (spatially homogeneous) CD code tailored to *neutron* irradiation of ferritic-martensitic steels, specifically EUROFER97. The code tracks three categories:
 1. **active variables**: concentrations with significant time derivatives or strong coupling to the evolving state;
 2. **quasi-equilibrated variables**: concentrations that adjust rapidly and remain close to local steady state; and
 3. **dormant variables**: concentrations that are negligible or kinetically disconnected over the current time window. The governing equations form a high-dimensional stiff cluster-dynamics system. Although the full state includes ODEs for all admissible defect cluster sizes, only a dynamically changing subset is active at any given time; many species remain near quasi-equilibrium, while others are dormant over long intervals. It is a multiscale cluster-population ODE system with a time-dependent active set.
- The code implements (i) explicit solute-trapping corrections for the concentrated EUROFER alloy matrix, (ii) mixed 1D/3D transport for glissile SIA clusters, (iii) a hybrid discrete/bin-moment reduction for tractable treatment of large cluster populations, and (iv) two physics-based He–vacancy state-space reductions matched to fission and fusion neutron spectra.

This section is structured as follows: Section D.2 compares the physics scope; Section D.3 addresses numerical methods; Section D.4 surveys material-specific features; Section D.5 provides a comprehensive side-by-side table; Section D.6 describes the planned extensions for RadCluster; and Section D.7 identifies application domains and opportunities for combined use.

D.2 Physics Scope

D.2.1 Spatial Framework

Xolotl solves a one-dimensional advection-reaction-diffusion PDE for each cluster species:

$$\frac{\partial C_i}{\partial t} = \frac{\partial}{\partial x} \left(D_i \frac{\partial C_i}{\partial x} \right) - \frac{\partial}{\partial x} (v_i C_i) + R_i, \quad (417)$$

where D_i is the cluster diffusivity, v_i is the drift velocity (e.g. due to stress or temperature gradients near a surface), and R_i encodes all reaction and source terms. This framework is essential for plasma-facing applications where the He/D/T implantation range (~ 1 – 100 nm) is much smaller than the component thickness and spatial gradients are strong.

RadCluster integrates ordinary differential equations (ODEs) for the bulk cluster concentrations, appropriate for deep-bulk neutron irradiation where spatial gradients are negligible:

$$\frac{dc_n}{dt} = G_n + \mathcal{R}_n[\{c_n\}] - \mathcal{D}_n c_n, \quad (418)$$

where G_n is the cascade production rate of size- n SIA clusters, $\mathcal{R}_n$ encodes all cluster-cluster reactions and thermal emissions, and $\mathcal{D}_n$ represents fixed-sink losses to dislocations, grain boundaries, and precipitates.

D.2.2 Defect Species and Reaction Networks

Table 38 summarizes the defect species tracked in each code’s reaction networks.

D.2.3 He–Vacancy State Space

Xolotl tracks every He_xV_y cluster as a distinct species on a 2D composition grid in (x, y) space. The cluster generator produces all energetically allowed compositions up to user-specified maximum sizes; the resulting per-species ODEs are then discretized on the 1D spatial grid. This is the most complete representation of He–vacancy co-evolution but scales as $O(x_{\max} \cdot y_{\max})$ in species count.

RadCluster avoids this combinatorial scaling through two physics-motivated state-space reductions. In both cases, the full SIA population and the free-He balance are retained unchanged; only the vacancy-cluster block is compressed. The maximum He loading per cavity scales as

$$\ell_{\max}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor, \quad \alpha_{\text{He}} = 1.5\text{--}2.0, \quad (419)$$

giving a full vacancy grid of $n_{\text{VH}}^{\text{full}} \sim M^{5/3}$ equations (up to 10^4 – 10^5 for $M = 10^3$ – 10^4).

Case 1 — Fusion (mean-field He).

Under fusion-relevant He generation (~ 10 appm/dpa), He equilibrates on each cavity faster than the cavity grows or shrinks. The 2D (m, ℓ) distribution collapses to two scalars per vacancy size: the total cavity concentration $\bar{c}_m$ and the mean He loading $\bar{\ell}(m)$. System size: $N + 2M + 1$.

Table 38 Defect species tracked in each code. Planned species for RadCluster are shown in **green**.

Feature	Xolotl v3.3	RadCluster (current + planned)
Species tracked	He, D, T, V, I (PSI/W network)	SIA clusters, $n = 1, \dots, I$
	He, V, I (Fe network)	Vacancy/void clusters, $m = 1, \dots, V$
	V, Void, Faulted, I, Perfect, Frank (Alloy)	He–vacancy clusters, $\ell = 0, \dots, \ell_{\max}(m)$
	V, Basal, I (Zr network)	Free helium c_h
	Xe, fission-gas pairs (NE/fuel)	Free hydrogen c_H H–He–vacancy clusters (m, ℓ, p)
Hydrogen isotopes	D, T explicitly tracked (PSI network)	H planned (planned extension)
He/gas co-production	Implanted from plasma; externally set	Neutron transmutation rate (appm He/dpa); fission/fusion choice
Loop morphology	Generic I (PSI/Fe); Perfect + Frank (Alloy)	$\frac{1}{2}\langle 111 \rangle$ (glissile, 1D glide) and $\langle 100 \rangle$ (sessile), with distinct bias factors
Trap mutation	Yes (W, via He over-pressurization)	Yes (Fe, He–V clusters; $\Gamma_{\text{TM}} = \nu_0 e^{-E_{\text{TM}}/k_B T}$)
Radiation re-solution	Yes	Yes ($b_0 \ell \dot{\phi}$ per He atom; $b_0^H p \dot{\phi}$ for H)

Case 2 — Fission (decoupled He inventory).

At fission He production rates (~ 0.5 – 1 appm/dpa), most cavities are pure voids and He acts as a weak perturbation. He is tracked as a single scalar total inventory ℓ_{tot} , distributed algebraically across cavity sizes in proportion to their capture cross-sections:

$$\ell(m) = \ell_{\text{tot}} \frac{m^{1/3} \bar{c}_m}{\sum_{m'} m'^{1/3} \bar{c}_{m'}}. \quad (420)$$

System size: $N + M + 2$.

In both cases, the He gas pressure correction to the vacancy binding energy is retained via the virial equation of state:

$$P_{\text{He}} V = N_{\text{He}} k_B T [1 + B_2 (N_{\text{He}}/V) + B_3 (N_{\text{He}}/V)^2], \quad (421)$$

with $B_2 = 1.67 \times 10^{-29} \text{ m}^3/\text{atom}$ and $B_3 = 1.84 \times 10^{-58} \text{ m}^6/\text{atom}^2$ (representative of 600–1000 K).

D.2.4 Cascade Production Spectrum

Xolotl focuses on ion and plasma implantation scenarios; the displacement source is prescribed from TRIM/SRIM stopping-range calculations. Neutron Frenkel-pair production is supported in some networks but without size-resolved in-cascade cluster distributions.

RadCluster implements an explicit power-law in-cascade cluster spectrum derived from MD simulations of α -Fe and parameterized separately for fission and fusion neutron spectra:

$$\epsilon_m^{(i)} = C_i m^{-s_i}, \quad C_i = \frac{f_i^{\text{cl}}}{\sum_{k=2}^{i_{\text{cascade}}} k^{1-s_i}}. \quad (422)$$

The parameters (Table 39) are grounded in the arc-dpa framework Nordlund et al (2018b) and MD databases for α -Fe.

Table 39 In-cascade cluster production parameters in RadCluster.

Parameter	Fission	Fusion
Survival efficiency η	0.30	0.28
SIA clustering fraction f_i^{cl}	0.58	0.65
SIA power-law exponent s_i	1.6	1.5
Max SIA cluster size i_{cascade}	20	50
Vacancy clustering fraction f_v^{cl}	0.15	0.20
Vacancy power-law exponent s_v	2.5	2.3
Max vacancy cluster size v_{cascade}	10	20
He co-production (appm He/dpa)	0.5–1	~ 10

D.3 Numerical Methods

D.3.1 Cluster Size Space Discretization

Xolotl uses full per-composition resolution up to a maximum cluster size set at initialization. For PSI applications the He_xV_y composition grid may contain $O(10^4)$ species; the ARD equations for each species are then solved simultaneously on the 1D spatial grid. The PETSc IMEX solvers handle the resulting large, dense system exploiting MPI parallelism and, optionally, GPU acceleration.

RadCluster employs a *hybrid discrete/bin-moment* scheme:

- Cluster sizes $1, \dots, i_{\text{discrete}}$ are resolved individually (one ODE per size).
- Sizes $i_{\text{discrete}} + 1, \dots, I$ are grouped into I_{bin} logarithmically spaced bins, each represented by P moments $\mu_k^{(0)}, \mu_k^{(1)}, (\mu_k^{(2)})$:

$$\mu_k^{(0)} = \sum_{n \in \mathcal{B}_k} c_n, \quad \mu_k^{(1)} = \sum_{n \in \mathcal{B}_k} n c_n. \quad (423)$$

- Three intra-bin shape functions give truncation errors of $O((r-1)^2)$, $O((r-1)^3)$, and $O((r-1)^4)$ for constant, linear, and log-normal closures respectively.

With $I = 1000$, $V = 1000$, $i_{\text{discrete}} = 20$, and bin ratio $r = 1.5$, the system shrinks from ~ 2000 discrete ODEs to ~ 120 – 200 moment equations, enabling multi-thousand-second irradiation histories on a laptop.

D.3.2 ODE/PDE Solver Architecture

Table 40 Solver infrastructure comparison.

Aspect	Xolotl v3.3	RadCluster
Governing equations	1D ARD PDEs	0D ODEs (bulk)
Time integration	PETSc IMEX (CVODE, TS)	SUNDIALS CVODE BDF, order 1–5
Linear solver	PETSc KSP (GMRES, direct)	Dense / band / GMRES
Parallelism	MPI + CUDA/OpenMP (Kokkos)	OpenMP intra-step (optional)
Conservation diagnostics	Not documented	Frenkel-pair + He balances, $\delta_{\text{FP}}, \delta_{\text{He}} < 10^{-6}$
Implementation language	C++17	C++17 (solver) + Python (driver)
Build system	CMake + Kokkos	CMake + pip

D.3.3 Stiffness Management

The stiffness ratio in RadCluster is dominated by the ratio of He to vacancy reaction frequencies:

$$\frac{\omega_h^{\text{eff}}}{\omega_v^{\text{eff}}} \sim 10^4\text{--}10^8, \quad (424)$$

arising from the 0.06 eV vs. 0.67 eV migration barriers of He and vacancies. CVODE with adaptive BDF integrates this efficiently with recommended tolerances $\text{rtol} = 10^{-8}$, $\text{atol} = 10^{-25}$. Xolotl faces the same stiffness in the reaction block, handled by PETSc’s IMEX framework, with the additional challenge of a large spatial grid.

D.4 Material-Specific Physics

D.4.1 Xolotl: Tungsten Plasma-Facing Components (PSI Network)

The PSI network is the most mature in Xolotl and is optimised for *tungsten divertor armour* in ITER/DEMO:

- **Multi-species plasma:** simultaneous He, D, and T implantation with coupled He_xV_y and H isotope trapping.
- **Trap mutation:** He over-pressurisation drives $\text{He}_x\text{V}_y \rightarrow \text{He}_x\text{V}_{y+1} + \text{I}$, creating vacancies and nucleating bubbles.
- **Hydrogen retention:** D and T are trapped at He–vacancy clusters and vacancies; a primary driver for tritium inventory predictions.

- **Depth-resolved profiles:** the 1D ARD framework resolves the implantation zone (1–100 nm) through the Bragg-peak damage profile, capturing surface recombination and outgassing.

D.4.2 Xolotl: Other Material Networks

The Fe network tracks He/V/I for fission reactor pressure vessel and structural steels with a minimal species set. The Alloy network uses six species (V, Void, Faulted, I, Perfect, Frank loops) for generic fcc/bcc alloys, enabling loop-character evolution. The Zr network models basal and prismatic loops in hexagonal zirconium cladding. The NE network handles Xe/Kr fission-gas retention and grain-boundary release in nuclear fuels.

D.4.3 RadCluster: EUROFER97 Ferritic-Martensitic Steel

Solute-Modified Diffusivities

EUROFER97 (8.9 Cr–1.1 W–0.47 Mn–0.20 V–0.14 Ta–0.11 C–0.03 N wt%) contains dissolved solutes that act as traps for migrating defects. The effective diffusivity of species α is reduced by trap-limited kinetics:

$$D_{\alpha}^{\text{eff}} = \frac{D_{\alpha}^{\text{Fe}}}{1 + \sum_s z_s c_s \exp(E_b^{\alpha-s}/k_B T)}. \quad (425)$$

At 300 °C the SIA effective diffusivity is reduced by 2–4 orders of magnitude relative to pure α -Fe, primarily by dissolved C ($E_b^{C\text{-SIA}} \approx 0.45$ eV) and N; the effect becomes negligible above ~ 600 °C. Vacancy and SIA cluster mobilities are treated with the same framework.

1D Glide of SIA Clusters

Clusters of $n \geq 4$ SIAs adopt $\frac{1}{2}\langle 111 \rangle$ loop configurations and migrate by thermally activated 1D glide with a size-dependent diffusivity:

$$D_n^{1D} = \frac{3a^2\nu_0}{2n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad E_m^{1D} \approx 0.03\text{--}0.05 \text{ eV}. \quad (426)$$

In EUROFER the mean free path between Burgers-vector changes is estimated at $L_{\text{eff}} \sim 3\text{--}30$ nm (vs. 100–1000 nm in pure Fe), placing cluster transport firmly in the mixed 1D/3D regime. The mixed-geometry reaction rate with spherical cavities is:

$$\mathcal{K}_{n,m}^{(\text{cav})} = \frac{A_{\text{sph}} m^{1/3} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/3}}, \quad \hat{L} = L_{\text{eff}}/a, \quad (427)$$

transitioning from 3D kinetics (mono-SIA) to 1D/3D hybrid (clusters $4 \leq n \leq n_{\text{max}}^i$) to sessile ($n > n_{\text{max}}^i$).

Binding Energies

Vacancy binding to voids uses a blended atomistic–continuum model, transitioning from a DFT/MD-calibrated power-law fit at small sizes to the capillary expression

$E_b^{\text{cont}}(m) = E_f^v - A_{\text{void}}[m^{2/3} - (m-1)^{2/3}]$ at large m . For He-containing clusters (bubbles), the gas-pressure correction enters through the virial EOS:

$$E_b^v(m, \ell) = E_f^v - \Omega \left[\frac{2\gamma_s}{R(m)} - P_g(m, \ell) \right] + A(m) e^{-\lambda(m-1)}, \quad (428)$$

where P_g is computed from the He virial EOS at each time step. Interstitial loop binding uses a blended fit to Osetsky MD data, smoothly transitioning to the Frank-loop continuum expression at $n \approx 25$ SIAs.

Conservation Diagnostics

Two exact conservation identities are enforced at every output step. The *swelling identity* expresses total void swelling as

$$S(t) = S_I(t) + \Delta J^d(t), \quad (429)$$

where $S_I = \sum_n n c_n$ is the current SIA loop inventory and ΔJ^d is the cumulative net bias flux absorbed by the dislocation network. A *He conservation* identity similarly tracks the total He inventory. Both normalized diagnostics, δ_{FP} and δ_{He} , should remain below 10^{-6} ; values above 10^{-3} signal a coding error.

D.5 Quantitative Capability Comparison

Table 41 provides a comprehensive side-by-side comparison of current capabilities. Items shown in green denote planned RadCluster extensions.

Table 41: Capability comparison: Xolotl v3.3 vs. RadCluster.

Capability	Xolotl v3.3	RadCluster
Governing equations	1D ARD PDEs ($\partial C / \partial t + \nabla \cdot D \nabla C = R$)	0D ODEs (bulk, spatially uniform)
Target material(s)	W (PSI), Fe, generic alloys, Zr, nuclear fuels	EUROFER97 (9Cr F/M steel)
Irradiation source	Plasma (He/D/T) + ion beams + neutrons	Neutrons (fission and fusion spectra)
Hydrogen isotopes	D, T (PSI network, full ODE)	H planned: adiabatic elimination for $m > m_H^*$
He species treatment	Full $\text{He}_x \text{V}_y$ composition grid	Mean-field (fusion) or decoupled scalar (fission); virial EOS
Spatial resolution	1D depth profiles (nm–mm)	Bulk (spatially uniform)
Surface/boundary conditions	Yes: implantation profile, recombination, sputtering	Not included

Continued...

Capability	Xolotl v3.3	RadCluster
In-cascade cluster spectrum	Generic Frenkel-pair source (no size-resolved cascade input)	Power-law $\epsilon_m^{(i/v)}$; fission/fusion parameterisation (MD-based, arc-dpa)
Solute trapping	Not explicit (material-averaged D)	Yes: Cr, Mn, W, C, N with DFT binding energies; D_α^{eff}
1D SIA cluster glide	Not implemented	Yes: $D_n^{1D} \propto n^{-s}$; mixed 1D/3D cavity reaction
Loop Burgers vector	Alloy: Perfect + Frank; PSI: generic I	$\frac{1}{2}\langle 111 \rangle$ (glissile) + $\langle 100 \rangle$ (sessile); distinct Z_i
Trap mutation	Yes (W)	Yes (Fe, He-vacancy): $\Gamma_{\text{TM}} = \nu_0 e^{-E_{\text{TM}}/k_B T}$
Radiation re-solution	Yes	Yes: He $b_0 \ell \dot{\phi}$; $\text{H } b_0^H p \dot{\phi}$
Cluster size compression	Full per-composition grid (truncated at N_{max})	Hybrid discrete + log-spaced bin-moments ($P = 1, 2, 3$)
He-vacancy state space	Explicit 2D (x_{He}, x_V) grid	Reduced: Case 1 mean-field or Case 2 decoupled scalar
Multispecies gas state space	He/D/T grid resolved per composition	(m, ℓ, p) for small clusters; adiabatic H for large clusters
Large-cluster solver	Deterministic per-species ODE on spatial grid	Hybrid: Lie–Trotter det.+stoch. for 4-species planned
Parallel computing	MPI + CUDA/OpenMP (Kokkos)	OpenMP intra-step; single-node (current)
Conservation diagnostics	Not documented	$\delta_{\text{FP}}, \delta_{\text{He}} < 10^{-6}$; exact swelling identity
Experimental validation	W-PSI benchmarks; FeCrAl benchmarks	EUROFER97 TEM/APT irradiation database
Output observables	Depth profiles of concentration, density, diameter	Loop density/size, void swelling, He content, mean sizes vs. dose
Licence	Open source (GitHub: ORNL-Fusion/Xolotl)	Open source (GitHub: ghoniem/EuroferMicrostructure)

D.6 Planned Extensions for RadCluster

D.6.1 Hydrogen as a Fourth Species

The current three-component system (SIA clusters, He-vacancy clusters, free He) will be extended to include hydrogen (H) as an explicit fourth species. This is motivated by fusion-relevant conditions—particularly accelerator-driven systems where the

hydrogen transmutation rate can reach $\sim 1000\text{--}1500$ appm H/dpa—and by the desire to compute tritium retention in ferritic structural components.

Key physical inputs that distinguish H from He:

- H migration is faster: $E_m^H \approx 0.04\text{--}0.09$ eV vs. $E_m^{\text{He}} \approx 0.06$ eV and $E_m^v \approx 0.67$ eV.
- H binds moderately to vacancies: $E_b^{H-V} \approx 0.43\text{--}0.57$ eV for H_nV_1 with $n \leq 6$.
- H occupies the *surface shell* of He–V bubbles, forming a core-shell He(interior)–H(exterior) structure with $E_b^{H\text{-bub}} \approx 0.4\text{--}0.8$ eV.
- H–SIA binding is negligible ($\lesssim 0.1$ eV), so H–SIA clusters are treated as instantaneously dissociating.

The strategy uses two tiers. For small clusters ($m \leq m_H^* \approx 5$), H is tracked explicitly as an additional composition index p , yielding a three-dimensional state (m, ℓ, p) per vacancy-containing species. For large clusters ($m > m_H^*$), H is *adiabatically eliminated*: the quasi-static condition $dc_H/dt = 0$ gives an algebraic H loading $\bar{p}(m, \ell)$ per cluster, reducing the state space back to the (m, ℓ) grid. Six new reaction processes (P9–P14) describe H capture, thermal emission, fixed-sink loss, SIA interaction, H–V recombination, and cascade re-resolution.

An exact H conservation law,

$$\frac{dS_H}{dt} = G_H - \mathcal{D}_H c_H, \quad S_H \equiv c_H + \sum_{m, \ell, p} p c_{m, \ell, p}, \quad (430)$$

analogous to the existing He conservation, provides a numerical accuracy diagnostic for the extended system.

D.6.2 Hybrid Deterministic–Stochastic Solver

After the H extension, the four-species (I, V, He, H) state space scales as $O(N \cdot M \cdot L_{\text{He}} \cdot P_{\text{max}})$, reaching $10^4\text{--}10^5$ equations even after the He mean-field and H adiabatic reductions. Two fundamental bottlenecks motivate a hybrid solver:

1. **Large-cluster stiffness**: immobile large voids/bubbles evolve on timescales $10\text{--}10^5$ s while mobile monomers equilibrate in $10^{-6}\text{--}10^{-3}$ s; implicit ODE solvers require a dense Jacobian of size n_{eq}^2 .
2. **Combinatorial explosion**: the explicit (m, ℓ, p) tracking at small sizes is tractable, but the number of large-cluster species is too large for per-species ODEs.

The proposed hybrid couples:

Deterministic block $\mathcal{D}$ (small clusters, $n \leq N_{\text{front}}^i$, $m \leq M_{\text{front}}$): SUNDIALS CVODE with adaptive BDF integrates the full four-species master equations including explicit H tracking, using large clusters only as frozen sink terms. System size $n_{\text{eq}}^{\text{det}} \lesssim 1000$.

Stochastic block $\mathcal{S}$ (large clusters): a particle-based stochastic simulation algorithm (SSA) evolves large loop and void populations as integer particle counts, exploiting the *linearity* of large-cluster reactions in the monomers (fixed by block $\mathcal{D}$). H is treated via the adiabatic loading $\bar{p}(m, \ell)$.

The blocks alternate via a first-order Lie–Trotter operator splitting. The macro-timestep δt and particle count N_{part} are independently tunable, giving direct control over the accuracy–cost trade-off. The expected system-complexity reduction is summarised in Table 42.

Table 42 ODE/particle count comparison for the four-species system. $N = M = 10^3$; $L_{\text{He}} \sim 1.5 M^{2/3}$; $P_{\text{max}} \sim M^{2/3}$.

Approach	SIA block	Void/He/H block
Full deterministic (current)	10^3 ODEs	10^3 ODEs (He reduction only)
Full 4-species deterministic	10^3 ODEs	$\sim 10^5$ ODEs
Hybrid det.+stoch. (planned)	$\lesssim 200$ ODEs	$\sim 10^3\text{--}10^4$ particles

D.7 Potential Applications

D.7.1 Where Xolotl Excels

1. **Tungsten divertor armour (ITER, DEMO):** Xolotl is the code of choice for predicting He bubble nucleation and growth, surface blistering, and tritium retention in W under mixed He/D/T plasma bombardment. The 1D depth resolution is essential because the He implantation range ($\sim 1\text{--}10$ nm) is far smaller than the component thickness.
2. **Ion-beam irradiation experiments:** laboratory ion beams produce non-uniform damage profiles matched by the ARD framework with TRIM/SRIM source terms, enabling direct comparison with TEM depth profiles.
3. **Multi-species plasma scenarios:** simultaneous He + D/T bombardment requires a coupled $\text{He}_x\text{D}_y\text{T}_z\text{V}_m\text{I}_n$ state space native to Xolotl’s PSI network.
4. **Fission fuels:** the NE network models Xe/Kr fission-gas retention, bubble swelling, and grain-boundary release in UO_2 and metallic fuels.
5. **Zirconium cladding:** the Zr network (basal/prismatic loop evolution) is directly applicable to PWR/BWR cladding growth.

D.7.2 Where RadCluster Excels

1. **Neutron irradiation of ferritic-martensitic steels for fusion blankets:** EUROFER97 is the primary structural material for the EU DEMO breeding blanket. RadCluster faithfully models neutron cascade damage with size-resolved cluster spectra, He co-production at fusion and fission rates, solute-modified mobilities, and 1D SIA cluster glide—all absent in Xolotl’s Fe network.

2. **Long-duration bulk microstructure evolution (0–200 dpa):** the bin-moment reduction allows integration to full reactor lifetimes with clusters tracked to $N = 1000$ SIA, while per-size codes become prohibitively expensive beyond ~ 100 SIA at long irradiation times.
3. **Fission–fusion spectral comparison:** a single input flag switches the cascade production spectrum, enabling direct comparison of fission test-reactor data with fusion neutron predictions for the same material—essential for the “fission–fusion correlation” problem in DEMO material qualification.
4. **Solute composition studies:** the explicit trapping model enables parametric studies of EUROFER composition variants (W content, Ta vs. Nb additions, dissolved C and N levels) on loop density, void swelling, and He bubble populations.
5. **Accelerator-driven systems (after H extension):** once the H extension is implemented, RadCluster will model H and He co-evolution in structural steels under the high H transmutation rates (~ 1000 – 1500 appm/dpa) of ADS targets and first-wall materials.
6. **Parameter calibration:** the Python driver with Jupyter notebooks supports systematic parameter scans and Bayesian calibration against the EUROFER97 TEM/APT irradiation database.

D.7.3 Complementary and Combined Use

The two codes address non-overlapping regimes and can be combined in multi-scale analyses:

- A *first-wall simulation* can use Xolotl near the plasma-facing surface (implantation zone, 1–100 nm) and RadCluster for the bulk structural region ($> 10 \mu\text{m}$) where neutron damage dominates. Large-cluster concentrations from Xolotl can seed initial conditions for the RadCluster bulk run.
- Xolotl’s Fe network provides spatially resolved He-bubble distributions near grain boundaries; these supply position-dependent nucleation rates for the homogeneous RadCluster calculation.
- The power-law cascade spectrum of RadCluster —parameterized separately for fission and fusion neutrons—could be incorporated into Xolotl’s Fe network to improve its neutron cascade fidelity beyond a simple Frenkel-pair source.
- RadCluster’s solute-trapping framework could inform composition-averaged effective diffusivities for use in Xolotl simulations of reduced-activation steels.

D.8 Summary

Xolotl and RadCluster are complementary codes occupying distinct but adjacent niches in the radiation damage modeling landscape. Xolotl’s primary strength is the coupling of 1D spatial resolution, surface physics, and multi-species plasma interaction —indispensable for plasma-facing material design in current and next-generation fusion devices. RadCluster’s primary strength is its faithful representation of neutron cascade damage spectra, solute trapping in concentrated alloys, long-time bulk

microstructure evolution, and exact conservation diagnostics—essential for structural material qualification for fusion blankets and fission reactors.

The planned extensions to RadCluster—hydrogen as a fourth species with adiabatic elimination for large clusters, and a hybrid deterministic–stochastic solver for the four-species (I, V, He, H) system—will extend its applicability to accelerator-driven systems and DEMO first-wall conditions where both He and H transmutation rates are high. Used together and connected by shared physical models, the two codes span the full range of neutron and plasma environments anticipated in ITER, DEMO, and ADS structural and divertor components.

E Experimental Data of Irradiated Eurofer

E.1 Data Sources

The *size distributions* of irradiation-induced defects are among the most fundamental quantitative outputs of post-irradiation TEM characterization because they encode the competing kinetics of nucleation, growth, and coarsening that govern the macroscopic consequences—irradiation hardening, embrittlement, and dimensional instability (void swelling). Reliable size-distribution data are indispensable for the validation of cluster-dynamics rate-theory models and for the calibration of the dispersed-barrier hardening model (Ghoniem and Cho 1979a; Ghoniem et al 1985b).

We compile here and analyze all experimentally reported size-distribution data for dislocation loops and voids in F/M steels and pure iron, covering the primary alloys of interest—EUROFER97, F82H, T91, HT9, Fe-9Cr model alloys, and pure α -Fe—irradiated by neutrons, self-ions, and dual-beam ions over a wide range of doses (0.1–650 dpa) and temperatures (60–600 °C). Section E.1 documents the data sources. Section E.5 describes the phenomenology of loop size distributions. Section E.6 covers the void/cavity distributions. Section E.6 compares neutron and ion irradiation data.

Table 43 summarizes the primary experimental studies from which size-distribution data were extracted. The materials, irradiation conditions, facilities, and characterization methods are specified for each dataset.

E.2 Dislocation Loop Density and Mean Size

Temperature Dependence at Fixed Dose

The most dramatic influence on both loop density and mean size is irradiation temperature. Figure 12 illustrates the characteristic non-monotonic behavior schematically, supported by data from EUROFER97 (16.3 dpa, BOR-60) (Gaganidze et al 2011), T91 (15.4 dpa, BOR-60) (Gao et al 2018), and T91 dual-ion (35 dpa) (Taller et al 2019).

The data establish the following trends across all F/M steels studied:

1. **Density maximum near 300 °C.** Loop density reaches a maximum in the range 280–320 °C. Below this temperature, the visible-loop density is artificially suppressed because most defects remain as sub-resolution “black dots” ($d < 3$ nm). Above this temperature, thermal diffusion accelerates coarsening, and large loops are absorbed into the pre-existing network dislocation structure. EUROFER97 at 16.3 dpa shows $N_d = 1.2 \times 10^{22} \text{ m}^{-3}$ at 300 °C, dropping to $0.5 \times 10^{22} \text{ m}^{-3}$ at 415 °C (Gaganidze et al 2011).
2. **Monotonically decreasing density with temperature above 300 °C.** In T91 at 15.4 dpa in BOR-60, N_d decreases from $1.0 \times 10^{22} \text{ m}^{-3}$ at 376 °C to $\sim 0.1 \times 10^{22} \text{ m}^{-3}$ at 460 °C, where network dislocations dominate (Gao et al 2018).
3. **Monotonically increasing mean size with temperature.** In EUROFER97, $\bar{d}$ rises from 2.5 nm at 250 °C to 12.5 nm at 415 °C (Gaganidze et al 2011). In

Table 43 Primary sources of size-distribution data compiled in this study. Abbreviations: N = neutron, SI = self-ion, DI = dual-ion, WBDF = weak-beam dark-field TEM, BF/DF = bright/dark-field TEM.

Material	Irrad.	Facility	Dose (dpa)	T (°C)	Method	Reference
EUROFER97	N fast	BOR-60	15, 32	330–335	WBDF	Klimenkov et al (2012a)
EUROFER97	N fast	BOR-60	16.3	250–415	BF/DF	Gaganidze et al (2011)
T91	N fast	BOR-60	15.4, 35.1	376–524	STEM	Gao et al (2018)
HT9	N fast	BOR-60	17.1, 35.1	377	STEM	Gao et al (2019)
HT9	N fast	ATR	4.02	300–600	STEM	Yan et al (2021)
Fe-9Cr	N fission	BR2	2.0	300	WBDF	Lambrecht et al (2014)
Pure Fe	N fission	Multiple	0.19	300	BF/DF	Lambrecht and Almazouzi (2009)
Pure Fe	N fission	Multiple	0.79	60	BF/DF	Eldrup et al (1999)
Pure Fe	N fission	Reactors	1.0	400	BF/DF	English et al (1982)
HT9	SI Kr	In-situ TEM	4, 20	420–470	In-situ	Zinkle et al (2018)
Fe-9Cr	SI Fe ⁺	Ion facility	2.0	300	WBDF	Lambrecht et al (2014)
Pure Fe	SI Fe ³⁺	Ion fac.	30	300	BF/DF	Tanaka et al (2018)
T91	DI Fe+He	Dual-ion fac.	14.6–35	406–570	TEM/STEM	Taller et al (2019)
T91	DI Fe+He	Dual-ion fac.	14.6–35	406–570	TEM	Taller and Was (2020)
HT9	SI Fe ²⁺	Self-ion	140–650	440–460	STEM	Getto et al (2017)
F82H	DI Fe+He	Dual-ion fac.	20 172	470	TEM	Yamamoto et al (2023)
F82H	DI Fe+He	Dual-ion fac.	≤ 82	500	TEM	Faney and Wirth (2022)

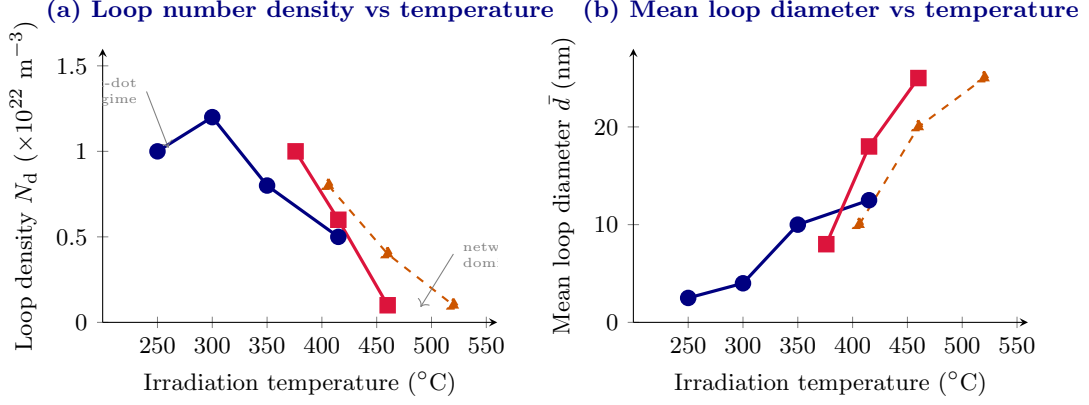


Fig. 12 Temperature dependence of (a) loop number density N_d and (b) mean loop diameter $\bar{d}$ at fixed dose. In both panels, blue circles denote EUROFER97 at 16.3 dpa (Gaganidze et al 2011), red squares denote T91 at 15.4 dpa under BOR-60 neutron irradiation (Gao et al 2018), and orange dashed triangles denote T91 under dual-ion irradiation at 35 dpa (Taller et al 2019). Both quantities show a clear monotonic trend: loop density decreases and mean size increases with increasing temperature, consistent with enhanced thermal diffusion accelerating coarsening and loop dissolution into the network dislocation structure. The peak loop density in EUROFER97 occurs at approximately 300 °C, above which coarsening dominates. The ion-irradiated T91 data are shifted by approximately +60–70 °C relative to the neutron data, as expected from the higher dose rate (Taller et al 2019).

T91, $\bar{d}$ grows from ~ 8 nm at 376 °C to > 25 nm at 460 °C (Gao et al 2018), reflecting coarsening by Ostwald ripening.

4. **Burgers-vector transition.** Below ~ 300 °C in EUROFER97, $\frac{1}{2}\langle 111 \rangle$ loops dominate. At and above 350 °C, $\langle 100 \rangle$ loops become dominant, consistent with their lower migration energy and higher thermal stability (Klimenkov et al 2012a; Gaganidze et al 2011). In T91 and HT9 under BOR-60 irradiation at 376–460 °C, $a\langle 100 \rangle$ loops dominate at all doses (Gao et al 2018, 2019).

Dose Dependence at Fixed Temperature

Figure 13 shows loop density and mean size as functions of dose at representative fixed temperatures.

The dose dependence separates into two regimes:

Accumulation regime ($< \sim 5$ dpa): Both density and mean size increase with dose. The growth of N_d follows a sub-linear power law $N_d \propto \phi^n$ with $n < 1$, reflecting simultaneous nucleation and clustering. The mean size grows as $\bar{d} \propto \phi^m$ with $m \approx 0.3$ –0.5, consistent with diffusion-limited coarsening. HT9 in-situ Kr data show the density peak at ~ 4 dpa (Zinkle et al 2018).

Coarsening/saturation regime ($\gtrsim 15$ –20 dpa): Density either saturates (EUROFER97 at 330 °C) or decreases as large loops absorb smaller ones and convert to network dislocations. Mean size continues to grow. T91 and HT9 (BOR-60) show density

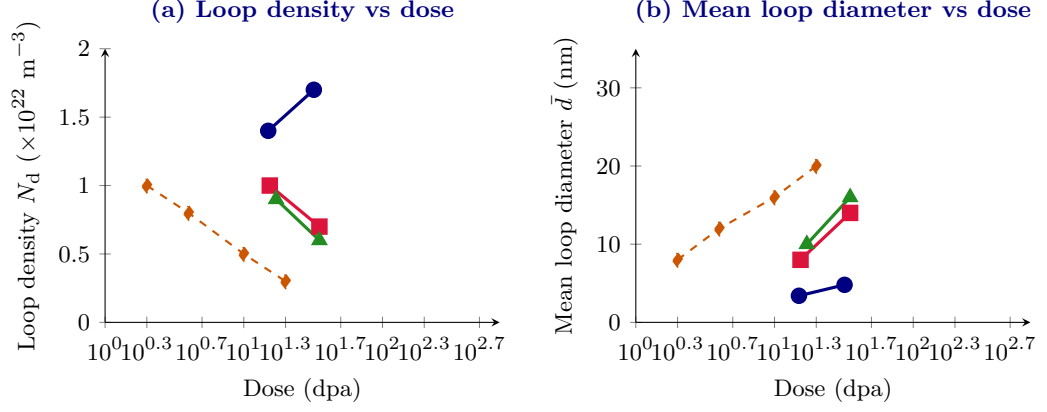


Fig. 13 Dose dependence of (a) loop number density and (b) mean loop diameter at fixed temperature. In both panels, blue circles denote EUROFER97 at approximately 330 °C under BOR-60 neutron irradiation (Klimenkov et al 2012a), red squares denote T91 at 376 °C (Gao et al 2018), green triangles denote HT9 at 377 °C (Gao et al 2019), orange dashed diamonds denote in situ Kr-ion irradiation of HT9 at 420 °C (Zinkle et al 2018), and the purple dotted pentagon in panel (a) marks the Fe-9Cr self-ion point at 2 dpa and 300 °C from Lambrecht et al. (Lambrecht et al 2014), which lies above the plotted vertical range. EUROFER97 at ~ 330 °C shows near-saturation behaviour: the density increases by only about 20% from 15 to 32 dpa, while $\bar{d}$ increases by about 1.4 nm. T91 and HT9 at 376–377 °C show coarsening: the density decreases while $\bar{d}$ grows by about 6 nm over the same dose range. The in situ Kr-ion data for HT9 at 420 °C reveal a density peak near 4 dpa followed by a monotonic decrease as loops convert to network dislocations.

decreasing from ~ 1.0 to $\sim 0.7 \times 10^{22} \text{ m}^{-3}$ and mean size growing from ~ 8 to ~ 14 nm between 15 and 35 dpa.

A key quantitative observation is the *product* $N\bar{d}$, which enters the DBH hardening model. In EUROFER97 at 330 °C, $N\bar{d}$ increases from $4.8 \times 10^{13} \text{ m}^{-2}$ at 15 dpa to $8.2 \times 10^{13} \text{ m}^{-2}$ at 32 dpa—a 70% increase in hardening potential despite only a 20% density increase, driven by mean-size growth.

Ion vs Neutron Irradiation

Self-ion irradiation of Fe-9Cr at 2 dpa/300 °C produces a loop density of $6.8 \times 10^{22} \text{ m}^{-3}$ and a mean size of 6.5 nm — a *homogeneous* field comparable to the near-dislocation values under neutron irradiation but 2–3 orders of magnitude higher than the matrix value under neutrons (Lambrecht et al 2014). The temperature must be raised by +60–70 °C for ion irradiation to match the neutron-irradiated microstructure (Taller et al 2019; Was et al 2014). The $\langle 100 \rangle$ -loop fraction is higher under ions ($\sim 40\%$) than under neutrons ($\sim 20\text{--}30\%$) at the same nominal conditions.

E.3 Void and Cavity Density and Mean Size

Bimodal Nature and the Critical Radius

Unlike dislocation loops, void/cavity microstructures in F/M steels exhibit a qualitative change in character with increasing dose: a *bimodal size distribution* develops once

the dose exceeds ~ 35 dpa at 376–420 °C. This bimodal structure consists of a high-density population of He-stabilized subcritical bubbles ($d < r_c \approx 2$ nm) coexisting with a lower-density population of supercritical growing voids ($d > 2$ –4 nm). Table 44 summarizes the key measured density and size values that delineate the transition.

Table 44 Key void/cavity density and mean-size data illustrating the unimodal-to-bimodal transition in T91 and HT9 at 376–377 °C under BOR-60 neutron irradiation.

Material	Dose (dpa)	Regime	Bubble population		Void population	
			N_b ($\times 10^{21} \text{ m}^{-3}$)	$\bar{d}_b$ (nm)	N_v ($\times 10^{20} \text{ m}^{-3}$)	$\bar{d}_v$ (nm)
T91	15.4	Unimodal	~ 10	1.2	—	—
T91	35.1	Bimodal	~ 6	1.3	~ 5	5.0
HT9	17.1	Unimodal	~ 3	1.4	—	—
HT9	35.1	Bimodal	~ 2	1.2	~ 5	7.0

References: T91 (Gao et al 2018); HT9 (Gao et al 2019).

Temperature Dependence

Figure 14 summarizes the temperature dependence of void density and swelling (which is proportional to $N_v \bar{d}_v^3$), showing the well-known bell-shaped swelling curve.

The key quantitative observations on the temperature dependence of void density and mean size are:

1. **Unimodal regime below ~ 415 °C.** In T91 and HT9 (BOR-60, 15–17dpa), only sub-2 nm bubbles are present with densities $\sim 10^{21} \text{ m}^{-3}$ and swelling $< 0.01\%$. EUROFER97 remains in this regime even at 32dpa/335 °C, with bubble density 0.11 – $0.14 \times 10^{22} \text{ m}^{-3}$ and mean size 1.6–2.6 nm (Klimenkov et al 2012a).
2. **Void density peaks in the bimodal regime.** The total cavity density (bubbles + voids) is highest near the peak-swelling temperature. At 460 °C (dual-ion T91), both bubble and void populations are most developed; swelling peaks here (Taller and Was 2020).
3. **Void mean size increases with temperature in the bimodal regime.** In dual-ion T91, the void-population mean diameter grows from ~ 5.5 nm at 406 °C to ~ 7 nm at 460 °C before the distribution collapses at higher temperatures (Taller et al 2019). In HT9 at 35.1dpa (BOR-60), the void-population mean is ~ 7 nm at 377 °C (Gao et al 2019).
4. **He/dpa ratio elevates bubble density and nucleates bimodal earlier.** Under mixed-spectrum neutron irradiation (HFR, Petten), EUROFER97 develops a higher void density than under fast-spectrum BOR-60 at the same dose and temperature, attributed to the higher helium production rate (Gaganidze et al

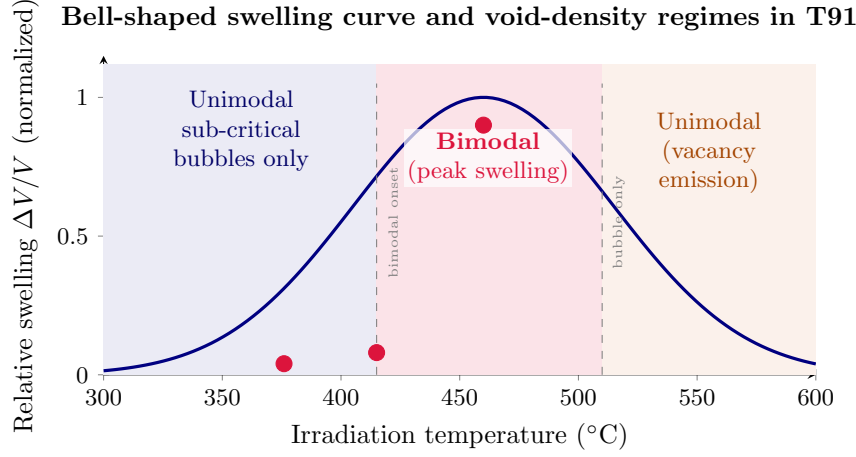


Fig. 14 Temperature dependence of void swelling in T91, showing the characteristic bell-shaped curve. The solid blue curve is a schematic representation of the relative swelling response based on dual-ion irradiation trends in T91 (Taller et al 2019; Taller and Was 2020), while the red circular markers denote representative BOR-60 neutron data points for T91 at 35 dpa (Gao et al 2018). Below approximately 415 °C, only a unimodal population of sub-critical He-stabilized bubbles is observed. Between approximately 415 and 510 °C, a bimodal distribution comprising bubbles and growing voids develops, with swelling peaking near 460 °C. Above approximately 510 °C, thermal vacancy emission dissolves the void population and the distribution collapses back to a unimodal bubble population.

2016). In dual-ion T91, higher He co-injection lowers the critical cavity radius and nucleates bimodal distributions at lower doses (Taller and Was 2020).

Dose Dependence

The dose dependence of void density and mean size differs fundamentally from that of dislocation loops. Three stages are identified:

Incubation ($\lesssim 20\text{dpa}$, $T \approx 380^\circ\text{C}$): Only sub-critical bubbles exist. Bubble density is high ($\sim 10^{21} \text{ m}^{-3}$) and mean size small ($< 2 \text{ nm}$). Swelling is $< 0.02\%$. The microstructure is effectively in the unimodal regime.

Transition and bimodal development ($\sim 20\text{--}50\text{dpa}$): A fraction of bubbles escape the subcritical bubble class and form supercritical voids. The total cavity density develops a bimodal character; swelling increases rapidly (T91: 0.01% at $15\text{dpa} \rightarrow 0.02\%$ at 35dpa ; HT9: 0.02% at $17\text{dpa} \rightarrow 0.07\%$ at 35dpa (Gao et al 2018, 2019)).

Steady-state swelling ($\gtrsim 100\text{dpa}$): Under self-ion irradiation, HT9 reaches a steady-state swelling rate of $\sim 0.2\%/ \text{dpa}$ at 460°C beyond $\sim 200\text{dpa}$ (Getto et al 2017). At 600dpa , the F/M HT9 reaches 2.6% maximum swelling (at 334 local dpa) (Kim et al 2021). Void mean diameter in this regime reaches $20\text{--}30 \text{ nm}$ and continues to grow.

Table 45 summarizes the swelling–dose relationship for the best-studied alloys.

Table 45 Summary of swelling and void size data at key dose levels in T91 and HT9.

Material	Source	Dose	T	Regime	$\bar{d}_v$ (nm)	Swelling (%)
T91	BOR-60 neutron	15.4dpa	376 °C	Unimodal	—	~0.01
T91	BOR-60 neutron	35.1dpa	376 °C	Bimodal	5.0	~0.02
HT9	BOR-60 neutron	17.1dpa	377 °C	Unimodal	—	0.02
HT9	BOR-60 neutron	35.1dpa	377 °C	Bimodal	7.0	0.07
HT9	EBR-II/FFTF	200dpa	420 °C	Steady-state	—	0.09–1.02
HT9	Self-ion (Fe^{2+})	650dpa	460 °C	Steady-state	~22	~0.2%/dpa
HT9 (F/M)	Self-ion	600dpa	450 °C	Steady-state	~30	2.6 (max)

References: BOR-60 (Gao et al 2018, 2019); EBR-II/FFTF (Garner and Toloczko 1994); self-ion (Getto et al 2017; Kim et al 2021).

Pure Iron

Pure α -iron provides a reference system free of alloying effects. At 60 °C (neutron, 0.79dpa), cavity densities exceed 10^{24} m^{-3} with sub-1 nm mean size—an extremely high density of athermal vacancy clusters (Eldrup et al 1999). At 400 °C (neutron, ~1dpa), voids develop a truncated octahedral morphology with $\{111\}$ facets and $\{100\}$ truncations; the mean size reaches ~5 nm and swelling peaks at 0.07% (English et al 1982). Above 400 °C, thermal vacancy emission suppresses void growth.

E.4 Scaling Laws and Empirical Correlations

Loop Density Saturation

The near-saturation of loop density in EUROFER97 at 330 °C between 15 and 32dpa is consistent with a balance between loop nucleation and loop-to-network conversion. If one assumes steady-state when the nucleation rate equals the network-absorption rate, the saturation density scales as (Ghoniem and Cho 1979a):

$$N_\ell^{\text{sat}} \propto \left(\frac{G_0}{\Omega_0 \mu b^3 / k_B T} \right)^{2/3} \quad (431)$$

where G_0 is the cascade defect production rate, Ω_0 is the atomic volume, and $k_B T$ is the thermal energy.

Power-Law Growth of Mean Size

In the coarsening regime, the mean loop diameter follows an empirical power law in dose:

$$\bar{d}(\phi) = \bar{d}_0 + A \phi^m, \quad m \approx 0.3\text{--}0.5, \quad (432)$$

where $\bar{d}_0$ is the as-nucleated size, A is a temperature-dependent prefactor, and m is the growth exponent. For T91 at 376 °C, fitting the two BOR-60 data points (8 nm at 15.4dpa and 14 nm at 35.1dpa) gives $m \approx 0.57$ and $A \approx 2.8 \text{ nm dpa}^{-0.57}$. For HT9 at 377 °C (10 nm at 17.1dpa and 16 nm at 35.1dpa), $m \approx 0.52$ and $A \approx 2.3 \text{ nm dpa}^{-0.52}$.

Swelling Incubation and Steady State

The swelling data for HT9 suggest a two-stage model:

$$S(\phi) = S_0 \left[\exp\left(-\frac{\phi_c}{\phi}\right) - 1 \right] + \dot{S}_{ss} (\phi - \phi_c), \quad \phi > \phi_c, \quad (433)$$

where $\phi_c \approx 20\text{--}30\text{dpa}$ is the incubation dose, $\dot{S}_{ss} \approx 0.2\%/ \text{dpa}$ is the steady-state swelling rate, and S_0 is a small transient offset. This is consistent with the measured values: 0.02% at 17dpa (before bimodal), 0.07% at 35dpa (bimodal onset), and $\sim 2.6\%$ at 334 local dpa (steady state) (Gao et al 2019; Kim et al 2021).

Summary of Quantitative Trends

Table 46 condenses the principal quantitative trends.

Table 46 Summary of quantitative trends in loop and void density and mean size as functions of temperature and dose in ferritic–martensitic steels. The symbols $\uparrow$ and $\downarrow$ denote increase and decrease, respectively.

Quantity	With temperature T at fixed dose		With dose ϕ at fixed T	
	$T < 300\text{ }^\circ\text{C}$	$T > 300\text{ }^\circ\text{C}$	$\phi < 5\text{ dpa}$	$\phi > 15\text{ dpa}$
Loop density N_d	$\uparrow$; black-dot regime	$\downarrow$	$\uparrow$	$\downarrow$ or saturation
Loop mean size $\bar{d}_\ell$	$\uparrow$; small loops	$\uparrow\uparrow$	$\uparrow$	$\uparrow$; coarsening
Void density N_v	Low; unimodal bubbles	$\uparrow$ then $\downarrow$	Low	$\uparrow$; bimodal regime
Void mean size $\bar{d}_v$	Very small, $< 2\text{ nm}$	$\uparrow\uparrow$	$< 2\text{ nm}$	$\uparrow\uparrow$
Swelling S	≈ 0	Bell-shaped; $\uparrow$ then $\downarrow$	≈ 0	$\uparrow$

The following quantitative anchors summarize the data:

- Loop density maximum: $\sim 1.2 \times 10^{22}\text{ m}^{-3}$ near $300\text{ }^\circ\text{C}$; decreases to $< 0.1 \times 10^{22}\text{ m}^{-3}$ at $460\text{ }^\circ\text{C}$ (T91).
- Loop mean size range: 2.5 nm (EUROFER97, $250\text{ }^\circ\text{C}$) to $> 25\text{ nm}$ (T91, $460\text{ }^\circ\text{C}$).
- Void density (unimodal regime): $\sim 10^{21}\text{ m}^{-3}$, mean $1.2\text{--}2.6\text{ nm}$.
- Void density (bimodal regime): bubble population $\sim 2\text{--}6 \times 10^{21}\text{ m}^{-3}$, mean $1.2\text{--}1.4\text{ nm}$; void population $\sim 5 \times 10^{20}\text{ m}^{-3}$, mean $5\text{--}7\text{ nm}$.
- Swelling: 0.02% at $15\text{--}17\text{dpa}$, 0.07% at 35dpa ; $\sim 2.6\%$ at 600dpa (HT9).

E.5 Dislocation Loop Size Distributions

Dislocation loops in bcc iron-based alloys form with two predominant Burgers vectors: $\mathbf{b} = \frac{1}{2}\langle 111 \rangle$ (glissile, $a/2$ type) and $\mathbf{b} = \langle 100 \rangle$ (sessile, a type). The relative proportion shifts with temperature and alloy composition: $\frac{1}{2}\langle 111 \rangle$ loops dominate below $\sim 300^\circ\text{C}$ in EUROFER97, while $\langle 100 \rangle$ loops become dominant at and above 350°C (Klimenkov et al 2012a; Gaganidze et al 2011). In T91 and HT9 under BOR-60 irradiation at $376\text{--}420^\circ\text{C}$, $a\langle 100 \rangle$ loops are dominant at all doses studied (Gao et al 2018, 2019).

The most dramatic variable controlling the loop size distribution is irradiation temperature. Figure 15 illustrates the progressive evolution schematically.

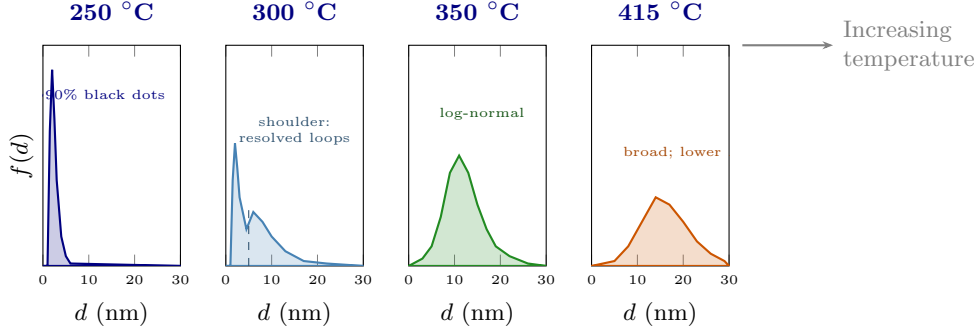


Fig. 15 Schematic representation of the temperature-dependent evolution of dislocation loop size distributions in EUROFER97 (16.3 dpa, BOR-60 neutron irradiation). At 250°C , the distribution is dominated by a narrow spike of sub-3 nm black-dot defects ($\sim 90\%$). At 300°C , a resolved-loop shoulder develops at 5–15 nm while the black-dot peak persists. Above 350°C , the distribution becomes a well-developed log-normal centred at 10–13 nm. At 415°C , the distribution broadens further and the total density decreases as large loops begin transforming into network dislocations.

For a fixed irradiation temperature, increasing the dose shifts the distribution rightward and narrows the relative spread. Figure 16 illustrates this for EUROFER97 (BOR-60) and HT9 (BOR-60).

Key experimental observations on the dose dependence are:

- In EUROFER97 at $330\text{--}335^\circ\text{C}$, the mean diameter increases only from 3.4 nm (15 dpa) to 4.8 nm (32 dpa), while the number density increases from $1.4 \times 10^{22} \text{ m}^{-3}$ to $1.7 \times 10^{22} \text{ m}^{-3}$ — an almost saturated state after 15 dpa (Klimenkov et al 2012a).
- In T91 at 376°C , the mean diameter grows from ~ 8 nm (15.4 dpa) to ~ 14 nm (35.1 dpa), while density *decreases*, indicating coarsening by the absorption of smaller loops into larger ones and into network dislocations (Gao et al 2018).
- In HT9 at 377°C , the mean diameter grows from ~ 10 nm (17.1 dpa) to ~ 16 nm (35.1 dpa) with a commensurate density decrease (Gao et al 2019).
- At very high doses under self-ion irradiation (>100 dpa), loop density first peaks around 4 dpa and then decreases monotonically as loops are absorbed into the network dislocation structure (Zinkle et al 2018).

Loop Size Distribution Evolution with Dose — HT9 at 377°C (BOR-60)

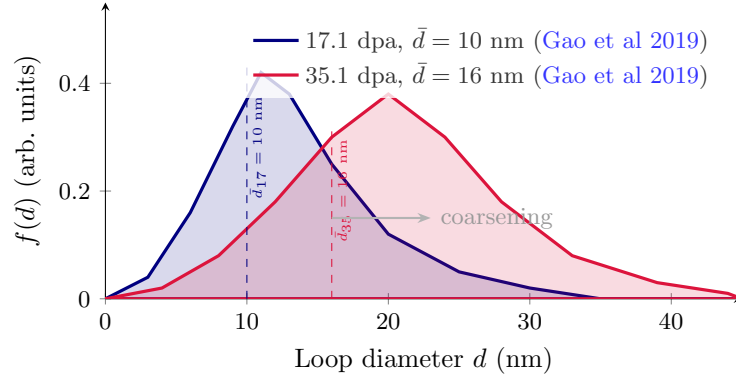


Fig. 16 Schematic dose evolution of the dislocation loop size distribution in HT9 at 377 °C (BOR-60 neutron irradiation). The distribution shifts rightward and the peak height decreases, indicative of coarsening: mean diameter increases from ~ 10 nm at 17.1 dpa to ~ 16 nm at 35.1 dpa while total loop number density decreases (Gao et al 2019).

Self-ion irradiation at the same nominal dose and temperature as neutron irradiation produces dislocation loop distributions that differ in two important respects:

1. The distribution is *more symmetric* and *more homogeneous spatially* —neutron irradiation produces a strongly heterogeneous distribution with high loop densities near pre-existing dislocation lines ($\sim 10^{23} \text{ m}^{-3}$) and sparse populations in the matrix ($\sim 10^{20} \text{ m}^{-3}$), while self-ion irradiation produces a uniform loop field at intermediate density (Lambrecht et al 2014).
2. The fraction of sessile $\langle 100 \rangle$ loops is larger under self-ion irradiation ($\sim 40\%$) than under neutron irradiation for the same Fe-9Cr alloy at 300 °C (Lambrecht et al 2014).

The temperature must be raised by +60–70 °C for ion irradiation to reproduce the same microstructure as neutron irradiation, due to the ~ 3 orders of magnitude higher dose rate (Taller et al 2019; Was et al 2014).

E.6 Void and Cavity Size Distributions

At doses below ~ 17 – 20 dpa in T91 and HT9 (BOR-60, 376–377 °C), and at all doses in EUROFER97 at 330–340 °C, cavity distributions are *unimodal* and confined to sub-2 nm diameters. These cavities are helium-stabilized bubbles that remain below the critical radius for spontaneous void growth. The distributions are narrow and right-skewed, with mean diameters of 1.0–2.6 nm and number densities of 10^{20} – 10^{21} m^{-3} (Figure ??).

The most distinctive feature of cavity microstructure evolution in F/M steels is the development of a **bimodal size distribution** once cavities exceed the critical radius

for void growth. This transition is observed at ~ 35 dpa in both T91 and HT9 under BOR-60 neutron irradiation at 376–377 °C (Gao et al 2018, 2019), and is reproduced across a wide range of parameters under dual-ion irradiation of T91 (Taller and Was 2020).

The bimodal distribution consists of:

- A high-density **small-bubble population** ($d < 2$ nm, density $\sim 10^{21} \text{ m}^{-3}$) of He-stabilized, sub-critical cavities.
- A lower-density **growing void population** ($d > 2$ nm, density $\sim 10^{19}\text{--}10^{20} \text{ m}^{-3}$) of super-critical cavities that have escaped He stabilization and grow by net vacancy absorption.

The two populations are separated by a valley near the critical radius $r_c \approx 1$ nm. Despite its lower density, the void population dominates swelling because swelling scales as $\Delta V/V \propto \sum N_i d_i^3$.

Figure 17 illustrates the transition from unimodal to bimodal and the temperature evolution of the bimodal distribution.

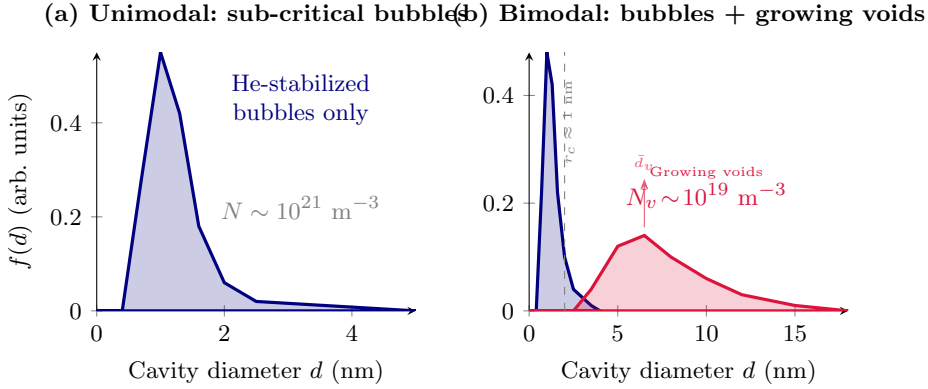


Fig. 17 Schematic comparison of (a) the unimodal cavity size distribution characteristic of low-dose or low-temperature conditions in F/M steels, where only He-stabilized sub-critical bubbles exist, and (b) the *bimodal* distribution that develops in T91 and HT9 beyond ~ 35 dpa at 376–377 °C, where a high-density population of small bubbles coexists with a lower-density population of growing super-critical voids. The critical radius $r_c \approx 1$ nm separates the two populations.

Comparison: Neutron vs. Ion Irradiation

Table 47 summarizes the key differences and agreements between loop and void size distributions measured under neutron and ion irradiation.

E.7 Implications for Cluster-Dynamics Modeling

The experimental size distributions described above impose several quantitative constraints on cluster-dynamics (CD) rate-equation models:

Table 47 Comparison of defect size distributions under neutron and ion irradiation in ferritic–martensitic (F/M) steels.

Observable	Neutron irradiation	Ion irradiation
Loop mean diameter	5–20 nm depending on temperature; heterogeneous spatial distribution	6–7 nm at similar temperature under self-ion irradiation; more uniform spatial distribution
Loop distribution shape	Right-skewed log-normal distribution; black-dot spike at low temperature	More symmetric and narrower distribution; fewer black dots
$\langle 100 \rangle$ loop fraction	$\sim 20\text{--}30\%$ at 300 °C	$\sim 40\%$ at 300 °C
Spatial heterogeneity	Strong: decoration of pre-existing dislocation lines dominates; density varies by three orders of magnitude	Weak: nearly homogeneous field; no significant dislocation-decoration effect
Void/cavity regime	Bimodal at >35 dpa and 376–420 °C	Bimodal across the investigated parameters at $T \approx 460$ °C
Temperature shift	—	+60–70 °C required to match neutron microstructure
G-phase / α'	Well developed under neutron irradiation at equivalent conditions	Suppressed under ion irradiation: less segregation and fewer Ni/Si clusters

1. **Log-normal loop distribution at steady state.** The distribution shape implies that growth and coarsening—not nucleation—control the loop microstructure above ~ 300 °C, favouring a Lifshitz–Slyozov–Wagner (LSW) coarsening mechanism or an Ostwald-ripening correction to the cluster-dynamics equations.
2. **Bimodal void distribution.** The bimodal form is a direct diagnostic of the bubble-to-void transition. A correct CD model must predict a *valley* in the cavity size distribution near $r_c \approx 1$ nm. This places requirements on the accuracy with which the model captures the critical cluster free energy including He-pressure effects.
3. **Nearly saturated loop density in EUROFER97.** The near-constant total loop density between 15 and 32 dpa in EUROFER97 at 330–340 °C ([Klimenkov et al 2012a](#)) constrains the sink-strength balance: it requires that loop nucleation and the rate of loop loss to network dislocations are nearly in equilibrium.
4. **Dose-rate sensitivity (ion vs. neutron).** The narrower, more homogeneous loop distribution under ion irradiation at higher dose rate requires that the CD model correctly implements recombination-dominant kinetics, which suppress spatial heterogeneity at high dose rates.

E.8 Summary

The principal conclusions from the experimental size-distribution database are:

- Loop distributions are right-skewed log-normals that narrow and shift to larger mean diameter with increasing temperature (250–415 °C) and dose (coarsening regime, >15 dpa in BOR-60 conditions).
- At low temperature (<300 °C), the distribution is dominated by a sharp spike of sub-3 nm black-dot defects; resolved loops form a secondary shoulder.
- Void distributions are *unimodal and sub-2 nm* at low doses or temperatures, transitioning to a *robust bimodal form* (small He-bubbles + growing voids) at ~35 dpa and 376–460 °C in T91 and HT9.
- The bimodal distribution is a fundamental feature of F/M steels in the peak-swelling temperature window and is reproduced across a wide range of dual-ion irradiation parameters.
- Ion irradiation produces narrower, more homogeneous loop distributions and requires a +60–70 °C temperature shift to match neutron-irradiated microstructures.

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